

Cambridge International AS Level
Information Technology 9626

Paper 1 Theory

Topical past papers

338 questions from 32 question papers, Feb/March 2022 to May/June 2026, sorted into the 11 AS Level topics.

Each topic has its questions first, in date order, then their mark schemes in the same order. Every question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, use the contents page to jump to a topic, or use the bookmarks panel.

A question that covers more than one topic appears under each of them. Topics follow the 2025-2027 syllabus.

Contents

1	Data processing and information	Questions p. 3	Mark schemes p. 86
2	Hardware and software	Questions p. 152	Mark schemes p. 204
3	Monitoring and control	Questions p. 248	Mark schemes p. 285
4	Algorithms and flowcharts	Questions p. 314	Mark schemes p. 350
5	eSecurity	Questions p. 379	Mark schemes p. 408
6	The digital divide	Questions p. 432	Mark schemes p. 447
7	Expert systems	Questions p. 457	Mark schemes p. 477
8	Spreadsheets	Questions p. 492	Mark schemes p. 505
9	Modelling	Questions p. 518	Mark schemes p. 537
10	Database and file concepts	Questions p. 554	Mark schemes p. 588
11	Video and audio editing	Questions p. 611	Mark schemes p. 624

Questions

Mark scheme

1989, Paolo, $\sqrt{\quad}$, ₹, $\neq$, OR

[4]

- 1
- 2
- 3

3

- 6 A school secretary is updating a database by copying data from some hard copy records. Here are three of the records:

First_name	Family_name	Student_number	Previous_school	Date_of_birth
Jonathon	Odoki	12036	St Mary's	07/12/2004
Sven	Johansson	09132	The Grange	13/10/2007
Manjit	Sangherra	10102	Bluestars	11/03/2007

The Student_number will be stored as text, with the first two characters representing the year group the student belongs to.

The Student_number and Date_of_birth are the only data that need to be validated as they are entered. A range check, limit check or presence check would not be appropriate for validating these data.

- (a) Describe, in precise detail, **three** appropriate validation checks that could be used to validate the data. Each validation check must be different.

1

.....

.....

.....

.....

.....

2

.....

.....

.....

.....

.....

3

.....

.....

.....

.....

1 The processing of customer orders is an example of batch processing.

Describe, using customer orders as an example, what is meant by batch processing.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

3 When personal data is sent across the internet, it is often encrypted.

Analyse the encryption of personal data when it is sent across the internet.

[8]

- 4 A supermarket uses a computer to automatically update stock levels. Each record in the **Product file** contains the fields shown.

Product file

Bar_code	Product	Number_in_stock	Reorder_level	Reorder_quantity	Supplier_ID
5022476130067	Oaters	131	130	250	B671841
0100108097858	Tea	182	130	310	A296593
0201570244453	Beans	317	212	450	C562148

When the Number_in_stock of a product item falls to the Reorder_level, that product must be reordered. The Reorder_quantity is the number of items of the product that must be reordered.

The **Supplier file** contains the contact details of all the suppliers.

Supplier file

Supplier_ID	Supplier	Contact
B671841	J Sullivan	js@cmail.co.ke
A296593	Bluants	bluants@hmail.ug
C562148	Golan	golan@smail.eg

Only some of the records in the Product and Supplier files are shown.

[7]

- 8 Ruritania has just passed a law which will allow supermarkets to purchase customer information from banks, providing they anonymise it. Joanne, the owner of a supermarket, wishes to analyse the spending habits of people in the area very near to the supermarket. She can either use direct data sources or obtain the information indirectly from local banks.

Discuss the advantages and disadvantages to Joanne of obtaining the information indirectly from local banks.

[8]

- 1 Some students have been asked by their teacher to carry out a study. They will collect data about the families living in five streets close to their school. They will carry out the study using direct data sources and will not use records from the 2020 census which are kept at the city hall.

Describe the suitability of using direct data sources rather than the census data for the study.

[5]

- 2** Encryption has many uses, including the encryption of hard disks.

Describe hard disk encryption.

[4]

- 3 Julia is setting up a hairdressing business. She has recruited several staff. She is creating a database of all her staff and has written down all their details ready for entering into a computer. She will need to verify the data when it is entered.

She has been told that she cannot use a parity check but should use visual verification and double data entry.

- (a) By describing what is meant by a parity check, explain why it might not be suitable in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) Compare the similarities and differences between the visual checking and double data entry methods of verification.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9 Many companies use computers to process their payroll.

A transaction file is produced once a week. It consists of the WorkerID and HoursWorked fields. The master file contains the WorkerID, RateOfPay and WagesSoFar fields as well as other details. The WagesSoFar field represents how much each worker has earned so far this year.

Each worker's wage is calculated by multiplying the HoursWorked by the RateOfPay. When the wage for that week is calculated, it is added on to the WagesSoFar field.

Every week the master file has to be updated to include the new wage earned so far this year.

(a) Complete the following pseudocode algorithm which shows the process of updating the master file. You can assume no other transactions are being carried out.

```

READ first record in transaction file
READ first record in old master file
WHILE not end of transaction file
    IF transaction file WorkerID = master file WorkerID
        THEN

```

[6]

1 Paulo is carrying out a study on the level of noise caused by traffic in the streets around his home. He will need data for his study.

(a) Describe what is meant by direct data.

.....
..... [1]

(b) Describe how Paulo could use direct data sources for his study.

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

6 Describe how the SSL/TLS protocol is used in client-server networks.

[6]

- 3 Most companies use transaction files to update master files in their payroll systems. One part of the updating process is to calculate weekly or monthly pay. Another part is to update workers' personal and work details.

Without referring to the calculation of pay, describe the three types of transaction that would take place when workers' details are being updated. You must give an example for each type of transaction.

Transaction type 1:

.....

Example 1:

.....

Transaction type 2:

.....

Example 2:

.....

Transaction type 3:

.....

Example 3:

.....

[6]

- 4 A customer at a supermarket checkout pays for their goods by inserting a card into a chip reader and entering their personal identification number (PIN). Electronic funds transfer takes place.

Describe what happens during the electronic funds transfer.

[6]

- 5** Encryption is a method used by organisations to keep data confidential. Symmetric or asymmetric encryption is used.

Explain what is meant by asymmetric encryption.

..... [5]

- 11 Explain what is meant by an indirect data source. In your explanation include descriptions of **two** different indirect data sources.

[4]

1 Here is a set of data.

>
<
=
>=
<=

Explain why this is just a set of data and describe how this particular set of data would become information.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 5 The owner of a book shop has created a database of her books. She will now test that it works. Some of the records are shown. The data shown contains the maximum and minimum values. There are only two types of book.

ISBN	Title	Author	Book_type	Copies
9781787632196	Blue Moon	Lee Child	H	4
9781858683416	Prelude to Foundation	Isaac Asimov	P	5
9780593072493	Inferno	Dan Brown	H	3
9781787461352	Revenge	James Patterson	P	2
9781529027488	One Good Deed	David Baldacci	H	5

- (a) Describe, in detail, the **three** most appropriate validation checks, apart from a check digit, type check or presence check, that could be used to validate the data. For each validation check, your description should include the field it could be used for. Each validation check and each field must be different.

1

.....

.....

.....

.....

.....

2

.....

.....

.....

.....

.....

3

.....

.....

.....

.....

.....

[6]

(b) The owner will also need to verify the data in her database.

Using examples from the three fields you used in part **(a)**, explain why verification is needed in addition to validation.

.....

.....

.....

.....

.....

.....

.....

..... [3]

8 Encryption is often used to protect data.

(a) Discuss the need for encryption.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Discuss the advantages and disadvantages of using symmetric encryption.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 1 Before companies make planning decisions, they need to collect a lot of data. The decisions they make will depend on the quality of the resulting information.

Apart from its age and its level of detail, describe the other factors which affect the quality of information.

[6]

- 2** A microprocessor controlled air conditioning system uses real-time processing.

Explain what is meant by real-time processing in this scenario.

[5]

- 1 Data is collected and processed to become information. Many factors affect the quality of the resulting information.

(a) Describe what is meant by accurate information.

..... [1]

(b) Describe how the use of interviews and questionnaires for collecting data may affect the accuracy of the resulting information.

[6]

5 Explain why encryption is needed.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

1 Encryption can be used to protect data.

(a) Define encryption.

.....

.....

..... [1]

(b) Explain how intercepted data could be misused if it was not encrypted.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

5 D'Oliveira Inc. is a company which sells stationery products to many stores. The company uses a transaction file to keep a record of the number of products it has sold each month.

Part of a transaction file is shown. It contains:

- ProductID
- NumberSold (the number of that product sold in a particular month)

Part of the master file used by the company is also shown. It contains:

- ProductID
- Product (description of product)
- NumberinStock (how many of that product are left in stock)

The NumberinStock on the new master file is calculated every month by subtracting the NumberSold from the NumberinStock on the old master file.

Transaction file		Master File		
ProductID	NumberSold	ProductID	Product	NumberinStock
2210	6500	1965	Folder wallets	9000
3356	7300	2210	Ring binders	12000
4268	2900	2976	Notepads	10100
5690	4600	3356	A4 paper	13450
		4268	A3 paper	8800
		5690	HB pencils	9200
		6528	Pencil cases	11150
		7649	Maths sets	8240

Fig. 5.1

(a) Identify the **two** processes that would have been performed on the transaction file before the master file could be updated.

.....

..... [1]

- (b)** List the steps involved when the master file is updated using the transaction file. Use the data shown in Fig. 5.1. Stop at the step where the third master file record is read.

[8]

- 11 Isabella is concerned about the amount of traffic in her local area. She is going to interview all the nearby families to discover if the amount of traffic is caused mainly by local families driving short distances. It will take her a week to collect the data.

Justify the use of direct sources such as interviews in this scenario.

[8]

- 1 A student has been asked by her sociology teacher to produce a study of the average number of family members in the local area.
- (a) Give reasons why using an Electoral Register as an indirect data source would not produce the data required by the study.

..... [5]

- (b)** Apart from the Electoral Register, give an example of another indirect data source.

..... [1]

2 Explain what is meant by symmetric encryption.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- 3** Many credit card purchases use online processing, whereas customer bills tend to be produced using batch processing.

Describe the differences between online processing and batch processing systems.

..... [5]

- 1 Describe what is involved in data logging.

[4]

2 Explain the purpose of the SSL/TLS protocol.

[5]

- 3 There are different types of processing including batch processing and real-time processing.
- (a) Without referring to master files and transaction files, describe what is meant by batch processing.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (b) Rocket guidance systems are used in space exploration.
- Explain why a rocket guidance system is an example of the use of real-time processing.

.....

.....

.....

.....

.....

.....

.....

..... [3]

2 Two methods of checking the accuracy of data are validation and verification.

Describe the differences between validation and verification.

[3]

4 Explain how HTTPS websites provide security of data.

[5]

- 11 A company receives orders which are entered into a computer system. The company has decided to use online processing to process these orders so employees enter orders as they are received.

Discuss the advantages and disadvantages of using online processing compared to using batch processing in this scenario.

[8]

- 1 A company secretary is creating a new database of workers to record:
 - their six digit ID number
 - the date they joined the company
 - their rate of pay per hour up to a maximum rate of \$50 per hour.

She has entered data from a source document into the database. There was an error in record 3 and an error in record 4 in the source document. The secretary also made an error when copying record 1.

Source document				Database		
Record	ID number	Date joined	Rate of pay	ID_number	Date_joined	Rate_of_pay
1	147563	13/06/2019	35	147563	13/06/2020	35
2	163459	09/10/2019	28	163459	09/10/2019	28
3	1827697	15/08/2020	31	1827697	15/08/2020	31
4	221524	23/12/2020	56	221524	23/12/2020	56
5	256817	27/07/2021	15	256817	27/07/2021	15

Using this data, explain, in detail, the need for verification and validation when entering data into a database.

[6]

- 5 Maria wants to use information about the families in her town for a study. She will get the information from an indirect data source, the town’s electoral register. This is kept at the local council office. She has decided **not** to use a questionnaire, which is a direct data source, to gather the information she needs.

Describe the suitability of using the electoral register rather than a questionnaire.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

11 When data is sent across the internet, it is usually encrypted.

Discuss the benefits and drawbacks of encrypting data which is transmitted over the internet.

[8]

- 1 Using a numerical example, explain the differences between data and information.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 4 A bank stores a database of all its customers. The first field in the database is the account number. The account numbers are different for each customer and consist of 10 digits, for example 7382690643.

New accounts are opened in branches of the bank on a regular daily basis.

- (a) Describe **three** errors that a person entering the account numbers of new customers might make. For each one describe an appropriate validation check that would trap the error. A range check or limit check would **not** be suitable. Each validation check must be different.

Error 1

.....

Validation check 1

.....

.....

Error 2

.....

Validation check 2

.....

.....

Error 3

.....

Validation check 3

.....

.....

[6]

When customers open new accounts, the details are gathered together in a transaction file. At the end of the day a branch transmits the transaction file to head office, using the internet.

(b) The transmission of the transaction file is verified using a hash total.

Describe this process, explaining why a hash total is used rather than a control total.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

4 An airline company uses online processing to allow passengers to book their seats.

(a) Describe how online processing is used in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe **two** benefits for passengers of using online processing when booking seats.

Benefit 1

.....

.....

.....

Benefit 2

.....

.....

.....

[2]

- 8 A secretary of a company has been asked to transfer all the staff records from the current paper-based system to a computer database. The secretary will use validation and verification to check the data.

Explain why both validation and verification are needed in this scenario.

[6]

- 9** Encryption can be used to protect data sent in an email.

Evaluate the use of encryption for this purpose.

[7]

1 (a) Describe, giving an example, what is meant by the term data.

Description.....

.....

.....

Example.....

.....

.....

[2]

(b) Give an example of information and explain why your chosen example is considered to be information.

.....

.....

.....

.....

.....

.....

[2]

2 Identify **four** factors which can affect the quality of information.

1

2

3

4

[4]

3 Describe the computer processing involved in an automatic stock control system.

[7]

10 Explain how each of these methods keeps data safe from unauthorised users.

(a) encryption

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) a firewall on a network

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 4 Data can be kept secure by using the Secure Socket Layer (SSL)/Transport Layer Security (TLS) protocols when it is transmitted over the internet.

Describe the use of SSL/TLS to ensure data is kept secure.

[8]

5 There are various methods of verification.

(a) Describe why checksum verification is used.

.....

.....

.....

.....

.....

..... [2]

(b) Describe a method of calculating a checksum.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

7 Data can be collected by using a direct or an indirect data source.

Discuss the advantages and disadvantages of using direct data compared to indirect data.

[8]

5 A college uses a database to store student records. Each student has a unique ten-digit student number.

The database stores the examination scores of each student. The maximum mark in each examination is 100.

The database already has presence, existence and integrity validation checks built in.

Describe **three** additional validation rules, using student number and examination scores as examples. For each rule, give a different reason why it can still allow inaccurate data to be entered.

Rule 1

.....

Reason 1

.....

.....

.....

Rule 2

.....

Reason 2

.....

.....

.....

Rule 3

.....

Reason 3

.....

.....

.....

[6]

5 A scientist is going to use direct and indirect data sources to study the effects pollution is having on farmland.

(a) Explain what is meant by direct data sources. Your answer must include examples linked to the scenario.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Explain what is meant by indirect data sources. Your answer must include examples linked to the scenario.

.....

.....

.....

.....

.....

.....

.....

6 Explain the need for both validation and verification to ensure data is accurate.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

2 The quality of information can affect the usefulness of decisions made by businesses.

Describe how the following factors can affect the quality of information and the quality of the decisions made.

(a) age of information

.....

.....

.....

.....

.....

..... [2]

(b) relevance of information

.....

.....

.....

.....

.....

.....

.....

..... [3]

8 A furniture company uses databases in its warehouse to help with stock control.

(a) The company has a Product database which contains the fields shown in the table.

Field Name	Data Type	Example of Data
Number_in_Stock		199 (Number_in_Stock can vary between a minimum of 0 and a maximum of 250.)
Location		07.44.03
Reorder_Level		250
Price		\$203.50
Supplier_Telephone_No.		07700900819

Complete the Data Type column in the table by writing the most appropriate data type for each field. For any numeric fields, specify the numeric data type. [5]

- [6]

- [5]

9 A reporter will use telephone interviews to gather information for an article.

Discuss the benefits and drawbacks of using telephone interviews.

[8]

4 Validation and verification can be used to check the accuracy of data.

(a) Describe the differences between validation and verification. Your answer must explain the purpose of each.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe each of the following:

(i) range check

.....

.....

.....

..... [2]

(ii) double data entry

.....

.....

.....

..... [2]

5 Describe the use of a public key and a private key in asymmetric encryption.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 Justify the use of online processing in stock control.

[8]

- 1 A college secretary has been given some data from a member of staff.

Chandra, Amruta, YR12538, 59, 75, 68, 04/07/2008

Explain, with examples, how these items of data can become information.

[5]

- 2** Staff at a historical building are producing leaflets so that visitors do not have to be accompanied by a guide.

Visual checking is used to verify that the data is the same as the source document.

Describe, by using an example, how visual checking could still result in errors in the leaflet.

..... [2]

- 5 Teachers at a school store records of their students and their examination scores on a computer system.
- Each examination is marked out of 120
 - Each student at the school has a unique student number, such as L35725 or Z83499

The system already uses presence checks on the student number and examination scores.

Justify additional validation checks that could be used on the:

- student number
- examination scores.

[8]

- 8 Describe the use of Internet Protocol Security (IPsec).

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 1 A teacher is collecting information from her students about what items they recycle.

Describe **two** factors that can affect the quality of the information gathered in this scenario.

Factor 1

.....

.....

.....

Factor 2

.....

.....

..... [2]

- 8 A customer pays for their shopping at a checkout by using their bank debit card. Funds are transferred electronically.

Describe the process of electronic transfer of funds.

[6]

1 A company is going to carry out an investigation into shopping habits. The company will get data from indirect data sources.

(a) Describe the benefits of getting data from indirect data sources.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe the drawbacks of getting data from indirect data sources.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 11 A college database contains details about students. Part of the database is shown.

StudentID	FirstName	FamilyName	PhoneNumber
AB37689	Mohammed	Abbas	016324908367
AH83965	Jonathan	Olsen	016328332901
BV72189	Suzanne	Usher	016325599100
DY64817	Hua	Owyang	016323652331
KT52964	Isabella	Mancini	016328899477
PS54200	Ben	Quansah	016329677901

Identify and describe **two** validation checks that are suitable to use on the StudentID field.

Validation check 1

Description 1

.....

.....

.....

Validation check 2

Description 2

.....

.....

.....

[4]

- 13** Discuss the benefits and drawbacks of using encryption when transferring data between web browsers and web servers.

[8]

1 Observation is one method of collecting direct data.

Identify and describe **two** other ways of collecting direct data.

Way 1

Description

.....

.....

.....

Way 2

Description

.....

.....

.....

[4]

2 Describe the use of TLS/SSL.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[5]

1 (a) Describe what is meant by data.

.....

.....

..... [1]

(b) Describe what is meant by information.

.....

.....

..... [1]

2 A hospital nurse records direct data about her patients every day.

(a) Describe the benefits of using direct data in this context.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe the drawbacks of using direct data in this context.

.....

.....

.....

.....

.....

.....

.....

..... [3]

10 A space rocket uses a guidance system.

Describe how this guidance system uses real-time processing.

[6]

1 A train timetable is published every six months.

State **three** factors that could affect the quality of this information.

1
2
3 [3]

2 Describe how a control total is used to verify data.

.....

.....

.....

.....

.....

.....

.....

..... [3]

7 A company wishes to gather some information from employees.

(a) Describe the process of organising face-to-face interviews.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Justify the use of a relational database to store the information gathered.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9 Justify the uses of encryption for data that is stored on a pen drive.

[6]

2 47, 22 and PXR5G are items of data.

Explain why this data is **not** information.

..... [1]

10 A list of some field names from a database is shown. For each field name, identify the most appropriate data type.

(a)

Field name	Most appropriate data type
employee_name	
hourly_rate_of_pay	
was_employee_born_in_france?	

[3]

(b) A presence check has already been run on each field of the database.

For each of the following fields, identify and describe **one** other type of validation that can be used. You should only use each type of validation once.

(i) hourly_rate_of_pay
.....
.....
.....
..... [2]

(ii) date_of_birth
.....
.....
.....
..... [2]

(c) Justify why a field holding the age of each employee has been set as a calculated field.

.....
.....
.....
.....
.....
.....
..... [3]

1 Data may be described as direct or indirect.

(a) Describe **two** differences between direct and indirect data.

.....

.....

.....

.....

.....

..... [2]

(b) Identify and explain **one** circumstance in which it would be better to use indirect data for a project.

.....

.....

.....

.....

.....

..... [2]

- 1 A businesswoman is opening a shop which will sell sports goods. She has decided to research into the spending habits of potential customers.
- (a) (i) Describe **one** source of direct data she could use for this purpose.
-
-
- [1]
- (ii) Identify **one** source of indirect data that she could use for this purpose.
-
-
- [1]
- (b) With reference to **both** relevance of data and age of data, explain why the information obtained from the direct data source may produce better quality information.
-
-
-
-
-
-
-
-
-
- [4]

The businesswoman has decided to use indirect data for her research into the spending habits of potential customers.

- (c) Identify and explain **two** possible reasons, other than relevance of data and age of the data, for her decision to use indirect data for this purpose.

reason 1

.....

.....

.....

.....

reason 2

.....

.....

.....

.....

[4]

- 4 In a school, a student needs to include sections from books in an essay he is writing.

Each section must be an exact copy of the section in the source material.

- (a) Compare the use of visual checking and double data entry to verify the sections used. You must include similarities and differences in your answer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

In a workplace, a file containing employee details, including employee ID numbers, is going to be transferred by File Transfer Protocol (FTP) to a manager's computer. The integrity of the file has to be checked after it has been downloaded.

- (b) Identify and describe **one** method of verification, other than visual checking and double data entry, that can be used to check the integrity of the file downloaded to the manager's computer.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 5 A school has paper-based records of all students that have attended the school since 1993.

These records include:

- home addresses and contact details
- the grade each student achieved in each of their end of year exams
- the years when they joined and left the school.

The school is going to create and use a single database to store these records.

- (a) Compare the use of flat file database structures and relational database structures to store these records.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

The database will include validation rules.

- (b) Identify and describe **one** validation rule that could be applied to the database. State why it is important.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 8** Compare the use of batch processing and real-time processing as methods of updating a record of daily sales in a supermarket that closes overnight.

[4]

- 1 Describe, with an example, the differences between data and information.

[3]

3 Personal data is held on a hard disk.

Explain **two** benefits of encrypting this hard disk.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

4 A shop owner needs to collect data about the shopping habits of potential customers. This data needs to be high quality.

(a) Identify and explain **one** reason why this data must be of high quality.

.....

.....

.....

.....

.....

..... [2]

(b) Evaluate the benefits and drawbacks of using this data to identify shopping habits of potential customers.

[8]

1 Data processing and information

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 1

[Back to question](#)

Question	Answer	Marks
1(a)	Four from: Data on its own has no meaning and there is no meaning to this data This data is just raw facts and figures Data has to be processed to become information and there is little evidence that any of this data has been processed as it has no meaning Data needs to have a context in order to become information/we don't know the context for this data... ...description using an example of why there is no meaning giving two alternative contexts Data can be in the form of numbers, characters, symbols as shown in the example	4
1(b)	Three from: 1989 could be a year, such as the year somebody joined a company Paolo could represent somebody's first name √ could be the square root symbol used in Maths ₹ could represent a currency symbol such as is used in India ≠ could be the not equal to sign used in Maths OR could be a Boolean operator used in IT/Computer Studies	3

Mark scheme · 9626/12 Feb/March 2022 · Question 6

[Back to question](#)

Question	Answer	Marks
6(a)	Three matched pairs (all must be different): A length check could be performed on the Student_number data Each data item must be exactly 5 characters long/no more and no less than 5 characters A format check could be performed on the Date_of_birth data Each date must consist of two digits followed by a slash, followed by two digits, followed by a slash, followed by four digits A length check could be performed on the Date_of_birth data Must be exactly 10 characters A consistency check could be performed on the Student_number and Date_of_birth data The first two digits/year group must relate to the Date_of_birth	6

Question	Answer	Marks
6(b)	Six from: Four from: Common errors when copying data are transcription and transposition errors Verification would pick up a transposition error, but most validation checks (except a check digit) would not Verification would pick up transcription errors, but validation might not Verification is checking that data that has been/is being entered into a computer has been <u>copied correctly</u> from the data source Validation is checking that the data entered is reasonable/sensible <u>Neither</u> method checks that data is accurate/correct Examples from: A format check might show that the Date_of_birth is in the correct format of dd/mm/yyyy but verification would pick up errors... ...Jonathon Odoki's Date_of_birth, for example, might have been incorrectly copied as 12/07/2004/month and day have been swapped A length check might show that the Student_number is the correct length but verification would pick up a transposition error... ...For example Sven Johansson's Student_number could have been copied as 09312/two digits have been swapped A length check might show that the Student_number is the correct length but verification would pick up a transcription error... ...For example Sven Johansson's Student_number has been copied as 09S32/text has been entered instead of a digit A consistency check might show that the Date_of_birth is consistent with the Student_number but Manjit Sangherra's Student_number may have been copied as 11102... ...and Date_of_birth as 11/03/2006 Allow follow through from part (a)	6

Mark scheme · 9626/11 May/June 2022 · Question 1

[Back to question](#)

Question	Answer	Marks
1	Four from: It is collecting the data and then processing it/processing it all in one go/all at once / stored until the system comes online to process the data in one 'batch' The customer purchases/repayments/orders are collected into batches and each batch/orders are processed as a whole Customer orders processed when the computing resources are less busy Batches/orders can be stored up during working hours and then executed/processed during the evening/weekend Transaction file is used to update master file Transaction file of customer purchases/orders is kept for later processing – weekly/monthly Master file of customer details is only updated weekly/monthly Transaction file is used with master file to produce customer orders at the end of the month	4

Question	Answer	Marks
3	Eight from: Encryption is the scrambling of data... ...it converts plaintext to ciphertext... ...into meaningless groups of symbols... ... can't be understood by unauthorised people/read without the decryption key... ...without encryption messages can be stored as plain text, making it easy for anyone to read and understand Encryption keeps personal data secure... such as credit card numbers and personal information from computer hackers If personal information is intercepted, but it can't be understood, it is of no use... ... the act of hacking in this case is pointless It does not prevent hackers intercepting personal data but it prevents them from understanding it Two types of encryption asymmetric and symmetric... ...symmetric and asymmetric use a private key but asymmetric uses a public key as well /symmetric uses the same key to encrypt and decrypt but asymmetric uses different keys... ...symmetric uses a private key to encrypt and decrypt /symmetric is where both sender and receiver use a private key... ...asymmetric uses a public key to encrypt and a private key to decrypt /sender uses a public key and receiver uses a private key Can only decrypt using a decryption key/only authorised users have the decryption key... ... Symmetric is a faster method than asymmetric... ... Symmetric is less secure than asymmetric Data is encrypted using an encryption key Must be a proper analysis to gain full marks Max. six marks if bullets/list of points Must have expansions to be a proper analysis	8

Question	Answer	Marks
4	Seven from: The computer searches for the bar code number 5022476130067 in the product file When the matching record is found, the number in stock, 131, is reduced by one to 130 The number in stock, 130, is now compared with the re-order level, 130 If the number in stock is equal to the reorder level... ... which it is, then the computer creates an order automatically The computer reads the reorder quantity, 250, needed for the product It reads the supplierID, B671841, for Oaters It searches the supplier file for the record corresponding to the supplier ID, B671841, read from the Product file The order/e-mail is automatically sent to the supplier... ...using the supplier's contact details from the Supplier file, js@mail.co.ke	7

Question	Answer	Marks
8	Eight from: <i>Advantages</i> It may allow a larger set of data to be examined... ...direct data gathering could have limitations due to the availability of the people being interviewed... ...using banks the sample could be increased giving rise to greater confidence in the findings Information can be of a higher quality as banking data will have already been collated and grouped into meaningful categories Joanne does not have to read through poorly written responses to questionnaires/her interview transcripts... ...to create the data source <i>Disadvantages</i> It may cost a lot of money to pay the banks to obtain the data... ... whereas people who are questioned will not need to be paid The various purposes for which data was collected originally may be different to the purpose of the current study... ... there may be much more data than needed... ...unnecessary data may need to be filtered out There may be <u>sampling</u> bias There may be coding difficultiesdifferent banks being used means there will be differences in the content ...there may be a lack of standardisation across the data required ...different banks will have different amounts of data If it has already been coded the coding may be difficult to understand Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
1	Five from: By interviewing the family, the students will obtain up to date information/census is two years old/census data may be out of date More/detailed information will be obtained as the census may not contain all the information required/has limited information/fewer details They are able to collect only as much information as they require/the census may include information they do not needmight take time to filter extra information out It may be cheaper to visit the streets than the school paying transport costs to get the students to city hall It may be too time consuming to collect direct data / to go to every house in these streets / only need to make one trip to city hall The study might be limited as it is only five streets/the census would allow them to extend the survey Residents might not be at home when the students call/nobody at home would not be a problem with the census Data may take time to collate and present in a meaningful way/census data is already collated and presented so it is easier to find the required information	5

Question	Answer	Marks
2	Four from: When a file is written to the disk, it can be automatically encrypted by encryption software Full disk encryption is when the data is encrypted as soon as it is saved to the hard disk When a file is read from the disk the decryption/encryption key/password is needed to decrypt it When a file is read from the disk, the software automatically decrypts it... ... while in some types of hard disk encryption leaving all other data on the disk encrypted (All) the files on the disk are encrypted/ It's the encryption of data on that disk... ...except in some cases the operating system Whether the disk stays in the computer or moved to another computer it is still encrypted Only the person/computer with the secret/decryption/encryption key/password can understand the data on the disk/data can't be read without the decryption/encryption key/unauthorised people cannot read the data	4

Question	Answer	Marks
3(a)	Four from: For even parity: Three max from When data is transmitted the sending device/computer counts the number of ones in each byte Each byte is checked individually The parity bit is used to make sure the number of ones is even If the number of ones is even it sets the parity bit to 0 (and adds this on to the end of the byte) If the number of ones is odd it sets the parity bit to 1 (before adding it on) Every byte of data should have an even number of ones The receiving device/computer checks each byte to make sure that it has an even number of ones If the number of ones is odd there has been an error during the transfer of data Two max from In this example data is not being transferred from one device to another/data is not being transmitted She is only entering data to the computer so will need to use an alternative verification method Accept answers which are odd parity answers (substitute odd for even in points above)	4
3(b)	Four from <i>Similarities</i> Both involve comparing data that has been/is being entered with another copy Both involve checking that data <u>has been entered</u> correctly not that it is correct/pick up errors made during data entry <i>Differences</i> Visual verification is carried out by a human/user/yourself Double data entry is carried out by the computer/ computerised Visual verification involves comparing the entered data with the original source Double data entry involves comparing two versions of the entered data Visual verification will enable the user to see mistakes if they have been made With double data entry computer will alert user to the fact that a mistake has been made and they decide which version has been copied correctly/freezes keyboard to prevent data being entered until error is corrected Must have at least one similarity and one difference to gain full marks	4

Question	Answer	Marks
9(a)	<pre> READ first record in transaction file READ first record in old master file WHILE not end of transaction file IF transaction file WorkerID = master file WorkerID THEN WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) WRITE updated master file record to new master file READ next record in transaction file ENDIF READ next record in master file ENDWHILE </pre> Six from: Correct calculation – RateOfPay*HoursWorked – 1 mark Correct calculation - WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) – 1 mark Correct update – 1 mark Read next transaction file record... – 1 mark ...immediately before ENDIF – 1 mark ENDIF– 1 mark Read next master file record... – 1 mark ... immediately after ENDIF – 1 mark ENDWHILE at end of algorithm – 1 mark	6

Question	Answer	Marks
9(b)	Eight from: <i>Advantages</i> Batch processing does not require constant use of the computer when it might be needed for other tasks... ...the payroll only needs to be run at the end of the week/month... ...whereas real time processing would require computer to be available all the time... ...batch processing/payroll can be run at convenient times for the company/when the computing resources are less busy/when there is little demand for computer resources/can be run overnight Real-time processing requires immediate processing of data which is unnecessary with payroll... ...as hours worked would only be known at the end of each day so no point in using real-time processing Real-time processing is not usually capable of processing large volumes of data such as running a payroll whereas batch processing is Batch processing systems only need lower specification computers... ...whereas real-time processing needs more complex computer systems Online processing requires more hardware making it more expensive than batch processing Batch processing only needs a few employees to enter the data... ...so total wage bill is less... ...whereas online processing systems can need employees to be constantly using the system In batch processing systems there are fewer/no transcription/update/data errors... ...than human operators would produce in an online system <i>Disadvantages</i> With batch processing there is a time delay between collecting the input data and getting an output/data is not processed until the time specified With batch processing the information is only up to date after the master file has been updated by the transaction file... ...whereas in real-time processing information is always up to date With batch processing systems errors are only revealed during processing/after batch process is complete which can be overnight and so cause delays/errors cannot be corrected until the batch process is complete... ...in online processing errors are revealed, and can be acted upon, immediately Users cannot terminate a process during execution/have to wait until execution is completed Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
1(a)	Data that is collected for a specific purpose or task/ for that purpose only/gives us data that is often called 'original source data'	1
1(b)	Four from: Questionnaires could be distributed amongst neighbours/local people asking about the level of noise Interviews could be carried out with neighbours/local people (asking about the level of noise) He could observe traffic, keeping a count of traffic coming down each street He could use sensors to detect the flow of traffic going down each street He could use sensors to measure the level of noise in each street	4

Question	Answer	Marks
6	Six from: Transport Layer Security (TLS) is used for applications that require data to be securely exchanged over a client-server network... ...such as web browsing sessions/ file transfers The server shows/sends its SSL/TLS/digital certificate to the client /the client requests server's SSL/TLS/digital certificate To open a SSL/TLS connection, a client needs to obtain the public key The public key is found in the server's digital certificate The SSL/TLS/digital certificate authenticates the server to the client/identifies the server The client then carries out a number of checks to make sure that the certificate was issued by a trusted CA... ...is in date and that the server is the legitimate owner of the public and private keys For the client to access the server the client and server must carry out a SSL/TLS handshake... ...the client tells the server what version of SSL/TLS it uses... ...and a list of the different types of encryption that it is able to use The client tells the server that it wants to set up a communications channel Handshaking occurs before the transfer of data can take place The server tells the client the type of encryption it has chosen from the client's list	6

Question	Answer	Marks
3	Up to six marks available for three matched pairs e.g. • Delete/remove a (master file) record (1) • To remove duplicates//Where a record is no longer needed (accept examples) (1) • Amend/change a (master file) record (1) • E.g. A worker moves department/gets a pay rise/changes name (1) • Add/append a (master file) record (1) • A new worker joins the company (1)	6

Question	Answer	Marks
4	Six from: • Chip is read/checked ... (1) ... to make sure it has not been reported stolen, is in date and is a valid card number (MUST be two of three) ... (1) ... If not message appears saying the card is invalid (and the transaction does not continue) (MAY BE AWARDED WHERE ONLY ONE OF PREVIOUS THREE EXAMPLES HAS BEEN GIVEN) (1) • PIN entered by the customer is compared with that stored on the chip (1) • If PIN is verified the transaction continues ... (1) ... otherwise (error) message will appear ... (1) ... customer is asked to re-enter PIN (1) • If after three attempts the PINs do not match the transaction is rejected (1) • Supermarket computer contacts customer bank (to find customer's records) to see if there are sufficient funds in the customer's account (1) • If there are sufficient funds in the customer's account transaction is completed (1) • If there are insufficient funds in the customer's account transaction is rejected (1) • Amount of the transaction is deducted from customer's bank account (1) • Amount of the transaction is credited to supermarket's bank account (1)	6

Question	Answer	Marks
5	Five from: • Often called public-key encryption (1) • Uses two different keys// one <u>public</u> and one <u>private/secret</u> (key) (1) • No need to transfer a key (with the message) (1) • The public key/key held by the person doing the encryption is used to encrypt the data (1) • The corresponding private/secret key is used to decrypt the data (1) • The public key is published to everyone (1) • The private key is kept secret (1) • Anyone with a copy of the public key can encrypt information (1) • Only the private key holder can read the information (1) • It is not possible to deduce the private key from the public key (1)	5

Question	Answer	Marks
11	Max TWO from: e.g. • Data has been collected for a particular/different purpose ... (1) ... but then that data is used for something else (1) Max TWO from: e.g. • The Electoral Register is a list of adults who are entitled to vote in a local or national election (1) • Customers in stores often provide personal information when they buy products (1)	4

Question	Answer	Marks
1	MAX four marks: Max three from: - Information is data which has a context/meaning/has been processed// this data can become information if meaning/context is added (accept any indication that meaning or context is key to data being information) (1) This data/it has no context/meaning//it doesn't explain what each symbol means (must clearly refer to the set of data here)//it is raw (1) - Any valid context//The context for this set of data could be that it can be used in search criteria in spreadsheets ... (1) ... the symbols would then be given meaning (1) Max two from: 1 mark for 2 correct meanings (1) OR 2 marks for 4 correct meanings (2) > means greater than < means less than = means equal to >= means greater than <u>or</u> equal to <= means less than <u>or</u> equal to	4

Question	Answer	Marks
5(a)	Three matched pairs: - A <u>length check</u> could be performed (on the data) in the ISBN field (1) Each data item must be <u>13</u> characters/numbers/digits long (1) - A <u>lookup check</u> can be performed (on the data) in the Book type field (1) Each data item is compared with the lookup list of P and H to see it exists//The code must be P or H only (1) - A <u>range check</u> could be performed (on the data) in the Copies field (1) Each value must not be less than the minimum/2 and must not be greater than the maximum/5//between <u>1 and 6</u> (1)	6

Question	Answer	Marks
5(b)	Three marks available: • Generic statement about why both required (no reference to fields). MUST be more than a statement about what each does. (1) • Verification would identify if P had been typed in the Book_type field for the first record//if it was relevant (accept any letter and book combination from the sample) (1) • A lookup check/validation would accept P or H (1) • Verification would identify if (e.g) 5/wrong number had been entered (into the first record) (1) • A range check/validation would accept 5 (1) • Verification would identify if two digits had been transposed in the ISBN field//the ISBN number was incorrect (1) • Length check/Validation would accept any 13 characters (1)	3

Mark scheme · 9626/12 Oct/Nov 2022 · Question 8

[Back to question](#)

Question	Answer	Marks
8(a)	Four from: • When a message/information is intercepted//accessed it is unreadable/ can't be understood ... (1) ... so, therefore, it is useless (1) ... to a hacker (accept alternative examples of 3rd party or just 'third party') (1) • It protects customers when they bank/shop online/any other suitable example from any area of use (MUST BE protecting data) ... (1) ... as the data cannot be used for identity theft (1) • Data can only be decrypted by the receiving computer if it has the private/decryption key (1) Max. three marks if bullets/list of points	4

Question	Answer	Marks
8(b)	Up to FOUR marks available: Advantages (MAX 3) • It is a faster process//quicker to encrypt compared to asymmetric (1) as it has less mathematical complexity (1) • It requires less computational power than asymmetric (1) Disadvantages (MAX 3) • Does not have a different key for encryption and decryption//sender and receiver use the same key (1) therefore only one key needs to be stolen (1) ... making it less secure (1) • The encryption key cannot be generally published (1) • The encryption key must be sent to the receiver of the message (1) • Key could be intercepted/stolen (by a hacker) whilst being sent ... (1) ... who can then decrypt any message encrypted by the sender (accept any reasonable implication) (1) Must have at least one of each to gain full marks Max. three marks if bullets/list of points	4

Question	Answer	Marks
1	Six marks available: - Identifies Accuracy as a factor (MUST be more than just the word 'accuracy') (1) - Identifies Relevance as a factor (MUST be more than just the word 'relevance') (1) <i>Accuracy</i> - Information must be accurate to be considered of good quality (1) - If the collected data is inaccurate then the information produced will also be inaccurate (1) - The method of collection can affect the accuracy of the collected data and thence information (1) - If questions in an interview/on a questionnaire are not phrased correctly this leads to inaccurate data being collected (then the information produced will also be inaccurate) (1) - If data is then entered incorrectly it will lead to inaccurate information being produced (1) - Example of inaccurate information/data (1) <i>Relevance</i> - The relevance of the data relates to the situation or problem being studied (1) - Relevant information is information which meets the needs of the user/relevant to its purpose (1) - It should not concentrate on one aspect of the problem (1) - It should not relate to a different area to the one being studied (1) - Should not have too much information ... (1) ... user will spend a lot of time searching to find relevant information (1) - In order for information to be useful it needs to be complete ... (1) ... if only some of the information required is present then it is very difficult to solve a problem/complete a task (1) - Example of irrelevant information (1) MUST be at least one for each side for full marks	6

Question	Answer	Marks
2	Five marks available: • A real-time system has to respond to/process inputs//inputted data instantaneously (1) • An air conditioning system has to deal with inputs continuously ... (1) ... unless the user switches the system off (1) • This type of processing is usually found in systems that use sensors (1) • The user interface of a real-time system may use specialist input devices to provide data input ... (1) ... such as a keypad/touchscreen (1) • Usually deals with small amounts of data//does not deal with a lot of data (1) • Real time systems are such that the output affects the input (1)	5

Question	Answer	Marks
1(a)	ONE from: - Information should be/that is free from errors/mistakes (1) - Information should be/that is true/correct (1) - Information that is provable/proven to be correct (1)	1
1(b)	SIX from: - People might not answer the questionnaire truthfully (1) - People might answer the interview truthfully (1) - People may be biased (1) - The questions might be badly phrased (1) - Interviews can clarify the answer (1) - Interviews can ask questions based on previous answers (1) - Questionnaires <u>cannot expand</u> on what the question means (1) - Interviews include open ended questions that are hard to quantify (1) - In a multi choice type question there may not be a sufficient number of alternatives//answers restricted to those provided (1) - There is the possibility that the people collecting the data may make errors when collecting it (1)	6
1(c)	ONE from: - User is in communication with a central computer (1) - Where a process is carried out on the internet/WWW (1)	1
1(d)	Five from: - Go to the website (1) - Log on (by entering username and password) (1) - Choose goods category (1) - Choose/select item (to examine the product) (1) - Add (items) to shopping basket/cart //click on buy button//pick up item (1) - Check good in basket/cart//make changes to goods in basket/cart (1) - Click on/proceed/go to checkout (1) - Confirm billing address (1) - Confirm/enter delivery address (1) - Choose method of delivery (1) - Select/use payment method (1) - Type in credit card/debit card number (1) - Enter card security code/last three digits of CVV/CVC/CSC (1) - Confirm order//click checkout button//click pay (1) (NO further marks to be given after this mark given, unless candidate clearly states that the action happens before this point)	5

Question	Answer	Marks
5	FOUR from: • (Using data encryption is) a way to keep their data secure//to protect their information from a hacker//to safely exchange data (1) • Information needs to be impossible to understand/unreadable/indecipherable (1) • Information needs to be useless to a 3rd party if intercepted/hacked/they should not have it (1) • Only authorised people/people with the key can understand the information (1) • Encrypted data needs to be decrypted which is hard/impossible (without the key) (1) • To make hacking not worthwhile (1)	4

Question	Answer	Marks
1(a)	One from: • Data is scrambled/jumbled up (1) • Converting information so that it is impossible to interpret/understand (1)	1
1(b)	Up to THREE marks: e.g. • Can be used for identity theft (1) plus one example (1) • Can be used for identity fraud (1) plus one example (1) • Company secrets could be sold by hackers to rival companies (1) • National secrets could be used by rival powers (1) • Secrets/personal data could be used for blackmail purposes (1) • Private details could be used to pass (bank) security tests (1st) – Accept any suitable extension for passing security tests (1) e.g. – And so access (e.g.) bank accounts (1)	3

Question	Answer	Marks
5(a)	Validation and sorting (1)	1
5(b)	Eight from: • First record//2210 in the transaction file (1) • First record in the master file/1965 is read/checked (1) • These (previously identified) two records are compared (ALLOW FT if attempt at identifying both records) (1) (Award 'the' instead of 'these' where MP1 AND MP2 given) • Records don't match (1st) – computer writes master file record to new master file (1) • Next record of master file/2210 in the master file is read/checked (1) • Transaction file record/2210 is compared with this record (1) • It matches so the Numbersold is subtracted from NumberinStock (1) • Using NumberinStock of 12000 from master file (1) • Using Numbersold of 6500 from transaction file (1) • NumberinStock is now 5500 (1) • Processed record/NumberinStock is written to new master file (1) • Next record/3356 in the transaction file is read/checked (1) • Next record/2976 in the master file is read/checked (1)	8

Question	Answer	Marks
11	Eight from: - Any correct definition of direct data sources, such as: - Direct sources are where Isabella asks people in person//primary source of data (must be an attempt to define) (1) - Can gain/gather qualitative AND quantitative data (1) - Data gathered is more up to date/recent (1) - Data is (guaranteed to be) relevant to the study/question (1) - Interviewer can base further questions on answer given (1) - Interviewer can ask a more in-depth/follow up questions (1) - Can interpret body language (1st) - such as (e.g.) facial expressions//nervous movement of hands (1) - Interviews tend to be taken seriously by people (1st) - whereas questionnaires may be ignored (1) - It ensures the data is relevant to the study//you get a better view of what is going on//get a better understanding (1st) - whereas indirect sources are not specific to the study (1) - You are in control of the sample size//can be as big/small as needed (1st) - whereas Isabella has no control over the amount of data collected from an indirect source (1) - The source of the data is known exactly (1st) - making it easier to judge its reliability (1) - Isabella could sell/pass/share the data on (to interested companies) (1st) - indirect data is not hers to sell/pass/share (1) MAX 6 marks for a bulleted list	8

Question	Answer	Marks
1(a)	Five from: • It would now be used for a different purpose to that for which it was originally collected (1) • Electoral register only shows adults (accept does not show children) (1) • Only shows people who are registered to vote (1) • The data on the electoral register might be out of date//not have been updated (1) people on the electoral register might no longer live in the area (1) ... people new to the area might not be included (1) ... new homes may have been built (1) ... people may have died (1) • Not all people who live at the same address are from the same family (1)	5
1(b)	Award ONE mark for ANY indirect data source: e.g. • Businesses collecting personal information when used by third parties (1) • Census (data) (1) • A newspaper (1) • An encyclopedia (1) • Medical records (1) • Museum archive (1)	1

Question	Answer	Marks
2	Five from: • One mark for a description of encryption (1) • Often called secret key encryption (1) • Uses a single/the same key (to decrypt/encrypt the data) (1) • Any awareness that it is a private key (1) • This (private/secret) key must be shared between the sender and the receiver (1) • The encryption key is sent to the recipient (1)	5

Question	Answer	Marks
3	Five from: MAX three if only batch processing OR online processing discussed. • Data is grouped into batches (in batch processing) (1) • With online processing, data is processed as quickly as possible (1) • With batch processing there is a delay between the collection of data and the actual processing (1) • With batch processing, the data is entered and processed altogether in one batch by the computer (1) • With online processing the user tends to be in direct communication with a central computer (1) • Online systems require more hardware/input devices/workstations than batch processing systems (1) • Batch processing uses very little processing time compared to online processing (1) • In online processing errors are revealed immediately (1) • (whereas) Errors in batch processing errors are revealed much later/only when processing takes place/overnight/when batch process is complete (1) • Batch processing can be done when the system is not busy//scheduled to a specific time (1) • (whereas) online processing has to be active for the whole day (1) • For batch processing, data is only accurate at the time of processing (1) • (whereas) online processing, the data is accurate all the time.	5

Question	Answer	Marks
1	Four from: • Automatic collection/storing of data (1) • Using a computer and sensors to collect data (1) • This data is analysed/processed (1) • The results are output in the form of graphs/charts/tables (1) • Data is collected over a period of time ... (1) ... continuously/at regular intervals (1) • The analysis/processing usually requires special software (1) • Two of the following examples of physical variables logged: temperature sound light pressure (1)	4

Question	Answer	Marks
2	Five from: • ONE of: SSL is Secure Socket Layer OR TLS is Transfer Layer Security (1) • Enable encryption in order to protect data (1) • To protect data whilst being transferred (accept close alternatives to 'transferred') (1) • To encrypt and decrypt data sent/received by web users (1) • Provides secure connection between web browser and website/client and server (1) • Authenticate the people/companies/users exchanging data (1) • Ensure the integrity of the data to make sure it has not been corrupted/altered (1) • Enable credit/bank card payments to be made securely ... (1) ... by setting standards for the storage/transmission/processing of bank card data (1) • Improve customer trust/more likely to do business with a company using SSL/TLS protocol on its website (1) • Enable users to securely complete online transactions (1) • Enable the creation of a secure connection between a web server and a browser (1) • Prevent eavesdroppers from trying to intercept data (1) • To provide security of communication in a network/secure transmission/sharing of data (1) • Encryption protocols enable credit card payments to be made securely (1)	5

Question	Answer	Marks
3(a)	Three from: - Data is grouped (1) - A group of data is processed all at once//in one go (1) - Jobs/batches are set up so they can be run to completion without human interaction (1) - Data is processed at a set time (1) Takes place when the computing resources are less busy (1) - With batch processing there is a delay between the collection of data and the actual processing (1)	3
3(b)	Three from: - The response time of the computer must be immediate (1) - Real-time responses are in the order of milliseconds, and sometimes microseconds (must be more than 'is fast', but this can be expanded upon)//real time processing is immediate (1) - Real time means that the inputs are processed and produce an output which in turn affects the input (1) - The processing involved in a rocket guidance system has to be continuous, so the process is never ending (1) - An example of consequences (1) - An example of benefit (1)	3

Question	Answer	Marks
2	Three from: - Validation is always carried out by a computer whereas verification can be carried out by a computer or by a human (1) - Verification is checking that data that has been/is being entered into a computer has been <u>copied correctly</u> from the data source (1st) - whereas validation cannot tell if the data has been copied incorrectly or not (1) - just checks that the data input is sensible/reasonable/follows a set of rules (1) - Validation is checking that data is sensible/reasonable/follows a set of rules (1st) - whereas verification cannot tell if the these rules have been followed/data is sensible/reasonable (1) - just checks that data input has been copied correctly (1) - Verification, (unlike validation) cannot check whether data is sensible or not (1) - Validation will pick up that data is invalid when collected but verification can only check that the data has been copied/transferred accurately (1) - Verification can pick up a transposition error but most validation checks (except a check digit) would not (1)	3

Question	Answer	Marks
4	Five from: • HTTP protocol with SSL/TSL encryption (1) • Any information transferred from the client to the server/server to the client is encrypted (1st) • enabling the secure transmission of (confidential) data (1) • As part of the handshake (between the web browser on the client computer) a session key is created (1) • This session key is encrypted using the public key and sent to the server (1) • All data sent between the client/browser and server is encrypted using the session key (1) • Session key generated using asymmetric encryption (1) • symmetric encryption is used to encrypt all further communications (1) • Each separate session requires a new session key to be created (1)	5

Question	Answer	Marks
11	Eight marks available: Advantages max 6 • Orders are/can be processed straight away (1st) • so process of filling/completing orders can start sooner (than for batch processing) (1) • With online processing errors are revealed immediately (1) • With batch processing systems if there is an error it is only revealed when processing takes place (1st) • Delay in finding the error (1) • If the database needs to be interrogated details can be retrieved immediately (1st) • Any example of why immediate interrogation would be required – e.g a sales invoice/order may be required when a customer queries a bill (1) • With online systems they are available whenever necessary (accept converse argument) (1) • Data is accurate all the time (1) Disadvantages max 6 • Extra hardware requirements/input devices/workstations of an online processing system can make it more expensive (than batch processing) (must clarify at least one process for the comparison) (1) • Online processing has to be active for the whole day (1) • There must be workers available (to enter data) at all times (1st) • more employees are needed (1) • company wage bills can be higher (1) • Batch processing systems only require a limited number of employees to enter all the data (1) • It is more expensive than batch processing as it uses much more computer processing time to update files (1) • With batch processing, processing occurs at a time convenient to the employees responsible for entering the data (1) • Batch processing only uses computers when the demand on the system is low (1st) • therefore entering data does not slow the network down (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
1	Max five from: • Verification (only) checks the data is copied correctly (1) • Validation checks/ensures that data conforms to a set of rules//is sensible/reasonable (1) • (Although copied correctly) the original data may have errors in it which could be picked up using validation (checks) (1) • (Although fitting the rules) the copied data may have been copied incorrectly, which would have been picked up by using verification (checks) (1) • Both checks are needed to cover mistakes that the other would not identify (1) Max four from: • (In record 1) 13/06/2019 has been copied incorrectly (as 13/06/2020) this would have been found using verification//example of verification (1) • Validation checks would not have picked up the error in record 1/identified error as it is in the correct format and length (1) • (In record 3) Validation/length check would have checked that the ID number was of the correct length//identified that the ID number had too many digits/was too long (1) • (In record 4) Validation/a range check on the Rate_of_pay field would have accepted only values between 15 and 50 (1) • (In record 4) Validation/Limit check accepting only values equal to and below 50 would have identified the error in record 4 (1) • In record(s) 3/4 the data has been copied correctly so verification would not have picked up this error (1) • Format check can be used on date joined field to check that it is dd/mm/yyyy (1)	6

Question	Answer	Marks
5	Five from: • Questionnaire needs to be written/produced/tested (1) • The Electoral Register is a single document//already gathered//immediately available (1) • Electoral Register data is already collated/processed(1) • Data from questionnaires may take <u>time</u> to collate/process (1) • May take too long to distribute questionnaires (1) • She may not have the time/resources available to send questionnaire to everybody/so many people (Award concept of saving time/resources if answer states how time/resources is saved) (1) • She will only need to make one trip to the council office (1) • It may be cheaper to make one trip to the council office rather than pay transport costs to the various parts of town (1) • Electoral Register only lists family members older than voting age//who are allowed to vote/does not include children (1) • Electoral register only holds limited information: name, address, age, gender (two of) (1) • Questionnaires would gather information about the whole family (including younger children) (1) • Questionnaires can ask specific/qualitative/open ended questions (1) • If using questionnaires, need to hire (e.g.) data entry clerks (which would increase costs) (1)	5

Question	Answer	Marks
11	Eight from: One mark for a description of encryption (e.g. data scrambled/converted to cyphertext) (1) Benefits MAX 6 • When data is sent across the internet in an encrypted form it cannot be understood/has no meaning/makes no sense (1st) • for third parties// unauthorised people//people who don't have of a decryption key (1) • and so is of no use (1) • Keeps (personal) data (more) secure/protects data//makes web browsing more secure (1) • Even if data is transferred to many devices/platforms it is still protected (1) • Enables IT departments/businesses to comply with data protection regulations (1) • Reduces the benefits of/motivation to hack(ing)/commit(ing a) crime (1) • Identification of TWO items of data that could be protected (DNA "banking information" – this is TV. Other examples from banking may be correct. (1) Drawbacks MAX 6 • It requires additional processing power to process encrypted data (1) • It takes longer to transmit encrypted data (1) • It takes longer to open/decrypt encrypted data (1) • The client/browser and server must send messages to each other several times before any data is transmitted (1st) • increasing the time it takes to load a web page (1) • It uses extra storage space for the client/the server (increasing costs) (1) • If the private key is lost, it is extremely difficult to recover the data//data may be lost permanently (1st) • data can be recovered by the reissuing of the digital certificate but takes time (1) • Makes it difficult for authorities to monitor criminal activities (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
1	Three from: • Data consists of raw figures/characters/numbers (which require processing for interpretation) (1) • Data does not have any meaning/context (1) • Information is data that has been given meaning/context (1) • Information is data that has been processed (1) One mark for any example. Example must cover BOTH data and information (1)	4

Question	Answer	Marks
4(a)	Three matched pairs from: • Might enter an account number with 9 or 11 digits/less than/more than 10 (1st) • Length check to make sure that the account number is exactly 10 characters (1) • Might enter a wrong digit (1st) • Use a check digit which creates a new digit by performing a calculation on individual digits of the account number and appending the resulting digit (1) • Might omit one of the account numbers (1st) • Presence check would make sure account numbers are entered. (1)	6
4(b)	Three from: • A key number would be calculated (1) • All the account numbers/numbers in the account number would be added together (1st) • To create a single number (1) • The data/file would be transmitted along with the hash total/key number (1) • The hash total is recalculated once received (1) • The original hash total is compared with the new one (1) • If they do not match an error has been made (1) One from: • A control total checks fields that are used for calculations whereas a hash total is used for fields that hold information (1) • The field holds data, and will not be used for calculations (1)	4

Question	Answer	Marks
4(a)	Four from: - Computers deal with each input and give (almost) immediate/real time responses//Booking is processed in real time/(almost) immediately - Online processing deals with one transaction at a time Once the transaction has taken place the database will be updated immediately/same time as the booking Once booked seat is shown as booked Once the transaction has taken place the computer sends a message/ticket to the user to inform him/her the seat has been booked	4
4(b)	Two from: - Flight won't be overbooked - Duplicate tickets will not be issued/ Seat cannot be booked by anyone else - Guaranteed that will be on the flight - Customers know immediately if they have successfully booked a seat//get immediate confirmation//get immediate advice if there is an issue - Seat is held as yours for a period of time//have time to think about whether to go ahead	2

Question	Answer	Marks
8	Six available: - These methods/verification and validation on their own/when not used together do not ensure that the data entered is (fully) accurate/correct - Example of both validation and verification in use to achieve accuracy, with clear explanation of why both are needed. Max five from: - Verification is checking that data that has been/is being entered into a computer has been copied correctly (from the data source)//correctly entered - Validation is checking that the data entered is reasonable/sensible/follows a set of rules - Although verification helps to stop users from making mistakes when entering data, it cannot check that the data was originally correct - Validation is needed because although the data might be copied correctly, the original data might be invalid - Therefore, both are needed to make sure that the data is both correctly copied and valid - Valid example of validation technique that could be applied (must be identified as an example of validation) - Valid example of verification technique that could be applied (must be identified as an example of verification)	6

Question	Answer	Marks
9	Seven from: One mark for a description of encryption (e.g. data scrambled/converted to cyphertext) Benefits Max six from: • Encryption makes data unable to be understood/accessed (1st) – for third parties// unauthorised people//people who don't have a decryption key (1) – and so is of no use (1) • Reduces the motivation for interception (1st) – Because data is protected (1) • Allows confidential/sensitive information to be transferred/shared/included in emails (accept converse argument about absence of encryption) • Enables IT departments/businesses to comply with data protection regulations Drawbacks Max six from: • Hackers can still intercept emails (1st) ...data can still be lost (1) • Encryption/decryption/private key must be transferred between sender and receiver • Anyone with the key can understand the email (1st) ...so gives the intended user a false sense of security (1) • If the same key is used repeatedly there is a security issue • Managing digital certificates can become complex and time consuming • Requires more computer processing power • It takes longer to transmit encrypted data • It takes longer to open/decrypt encrypted data	7

Question	Answer	Marks
1(a)	One from: - It has no meaning (1) - Raw/unprocessed values (1) One from: - any correct example (1)	2
1(b)	One from: - Any example. Must clearly include context or meaning (1). May simply be a description of a type of information. One from: - Processed data (1) - Data + meaning (+ structure) (1)	2

Question	Answer	Marks
2	Four from: - Accuracy (1) - Relevance (1) - Age (1) - Level of detail (1) - Completeness (1)	4

Question	Answer	Marks
3	Seven from: Sale • Computer reads the barcode number (1) • Computer searches for this barcode number in the product file/database (1) • If this barcode number is found, then number in stock is reduced (1) Order • Computer compares the number in stock to the re-order level/set value (1) • If the number in stock is not equal (or lower) to the re-order level/set value another barcode is input//systems returns to the start (1) • If the number in stock is equal/lower than the re-order level//a threshold/set value (1st) – the computer creates an (automatic) order (1) • Item is flagged to be reordered (1) • Computer looks up/finds the re-order quantity (for that product) (1) • Computer looks up/finds the supplier (reference) number (for that product) (1) • Computer searches/checks (the supplier file) for the record of the supplier (1) • Computer orders automatically (1st) – using the supplier's contact details. (1)	7

Question	Answer	Marks
10(a)	Three from: • Data is scrambled (1st) – so it is not understandable (1) – of no use to a hacker/third party (1) • Can only be unencrypted with a key//need the key to unencrypt//access the data (1) • Use of/access to encryption key/software known only to authorised user (1)	3
10(b)	Three from: • Acts as 'barrier' between internal and external networks/internet (1) • Reads (packet) sender's IP address/reads domain of sender (1st) – checks against database/list of IP addresses (1) – checks against database/list of network ports (1) • Permits access/passage of packet if not forbidden/if on allowed list/rules (1) • Drops packet if on forbidden/not on allowed list (1) • Can create logs for analysis (1) • Can create alerts. (1)	3

Question	Answer	Marks
4	Eight from: Data answers - Transmitted data/file is encrypted (1) so that a third party cannot understand the data//data/file is useless to a hacker (1) - Only the user's computer and the secure/receiving server can recognise the data/file (1) - SSL/TLS use two keys//private and public keys (1) Certificate/process answers (may be answered in any order) - Websites often use (SSL/TLS) certificates to verify their authenticity (1) - Web server is asked to identify itself (1) - The web server sends a copy of the (SSL/TLS) Certificate (1) Certificate Authorities are used to ensure the certificate is real/authentic of the certificate/provides a list of approved certificates (1) - The browser checks to see if the SSL/TLS Certificate is trusted (1) - If the SSL/TLS Certificate is trusted, then the browser sends a message to the web server (1) - If the SSL/TLS Certificate is not trusted, the browser will refuse to open the website (1) and will display a message to the user (1) (IF TRUSTED) The server then responds to the browser with a digitally signed acknowledgement (1) to start an SSL/TLS encrypted session (1) - Session keys are set up (1) - SSL/TLS requires a handshake to be carried out/the browser and the website carry out a (SSL/TLS) handshake. (1)	8

Question	Answer	Marks
5(a)	Two from: - Confirm/check that a copy of a file is identical to the original file (NOT source) (1) - Used to check/validate that data has been transmitted accurately (1st) – from one device to another/over a network/over the internet (1) - It is used for the whole files of data (1) - Provide a unique identifier for data (1)	2
5(b)	Three from: - Calculated by algorithms (1) - Use a hash function (1) - Adds number of bytes in a file /accept alternative of this process (1) - Checksum/number transmitted at the end of the file (1)	3

Question	Answer	Marks
7	Eight from: Simple statement of the difference between the two (1): e.g. • Direct data is a primary source, indirect data is a secondary source. • Description of how primary and secondary are collected. Advantages MAX SIX • Direct data source/direct data is reliable/trusted (1st) – because we collected it (1) • Data needs to represent the views of a specific target audience (accept other terminology) (1st) – we can make sure a cross-representation of the group is sampled/included in the sample (1) – and so avoid sample bias (1) • ONLY Specific/relevant data is collected (1) • Indirect data includes irrelevant data (do not allow opposite) (1) • Person/people collecting the data only need to collect as much/or as little (quantity) as is needed//to suit your needs (1st) – whereas with indirect data sources, the original purpose may have not collected enough//collected too much data (1) • Collected data could be sold on to other organisations (1st) – thereby helping to recoup some of the expenses/money spent (1) • Direct data is (more) up-to-date (1) Disadvantages MAX SIX • The sample size may be small (1st) – whereas indirect data sources tend to be larger sets of data (1) • Some people/groups may not be accessible by the person collecting the data (1st) – whereas with indirect data sources, would allow data from these groups to be collected/gathered (1) • With direct data collection, people may not be at home/available to be interviewed (1st) – can result in a smaller sample size (1) • Original data may not be possible to collect because of the time of year (1) • Indirect data allows historic data to be used (1) • Direct data can take a long time to collect (1) • With direct data, data needs to be collated (NOT “processed”) (1) • Collecting direct data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)	8

Question	Answer	Marks
5	Six available. First mark : Describes the type of check and the data that is being checked (student ID or Student examination score). Dependant mark: Candidate states why inaccurate data could still be entered. Three matched pairs from: - Length check on student number (1st) ONE OF - won't identify if the number is wrong (1) - Won't check invalid characters//or example (1) - Won't check if value is unique (1) - Invalid character/Type check on either (1st) ONE OF - Type check on student number won't trap fewer/more than 10 digits (1) - Type check on exam mark might still allow abnormal data to be entered (1) - Won't identify if the number is wrong (1) - Won't check if value is unique (1) - Range check on either (1st) ONE OF - Range check will only check the number is in range//will not check if number incorrect (1) - Won't identify if the number is wrong (1) - Limit check on either (1st) ONE OF - Range check will only check the number is below a higher number//will not check if number incorrect (1) - Range check only checks for exceeding the maximum value/score (1) - Won't identify if the number is wrong (1) - Check digit on student number (1st) - won't identify if the number is wrong (1)	6

Question	Answer	Marks
5(a)	THREE marks available: Max TWO • Data is collected for a specific purpose/task (1) • Provides original data/first hand data (1) Max TWO e.g. • Observation could be used, by the scientist going to the farmland and looking (1) • Questionnaires could be distributed amongst farmers/3rd party linked to scenario (1) • Interviews could be carried out with farmers/3rd party linked to scenario (1) • Data logging/sensors could be used to gather pollution data (1).	3
5(b)	THREE marks available: Max TWO • An indirect data source is when data is obtained from a third party/is second hand (1) • Means data that was collected for a particular reason but is then used for another purpose (award concept of data being used for a different purpose) (1) Max TWO • Could collect data from local government agencies (1) • Could use information presented in scientific journals (1)	3

Question	Answer	Marks
6	FOUR from: • Verification is checking that data has been copied correctly from the data source (1) • Validation is checking that the data entered is reasonable/sensible/follows rules (1) • Verification will not pick up the fact that the original data might be invalid (1) • Verification would pick up a transposition error/most validation checks (except a check digit) would not (1) • Neither validation nor verification on their own will pick up all errors//need both to cover all possible errors. (1)	4

Question	Answer	Marks
2(a)	ONE from: - New information is of a higher quality (accept converse argument) (any awareness shown) (1) ONE from: - Up-to date information leads to better decisions (1) - Out of date information leads to poor decisions (1) - Any relevant example used to exemplify out of date information. (1)	2
2(b)	THREE from: Max TWO from: - Information needs to be relevant to the reason why/purpose it has been collected//must meet user needs (1) - Relevant information is good (1) - Irrelevant information is not useful (1) Max TWO from: - Irrelevant information leads to poor decisions/Relevant information leads to good decisions (1) - Wastes time as has to be sifted (1) - Irrelevant information can distract from the correct analysis (1) - Any relevant example used to exemplify irrelevance. (1)	3

Question	Answer			Marks
8(a)	Field Name	Data Type	Example of Data	5
	Number_in_Stock	Integer (1)	199	
	Location	String/alphanumeric/text (1)	07.44.03	
	Reorder_Level	Integer (1)	250	
	Price	Currency (1)	\$203.50	
	Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819	

Question	Answer	Marks
8(b)	SIX from: Three matched pairs: • Number_in_Stock – Type check (1st) must be numeric (1) • Number_in_Stock – Range check (1st) between 0 and 250 (1) • Price – Limit check (1st) must be more than zero (1) • Price – Type check (1st) must be numeric/currency (1) • Location – Format check (1st) Must be three pairs of digits separated by a full stop/NN.NN.NN (1) • Location – Length check (1st) must be 8 characters (1) • Reorder_Level – Type check (1st) must be numeric (1) • Supplier_Telephone_No – Length check (1st) must be 11 characters/digits (1).	6
8(c)	FIVE from: • Any awareness of normalising data (1) • Import the (two) tables into the relational database structure (1st) as separate tables (1) • Create/use a linking table (1) • Create/use/it needs Primary Keys for both/all tables (1) • Create/use/it needs Foreign Keys. (1st) These are needed to act as the link field between the tables (only award if BOTH of the previous two answers has been given) (1) • In the linking table (1) • Link the Primary Key(s) to the Foreign Key(s) (1) • Any awareness of one-to-many relationships (1)	5

Question	Answer	Marks
9	EIGHT from: - This method gathers direct data, which is more up to date (1) Benefits max 6: - There is no significant time delay between question and answer (1) - Reporter can more easily ask extension questions based on answers (1st) that can add to the verbal answer from the interviewee (1) - People from (e.g.) all over the country (any awareness of wide location between two parties) can be interviewed (1) - Enables interviewers/reporters to contact people in restricted/remote areas (1) - No need to travel/no travel costs for reporter (1) - No time wasted/used by reporter in travelling to interview (1) - Interview can easily be recorded for later transcription/reference (1) Drawbacks max 6: - The reduction of social cues/cannot see interviewee (1st) so e.g. body language/eye contact cannot be used as a source of extra information (1) - The interviewer/reporter has no knowledge of the situation in which the interviewee is situated (1st) so less able to create a good interview ambience (1) - Telephone line/mobile signal quality can affect responses/interview (1) - Easier for interviewee to find excuse/make up excuse/find reasons to terminate the interview (1) - Can only interview one person at a time (1) - Interviewee may not pick up the phone (1) - Sample is restricted to those people who have a telephone (1) - No way to visually verify who you are interviewing (1)	8

Question	Answer	Marks
4(a)	THREE from: • Verification is checking that data that has been/is being entered into a computer has been copied correctly (1). • Validation is checking that the data entered is reasonable/sensible/valid/follows a set of rules. (1) • Verification will not pick up the fact that the original data might be incorrect. (1) • Validation can not tell if the data has been entered incorrectly (1) • Verification would pick up a transposition error. (1) • Verification would pick up a transcription error. (1)	3
4(b)(i)	TWO from: • It is a validation method (1) • It checks that data is possible/sensible for the context (1) • It checks that data is between an upper and lower acceptable value (1) • Sensible example (1)	2
4(b)(ii)	TWO from: • It is a verification method (1) • It is used to avoid/pick up transcription errors (1) • Two versions of the data are entered and compared (1st) – If a discrepancy between the two versions is found, user will be alerted (1) • Sensible example (1)	2

Question	Answer	Marks
5	FOUR from: • The public key is used to encrypt the data (1) • The private key is used to decrypt the data (1) • The public key is published to everyone (1) • The private key is sent separately to the receiver (1) • The private key is kept secret (1) • Anyone with a copy of the public key can encrypt information (1) • Only the private key holder can decrypt the information (1) • It is not possible to decrypt the data from the public key (1)	4

Question	Answer	Marks
10	EIGHT Marks available: Definition of online processing: ONE from: • User is in direct connection with a central computer (1) • A real time relationship between user and seller via the internet (1) Max EIGHT: • Give a positive impact (1st) • Give an extension that is a positive implication on a customer or business of the identified justification. (1) e.g. • Speeds up the process of an individual order (1st) results in better customer satisfaction (1) Other areas of focus • Speeds up the process of a large number of orders • Allows online purchases • Allows online financial transactions • Helps to keep track of your stock levels • Continues to collect data, therefore more accurate • Reduces the time between ordering and delivery • Automated reordering • Integration with other online systems.	8

Question	Answer	Marks
1	One mark per bullet point to a maximum of five marks. • Information is data with context (1) and meaning (1) • This could be about e.g. a student/lecturer/employee/payroll system (1) Any examples of the data presented with context or meaning (interpretation) (max 5 – each item of data may only be used once) e.g. • Chandra could be the first name. (1) • 59 could be their age. (1)	5

Question	Answer	Marks
2	One mark per example and reasonable expansion. - First mark – any reasonable reason why a checker may miss issues when comparing two documents (1st) - Second mark – example of the error in practice (1) Answer must be in the context of comparing two documents and not be about proof reading.	2

Question	Answer	Marks
5	One mark per valid validation check identified (max 5) and justification given to a maximum of eight marks. Max six if only one of student number or examination score used. - A format check (1st) - To ensure student numbers match the expected format/To check/while checking if student numbers starts with letter followed by 5 digits. (1) - A length check (1st) - To ensure student numbers have the correct/expected number of digit (1) - A check digit (1st) - To help prevent/detect typos/errors during entry of student numbers/As a redundancy check for student numbers (1) - A lookup check (1st) - To help verify that a student number exists in a master list// a/the student number belongs to an actual student enlisted/registered at the school. (1) - A consistency check (1st) - To ensure no duplicate/multiple records exist for the same student number against the same examination (1) - A range check (1st) - To ensure that the exam score falls within the valid/expected range of 0 to 120 (1) - A limit check (1st) - To ensure lowest exam score of 0/highest exam score of 120 (1) - A type check (1st) - To ensure that the student number field had data entered as text only (1) - A type check (1st) - To ensure that the exam score field had data entered as numeric only (1)	8

Question	Answer	Marks
8	One mark per bullet point to a maximum of three marks. • Authenticates packets of data (1) • Encrypts data at source (1) • Decrypts data at destination (1) • Provides secure communication between two computers/across the internet (1) • IPsec is used in a virtual private networks (VPNs) (1) • Negotiation of cryptographic keys to use during a session. (1)	3

Question	Answer	Marks
1	One mark per bullet point to a maximum of two marks. • Age//possibly too old to be of any use (1) • Relevance//may not be about recycling (1) • Accuracy//the figures/information may be incorrect (1) • Completeness// the information may be too vague (1) • Level of detail//there may be too much or too little detail (1)	2

Question	Answer	Marks
8	Six marks from: • Customer taps card/card is read (1) • Supermarket computer contacts customer bank (1) • Checks to see if there are sufficient funds (1) • If there are sufficient funds transaction is authorised/made//money is transferred (1) • If there are insufficient funds transaction is rejected/not made//money is not transferred (1) • Amount of the transaction is deducted from customer's bank account (1) • Amount of the transaction is credited to supermarket's bank account (1) • The transaction is closed//shown as completed. (1)	6

Question	Answer	Marks
1(a)	Four marks from: • A larger sample size (1) • A larger/large set of data to be examined (1) • It allows data to be gathered from subjects to whom the data gatherer does not have physical access (allow concept) (1) • Using less time (to collect the data)//data is readily available//already exists (1) • Data collected indirectly has already been collated/grouped (1) • Data collected indirectly has already been processed (1) • Information can be of a higher quality. (1) • Costs of (e.g.) travel//printing//employing interviewers avoided (must state the cost saved) (1)	4
1(b)	One mark per bullet point to a maximum of four marks. • The data may be out of date (1) • The data may be costly to purchase (1) • Inappropriate//irrelevant sample (who is asked//size) (1) • Irrelevant questions asked//methods used//data gathered (1) • The data may not meet the company's needs//different purpose (1) • The data may not be specific enough//lacking detail. (1)	4

Question	Answer	Marks
11	Max FOUR marks, TWO per paired answer • Length check (1st) – Checks that the length of the data is exactly 7 characters (1) • Format check (1st) – Checks that each item of data begins with two letters followed by 5 digits (accept exemplification) (1) • Presence check (1st) – Checks that there is data present in this field (1).	4

Question	Answer	Marks
13	One mark from: - Definition of encryption (1) Max 6 from: <i>Benefits:</i> - Content of the web page cannot be understood by any third-party (accept any awareness) OR Can only be understood when decrypted//cannot be understood without decryption key (1) - Only access by authorised users (1) - Encrypted connections make man-in-the-middle attacks ineffective (1) - Reduces the impact of would be/potential hackers (1) - Dissuade/deters hackers (1) - Content of the web page (being transferred) is kept private (1) - Gives users confidence (1) Max 6 from: <i>Drawbacks:</i> - Encryption uses a lot of server resources (1) - Requires processing power/memory (1) - Encryption introduces latencies/delays//takes more time to transfer data (1) - Connections take longer to set up (1) <u>More</u> data exchanges as keys have to be exchanged (1) - Need to buy an SSL certificate (1) issued by certificate authorities (1) - There are proxy caching problems. (1)	8

Question	Answer	Marks
1	Max 2 for each method (max two methods): Descriptions <u>must</u> match the method. • Questionnaire (1st) – A remote method of gathering data (1) – Is a fixed set of questions that cannot be changed (1) – Completed in respondent's own time/location (1) – Can be online or printed on paper//in physical form (1) • Interviews (1st) – The interviewer(s) asks the questions//questions are asked face to face (1) – Answers are gathered in real time (1) – In structured interviews the questions are asked in the same order for each interviewee and with a choice of answers (1) – Unstructured interviews can be different for each interviewee//interviewees can expand on their answers/usually no pre-set list of answers (1) – Interviewer can ask follow-up questions (1) • Data logging (1st) – Uses sensors to collect data (1) – Can be used to collect data over a period of time (1) – Either continuously or at set/regular intervals. (1) • Document analysis (1st) – Examining existing documents (1) to create new information (1)	4

Question	Answer	Marks
2	One mark per bullet point to a maximum of five marks. • SSL stands for Secure Socket Layer and TLS stands for Transport Layer Security (1) • They provide network security (1) • TLS/SSL are used in web browsing/email/Internet faxing/instant Messaging/Voice over IP/VoIP (must have at least two examples) (1) • Client-server applications use TLS in a network to try to prevent Eavesdropping (1) • Encrypts data sent from users to website//data sent from websites to users (1) • Certificates are sent (1) and confirmed with a trusted authority (1) • They enable credit card/online payments to be made securely (1) • SSL/TLS requires a handshake to be carried out. (1) • SSL/TLS is used by the presentation layer//SSL/TLS operate/work between the application layer and the transport layer (1)	5

Question	Answer	Marks
1(a)	One from: • Data is raw facts (1) • Data has no meaning (1) • Data is unprocessed/unorganised (1) • No context (1)	1
1(b)	One from: • Information is data with meaning (1) • Information is data in context (1)	1

Question	Answer	Marks
2(a)	Three from: • She will be able to provide a better quality of service (1) • She will have better quality information about her patients (1) • She can get exactly the data she wants//can ask follow-up questions//concept of relevancy (1) • Data is more reliable/trustworthy//she knows where it came from (1) • Data is more up to date (1) • Data only relates to the sample/patient that she want/collects from (1)	3
2(b)	Three from: • No organisation of data (1) • No pre-analysis of data (1) • Requires access to the source//patient is needed (1) • Prone to human error (1) • May suffer from bias (1) • May require specialised equipment (1) • No peer review (1)	3

Question	Answer	Marks
10	Six from: • It uses sensors and output/actuator (1st) – A valid example of input e.g. proximity sensors (1) – A valid example of output e.g. thrusters (1) • Real time processing occurs in a (micro-)processor (1) • It uses a process called feedback (1) • Output from the system affects the input/sensors/environment (1st) – Input from the sensors is used to alter the output (1) – Valid example of action e.g. trajectory correction (1) • Description of the output/input system (as a whole) in action e.g. sensor detect object in path of rocket and fires thrusters to avoid (1) • Real time processing produces an extremely/fast response(s) (to avoid critical incidents) (1) • Real time processing runs continuously (1) • Real time processing can run autonomously (1)	6

Question	Answer	Marks
1	Three from: • Accuracy/free from error (1) • Completeness (1) • Level of detail (1) • Relevance (1) • Time/age/out of date (1)	3

Question	Answer	Marks
2	Three from: • Used to check the file has been transmitted correctly (any awareness of verification/checking of file transfer) (1) • The control total is done only on numeric data (1) • The control total is a calculated value/any awareness of calculation to achieve a total (1) • The control total is added to the file (to be transmitted) (1) • The control total is recalculated when the file has been transmitted (must have concept of “after transmission” linked to the recalculation) (1) • Statement about outcome based on comparison of two totals (1)	3

Question	Answer	Marks
7(a)	Four from: - Decide on what you want to learn/know (1) - Identify/prepare questions (1) - Decide on interviewer (1) - Identify population/decide which group of staff to interview (1) - Choose suitable date/time (1) - Choose suitable location (any indication of choice OR criteria) (1) - Select/invite attendees/employees (1) - Choose suitable timetable/length (1) - Prepare suitable recording methodology (1)	4
7(b)	Four from: Analysis - You can create complex reports (1st) - allows to combine data from multiple tables (1) - You can create complex queries (1st) ... (1) - You can create dynamic queries (1st) ... (1) Organisation Quicker to search (1st) - Because data is held in tables (1) Consistency - Data is not repeated/reduced redundant data/only entered once (1st) - Quicker/easier to modify the data (1) - File size can be smaller/frees up (NOT “creates”) storage space (1) - Reduced chance for errors (1)	4

Question	Answer	Marks
9	Six from: - Definition of encryption (1) - Protects data/data “is safe”/improves data security (1) - Data remains protected if pen drive is lost (1) - Proves that the data has maintained its integrity (1) - Files cannot be understood unless decrypted (must have concept of encryption/decryption) (1) - Files cannot be understood by an unauthorised user/anyone without the decryption key (1) - Remains encrypted regardless of where it is moved to (1) - Reduces motivation to steal pen drive (1) - Increased confidence in saving sensitive data/files on the pen drive (1)	6

Question	Answer	Marks
2	ONE mark available: • They do not have meaning (1) • They are not in context (1) • They have not been processed (1) • They are raw data (1)	1

Question	Answer	Marks								
10(a)	THREE marks available: <table><tr><th>Data title</th><th>Most appropriate data type</th></tr><tr><td>employee_name</td><td>Text/alphanumeric (1)</td></tr><tr><td>hourly_rate_of_pay</td><td>Currency/numeric (1)</td></tr><tr><td>was_employee_born_in_france?</td><td>Boolean (1)</td></tr></table>	Data title	Most appropriate data type	employee_name	Text/alphanumeric (1)	hourly_rate_of_pay	Currency/numeric (1)	was_employee_born_in_france?	Boolean (1)	3
Data title	Most appropriate data type									
employee_name	Text/alphanumeric (1)									
hourly_rate_of_pay	Currency/numeric (1)									
was_employee_born_in_france?	Boolean (1)									
10(b)(i)	TWO marks available: First mark for identifying a suitable validation rule/test (1st) Second mark for a description of that test (1) • Range (1st) – to check value is BETWEEN two figures (1) • Type – to check data is in numeric/currency • Format check – to check entered as (e.g.) \$3.14//description of • Lookup check – to check against a list of values • Limit check – to check that is not below//above a certain figure • Character check – to check that is a decimal/real	2								

Question	Answer	Marks
10(b)(ii)	TWO marks available: First mark for identifying a suitable validation test (1st) Second mark for a description of that test (1) • Range – to check value is BETWEEN two figures • Type – to check data is in numeric/date • Format check – to check entered as (e.g.) dd/mm/yyyy • Limit check – to check that is not below/above a certain figure • Character check – to check that is an integer	2
10(c)	THREE marks available: • Age is based on the calculation between date of birth and today (1) • Age constantly increases with time//age changes every year (1) • A calculated field allows for a changing/dynamic figure (1) • The result shown//age is constantly accurate (1st) Reduced chance of human error resulting in the wrong number being stored (1)	3

Question	Answer	Marks	Guidance
1(a)	TWO marks available, one per complete difference: - Direct data is used for the same purpose for what it was collected.//Indirect data is used for a different purpose to what it was collected for. (1) - Direct data is collected by/for you/by the user.//Indirect data is obtained from someone else. (1) - Indirect data is from a secondary source//direct data is from a primary source (1) - Direct data is collected first hand//secondary data is second hand data (1)	2	Do Not Award - Direct data may be free, indirect data you may have to pay for - Swapping used and collected
1(b)	TWO marks available: - Candidate identifies a suitable circumstance (1st): e.g. Project needs a lot of data/named example. Project needs to be completed very soon/named example. Direct data would be too expensive to gather/named example. Direct data would be dangerous to gather/named example. Data to be gathered refers to something in the past/named example. Data may be seasonal and it's the wrong season Data is available now Direct data does not exist - Candidate explains why this circumstance would be better suited to use indirect data (MUST match circumstance) (1) e.g. Because collecting a lot of data would take time/beyond the scope of the person who needs it to collect. Because data would be gathered already. Because data is being gathered from a warzone/identified place or circumstance of danger. Because situation no longer exists for the data to be gathered.	2	

Question	Answer	Marks	Guidance
1(a)(i)	ONE from: - Questionnaire (e.g.) where questions are asked//customers are asked their opinions (1) - Interview is (e.g.) a face to face meeting/questions are asked (1) - Observation where situations are watched (1) - Surveys (e.g.) where questions are asked (1)	1	Marks may only be awarded where the method has been stated. Answer MUST do more than identify a method
1(a)(ii)	ONE from: e.g. - Textbooks (1) - Journals (1) - Websites//research published online (1) - Someone else's (e.g.) questionnaire (1) - Market research DONE FOR someone else (1) - Government statistics (1) - Newspapers/magazines (1)	1	Do Not Award: - Weather data - Census data - Electoral register - Any source that would not include expenditure patterns - Any source where it is not clearly someone else's

Question	Answer	Marks	Guidance
1(b)	Four marks available: TWO from relevance: • Reduces the chance of excess/unnecessary data (accept converse) (1) • The businesswoman has control over what information is gathered (accept exemplification) e.g. ask questions about sports goods (1) • The businesswoman can design the method/stated method, so can ensure that the (e.g.) questions asked gather the information she needs (1) • Can ensure data is gathered from the town where she will open the shop (1) • Data is obtained/collected with the set purpose in mind (1) TWO from age: • The questions are being asked now (1) • The age of the data is a measure of how long ago it was collected (1) • Use of a direct data source means that data can be/is more up-to-date (1) • Patterns of behaviour/attitudes change over time//what was correct (e.g.) twenty years ago is not necessarily correct now (accept converse about direct data) (NOT 'things could have changed' – TV (1)	4	

Question	Answer	Marks	Guidance
1(c)	Four marks available. Two per matched explanation: Identify reason (1) and state why it is a reason/why beneficial (1) Areas for consideration (Note: may also be used as expansion. Mark as candidate's intension) • Larger sample size (must be the concept of more data/more people asked) • Less biased/more accurate/more reliable NB accuracy or reliability could be used as an extension of (e.g.) less biased for two marks • Wider variation/type of people • Seasonal nature of some sports • Convenience • No time to collect data so use someone else's • Avoidance of a cost (1) such as (1) • Quicker to achieve results (of analysis) Exemplar answers • The sample size may be larger (1) therefore the results gathered are less likely to be biased (1) • She may not have the time to collect the data (1) and so uses data gathered by someone else (1)	4	

Question	Answer	Marks	Guidance
4(a)	FOUR from: • Both methods check that the information entered matches the <u>original/the book/the source</u>//has been inputted correctly (1) Accuracy • Neither method checks if the source data is correct (1) • Double data entry is more accurate (accept converse) (1) • Visual checking may be affected by human error//checker may miss error//double data entry does not have error (when checked) (1) Who does it • Visual checking is carried out by a person/user/manually//For double data entry, it is done by the computer (1) • For double data entry, the user is only involved if an issue is found (1) Speed • Double data entry is a slower process (accept converse) (1) General • For visual checking, the data is only entered once//for double data entry the data must be entered twice (1) • For double data entry, a second copy is temporarily created (1)	4	

Question	Answer	Marks	Guidance
4(b)	THREE marks available for ONE identified method and suitable description - Parity check (1st) Counts the numbers of 1s in (each byte) of data and adds a 1 if it is an uneven number/adds a zero if it is an even number (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Checksum (1st) Uses a hash function (1) Counts the number of bytes/bits in a file//takes file size and produces a number (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Hash total//hash sum (1st) A number is calculated from the content of each field of data. (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1) - Control total (1st) A number is calculated from the fields holding numerical data. (1) Concept of recalculating and checking again (1) Reports an issue if the recalculated number does not match the original//if the recalculated value matches the original there is no issue with the transfer (1)	3	Do not accept: - Visual checking - Double data entry

Question	Answer	Marks	Guidance
5(a)	FOUR from: STRUCTURE - In a flat-file database, there will be one table// in a relational database there will be more than one table (1) ACCURACY (accept converse for each) - A relational database is less likely to have data entry errors (1) (For the same data,) a relational database will be smaller//save storage space (1) THE DATA - A relational database will have less duplicated data/redundant data (1) OR - a flat-file database can have the same data repeated (many times)//have lots of redundancy (1) - Each record in a relational database only has to be entered once//in a flat-file database, the same data must be entered every time it is needed (1) - Any mention of data integrity being applied to a relational database (1) TIME (accept converse for each) - Relational databases take less time to maintain/edit the data (1) - Relational database takes longer to edit the structure (1) - Querying a relational database is faster than querying a flat-file databases (1) COMPLEXITY (do not accept converse) - Relational database is more complex (1) CAN ALSO AWARD IF STATED SEPARATELY - Relational database can create complex queries (1) - Relational database can create complex reports (1)	4	Do Not accept - Less time/more time to plan/make - Easier to create/update - Any reference to files rather than tables

Question	Answer	Marks	Guidance
5(b)	Three marks available: Structure: - Identifies the validation type (1st) - Describes a rule (1) - States why it is important//why is it used//what error does it stop (1) e.g. - Presence check (1st) Each field has to contain an item of data//cannot be left empty (1) To check that data has been entered into the field//because the data is necessary (1)	3	

Question	Answer	Marks	Guidance
8	FOUR from: Batch processing • All processing is done at the same time//processing is a single process//all at once (1) • Done at a fixed time (1) • Likely to be at the end of the day/when the supermarket has closed (1) • No action occurs until sometime after the sale (1) Real time processing • Real-time processing is done immediately (1) • As soon as an item is sold (1) • Supermarkets can order immediately stock levels become low//allows just in time stock control (1) General • Information for real-time processing is (more) up to date (1) • If the supermarket does not sell out before the close each day, both methods are equally effective (1) • If the supermarket sells out before the close each day, real-time processing allows items to be ordered immediately (and restocked that day) (1) • Errors are not identified immediately with batch processing (accept converse) (1) • Data may be lost (between sale and processing) for batch processing//data is less likely to be lost (1)	4	Accept converse arguments for the general section ONLY

Question	Answer	Marks	Guidance
1	THREE marks available: TWO from: - Data consists of raw facts and figures which require processing for interpretation (1) - Data does not have meaning (1) - Information is data that has a context (1) Allow converse for the above ONE mark for: - Use of ONE example to describe BOTH data and information (clearly stated as such) (1)	3	

Question	Answer	Marks	Guidance
3	FOUR from: Identify a benefit and state why it is a benefit OR cause of benefit Benefit e.g. - Data is safer/protected/secure (1st) - Stolen data cannot be understood (1st) - Reduced desire to hack (1st) - Data is unusable if stolen/accessed (1st) Explanation e.g. Any suitable explanation	4	

4(a)	TWO marks available: • High quality data is (more) accurate/reliable/detailed/relevant/up-to-date/complete//data needs to be/should be accurate (1) • and decisions based on them are more likely to be correct//decreases risk of making an incorrect decision (1) • provides better insights into shopping habits (1) Answer may be focused on one factor: e.g. • Data must be up-to-date (1) because trends change over time /you know what is popular now (1) ACCEPT CONVERSE of these arguments where applicable	2	
------	---	----------	--

Question	Answer	Marks	Guidance
4(b)	EIGHT from: - The report holds information that may be useful, but there are reasons why it may not be (1) Three marks may be awarded where relevant issues are identified, but not explained, where no other marks would be awarded overall other than Mark Point 1 (above). Benefits (Max six) e.g. - It is data that has been collected for the similar purpose (1) - The data already exists (1) so there is no time required to gather the data (1) decision can be made more quickly (1) - The data has already been organised (1) therefore it is not raw data, so no organisation required (1) - Data will be complete/lots of detail (1) - It is from a larger sample (1) removes bias (1) - Allows access to data that may be (currently) impossible/difficult to gather (1) COSTS - The costs of gathering the data (to the shop owner) are zero/minimal/reduced (1) - Identified saving cost of gathering data (1)	8	Answers must be linked to using the data to identify shopping habits. Do Not Award - Data is untrustworthy - Data is up-to-date

Question	Answer	Marks	Guidance
4(b)	Drawbacks (Max six) • The data sample is national (1st) therefore not focussed enough//irrelevant (1) • The data is old// is now out of date (1) so trends//shopping habits may have changed (1) • Report may be biased (NOT 'false') (1) Conclusion One mark for a value judgement/conclusion based on two arguments (positive or negative or both) (1)		

2 Hardware and software

Questions

9626/12 Feb/March 2022 · Question 2

[Mark scheme](#)

- 2 An interpreter is a type of system software.

Describe how an interpreter operates.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9626/12 Feb/March 2022 · Question 3

[Mark scheme](#)

- 3 Mainframe computers are particularly suited to carrying out the transaction processing of online banks.

(a) Describe the features of a transaction processing system.

.....

.....

.....

.....

.....

.....

.....

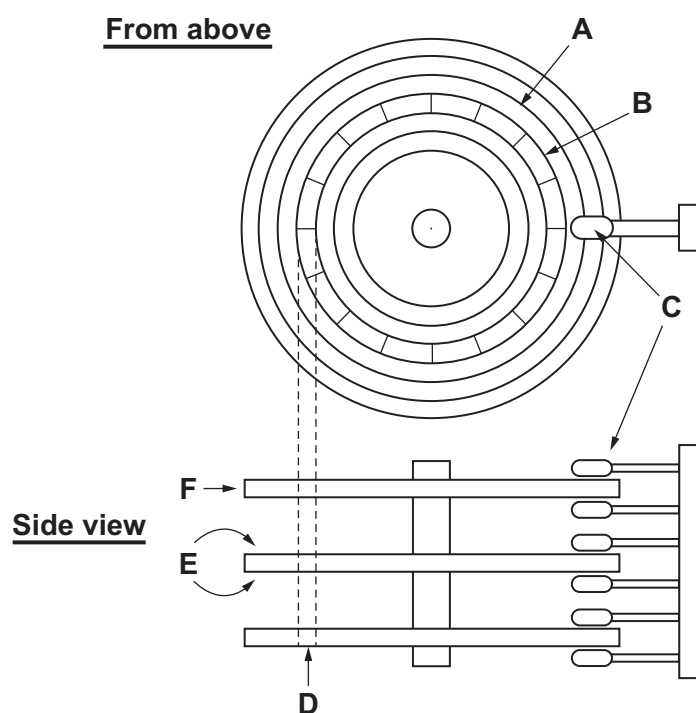
.....

..... [3]

[8]

[8]

5 Here is a diagram showing the component parts of a hard disk drive.



(a) Describe the component parts of a hard disk drive. Refer to the labelled diagram in your answer.

[6]

(b) Explain what is meant by disk defragmentation and why it is needed when using hard disks.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

11 A gesture based interface is a type of user interface.

Describe the advantages of using a gesture based interface rather than other types of interface.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- 4 The performance of mainframe computers and supercomputers is often measured using performance metrics.

Compare, by describing the similarities and differences between them, the different types of performance metrics.

[6]

- 5** When a high level language program is written, it has to be translated so the computer can run it. This can be achieved using a compiler or an interpreter.

Discuss the advantages and disadvantages of using a compiler rather than an interpreter.

[8]

- 2 Explain how high level language programs can be translated to run on different computer systems, including any difficulties which may result from this.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 4 The directors of a company wish to create a database of customers. They are not sure whether they should ask a computer programmer to produce custom written software or they should purchase off-the-shelf software.

(a) Describe the features of custom written software.

[4]

(b) Describe the features of off-the-shelf software.

[4]

- 5** A doctor drives to work. While at work and while travelling, the doctor uses a number of IT-based devices. She uses a dialogue interface to control these devices.

Discuss the advantages and disadvantages of the doctor using a dialogue interface rather than a graphical user interface (GUI).

[8]

- 1 Computers often have utility software supplied with the operating system.

Explain what is meant by utility software and why it is needed.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 8 Maria has been asked to write a word-processed report for her homework. She needs to choose between using proprietary software and open source software.

Describe the differences between these two types of software.

[6]

- 2** Describe what is meant by the term RAS (Reliability, Accessibility and Serviceability) when applied to mainframe computers.

..... [5]

- 6** Explain why supercomputers are used for weather forecasting.

[4]

3 Back-up software and anti-virus software are both types of utilities that are often provided with operating systems.

(a) Without detailing the different methods of detection, describe the features of anti-virus software.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Without referring to the different types of back-up, describe the features of back-up software.

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 6** The IT department manager in a company has decided new software is needed to run the payroll system. She will make a choice between using off-the-shelf software and custom written software. She is already aware that there are differences in the level of support offered and the amount of testing that will have been carried out on the software.

Discuss the other advantages and disadvantages of choosing off-the-shelf software compared to choosing custom written software.

[8]

- 2** A linker is a type of system software used during the process of translating high level language programs to machine code.

Discuss the benefits and drawbacks of using a linker.

[6]

- 3** Most computer systems have a variety of utility software including anti-virus software.

Discuss the benefits and drawbacks of anti-virus software.

[6]

- 3 Explain why a compiler may not be suitable for translating a high level language program for use on different computer systems.

[3]

- 6** Back-up software is a type of utility software. A network manager wants to back up the data stored on a server.

(a) Describe, with reasons, the steps the manager takes when setting up this process.

[5]

- (b)** Some of the data to be backed up consist of audio files. The network manager will compress these before starting the back-up process. Different methods of compression can be used.

Describe these methods giving the effects on the audio files of using each method.

[4]

5 A printer driver is a type of device driver.

Explain, using a printer driver as an example, what is meant by a device driver and why it is needed.

[5]

- 4 One of the functions of an operating system is to maintain the security of a computer system.

Describe the **other** functions of an operating system.

[8]

- 9 Discuss the advantages and disadvantages of using a gesture-based user interface rather than a graphical user interface to control devices.

[8]

3 Companies have the choice of using custom written software or using off-the-shelf software.

Explain the differences between custom written software and off-the-shelf software in terms of:

(a) cost

.....

.....

.....

.....

.....

..... [2]

(b) level of support

.....

.....

.....

.....

.....

..... [2]

(c) degree of testing.

.....

.....

.....

.....

.....

..... [2]

12 Analyse the effectiveness of supercomputers.

[8]

2 Mainframe computers are used in banking. Supercomputers are used in climate research.

(a) Explain what is meant by a mainframe computer and by a supercomputer with reference to the types of processing and the processors they each use.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Explain how supercomputers are used in climate research.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

3 Give brief descriptions of the following types of user interface.

(a) command line interface
.....
.....
.....
..... [2]

(b) gesture based interface
.....
.....
.....
..... [2]

(c) dialogue interface
.....
.....
.....
..... [2]

- 3 Mainframe computers and supercomputers are specialised types of computer with some similar characteristics.

Explain what is meant by each of the following characteristics.

(a) fault tolerance

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) heat maintenance

.....

.....

.....

.....

.....

..... [2]

Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census.

(c) Explain why mainframe computers are used to process the data from a census.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 10** The manager of the payroll department within a company has commissioned a computer programmer to write software to run the payroll.

Justify this decision to use custom written software.

[8]

Please turn over for Question 11.

3 Keith wishes to ensure that his computer is at its optimum performance.

Identify and describe **three** different types of utility software he should use.

Utility 1

.....

.....

.....

.....

.....

Utility 2

.....

.....

.....

.....

.....

Utility 3

.....

.....

.....

.....

.....

[6]

6 Some computer systems have a command line interface (CLI).

Evaluate the use of a CLI.

[6]

7 The IT manager of RockICT can purchase either off-the-shelf software or custom written software.

(a) Describe **three** advantages for RockICT of purchasing off-the-shelf software.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe **three** advantages for RockICT of purchasing custom written software.

.....

.....

.....

.....

.....

.....

.....

..... [3]

7 Compilers and linkers are types of system software.

(a) Describe the role of a compiler.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe the role of a linker.

.....

.....

.....

.....

.....

.....

.....

..... [3]

2 Describe how supercomputers are used for weather forecasting.

[6]

- 9** Discuss the advantages and disadvantages of using a dialogue interface rather than a gesture-based interface.

[8]

1 Utility software is needed to maintain a computer system.

Explain why each of the following tasks is needed and how utility software performs each task:

(a) disk formatting

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) data compression

.....

.....

.....

.....

.....

.....

.....

..... [3]

(c) disk defragmentation

.....

.....

.....

.....

.....

.....

.....

2 Describe the following characteristics of mainframe computers:

(a) volume of input, output and throughput

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) security

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 7 Discuss the benefits and drawbacks of the use of a compiler to convert a computer program in a high level language into a low level language.

[8]

9 Utility software is a type of system software designed to help optimise, manage and maintain the performance and health of computer systems.

(a) Explain how the delete utility works.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Explain why files can still be recovered even after they have been deleted.

.....

.....

.....

.....

.....

..... [2]

- 5** A manager can either choose off-the-shelf software or custom-written software for their company.

Explain why the manager would choose off-the-shelf software.

[6]

- 11 Discuss the benefits and drawbacks of using an interpreter to convert a program written in a high-level language into a low-level language.

[8]

- 4 A company finance director could use either off-the-shelf or custom-written software to help manage the company's finances.

Contrast these two types of software.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 10** A person uses a dialogue interface to control multiple devices in their home.

Evaluate the use of a dialogue interface in this context.

[6]

- 12** Banks process a large number of transactions simultaneously. They use mainframe computers for transaction processing.

Explain why mainframe computers are used by banks for this type of processing.

[3]

8 Interpreters and device drivers are types of system software.

(a) Describe the functions of device drivers.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Justify the use of an interpreter.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 A student is considering defragmenting their hard disk.

(a) Describe the benefits of defragmentation.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe the drawbacks of defragmentation.

.....

.....

.....

.....

.....

.....

.....

..... [3]

6 Describe what utility software is and why it is needed on a computer system.

.....

.....

.....

.....

.....

.....

.....

..... [3]

9 A programmer is developing a new piece of software. The programmer has decided to write code that needs to be interpreted so that it will run.

Justify the programmer’s decision.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

11 Describe **four** characteristics of a supercomputer.

- 1
-
-
- 2
-
-
- 3
-
-
- 4
-
-

[4]

12 Describe what is meant by the term MIPS.

-
-
-
-
-
-
- [2]

10 Describe **four** characteristics of a mainframe computer.

- 1
-
-
- 2
-
-
- 3
-
-
- 4
-
-

[4]

11 Describe what is meant by the term FLOPS.

-
-
-
-
-
-
- [2]

12 Describe how an operating system manages data on a hard disk.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

3 A consultant has suggested that a gesture-based interface could be used as part of a security system on the door of a government office. The office is only used by staff.

(a) Describe how a gesture-based interface can be used to interact with the security system.

.....

.....

.....

.....

.....

..... [2]

(b) Evaluate the use of a gesture-based interface as part of the security system for this office.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 4 Mainframe computers and supercomputers have differing characteristics.

Using **three** different characteristics, describe differences between mainframe computers and supercomputers.

characteristic 1

.....

difference

.....

.....

characteristic 2

.....

difference

.....

.....

characteristic 3

.....

difference

.....

.....

[6]

- 9 Wei has recently started an accountancy business. She is considering using either custom-written or off-the-shelf software in the business.

Discuss the suitability of each type of software.

[8]

- 9** An international airport has installed a computer system to record customer satisfaction. The system uses a graphical user interface (GUI) involving the use of graphical icons.

(a) Justify the use of the GUI for this computer system.

[6]

[6]

- (b)** Identify and explain **two** reasons why a dialogue interface may **not** be suitable for this system.

[4]

[4]

- 9 One type of utility software is anti-virus software. This detects viruses and malware that may damage files on a computer system.

(a) Define the term utility software.

..... [1]

(b) Identify **one** other type of utility software that protects files on a computer system.

..... [1]

- 5** Supercomputers can be used for weather forecasting.

Discuss, with the example of weather forecasting, the benefits and drawbacks of using a supercomputer.

[6]

6 Disk defragmentation is a type of utility software.

(a) Describe what is meant by disk defragmentation.

.....
.....
..... [1]

(b) Explain why disk defragmentation is less likely to be used in modern computers.

.....
.....
..... [1]

Back-up and file deletion are also types of utility software.

(c) Other than disk defragmentation, back-up and file deletion, identify and describe the specific purpose of **two** types of utility software.

type 1
.....
.....
.....
.....
.....
.....
type 2
.....
.....
.....
.....
.....
.....
.....
..... [4]

2 Hardware and software

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Four from: Translates a high-level language program... ...into an intermediate stage Translates a line of source code and then executes that line/statement Translates instructions one at a time Reports on errors as lines of source code are processed An interpreter has to be resident in memory in order for the program to run Only a few lines of the program need to be in memory at any one time	4

Mark scheme · 9626/12 Feb/March 2022 · Question 3

[Back to question](#)

Question	Answer	Marks
3(a)	Three from: A transaction processing system's performance is measured by the number of transactions it can process in a given period of time The system must be continuously available The system must be able to maintain the integrity of the data and overcome hardware or software problems It must also be possible to individually upgrade hardware and software components without shutting down the system There should be controlled access, allowing only authorised users to use the system	3

Question	Answer	Marks
3(b)	Eight from: <i>Advantages</i> They can perform hundreds of MIPS/have faster processing than PCs, laptops and other devices. They have a high value of mean time between failures (MTBF) so usually do not have any time where the system is downPCs/laptops/other computing devices can crash more often They have greater fault tolerance than other systems... ...they can transfer processing from one core to another... ...so hardware and software upgrades can occur while the system is still in operation They can run different operating systems so can handle different types of database They are more secure than most other types of computer as they use complex encryption systems. <i>Disadvantages</i> They are very expensive to buy/installsome organisations are changing to using the cloud rather than buying a mainframe... ...there is a higher cost for the people needed to manage them... ...which is not the case with cloud-based computing The software required to run a mainframe is more expensive to buy than using cloud computing Mainframes need large rooms to accommodate them They generate a lot of heat... ...cooling systems are expensive to buy, install and run Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
5(a)	Six from: A is a track, B is a sector, C are read-write heads, D is a cylinder, E are surfaces, F is a platter (3 marks for all 6 correct, 2 marks 5 correct, 1 mark for 3 or 4 correct) A hard disk drive consists of several platters which are individual disks Each surface of a platter has its own read/write head The read/write heads move across the platters/surfaces stopping only to read <u>data</u> from or write <u>data</u> to the surface... ... never touching a disk surface each of which is used to store data Each surface is divided into several tracks which are in the same position on each disk The track on the top platter together with the tracks exactly below it, form a cylinder Each track is divided into sectors	6
5(b)	Four from: Organises the contents of the disk into the smallest number of contiguous regions/fragments... ...by moving the data blocks of a file/defragmented files around to bring whole/all the parts of a file together Attempts to create larger regions of free space Some defragmentation utilities try to keep smaller files within a single directory together It is the rearranging of files stored on a disk The movement of the hard drive's read/write heads over different areas of the disk when accessing fragmented files is slower... ...compared to accessing the entire contents of a non-fragmented file sequentially... ...data retrieval is made easier and quicker	4

Question	Answer	Marks
11	Five from: Gestures may be a quicker way of initiating a response from a device People who have speech impediment can find a gesture based interface useful rather than dialogue Gestures will still be reliable if there is background noise or user has a cold/strong dialect/accent which would affect a dialogue interface Dialogue interface often requires a training session with user whereas gestures can be taught through manuals Users with physical handicaps may find it easier to make gestures rather than gripping a mouse Don't have to learn so many commands as with a CLI For reasons of hygiene doctors/health workers/restaurant chefs may not be allowed to touch a display or device using a GUI... ...appropriate gestures are a more hygienic way to control the device	5

Question	Answer	Marks
4	Six from: <i>Similarities</i> Both are used to measure performance speeds of computers/processors Both can be unreliable as the complexity of instructions can vary according to which <u>benchmark</u> program/test is used/complex instruction takes longer to process than a simple one <i>Differences</i> MIPS are used with mainframes (M)FLOPS used with supercomputers MIPS can be used with integers (M)FLOPS with real/floating point numbers MIPS measured in millions (M)FLOPS measured in quadrillions and higher MIPs is millions of instructions per second, FLOPs is floating point operations per second (must have both) MIPS used when measuring performance of computers involved in running application software... ... whereas FLOPS used where scientific/complex calculations need to be carried out MIPS do not take into account input/output speeds... ... whereas FLOPS do not take into account CPU's clock speed , bus speed, the amount of RAM available Must have at least one similarity and difference to gain full marks	6

Question	Answer	Marks
5	Eight from: <i>Advantages</i> Once it is compiled a program does not have to be translated again A compiled program can run faster than interpreted software Compiling a program is a much quicker process /translates it faster than interpreting... ... as whole program is translated all in one go /all at onceunlike interpreter translates one line at a time Once the program is already compiled into machine language, there is no translation software that the user has to keep up to date Compilers also improve security for programmers... ...makes it much harder to copy code/machine code is difficult to understand and alter... ...so fewer risks of copyright infringement... ... with an interpreted program, the source code can always be seen so greater risk of software copyright infringement <i>Disadvantages</i> Can only be used on one operating system ... source code can be interpreted on any operating system provided interpreter is present... ...will need to use an interpreter if program is to be shared... It is slow/(more) difficult to debug the program/find errors... ...the whole program needs to be compiled again after an error is corrected... ...error messages are output at the end... ...whereas with an interpreter error messages are output <u>as they are found</u> It uses up more memory than an interpreter... ...as whole program has to be loaded... ...interpreter only requires a few statements of the program to be in memory Unlike an interpreter, does not allow small pieces of code to be tested to make sure they work before continuing with the rest of the program More likely to crash the computer as it is running directly on the CPU Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
2	Four from: An interpreted program will still <u>be in its original source code</u> so it will work on <u>any system</u>... ...but must have an appropriate interpreter Only use high level languages which can be interpreted With languages which are compiled can use a cross compiler... ...so programs can run on a computer with a different operating system/target computer... Cross compilers can be a smaller version of the compiler which is normally used on the host computer/native compiler A problem with using a cross compiler is that the compiled program will no longer run on the computer the program was written on/host computer The cross compiled code can run more slowly than if it had been originally compiled on the target machine A cross compiler produces more errors and mistakes than a native compiler	4

Question	Answer	Marks
4(a)	Four from: Software that is specially developed for a specific company/business It is made to accommodate that customer's particular preferences and needs Written by programmers to solve <u>specific problems</u> Owned by the business that commissions it It is made from scratch Support is obtained directly from the creators Development costs have to be paid for by client for whom its written/ commissioning company/business	4
4(b)	Four from: Software that is ready-made/ready for use and already exists/can be used immediately It is available to all businesses and companies/general public/everyone It is owned by the company that created it/users only have a licence Has to be adapted to fit the business that has purchased it Development costs are spread amongst several customers Support is obtained from help desks/call centres/user forums run by the producers of the software	4

Question	Answer	Marks
5	Eight from: <i>Advantages</i> For reasons of hygiene a doctor may not be allowed to touch a display or device... ...speaking into a microphone is a more hygienic way to control the device Dialogue interface allows hands free control / ensures safety when driving... ...whereas GUI would need at least one hand to leave the steering wheel Dialogue interface does not require doctor to be sat in front of computer/device so can operate computer/device remotely/allows doctors to be doing other things/tasks at the same time/multitask Users with physical disability may not be able to use a keyboard or mouse... ...or control their limbs accurately... ...could speak into a microphone to control a device <i>Disadvantages</i> Some background conversation may result in a reaction from the device... ...GUIs more likely to require actions to be deliberate Doctors who have a speech impediment may find a dialogue interface too difficult to use Dialogue interface may not operate reliably if there is background noise... ...commands might be misinterpreted/misheard Dialogue interface may not operate reliably if doctor has a cold/strong dialect/accent... ...while GUI will still be reliable Dialogue interface often requires a training session with user/doctor... ...use of GUI can be taught through manuals Dialogue interface can be more expensive to develop compared with a GUI Dialogue interface is not suitable for safety dependant commands such as 'stop' as it would not be as easily understood as GUI Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
1	Three from: • A program that helps perform maintenance (in a computer system) (1) • A program that performs a specific task ... (1) ... usually related to managing system resources (1) • Operating systems can contain a number of different utilities/types (for managing files and storage devices) (1) • Identified suitable example (e.g. file management/storage device) (1) One from: • Generic statement of why utility software is required (1) e.g. • It is needed to keep the computer free from viruses (1) • It is needed to make the files contiguous (1) • It is needed to improve the computer's performance by managing the allocation of computer memory (1)	4

Question	Answer	Marks
8	Six from: • Proprietary software is owned by the copyright holder//not owned by the user (1) • Proprietary software users have to purchase a licence/pay to use it ... (1) ... open source software is usually free to download// includes a request for a donation to charity (1) • There are almost always restrictions on the use of proprietary software ... (1) ... open-source software can be used/accessed/implemented by anyone (1) • The specific terms of use are included in an end-user license agreement (1) • The source code of proprietary software is a secret ... (1) ... open-source software source code is open to inspection (1) • Usually covered by copyright which provides a legal basis for its owner to establish exclusive rights ... (1) ... open-source software code can sometimes be amended without breach of copyright legislation (1)	6

Question	Answer	Marks
2	Five from: - Describes the robustness/durability of mainframe computers (1) Reliability is the chance of the system behaving as intended (1) Accessibility refers to the mainframe being operational at all times//the operating time as a percentage of total time (1) Serviceability means it can be easily fixed within a short period of time (1) - Mainframes are reliable as the system will use alternative methods to carry out tasks (if there is a failure) (1) - The system's software is reliable as it is extensively tested and updates are made quickly to overcome problems (1) - Mean time between failures (MTBF) is quite long for mainframes (1) - Mainframes give months/years between system downtimes (1) - Even when unavailable (due to failure) a mainframe is unavailable for a very short time (1) - Spare components/CPU are often included in mainframes so that if one component/CPU fails another takes over (1) - The mainframe can determine why a failure occurred (1) - This enables hardware and software components to be replaced without affecting the mainframe's operations (1) - Can be serviced while the system is still running (1)	5

Question	Answer	Marks
6	Four from: - A supercomputer is fast/powerful/run at several petaflops enough to carry out the calculations that are required for weather forecasting (1) - A supercomputer can process huge amounts of data// a supercomputer is needed because a huge amount of data is collected (1) - Weather forecasting is based on the use of very complex/scientific computer models (which only supercomputers can handle) (1) - A sophisticated/three-dimensional model of the earth's atmosphere/ocean/space has to be created (which needs the processing power of a supercomputer) (1) (A supercomputer is needed) because a large number of calculations have to be carried out//a supercomputer can do a large number of calculations (1) - Needs to do complex/scientific calculations (1) Max two marks for the data that has to be collated: e.g. - records of previous weather conditions (1) - weather measurements from weather stations (candidate may exemplify) (1) - observations from radar/satellites (1) - information from ships /aircraft (1)	4

Question	Answer	Marks
3(a)	Three from: • Scans the computer to make sure it isn't infected with a virus/to find viruses (1) • Can scan whole disk or individual files (1) • Can be set up to scan automatically//run at a set time (1) • Compares with a database/existing viruses (1) • Detects changes in behaviour of files (1) • Quarantines the infected programs (1) • Reports results of scan to user (1) • It will ask the user whether or not they want to delete the infected programs (1) • Does background scans of downloads and attachments (1) • Automatically update itself (1)	3
3(b)	Three from: • Creates additional copies of files/data/databases/hard disks/ network servers (accept 'duplicates') (1) • Backed up//copy of (e.g.) files are still available in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) • Can use these copies to restore the original contents in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) • Asks user to enter type of back-up (1) • Can happen automatically (accept reverse argument) (1) • Asks if you wish to restore the back-up (1) • Asks if you wish to verify the back-up (1) • Asks when back-up is to take place/frequency of backups (1) • Ask which medium the user wishes to store/restore backup (1)	3

Question	Answer	Marks
6	Eight from: Advantages <i>Max six from</i> - Off-the-shelf software is cheaper to buy (1) ... as it is mass produced/cost is spread between customers (1) ... whereas with custom written software it costs more to pay programmers to write code specifically for the task/user's needs ... (1) ... written for one purpose/don't share development costs with other users (1) - Off-the-shelf software is already available//available <u>straight away (or very near equivalent)</u> ... (1) ... whereas with custom written software client would need to wait whilst the software was being developed (from scratch) (1) Disadvantages <i>Max six from</i> - Off-the-shelf may need to be adapted for use which can be difficult ... (1) ... there may be settings within the software that cannot be changed ... (1) ... whereas custom written software is designed specifically for the task/company requirements (1) - Off-the-shelf software may not have all the features//may not meet needs of the/ required by the company ... (1) ... whereas with custom written software the programmer can ensure the software will have all the features that are required (1) - Off-the-shelf software may have unnecessary features ... (1) ... whereas the programmer will make sure that custom written software has <u>only</u> the features required (1) - Off-the-shelf software may not be fully compatible with the company's computer system ... (1) ... whereas the programmer will have taken the computer system into account making sure that the software is compatible with it (1) Max six marks if bullets/list of points Max six marks if no attempt at expansion	8

Question	Answer	Marks
2	Up to Six marks available: Benefits Max 5 - Programs can be written in modules (1) ...which requires less RAM (1) ... so saving cost of memory (1) - Whole program and compiler don't need to be in memory at the same time (1) ...which requires less RAM (1) ... so saving cost of memory (1) Both extension marks above may be given twice (once per first mark item) where used appropriately. - A number of programmers can be used to write separate modules ... (1) ...which saves time compared to one person writing the whole code (1) - If there is an error in the code only that module has to be corrected (1) Drawbacks - Variable names can cause problems... (1) ...the same variable may have been given different variable names in different modules (1) - Documentation has to be more detailed ... (1) ... so takes longer to write (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
3	SIX marks available: Benefits (MAX 4) - Protects from computer viruses/malicious code (accept either 'finds' or 'stops new infections' (1)) - Prevents data/files (in files) becoming corrupted/deleted (1) - Can prevent passwords from being changed (1) - Can prevent user's email account sending out spam (1) - Can prevent computer from crashing (1) - User can select how regularly the anti-virus scan is to take place (1) - User can select which files to scan (1) Quarantines any infected files (removes BOD) (1) Drawbacks (MAX 4) - Anti-virus software can cost//has to be purchased (any awareness of cost or expense) (1) - Only purchase for one year (1) - Anti-virus needs to be kept up to date ...(1) ...because there always new virri to be countered (1) - Anti-virus can only detect a virus if it is listed in the AV database (1) - Anti-virus background scans can slow a computer down//slow down download process (1) - Can give a false sense of security/does not give 100% coverage (1) - Can result in false positives (award as a concept, not necessarily these words) ...(1) ...what appears to be suspicious activity turns out to be harmless (1) - Needs expertise to personalise (1) Accept examples such as – "Can take a long time to customise account/settings". - Takes up space on the hard drive (1) - Can only delete the whole file that is infected (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
3	Three from: - A compiled program will only run (on a computer with) the same operating system (1st) as it was originally translated on (1) - The (other) computer system it is transferred to may not have the same operating system (1) - Cross compilers can be used but the compiled program will no longer run on the host computer/computer the program was written on (1) - Cross compiler produces more errors and mistakes than a native compiler (1) - The cross compiled code can run slower than if it had been originally compiled on the target machine (1)	3

Question	Answer	Marks
6(a)	Five from: • Manager checks the space on the target drive/location (1st) – to make sure there is sufficient space for the data (that will be moved) (1) • Manager selects the time they want the back-up to take place (1st) – manager can select to carry out the back up immediately... (1) – BECAUSE the back-up takes place at a time convenient to the manager/when the system is quiet (may explain why timing is convenient) (1) • Manager selects type of back-up (1st) – backing up only the specified data/files (1) – BECAUSE may want to back up whole device OR new files (1) • Manager selects how regularly the back-up is to take place (1st) – BECAUSE to minimise data loss (1) • Manager will decide where back-up is to be stored (1st) – BECAUSE to make sure back-up will be secure (1) • Manager selects whether to verify the back-up or not (1st) – BECAUSE to make sure the backup has been completed correctly (1) • Manager chooses whether or not to encrypt the back-up (1st) – BECAUSE adds an extra level of security (1) Must have at least TWO reasons to obtain full marks	5

Question	Answer	Marks
6(b)	MAX three from: • Can be either lossy or lossless. (1) • Lossless compression reduces number of bits (1st) – ...by identifying repeated patterns of data. (1) • No information is lost in <u>lossless</u> compression. (1) • <u>Lossy compression</u> reduces number of bits (1st) – ...by identifying unnecessary information. (1) Effects max three: • Quality of audio is the same when using <u>lossless compression</u> (1) • Quality of audio is lower when using <u>lossy compression</u> (1) • File size is smaller for lossy than lossless (accept converse) (1)	4

Question	Answer	Marks
5	Five from: Four max from: • A device driver is a piece of (system) software (1) • It operates/controls a device attached to the computer (1) • Allows communication between printer and computer (1) • A device driver tells the operating system/application software how to communicate with a hardware device (1) • A printer driver acts as an interface between the operating system/application software and the printer (1) • When a document is to be printed the application tells the printer driver and the printer driver tells the printer (1) • The user appears to have control of the device (1) • Upon installation it detects and identifies the peripheral device (1) • Handles the translation of requests between a device and the computer (1) • Defines where outgoing data must be stored before it can be sent (1) Two max from: • Without the required (printer) driver the printer fails to work (1) • The software used by a computer tends to be created by different companies to those that manufacture printers (1) • Computers (may) use software which uses a different type of language/instruction set to the printer (1) • The printer driver is needed to convert the language/instruction set so that the software is able to communicate with the hardware (1)	5

Question	Answer	Marks
4	Eight from: • Manages/operates/controls computer hardware resources (1) • Manages software resources (1) • Interacts with device drivers (1) • Responds to input devices/recognises inputs (1) • Sends data/instructions to printers/output devices (1) • Manages files/directories (1) • Responds to inputs (1)// interface between the user and the computer (1) • Allocates memory to software (1) • In multi-tasking/multi-programming systems allocates equitable processing time to each task/program (1) • Supplies utilities for managing the computer – (may state or exemplify) (1) • Outputs to hardware//interface between an application program and the computer hardware (1) • Knows the names of each file and (exactly) where they are stored (on the storage media) (1) • It keeps a record of any empty spaces on the medium to know where new files can be stored (1) • Displays error messages to the user if an error occurs (1) • Opens and closes files (on storage devices) (1) • Allows communication between attached hardware (1)	8

Question	Answer	Marks
9	Eight from: Advantages of GBI/disadvantages of GUI • Can be advantageous to disabled users ... (1) ... who maybe cannot control their hands accurately//have limited mobility (1) ... could make gestures using their limbs/eyes (1) • For reasons of hygiene a doctor/food worker may not be allowed to touch a display or device ... (1) ... gestures control the device/ allow them to control devices without touching anything (1) • Gestures may be a quicker way of initiating a response from a device (1) • Some gestures may be unintentional but still initiate a reaction from the device (1) • Gesture based interface does not require person to be sat in front of computer/device so can operate computer/device from a distance (1) • Enhanced gaming experience (1) Disadvantages of GBI/advantages of GUI • GUIs tend to be more accurate than gestures (1) • Certain gestures may not be socially acceptable/user could feel uncomfortable making gestures//reduced privacy (1) • Some gestures may be unintentional but still initiate a reaction from the device//GUIs require actions to be deliberate (1) • Gesture based is less effective when several users/background movement involved (1) • In certain lighting conditions gesture-based interface may misinterpret/fail to register a gesture (1) • GUIs can be used in any lighting/dark situations (1) • Increased danger of injury (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
3(a)	Two from: - Companies have to pay the programmers to write the programs which could cost more (than buying off-the-shelf software) (1) - Off-the-shelf software is cheaper to buy because it is mass-produced (1) - With custom written software one company pays for the development because they commissioned it (1st) - with off-the shelf software development costs are spread amongst several customers because it is written for a mass market/generic software (1)	2
3(b)	Two from: - With custom written software there may be only one programmer/small team of programmers so companies find it difficult to get support (1) - With custom written software, help can be obtained directly from programmers so can be more helpful (1) - Help desks/call centres/user forums websites for customers are readily available because software is generic//used by a mass audience (1)	2
3(c)	Two from: - Custom written software may not have been tested as thoroughly/rigorously as off-the-shelf software as fewer users/shorter lead time (1) - The tests that are carried out on custom written software are those which the programmer thinks are necessary based on how they think the software will be used which not necessarily be how it is eventually used (1) - Off-the-shelf software will have been used many times before companies buy it so any bugs will have been identified and fixed (1)	2

Question	Answer	Marks
12	Eight from: • Have a limited lifespan of just a few years (1st) • and will need replacing/repairing (1) • Carries out complex calculations (1) • Can use complex mathematical formulae (1) • Can handle a large amount of items of data at the same time (1) • Are the fastest data processing computers (1) • Are the most expensive computers to buy/install (1) • Each/processor core(s) each have their own operating system (1) • Use massively parallel processing (1) • Are more powerful compared to mainframes (1st) • can be up to a 1000 times faster (1) • Can include scientific rules (1) • Some tasks do not require super computers (1st) • as the task is not complex, so it would be a waste of e.g. resources (1) • Very likely to overheat (1) (and) difficult to keep cool (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
2(a)	MAX two from: • A mainframe computer can have hundreds of (processor) cores (1) • All cores share//has one Operating System (1) • Mainframes use parallel processing (1) • Can process a large number of small tasks at the same time (1) MAX two from: • Supercomputers can have more than 100 000 (processing) cores (1) • Supercomputers use massively parallel processing (1) • Each/Processor core(s) each has their own operating system (1) • Supercomputers use more than one GPU (1)	3

Question	Answer	Marks
2(b)	Four from: - Many (types of) variables are collected (1) - Complex calculations carried out (1) - Identifies TWO items measured by sensors for this purpose (1) - Large quantity of data is collected/handled/processed (for each variable) (1) - Shows the (historical) trends of different variables over several decades/long period of time (1) (Complex) mathematical formulae are used to describe how the different parts of a climate system interact (1) - Computer models are used to describe how the climate will change (1) - Supercomputer models will include several (scientific) rules (1) Model include so many uncertainties//are so complex that there is no known exact solution//answer is a “best guess” (1)	4

Mark scheme · 9626/12 Oct/Nov 2023 · Question 3

[Back to question](#)

Question	Answer	Marks
3(a)	Two from: - User types in text/instructions//command must be typed in//is a text based interface (1) - Prompt appears in screen (after which command is typed in) (1) - Requires fewer resources than GUI (1) - Requires a keyboard and screen (1)	2
3(b)	Two from: - User moves body/identified body part (to communicate with computer//give instructions) (1) - Requires a camera/sensors (to detect the movements) (1) - Computer matches body movements it has stored (in a database) (1) (Stored) body movements relate to (specific) commands (1) - One mark for any identified body movement (1) e.g. finger pointing, head nodding, hand waving/raising, eyes rolling/blinking	2
3(c)	Two from: - User uses their voice/spoken word (to communicate with the computer/give instructions) (1) - Speech recognition software is used to recognise speech (1) - Computer matches speech it has stored (in a database) (1) - Speech is converted into commands//example of command and action that follows (1) - Requires a microphone (1)	2

Question	Answer	Marks
3(a)	Three from: - Whether a computer will repair itself if an error occurs (1) - Whether a computer can continue to operate even if one (or more) of its components has failed (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether a computer's operating quality can reduce, but does not fail completely (1) - Whether when a hardware component/processor fails, another takes over/elements are hot swappable (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether computers have two different instances/installations/versions of software (1st) - if there is a problem with the first instance/installation/version, the other version is automatically run (1) - this (characteristic) does not apply to mainframes (1)	3
3(b)	Two from: - Computers generate heat/become overheated (1). - When a computer overheats, it will address this issue (1). - Overheating of mainframes and supercomputers is a problem (1st) - due to the quality and quantity of processors generating a lot of heat (1) - hence the need for cooling systems (1) - Air cooling systems have been used but uses up a lot of power/are very expensive to build and operate (1) - Water cooling systems are more cost-effective/more efficient (1) - in removing any excess heat (1) - Many data centres are now being built in cooler parts of the world (1)	2
3(c)	Three from: - The amount of data that has to be processed is very large (1st) as countries tend to have a lot of people (1) - The amount of data is increasing (1st) as each census produces more data (to add to what already exists)/world populations are rising (1) - All data needs to be processed in a short period of time (1st) - and mainframes have a high speed of processing data (1) (As more data is added), more powerful machines are needed (1st) - and mainframes are easily scalable (1) - Mainframes are very reliable (1st) MAX two marks if answers are all only first marks.	3

Question	Answer	Marks
10	Eight from: • (This) custom written software is created for a defined/this purpose/running a payroll/commissioned by an individual company/business (1) • Does not need to be adapted for use (1) • (Any) settings built into the software can be changed (1st) • if the software doesn't initially meet the manager's requirements (1) • such as ability to copy software on to several devices (1) • Software will have/Programmer can ensure the software will have all the features that are required (1) • Software will be/Programmer will ensure that it is compatible with the existing system and software (1) • Custom written software will not have any unnecessary/distracting features/bloat ware (1) • The company will own the copyright of the custom written software (1st) • so can sell it to others (1) • Usually takes up less storage space than off-the-shelf software (1st) • so less expensive media/data storage costs (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
3	Max six overall. Max two for each suitable utility software. Description may be what the utility does, or how this is achieved, or the impact on optimum performance. e.g. Device driver (1st) e.g. ...allows computer to interact with a device (1) ...manages hardware components/peripherals (1) Data compression tools (1st) e.g. ...to free up storage space (by compressing files)//reduce file size (1) Anti-virus/anti-malware (1st) e.g. ...checks for/DETECTS malicious software (1) ...prevents viruses from being downloaded (1) ...quarantines any virus it finds (1) ...Stops your computer from being infected by a virus (1) File management tools (do not accept examples as first mark) (1st) e.g. ...to organise directories (1) ...delete unnecessary/unused files (1) Searching (1st) e.g. ...to find file duplication (1) ...to find files by name or another criterion (1) (Disk) defragmentation (1st) e.g. ...reassemble files into (contiguous) clusters (1) ...files are able to be opened more quickly (1) Create/resize (logical) volumes (1st) e.g. ...the organisation of disk structure (1)	6

Question	Answer	Marks
6	Six from: • A CLI is one where the user interacts by typing text commands • Users with physical disabilities may find CLI hard to use (1) as not able to use a keyboard (1) • Instructions have to be learned/memorised • CLI commands are complicated • CLI commands need specific structure • If the syntax is incorrect there maybe unforeseen circumstances//will not work • Increases possibility of human error • A slower way of initiating a response from a device • CLI commands may be processed more quickly (once typed in) • CLI uses less memory than other interfaces • Requires specific training • Does not allow multiple windows/multitasking • More flexible	6

Question	Answer	Marks
7(a)	Three from: • The software is available now//any awareness that software is more readily available • They can choose between different manufacturers of software • The software is more likely to have been fully tested • Less likely to have errors in the code • More support available from books/manuals/ company staff/user groups • The development cost of the software is spread between multiple customers • Off-the-shelf software tends to be cheaper than custom written software • Updates/patches to the software can be more frequent/often/regular	3
7(b)	Three from: • The software is more likely to be written to work with the existing software • The software is more likely to be written to work with the existing hardware • Can get (direct) support from the developer • The evolution of the software meets changing circumstances within RockICT • Can be written so that less computer resources are used • Only includes the required features • RockICT will own the software and could legally sell it on to others • RockICT has more control of the writing of the software • RockICT will be able to specify exactly what it wants/fulfil the exact purpose for which it has been created.	3

Question	Answer	Marks
7(a)	Three from: • Translates the whole program as one complete unit/at once (1) • Translates the program into machine code/language (1) • Creates an executable file (1) • Produces a report of errors in the code after compilation (1) • Can optimise source code (to run as fast or as efficiently as possible) (1).	3
7(b)	Three from: • A linker combines object files/modules (1st) – that have been created using a compiler (1) – into an executable file (1) • A linker can link modules from a common library file into the executable file (1) • Replaces symbolic addresses with real addresses (1).	3

Question	Answer	Marks
2	Six from: • Complex models are created//uses complex calculations (1) • Data is gathered/collects/used/records by sensors (1st) – Identifies one suitable sensor//variable gathered (1) • Vast amount of data is captured/collected (1) • Data is input into the model (1) • Vast number of calculations carried out (1) • Data is collected from around the globe (1) • Searches for a match in past conditions (1) • The software/computer makes prediction(s) – based on past data/past patterns/past conditions (1) – (and) based on current data/current patterns/current conditions (1) • Selects most likely prediction (from a range of possibilities) (1) • Creates output charts (to screens) (1) • Uses plotters to print out charts/maps of pressures/wind speeds. (1)	6

Question	Answer	Marks
9	Eight from: Advantages MAX SIX: e.g. • Users can use speech to interact with the computer/do not need to learn the gestures (1) • It is hands free (1st) – can be used in vehicles/telephone systems (1) ○ is safer than gesture based (ONLY award when this fits the given context in the answer) (1) • More flexible, can use many different commands (1) • Do not need direct line of sight/close proximity (1) • Works in low visibility/ Not affected by background movement (1) • Useful for people with sight loss (1) • Useful for people with mobility issues//disabilities (1) • Does not involve unacceptable gestures (1) Disadvantages MAX SIX: e.g. • Potential loss of privacy (1st) – always listening (1) • Verbal input can be heard by others (1st) – So exposes private information (1) • If they are too close together they can respond to unintended commands (1) • Requires ability to speak//speak clearly speech to be able to follow instruction (1) • Needs to learn the user's speech pattern (1) • Could have problems if there is any background noise (1) • Not suitable for safety/critical instructions (1) • Need to use commands in a certain format/pattern (1) • Can be language dependent//impacted by a dialect//person may speak unsupported language (1)	8

Question	Answer	Marks
1(a)	THREE marks available Max TWO marks: - To prepare a data storage medium for initial use (1) - The purpose of low-level formatting is to prepare the disk surface to receive data/allow the user to save data/files (1) - High-level formatting creates a file system on the disk (1) - Allows the operating system to use the disk space to store and access files (1). Max TWO marks: - Low-level formatting divides the disk surface into tracks/sectors/cylinders (1) - Write heads magnetise the disk areas (1) - Tracks are numbered starting from 0 (1) - A gap is left between tracks (1) - Each track is organised into numbered sectors, starting at 1 and separated by gaps (1)	3
1(b)	THREE marks available: - Identifies Lossless and identifies Lossy compression (1) Max TWO marks - Reduces the overall number of bits/bytes (in a file)//reduces file size (1) - To increase available storage space on a hard drive (1) - File compression programs remove redundancy (1) Max TWO Marks - A lossless file compression program uses codes to represent repeated Information (1) - Lossless compression removes meta data (1)	3
1(c)	THREE marks available: Max TWO marks: - It speeds up the access time to a file (1). - To create larger regions of (contiguous) free space (1) Max TWO marks: - Organises the contents of the disk into the smallest number of contiguous regions/fragments (1) - The data blocks of a file/defragmented files are moved around to bring whole/all the parts of a file together (1)	3

Question	Answer	Marks
2(a)	THREE from: • Mainframes have peripheral processors that deal with all input and output operations (1st) • this leaves the CPU to concentrate on the processing of data (1) Volume of Input (Max 1) • Mainframe computers deal with a very high level of input (1) Volume of Output (Max 1) • Mainframe computers deal with a very high level of output (1) Throughput (Max 1) • This refers to how many processes can be carried out per day (1) • Mainframes can carry out many transactions per day//have a fast rate of processing (1) • Throughput is a discrete mainframe characteristic. (1)	3
2(b)	THREE from: • Mainframe computers are more secure than other types of computer systems (1) • Mainframe has many layers of security (1) • Uses different levels of access (1st) • depends on a user's level of security as to which sets of data they can access (1) • Data that is transmitted and data within the system are encrypted//mainframe computers use encryption to protect data (1) • System is continually monitored for unauthorised access attempts (1)	3

Question	Answer	Marks
7	EIGHT marks available e.g. Benefits max 6 • Compilation process only needs to be done once (1) • Once the program is already compiled into machine language, there is no translation software that the user has to keep up to date (1) • Compiled software can run faster than interpreted software (1) • Compiling a program is a much quicker process than interpreting the code (1) • Compilers compile the code all at once/interpreters converts the code line by line (1) • Do not need to share source code (1) • Compilers also improve security for programmers (1st) by making it much harder to copy the code (1) Drawbacks max 6 • Can only be used on the operating system it was compiled upon (1) • Only produces error list once whole process has been completed (1) • It is slow/more difficult to debug the program (1) • The whole program needs to be compiled again after an error is corrected (1) • It uses up more memory than an interpreter as the whole program has to be loaded (1) • Once compiled, any changes to the original high-level code will require the program to be compiled again (1).	8

Question	Answer	Marks
9(a)	One mark per bullet point to a maximum of three marks. • Does not actually erase/delete the data (1). • Removes the reference from the directory table//index in the HDD/SSD. (1st) – so that the file can not be found//search facility would not work (1) • Data is still written/remains on the storage device (1st) – until overwritten by new information. (1)	3
9(b)	Two marks for a full explanation. • Because data has not been removed/deleted from storage device (1) instead space it has been made available for the OS. (1) • Because while no new information overwrites old/previous/deleted data (1) it can potentially be recovered/accessed using specialised software. (1) • Because they are sent to the recycle bin (1)	2

Question	Answer	Marks
5	One mark per bullet point to a maximum of six marks. • Can be available quicker than custom written software//is readily available (1) • Costs to buy are less than custom written software (1) • Has been fully tested before launch (1) • More (informal) support available (1st) – Example of support other than direct support from the company (1) • Company provides a help desk (1) • May offer more features (1) • Increased flexibility//not task specific (1)	6

Question	Answer	Marks
11	Max 6 marks: <i>Benefits</i> • Program code will run on different operating systems (1) • Only need to maintain one version of the program code/source code (1) • Less time will be spent on program code maintenance (1st) – When updates are needed (1) • Errors are highlighted/user is alerted as they are encountered (1) it is easy to debug the program (1) • As only a few lines of program code are in memory (1) uses less memory than a compiler (1) • Code can run in a virtual mode (1st) – so is less likely to crash the computer (1) Max 6 marks: <i>Drawbacks</i> • The interpreter software needs to be kept up to date (1) • If the interpreter is updated then the program code may need to be changed to work with it (1) • Code has to be interpreted every time it is run (1) • Interpreted programs run slower than compiled programs (1) • It stops every time an error is encountered (1st) – interpreting a program is a much slower process than compiling (1) • The source code must always be shared (1) • Source code is easier to convert by fraudulent users (1) • Easier to copy the program. (1)	8

Question	Answer	Marks
4	One mark per heading to a maximum of four marks. Cost: valid comment on cost e.g. • Custom written software can be more expensive (than off-the-shelf software) (1) Ownership: valid comment on ownership e.g. • Custom written software is owned by the company that commissions/creates/buys it (1) • off-the-shelf software means the company only has a licence (1) Degree of testing: valid comment on degree of testing e.g. • Custom written software is tested by the users as part of its development (1) Level of support: valid comment on level of support e.g. • Custom written software means that support is available directly from the creators (1) • Off-the-shelf usually provide call centres which may not be able to solve problems specific to the company (1) Customisation: valid comment on customisation e.g. • Off-the-shelf software may have several unnecessary features that the company may not need/custom written software has only the features the user needs (1) • Off-the-shelf software has to be adapted to fit the company that has purchased it (1) Availability: valid comment on availability e.g. • Off-the-shelf software is available to all businesses and companies/general public not just a specific company/business (1) • Off-the-shelf software is ready-written/ready for use and already exists/there is often a delay between commissioning and receipt of custom written software. (1) Compatability: valid comment on compatabilty e.g. • Off-the-shelf software may not work with your current hardware (1)	4

Question	Answer	Marks
10	Six marks from: - No need to physically interact (may exemplify) (1st) - Can be more hygienic (1) - Can undertake other tasks at the same time (1) - Can operate the device remotely (1) - Can be triggered without meaning to (1) - More accessible to disabled users (may refer to device itself or devices around the home) (1) - The dialogue interface relies on speech/sound (1st) - User has to be in range (1) - Can undertake other tasks at the same time (1) - Can operate the device remotely (1) - External noise may result in the dialogue interface not understanding the command (1) - If a person has a speech impediment, may find dialogue interface difficult to use (1) - Dialogue interface may misinterpret what is said (1) - More accessible to disabled users (may refer to device itself or devices around the home) (1) - Person may not speak supported language/dialect (1) - Audible acknowledgement (1st) - Know that instruction/command has been actioned (1) - One mark for a valid conclusion (1)	6

Question	Answer	Marks
12	One mark per explanation to a maximum of three marks. Banks require 24/7 operation and mainframe computers have a high level of Reliability Availability and Serviceability (RAS) to keep the system running (1) Banks require 24/7 operation - ONE FROM - and it must be possible to upgrade/replace/maintain the system without shutting down the system (1) - Mainframe Computers have a high level of serviceability//avoid downtime (1) Banking transactions need to be highly secure and mainframe computers have high levels of security (1) Banks hold sensitive information so high fault tolerance of mainframe computers means that data integrity is maintained (1) - Mainframes have the ability to transfer processing from one core to another so that processing is uninterrupted if it breaks down and so bank service is not interrupted//needs to be 24/7 (1)	3

Question	Answer	Marks
8(a)	One mark per bullet point to a maximum of four marks. • It operates/controls a device attached to the computer (1) • It provides an interface between operating systems/applications software and the device (1st) – Which handles the translation of requests between a device and the operating system (1) • It allows an operating system to detect/identify the peripheral device (1) • Activates the device when it is needed and deactivates it when it is not needed (1) • Logs and reports errors (1) • Defines where data must be cached (1)	4

Question	Answer	Marks
8(b)	One mark per bullet point to a maximum of four marks. • Translates a high-level language into machine code (1) • Creates an executable file (1) • Identifies errors in code (1) • Stops interpreting code when an error is found//shows errors as it translates (1) • Errors are dealt with as they are found//immediately (1) • Uses less RAM as only one line at a time is stored/processed (1)	4

Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of three marks. • Optimises space on the disk (1) • Files are better organised (1) • Increases reading and writing speeds (1) • Related files are kept together (1) • Files are contiguous (1) • Data retrieval is quicker//files are opened more quickly (1) • Reduces the movement of the read/write heads (1) • Increases longevity/life span of the disk. (1)	3
10(b)	One mark per bullet point to a maximum of three marks. • It can take time to do (1) • It can use a lot of computer memory (1) • It can use a lot of computer processing (1) • It can make the computer less responsive (1) • It can lead to file corruption (1) • The power must be on/uninterrupted throughout the whole process. (1) • Shortens the life of SSDs (1)	3

Question	Answer	Marks
6	Max one for: - It is a type of system software/Software used by the operating system (1) Max two from: - Used by the operating system to maintain a computer/file system/add functionality (1) - It improve/optimises the computer's performance (1) - Without utility software some devices/peripheral devices would not be available/accessible to the operating system (1) - Describing a clear specific need that is solved by a named/example of utility software (1)	3

Question	Answer	Marks
9	Four from: - Programmer can write in high level language (1) - Programmer does not need to code in machine code (1) - The program code will run on different computers/operating systems (1) - Only need to maintain one version of the program (1) - It is easier to debug the program//It is easier for the programmer to spot mistakes in the code (1) - An interpreter will highlight mistakes as they are found (1)	4

Question	Answer	Marks
11	Four from: - They have a short lifespan (1) - They perform consistently and reliably (concept of reliability) (1) - They have very little downtime (concept of availability) (1) - They are no more complex to maintain than other computer systems (concept of serviceability) (1) - They can sacrifice security for faster processing//they might be more liable to attacks because of the data they contain (concept of increased security) (1) - They have high fault tolerance (1) - They achieve very high temperatures//They require heat maintenance (1) - They have a very large number of processors/multi-core processors running in parallel (1) - Their processors are close together (in a 'computer cluster') (1) - They can run multiple operating systems at the same time (1) - They carry out multiple processes at the same time (1)	4

Question	Answer	Marks
12	Two from: - Stands for millions of instructions per second (1) - Are a unit of measurement of processing power (1) - Are used to measure mainframe performance (1)	2

Question	Answer	Marks
10	Four from: - They have a long lifespan (1) - They perform consistently (1) - They have very little downtime (1) - They are easier to maintain/hot swap (1) - They have stronger security (1) - They have high fault tolerance (1) - They achieve very high temperatures/They require heat maintenance (1) - They have a large number of processors/multi-core processors/uses parallel processing (1) - They can carry out multiple processes at the same time - They can run multiple operating systems at the same time (1) - They have a high degree of backwards compatibility (1)	4

Question	Answer	Marks
11	Two from: - FLOPS is floating point operations per second (1) - Are a unit of measurement of processing power (1) - Are used to measure supercomputer's performance (1)	2

Question	Answer	Marks
12	Four from: Organisation e.g. • It organises the arrangement of the files on the disk (1st) Max one from: – uses indexes to determine where to put data (1) – organises data by (two from) cylinder, head, sector (1) – organises data by memory location/index (1) Maintenance e.g. • Maintains / creates directory structures (1) • Random access/serial access to files (1) • Disk formatting tools/Defragmentation tools (any mention) (1) Access e.g. • Controls access / implements access rights / implements password protection / Makes file sharing possible (1) • Specifies tasks that can be performed on a file (two from e.g. open, close, delete, copy, create, move) (1)	4

Question	Answer	Marks
3(a)	TWO marks available: • (Employee/user) moves part of body (may state part e.g. head/wave of hand) (DO NOT accept 'gestures are made') (1) • Movements are captured by a camera (1) • (Movements/gestures are) compared to a database (of stored movements gestures) (1st) Associated commands are triggered/enacted (may exemplify) (1)	2
3(b)	FOUR marks available: Positives Max 3 • No need to have a pass card/word/code (1st) So fewer things to lose//can't get in by finding a card (1) • No need for physical contact//operate from a distance (1st) Therefore reduced spread of diseases (1) Negatives Max 3 • Gestures may be seen from a distance and copied by others (1st) Therefore reduces the security of the system (1) • Gestures may take time to complete (1st) Therefore are slower//impact of being slower (security risk if trying to avoid someone etc.) (1) • Gestures may be misinterpreted//possibility for false positive//high degree of accuracy required (1st) So incorrect action occurs (accept any realistic action) (1) • System may be inaccessible to disabled staff who cannot make the gesture//if carrying something (1st) Therefore they can not get into the office (1) • Only works when visible to the camera (1st) Delays ability to open door (1) • Gestures may be embarrassing/offensive (1) Conclusion (may only be awarded if argument(s) for and/or against are included) • One mark for a value judgement/conclusion (1)	4

Question	Answer	Marks
4	SIX marks available: ONE mark per identified characteristic, ONE mark for the difference: • Longevity (1st) Mainframe computers last longer than supercomputers (1) • RAS/Reliability, availability and serviceability (1st) Mainframe computers are more reliable/available/serviceable than supercomputers (1) The term RAS is NOT used to assess supercomputers but does refer to mainframe computers (1) • Security (1st) Mainframe computers are more secure than supercomputers (1) • Performance metrics//any statement of processing (1st) Supercomputers are measured in FLOPS/TEPS and mainframes are measured in MIPS (1) Supercomputers are more powerful computers than mainframe computers (1) • Volume of input, output and throughput (1st) A mainframe has a higher rate of throughput//can deal with a large number of simultaneous transactions (1) • Fault tolerance (1st) Mainframe computers have a high level of fault tolerance (1) Mainframe computers allow hot swapping (1) Mainframes have a high mean time between failures (MTBF) (1) • Operating system (1st) Super computers tend to only use Linux (derivative) operating systems (1) Mainframe computers can have multiple operating systems (1) • Number of processors (1st) Mainframe computers have multiple processors (1) • Cores (1st) Supercomputers have more cores than mainframe computers (1)	6

Question	Answer	Marks
9	EIGHT marks available: Areas for consideration (one mark each) • Ability to dictate how the product is tested//Extent to which testing has occurred//Potential for existence of bugs (1) • Extent to which user is/can be involved in testing (1) • Extent to which software has features as required//meets the requirements//tailored for the business (1) • Extent to which software includes unnecessary features//ONLY required features (1) • Compatibility with current system (1) • Copyright ownership//ability to sell on (1) • Who pays/paid development costs (1) • Whether immediately available for use or not (1) • Availability of support (online forums etc.) (1) • Updates (costs of or availability of) (1) • Reliance on continued existence of vendor going forward//product ceases to be available (1) • Scalability (1) Marks may only be awarded where the point made is correctly linked to the type of software. MAX six marks where only custom-written or off-the-shelf are considered.	8

Question	Answer	Marks	Guidance
9(a)	SIX marks available: Effect of icons • Avoids the need to read (1) • Graphical icons engage on an emotional level//are more appealing//visually appealing (1st) Therefore, more likely to be completed (1) Results are more reliable/accurate (1) • Less time is required to record customer satisfaction (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Graphical icons are simple to understand (1st) Therefore people will not need to queue/stop to be able to read them (1) Results are more reliable/accurate (1) • Graphical icons are universal/international/not language dependant (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Icons will be more widely understood (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) Languages • No need to use a 'select language button' (1st) Therefore people will not need to queue/stop to be able to read them (1) Therefore more likely to respond (1) • No guarantee that everyone speaks the same language//the language of the country in which the airport is (1) • In an airport, there will be a range of languages spoken (1)	6	Only award answers that support the use of a GUI as a method Do not award • repetition of explanations • answers about the system finding icons easier to process//.impact of use of icons on processing etc. • Users of the airport may be in a rush • GUI more easy to understand • Easier to navigate/use

Question	Answer	Marks	Guidance
9(b)	FOUR marks available: - Lots of noise (may interfere with vocal/vocal may not be clear) (1st) Therefore the computer may misinterpret what has been said (1) - Can not be used by certain disabilities (1) because these users cannot (e.g.) speak (1) - Would need to support all/many languages//may not support a specific languages (1st) Would need a very large database of languages/dialects (1)	4	Mark first two answers only. Do Not Accept: Will have difficulty with accent

Mark scheme · 9626/12 May/June 2026 · Question 9

[Back to question](#)

Question	Answer	Marks	Guidance
9(a)	ONE mark available: Software that helps to manage/maintain/optimise a computer system (1)	1	
9(b)	ONE mark available: - Back-up software (1) - Encryption software (1) - Disk checking software (1)	1	

Question	Answer	Marks	Guidance
5	SIX from: MAX 5 if no reference to weather forecasting Benefits (Max four) • Very fast processing//has a high FLOPS (1) • Weather forecast can be produced quickly (1) • Can handle a great deal of information (NOT store) (1) • Produce more/very detailed forecasts (1) • Consider a wide range of possible scenarios (1) • Decision is data driven (1) Drawbacks (Max four) • Supercomputers struggle to predict extreme events//working with extreme data (1) • Lacks human intuition (1) • Supercomputers have a limited lifespan (1) • Supercomputers are expensive to buy (1) • Supercomputers require specialist maintenance teams (1) • Supercomputers are very large//take up a lot of space (1) • Supercomputers require a lot of heat management//generates a lot of heat (1)	6	Do Not Accept: • ...and supercomputers can do this Some marks are awarded based on the use of <u>superlatives</u>. Candidates MUST have a clear awareness of the absolute vastness of the processing speed, or of the amount of information able to be handled. Therefore, if this is established, words such as 'a lot' or 'a great deal' can be awarded.

Question	Answer	Marks	Guidance
6(a)	ONE from: - The rearrangement/reorganisation of data/files (on a storage medium) to ONE OF (1) - create contiguous files - ensure efficient storage/access	1	Do Not Award Move or put together
6(b)	ONE from: - Modern computers use SSD storage (1) - SSD storage does not spin/is static (1) - SSD storage does not fragment files (1) - Defragmentation of an SSD reduces its (effective) life (1)	1	

Question	Answer	Marks	Guidance
6(c)	FOUR marks available (TWO per fully described type of utility software): e.g. • Anti-Virus (1st) Detect/remove viruses/malware (1) • Data compression (1st) To create extra storage/decrease the file size of files (1) • File copying (1st) To make another/ a second version of a file (1) (do not accept 'to copy the file') • Disk formatting (1st) To prepare a disk for first use//to return to disk to factory settings (1) Verify all tracks/sectors are ready for recording data (1) • Disk cleaners (1st) Remove unwanted files from the hard disk (1) • System optimisers (1st) Manage system resources so that computer's performance is improved (1) • Disk encryption (1st) Converts data on the disk into an unusable format (1) • Firewall (1st) Monitors/controls network traffic (1) Acts as a barrier between the network and the outside world (1)	4	Do Not Accept: • Back-up//recovery • File deletion • Disk defragmentation

3 Monitoring and control

Questions

9626/12 Feb/March 2022 · Question 4

[Mark scheme](#)

- 4 Computerised weather stations are used to monitor a number of analogue atmospheric variables.

Describe how **three** of these variables are collected and what needs to happen to the data before it can be used by a computer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 6** Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
    IF temperature < preset
        THEN
```

..... [6]

- 10** A farmer is concerned that the chemicals he is using to improve his crops are seeping into the local river. He has asked a local conservation group to investigate if this is the case. They will be using a computer and one set of sensors to monitor the level of pollution.

List examples of sensors the group will use and describe how they will be used with the computer to monitor the pollution level.

[6]

- 6** Sensors are used in monitoring systems to measure physical variables. It is important that sensors are calibrated before use.

- (a)** Describe, in detail, these sensors and give an appropriate use for each. Each use must be different.

Humidity

.....

.....

.....

.....

Sound

.....

.....

.....

.....

[4]

- (b)** Explain the importance of calibration.

.....

.....

.....

.....

.....

..... [2]

(c) Describe the differences between one point calibration and two point calibration.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- 8 A microprocessor-controlled car park barrier is used to allow cars to enter a car park. When a car approaches the barrier, the barrier automatically rises to allow the car to enter. It does not lower until the car is safely clear of the barrier system. There is a light sensor and light beam source immediately before the barrier. No tickets are issued by the system. The system operates continuously while it is switched on.

Complete this pseudocode algorithm to describe the processing involved. Your pseudocode algorithm should include IF...THEN statements.

```
REPEAT
    INPUT reading from induction loop
```

..... [7

- 9 Jose has a greenhouse to grow his plants. He lives in a country which has a warm climate so the greenhouse does not need a heater but has microprocessor controlled windows and a temperature sensor.

- (a) Describe how the sensor represents the temperature so the computer can read the data.

..... [1]

- (b)** Complete this pseudocode algorithm to show the processing which takes place to control the temperature of the greenhouse.

```

WHILE system switched on
  INPUT temperature
  IF temperature > preset
    THEN
      IF window closed
        THEN

```

[8]

- (c)** Discuss the advantages and disadvantages, to Jose, of using microprocessor control in this scenario.

[8]

- 10** Schools often use computerised weather stations to collect data, rather than requiring their students to do it manually.

Discuss the advantages and disadvantages of schools using monitoring technologies rather than students taking readings manually.

[8]

- 2** A microprocessor controlled air conditioning system uses real-time processing.

Explain what is meant by real-time processing in this scenario.

[5]

7 Describe the inputs, processing and outputs in a microprocessor controlled burglar alarm system.

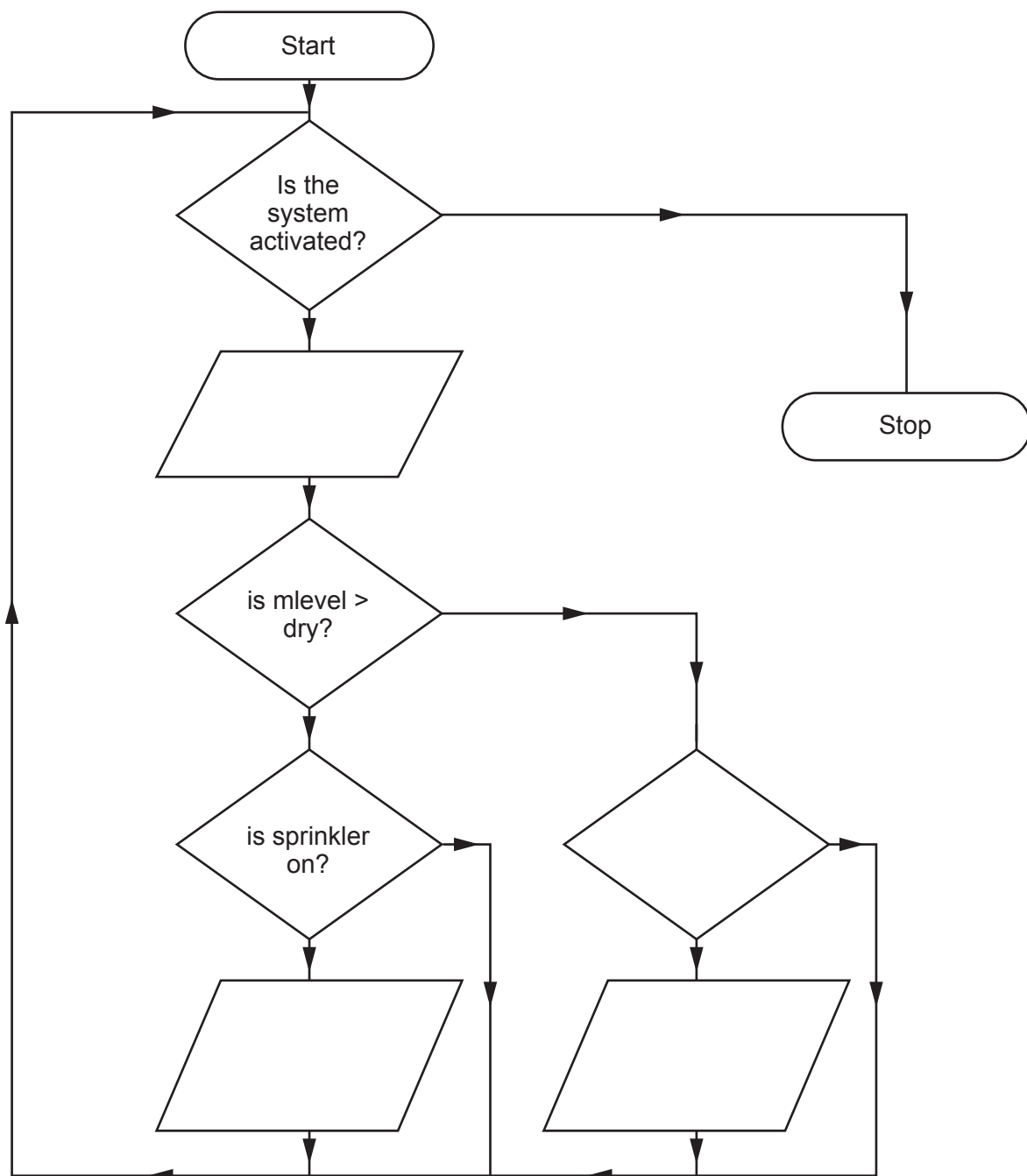
[6]

- 4 Mark has a small glasshouse. He lives in a warm country where plants can die if they are left without water for too long. The glasshouse has a microprocessor controlled sprinkler system and a moisture sensor in the soil.

Complete this flowchart to show the processing required to control the level of moisture in the soil of the glasshouse. You may assume that a pre-set value, **dry**, has already been input so that the system can test if the amount of moisture in the soil has reached or fallen below a certain level.

The user activates the microprocessor system by pressing a switch.

- (a) Complete the flowchart which inputs the moisture reading in the glasshouse and switches the sprinkler on and off according to this value. It also checks if the sprinkler is already on or off before attempting to switch it on or off.



[6]

(b) Flowcharts often contain subroutines.

(i) Draw the symbol used to represent a subroutine in a flowchart.

[1]

(ii) State what you would expect to see in the first box of a subroutine.

.....

..... [1]

6 Sensors are used to measure physical variables.

(a) Describe these sensors and give an appropriate use in a monitoring system for each.

(i) Light
.....
.....
..... [2]

(ii) Temperature
.....
.....
..... [2]

(b) Sensors need to be calibrated before use.

Evaluate the different forms of calibration.

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [6]

2 Describe what is meant by a smart home.

[4]

7 A computer system is used to monitor the level of pollution in the air.

Describe how the system would collect, process and display the data.

[6]

6 Sensors are used in monitoring and control to measure physical variables.

(a) Describe these sensors. For each sensor, explain how it could be used in a control system. Each system must be different.

Infrared

.....

.....

.....

.....

Ultrasonic.....

.....

.....

.....

.....

[4]

1 Describe what is involved in data logging.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

5 Justify the use of Information Technology for monitoring the environment.

[8]

- 6** A computer-controlled traffic light system controls the flow of motor vehicles and pedestrians.

Describe the inputs, processing and outputs involved in this system.

[8]

6 Most control technology systems use actuators and sensors.

(a) Describe the purpose of an actuator.

.....

.....

.....

.....

.....

..... [2]

(b) Describe how an electromagnetic field sensor works when used in a car park control technology system.

.....

.....

.....

.....

.....

..... [2]

(c) Describe how an ultrasonic sensor works when a driver is parking a car.

.....

.....

.....

.....

.....

..... [2]

- 4 (a) Describe how touch sensors are used as part of a control system in nuclear power plants.

.....

.....

.....

.....

.....

..... [2]

- (b) Describe how infrared sensors are used in burglar alarm systems.

.....

.....

.....

.....

.....

..... [2]

- 5 It is essential to calibrate sensors used in any monitoring technology.

Describe what is meant by:

- (a) two point calibration

.....

.....

.....

.....

..... [2]

- (b) multipoint calibration.

.....

.....

.....

.....

..... [2]

5 An apartment has a controlled central heating system. This system includes a microprocessor.

Explain how the temperature in the apartment is controlled by the system.

[6]

- 6** The environmental conditions around a chemical factory need to be monitored to make sure the environment is safe from pollution.
- (a)** Describe how data logging could be used to help the factory owner monitor the environmental conditions around the factory.

[6]

- (b)** Discuss the advantages and disadvantages of monitoring the environmental conditions around the factory using a computer system rather than using manual methods.

[8]

- 9 A car park barrier opens automatically when a vehicle approaches and closes when the vehicle is clear of the barrier.

Describe how a microprocessor could be used to control the operation of the barrier.

[6]

8 (a) Describe the purpose of **three** sensors that are needed in a microprocessor-controlled burglar alarm system.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Explain how a microprocessor-controlled burglar alarm system works.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 9 A computerised system will be used to collect and monitor environmental data in mountainous areas.

Evaluate the use of monitoring technologies in this scenario.

[8]

3 Control technologies can be used in traffic management systems.

Evaluate the use of control technologies in traffic management systems. You must identify specific control technologies and evaluate their use in towns and on motorways.

[8]

- 11** It is important that sensors are calibrated to ensure their readings are accurate.

Describe one point calibration.

[4]

- 4 Describe the drawbacks of using microprocessors to control devices and appliances in a smart home.

[6]

7 Sensors can be used to collect data about physical variables over a period of time.

(a) Describe the use of different sensors to collect data about the weather.

[6]

(b) Describe how the data from the sensors can be processed.

[6]

9626/12 May/June 2025 · Question 5

Mark scheme

5 Describe **five** advantages of using computers rather than a human to monitor water pollution.

[5]

8 Describe what is meant by an actuator. You must include an example of an actuator in your answer.

.....

.....

.....

.....

.....

..... [2]

9 Describe how a microprocessor-controlled car park barrier system makes use of an induction loop.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

5 (a) Explain the reasons why sensors must be calibrated before they are used.

[2]

(b) Compare one-point calibration with multipoint calibration.

[5]

5 Pressure sensors and humidity sensors may be used in a control system.

(a) For each sensor, describe how it gathers data.

(i) pressure sensor

..... [1]

(ii) humidity sensor

..... [1]

A car park management system has a pressure sensor in each individual parking space. These pressure sensors are used to find out how many free spaces are available.

(b) Describe how this system works.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [8]

(c) (i) Define the term calibration.

.....
.....
..... [1]

(ii) Explain why it is important to calibrate pressure sensors.

.....
.....
.....
.....
.....
..... [2]

9626/11 May/June 2026 · Question 8

[Mark scheme](#)

8 A combined heating and air conditioning system has been installed in a house. The system measures the temperature in the house and reacts as programmed. The user can override the temperature settings of the system to suit their needs.

(a) Identify **two** input devices that will be used as part of this system.

input device 1
input device 2 [2]

(b) Explain, with the example of a pump, the purpose of an actuator in this system.

.....
.....
.....
.....
..... [2]

The system has the following rules:

- If the temperature exceeds 28 degrees Celsius, a fan will turn on and the heating system will turn off.
 - If the temperature falls below 19 degrees Celsius, the fan will turn off and the heating will turn on.
- (c) Complete the pseudocode algorithm to show the processing that takes place in this combined system. The variable `temperature` represents the temperature of the house.

```
REPEAT
    INPUT temperature
```


3 Monitoring and control

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 4

[Back to question](#)

Question	Answer	Marks
4	Three from Temperature sensors measure ambient temperature Pressure sensors measure atmospheric pressure Humidity sensors measure absolute and relative humidity Light sensors measure sunlight/light levels Reed switch and tipping bucket measure rainfall Analogue data needs to be converted into digital data/data that a computer understands – 1 mark	4

Mark scheme · 9626/11 May/June 2022 · Question 6

[Back to question](#)

Question	Answer	Marks
6	Six from: <pre>REPEAT INPUT temperature IF temperature < preset THEN IF pump switched off THEN send signal to switch pump on ENDIF ELSE IF pump switched on THEN send signal to switch pump off ENDIF ENDIF UNTIL system switched off</pre> 1 mark for correct 1st IF statement 1 mark for correctly positioned first action 1 mark for correctly positioned 1st ENDIF 1 mark for correct 2nd IF statement 1 mark for correctly positioned second action 1 mark for correctly positioned 2nd ENDIF 1 mark for correctly positioned 3rd ENDIF 1 mark for correct UNTIL statement... ...1 for correct position of correct UNTIL statement	6

Question	Answer	Marks
10	Six from: Temperature, light, turbidity, pH, O₂, CO₂, nitrite, nitrate (must have two) Place sensors in the river next to the farm The sensors would be connected to ADC/computer Sensors send data to analogue to digital convertor ...enables the computer to understand the data Computer receives data from ADC Readings from sensors are compared with normal river values <u>by the computer</u> Differences/results are printed out/displayed on screen Graphs are <u>automatically</u> produced by computer showing values from river and normal values Computer stores readings in a table/database/file/spreadsheet ready for further processing	6

Question	Answer	Marks
6(a)	Humidity sensor combination of moisture sensor and temperature sensor/measures how much moisture is in atmosphere together with temperature – 1 mark Used in weather stations to measure relative/absolute humidity /air humidity /any suitable example – 1 mark Sound sensor converts sound waves into voltages/electrical signals – 1 mark Used in environment monitoring systems to measure noise pollution /any suitable example – 1 mark	4
6(b)	Two from: The accuracy of all sensors reduces after a period of time... ...due to constant use in/exposure to the atmosphere/liquids that are being used Regular calibration helps to maintain the accuracy of the sensors/helps to make sure sensors produce accurate/correct/reliable readings Other devices within a monitoring system may deteriorate over time resulting in a need for recalibration	2
6(c)	Five from: Only one measurement point/reading with one point calibration whereas with two point calibration at least two readings are taken One point calibration is appropriate for sensors used to measure a value that is constant/never changes... ...whereas two-point calibration is used with sensors that measure a constantly changing variable With one point calibration the offset only is calculated/ the sensor reading is subtracted from the known value only... ...the offset is then added to every subsequent reading With two point calibration sensitivity needs to be included not just an offset Two point calibration is used where there is a linear relationship between the two readings The slope of the line connecting the two readings is compared with the slope of the line of the pre-calibrated sensor/the range values of the sensor is compared with the range values for the standard Two point calibration compensates for both offset errors and sensitivity/slope errors	5

Question	Answer	Marks
8	<i>REPEAT</i> <i>INPUT reading from induction loop</i> IF reading = pre-set value THEN send signal to raise barrier REPEAT IF light sensor activated THEN send signal to lower barrier ENDIF UNTIL light sensor activated ENDIF UNTIL system switched off Seven from: Appropriate IF statement (comparison of reading) Send signal to raise barrier after first IF Appropriate other IF statement (checking if light sensor activated) Send signal to lower barrier after 2nd IF Appropriate position and matching 1st ENDIF Appropriate position and matching 2nd ENDIF Appropriate UNTIL system switched off statement Appropriate loop checking that light sensor has been activated	7

Question	Answer	Marks
9(a)	One from: - Electrical resistance is changed/voltage is generated (depending on the temperature) (1) Electrical signals are generated (which are converted into values (to represent temperature)) (1).	1
9(b)	<pre> WHILE system switched on INPUT temperature IF temperature > preset THEN IF window closed THEN send signal to motor to open window ENDIF ELSE IF window open THEN send signal to motor to close window ENDIF ENDIF ENDIF ENDWHILE </pre> - send signal to open//open window (NOT 'output') (1) - send signal to open//open window as 1st statement (1) - ENDIF closes the IF statement (1) - ALL three ENDIFs in same vertical plane as relevant IFs (within reason) (1) - ELSE (1) - ELSE in same vertical plane as THEN (1) 'IF window open' (or equivalent) ... (1) ... IF window open condition closed with ENDIF (1) - Close window instruction (1) - Send signal to motor to close window appropriately positioned (1) - IF temperature > preset value condition closed with ENDIF (1) - Whole of IF Window open section indented correctly (allow follow through on close window instruction position) (1) - ENDWHILE as final statement (1)	8

Question	Answer	Marks
9(c)	Eight from: Advantages max 6 • Reduced chance of plants dying/wilting etc (1) • Jose could forget to (e.g.) open window//be so busy that he is unable to (e.g.) open window ... (1) ... computers can be permanently switched on and take action regularly/will not forget (1) • Jose can do other tasks (1) • Computers can take action immediately if the temperature gets too hot ... (1) ... there would be a delay before Jose can take action (1) • Readings taken by computers tend to be more accurate ... (1) ... and action taken is therefore more likely to be the correct one (1) (ACCEPT CONVERSE ARGUMENTS ABOUT JOSE'S INACCURACY) • If Jose was physically disabled, he may find it difficult to open window ... (1) ... whereas computer can do it automatically (1) Disadvantages max 6 • Computers can be expensive to buy ... (1) ... whereas it would not cost Jose any money to look after the plants himself (1) • Computer maintenance costs can be expensive ... (1) ... whereas with Jose that is not a problem (1) • It is quite difficult to program computers to take correct action ... (1) ... but Jose would know instinctively//will have expertise (1) • Sensors can deteriorate//become inaccurate ... (1) ... whereas Jose's actions will tend to be more consistent (1) ... and will need recalibration (1) • A computer-controlled system will not be able to function if there is a problem with the computer/there is a power cut/load shedding (1) Max. six marks if bullets/list of points Must have at least expansions or comparisons to achieve six marks in either section	8

Question	Answer	Marks
10	Advantages Max 6 marks e.g - Computers are able to take readings at more frequent intervals ... (1) ... as students take time writing down readings/values (allow once only)(1) - Taking readings manually takes time//time consuming (not slower) ... (1) ... because students have to write down the readings/values (allow once only) (1) - Computers can read several physical variables at the same time ... (1) ... whereas students would only be able to read one at a time (1) - Computers can take continuous readings ... (1) ... whereas students find it difficult to continually take readings/students may need to take breaks (1) - Students wouldn't be able to take readings during school holidays ... (1) ... but computers can take readings every day//computers work 24/7 (1) (Readings taken by computers) tend to be more accurate ... (1) ... and so are more reliable//students can make mistakes when tired/due to human error (1) - Training is not required ... (1) ... to collect accurate data (1) Disadvantages Max six marks e.g. - Monitoring systems/computers can be expensive to buy ... (1) ... but students do not have to be paid (1) - Monitoring systems/computer maintenance costs are expensive ... (1) ... whereas there is none with students (1) - Sensors can lose accuracy over time ... (1) ... which would lead to inaccurate readings being taken (1) - Sensors may not be calibrated properly ... (1) ... so any measurements will be inaccurate (1) Max. six marks if bullets/list of points Max six marks if no attempt at expansion	8

Question	Answer	Marks
2	Five marks available: • A real-time system has to respond to/process inputs//inputted data instantaneously (1) • An air conditioning system has to deal with inputs continuously ... (1) ... unless the user switches the system off (1) • This type of processing is usually found in systems that use sensors (1) • The user interface of a real-time system may use specialist input devices to provide data input ... (1) ... such as a keypad/touchscreen (1) • Usually deals with small amounts of data//does not deal with a lot of data (1) • Real time systems are such that the output affects the input (1)	5

Question	Answer	Marks
7	These answers are based on the use of devices that take measurements. Follow a similar procedure for devices such as light beams, or electronic contacts Six marks available: <i>Identification of inputs and purpose</i> e.g. • A keypad to switch the system on/select zones (1) • Infra-red sensors to detect heat from body (1) • Sound sensors to detect the level of sound (1) • Pressure sensors/pads to detect an increase in weight (1) <i>Processing of signal and decision making</i> e.g. • <u>The microprocessor</u> compares the input reading with the pre-set value/criteria ... (1) ... which would have been programmed in (1) ... to check if reading is above pre-set value (1) <i>Outputs as a result of processing</i> e.g. • Instruction/signal/activation sent to actuator//output device//named suitable device (1) • Relevant output from named suitable device (1) • Signal (NOT 'an output') sent to the police//monitoring service (1). Must have at least one from each to get full marks	6

Question	Answer	Marks
4(a)	<pre> graph TD Start([Start]) --> D1{Is the system activated?} D1 -- no --> Stop([Stop]) D1 -- Yes --> I1[/Input mlevel/] I1 --> D2{is mlevel > dry?} D2 -- no --> D3{is sprinkler off? R} D2 -- yes --> D4{is sprinkler on?} D4 -- yes --> O1[/send signal to actuator to switch sprinkler off/] D4 -- no --> D3 D3 -- yes --> O2[/send signal to actuator to switch sprinkler on/] D3 -- no --> D1 O1 --> D1 O2 --> D1 </pre> SIX marks available: • Yes/No in correct position off first decision box (1) • Two other Yes/No used (off a decision) (1) – A further Yes/No used (off a decision) (1) • <u>Input MLevel</u> in correct place (1) • <u>Is sprinkler on/is sprinkler off</u> added in correct place (1) MLevel>dry NO branch: • Is Sprinkler OFF – YES – switch <u>Sprinkler ON</u> in correct place (1) OR • Is Sprinkler ON – NO – switch <u>Sprinkler ON</u> in correct place (1) MLevel >dry YES branch • Is Sprinkler ON -YES- switch <u>Sprinkler OFF</u> in correct place (1) OR • Is Sprinkler OFF – NO – switch <u>Sprinkler OFF</u> in correct place (1) For final four mark points, marks may only be awarded where the logic leading up to the decision matches that given here and is shown in full.	

Question	Answer	Marks			
4(b)(i)	<table border="1"> <tr> <td></td><td></td><td></td></tr> </table>				1
4(b)(ii)	ONE from: - The name of the subroutine (1) - Start (1)	1			

Mark scheme · 9626/12 Feb/March 2023 · Question 6

[Back to question](#)

Question	Answer	Marks
6(a)(i)	One mark for description, one mark for example: - Measures/detects light <u>intensity</u>//<u>how much light hitting a surface</u>//level of light (1) - Any suitable example of use - e.g. Used in weather stations to measure the amount of sunshine - Test match cricket to see if its bright enough to play (1)	2
6(a)(ii)	One mark for description, one mark for example: - Measures temperature <u>value</u> (1) - Any suitable example of use - Used in weather stations to measure the temperature (1)	2

Question	Answer	Marks
6(b)	SIX from: One point calibration Max 2 - Is the easiest method to carry out (1) - Only one point of data is used (1) - Can only be used when measuring set/constant physical variables... (1) ...therefore has a limited use (1) Two-point calibration Max 3 - Is more complex than one point calibration (accept less complex answer)... (1) ...because is needed when a range of values is being monitored (1) - Two points of data are used (1) - Needs to be used where a linear relationship between sensor readings and the variable being monitored exists... (1) ...therefore, cannot be used when the relationship is a curve (1) Multi point calibration Max 3 - Multi-point calibration is the most complex method... (1) - used for non-linear, multipoint relationships (1) - Multiple points of data are used (1) - Multi-point calibration is the most complex method... (1) ...greater knowledge of maths is needed (1) Max. five marks if all three methods not covered Max. five marks if bullets/list of points	6

Question	Answer	Marks
2	Four from: - Aspects of the home are controlled from Smartphone/Voice control device/computer system/remotely (1) - Devices/appliances in the home are connected (1) - TWO types of devices (in the home) that can be controlled: e.g. - lighting, - heating - air conditioning (accept "temperature" once only) - television - kettle - plugs (1) Controlled by issuing (voice) commands/setting up routines (1) - The commands can be given by voice/sound/ remote control/ tablet/ smartphone (1) - Uses Wi-fi/home network (1) - Uses internet to access devices when outside the home (1)	4

Question	Answer	Marks
7	Six from: Collection • Sensor(s) are used to gather data (DO NOT ACCEPT ‘COLLECT’) (1) • Naming of TWO physical values that would be gathered: – temperature – pH – CO₂ – O₂ – Gas (BOD) – <u>UV</u> light – infrared (1) Note: where candidate states (e.g.) temperature sensor, this may be taken as identification of a physical variable. Processing • (Collected) data is sent to ADC (1) • Analogue data is converted into digital (data) (1) • Digital(ised) data is sent to microprocessor/computer/CPU • Computer stores readings in a table/database/file/spreadsheet (1) • Naming of TWO results being calculated. E.g. – Max temperature – Min temperature – Average temperature (1) • ONE mark for any description of how one result has been calculated. (1) Display • Computer plots graphs showing trend of pollution variables ...(1st) – ...over a period of time (1) • Computer plots graphs automatically (1) • Computer outputs the results/graphs on screen/printer (1) • Results may (also) be shown on a website (1)	6

Question	Answer	Marks
6(a)	Infrared • Detects heat (1) • Turns infrared electromagnetic waves into an electrical signal (1) e.g. • Used in a burglar alarm – accept any reasonable use (1) Ultrasonic • Detects sound (1) • Turns sound waves into an electrical signal (1) e.g. • It is used in automated car parking//reversing system (1) • Measure distance (1)	4

Question	Answer	Marks
6(b)	Advantages MAX five from • Increased employment for technicians needed to maintain the systems//maintenance (1) and programmers to program the systems (1) (May also be used as a disadvantage - Award ONCE only) • Data is processed more quickly (1) • Reaction times of control technologies is much faster (1) • Computer-controlled traffic lights tend to create fewer traffic jams (1) • Computer-controlled car park barriers lead to fewer/shorter queues (at car park entrances) (1) • Any indication of reduced day to day costs (may be exemplified – e.g. ‘no need to pay anyone to do the job’) (1) • System can operate 24 hours a day//the whole/all day//continuously ... (1) ... without needing to pay overtime (1) • The readings/inputs taken by computers are more reliable/accurate than human readings ... (1) ... the number of car park spaces are more accurately calculated (1) Disadvantages MAX five from • Has increased unemployment for car park attendants (1) (Converse answer may also be given as an advantage – Award ONCE only) • The number of new jobs created tends to be far fewer than the number of old jobs lost (1) • (Control systems) can be expensive to install/purchase (1) • A computer-controlled system will not be able to function if there is a problem with the computer/power outage ... (1) ... leading to possible chaos at traffic lights/traffic jams (1) • ... causing longer queues at car park entrances (1) • Computer maintenance costs can be expensive (1) • Sensors can deteriorate after a period of time //suffer wear and tear (1) ... whereas this would not be a problem with car park attendants (1) Max. six marks if uses a bulleted list.	8

Question	Answer	Marks
1	Four from: • Automatic collection/storing of data (1) • Using a computer and sensors to collect data (1) • This data is analysed/processed (1) • The results are output in the form of graphs/charts/tables (1) • Data is collected over a period of time ... (1) ... continuously/at regular intervals (1) • The analysis/processing usually requires special software (1) • Two of the following examples of physical variables logged: temperature sound light pressure (1)	4

Question	Answer	Marks
5	Eight from: • Computers are able to take readings at more frequent intervals ... (1st) ... as humans take time writing down readings (1) • Computers can read several physical variables at the same time ... (1st) ... whereas humans would only be able to read one at a time (1) • Computers continuously take readings ... (1st) ... whereas humans take breaks (accept any reason why an individual human would not take readings continuously) (1) • Computers take readings as planned (as planned is an example) ... (1st) ... whereas humans may forget (1) • Readings taken by computers tend to be more accurate (1) • Computers can produce accurate graphs automatically/instantaneously ... (1st) ... whereas humans take time to draw them (1) • Tables of results are produced automatically after the readings are received by the computer ... (1st) ... unlike a human who would take a lot of time to write them down neatly (1) • Computers can work in dangerous conditions (1st) plus one mark for an example of a dangerous condition (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
6	Eight from: <i>Inputs (accept device OR data) (Candidate has to be clear they are discussing input)</i> • A button for a pedestrian to press//to indicate they wish to cross (1) • Signal from button (1) • Preset value(s) for delay between request and light change ('how long a light stays on' on its own is TV) (1) <i>Processing</i> • System waits for a preset time (or similar description (not 'when time is up')) (1) • Computer sends a signal to the actuator/control box... (1st) ...to change the traffic lights from green to red (1) • Computer sends signal to actuator to change pedestrian control light to safe (1) • Computer initiates count down (1) • After another pre-set time sends signal to actuator to change pedestrian control light to unsafe (1) • Computer sends a signal to the actuator/control box to change the traffic lights from red to green (1) <i>Outputs – (Candidate has to be clear they are discussing outputs)</i> • Display above pedestrian button indicating don't cross/red man/indicating safe to cross/green man (1) • Beeping sound (1) • Red/amber/green light that drivers can see (1) • Count down (1) Must have at least one from each group to get full marks	8

Question	Answer	Marks
6(a)	Two from: • An actuator provides the output from a control system (1) • Is the mechanical part of a control system (1) • An actuator controls/operates a device (1st) • such as opening a valve/activating a motor/switch (1) • It turns an electrical signal (from a microprocessor) into movement/action/physical (1)	2
6(b)	Two marks from: • Measures the change in a magnetic field(1st) • caused by the presence of a metallic/ferromagnetic object/body of the vehicle (1) OR • Identifies a magnetic field from a car (1st) • caused by the car being metal (1) Generic answers: • Are used (at the entrances to car parks) to control barriers (1) • Are used to detect the number/position of spaces available//whether a (parking) space is occupied (1)	2
6(c)	Two from: • A (device in the) car sends out sound waves (1st) • and the sensor receives the sound waves //after they are reflected back (1) • The device calculates the distance from an object (1st) • by measuring the time between emission and reception of a sound wave/signal (1)	2

Question	Answer	Marks
4(a)	Two from: • Touch sensor is used for measuring fluid levels (1st) • such as the cooling water/coolant level of the reactor (1) • A capacitive touch sensor measures the capacitance between two conductors separated by an insulating plate (1st) • One of the conductors will be the fluid (whose level is being measured) (1) • Detects when the fluid is touching the conductor (1)	2
4(b)	Max Two from • Used to detect (presence/movement of) humans/living beings (1) • Human bodies emit heat/thermal energy (1) • By detecting heat/radiation (from heat) (1) OR Max Two from • Used to detect <u>movement</u> (1) • (Infrared) beams can be beamed across (e.g.) the entrance to the room (1) • The infrared sensor detects the beam/detects if it is broken (1)	2

Question	Answer	Marks
5(a)	Two from: - Two readings/measurements need to be taken (using the sensor) (1) - Can correct slope/offset errors (1) - Best used where readings are linear (1) - Reading taken at the low end of the range and the high end of the range (1) - Readings taken by a check/reference instrument/uses (previously) known readings (1) - Value for offset is calculated (1st) - by comparing to reference readings/known readings (1)	2
5(b)	Two from: - Uses many/more than two readings/measurements (from the sensor) (1) - Readings taken at a high value, a middle value and a low value (1) - Best used where readings form a curve/non linear (1) - Readings taken by a check/reference instrument/uses (previously) known readings (1) - The calibration cannot be carried out by just adding an offset value/by allowing for the slope (1) - To calibrate the sensor the algorithm needs to include 'curve fitting' (1) - Uses a formula to calculate the correction (1) - Uses a quadratic function such as $y = ax^2 + bx + c$ (1st) - where y is the reading from the standardised sensor/actual value x is the reading from the sensor needing calibration (1)	2

Question	Answer	Marks
5	Six from: - The required temperature is set by the user - The required temperature is input using a remote control/touchscreen/number pad - A sensor is used to read/gather/monitor the (current) temperature - The data collected by the sensor is analogue (shows clear awareness that original data is analogue) - The data is converted to digital via an ADC Microprocessor compares this data/temperature to the required/input value (this must be stated and not implied) - If temperature is, nothing happens - If the data is different than the required value, the microprocessor sends a <u>signal</u> ... (1st) ...to the actuator (1) If lower, the microprocessor/actuator switches the heater on If higher, then the microprocessor/actuator switches the heater off - Process is constantly repeated	6

Question	Answer	Marks
6(a)	Six from: • Sensor used to gather data (1) • Identifies ONE suitable sensor for collecting information on presence OR absence of pollution (1) • Data collected is analogue (any awareness) (1) • Analogue to Digital Converter converts data to digital (1st) • So can be used/understood by the computer (1) • Computer analyses/processes data (1) • Results presenting in appropriate format/tables/graphs/charts (1)	6

Question	Answer	Marks
6(b)	Eight from: <i>Advantages MAX SIX:</i> • Data is continually monitored//no reason for monitoring not to occur (may exemplify) (1st) • if an event happens it should not be missed (1) • Computer systems can monitor multiple sensors simultaneously (1) • The logged data is already in electronic form (1st) • no transcription errors/no human error (1) • Does not need to be entered/typed in (1st) • which saves time (1) • Personnel can do other useful tasks (1) • Data is immediately sent to the computer (1st) • The data can be displayed immediately/in real time (1) • The data can be analysed immediately/in real time (1) • Can monitor 24/7 (1) • No risk to humans/equivalent answer (1) • Can be monitored remotely (1) <i>Disadvantages MAX SIX:</i> • Any interruptions to the power supply could cause sensor to stop working//collection of data to be missed/not happen (1) • Any damage/failure of the equipment could cause collection of data to be missed/not happen/be inaccurate (1) • The maintenance of equipment can be expensive/dangerous/time consuming//requires expertise/training. (1) • Data logger can not react to emergencies//unusual circumstances (1)	8

Question	Answer	Marks
9	Six from: • Sensor identifies approach/presence of vehicle (1) • Identifies ONE suitable sensor (Induction loop/pressure sensor/light sensor) (1) • Microprocessor registers presence of vehicle (1) • Microprocessor checks if barrier is raised/lowered (1) • If barrier down it needs to be raised (1) • Microprocessor sends a signal to an actuator/motor (1st) • which then raises the barrier (1) • If barrier up microprocessor does nothing (1) (if mark not awarded for MP 5, a mark may be awarded for “If barrier up”) • Computer checks to see whether vehicle is present/has gone (1) and if vehicle not detected then the barrier comes down (1)	6

Question	Answer	Marks
8(a)	THREE marks available e.g. • Motion sensor detects movement (1) • Infrared sensor to detect heat (1) • Sound sensor to detect level of sound/unexpected sound (may exemplify) (1) • Pressure sensor (placed under a rug/carpet) to detect increase in weight by burglar standing on it (1) • Light sensor to detect a break in a light beam. (1)	3
8(b)	THREE from: • Microprocessor may be programmed to have (a range of) acceptable levels (1) • If sensor is activated/sensor readings are outside the expected levels/range (1) • Data is sent from sensor to Analogue to Digital Converter (ADC) (1) • ADC converts the data to digital (1st) – so can be understood by the microprocessor (1) • It sets off an audible alarm sounds/activates lights/alerts police or alarm company or key holder. (1)	3

Question	Answer	Marks
9	EIGHT marks available EIGHT from: Pros max SIX: - Computers never forget to take readings (1) - They take readings constantly (accept 24/7) (1) - No need for human intervention to take readings (1) - The readings are more accurate//can detect readings that humans may not (1) - Readings of several variables can be taken simultaneously (1) - Computers can take readings in dangerous areas for humans to be (1) - Transmission of data can be in real time (not quicker) (1) - Data can be stored for later collection to save electrical power/to reduce human impact (1) Cons max 6: - The monitoring equipment will have to be placed by humans (1) - Sensors/computers have to be repaired/maintained by humans (1) - Environment can affect the signal (1) - Geographical events can affect or move the equipment (1) - The presence/maintaining of the system can affect pollution itself (1) - Humans still have to plan/decide where to place sensors and which sensors to use/which variables to monitor. (1) Conclusion - Valid conclusion based on points made (1)	8

Question	Answer	Marks
3	EIGHT marks available: Identification of control technology (1st mark) (Max FOUR) e.g. • (Smart) traffic lights (1st) • Smart/reactive motorways (1st) • Pedestrian/user controlled lights (1st) • Pollution speed limits (1st) • Lane speed/closure displays/reactions (1st) Evaluation of (2nd mark) (Max SIX) e.g. (Smart) traffic lights • Fewer traffic jams (than preset controlled) (1) • Reduces waiting time (1) • Reduces accidents (1) • Drivers more likely to respect the lights (1) • Cost advantage/disadvantage (1) • Dependant on camera (1st) • To track traffic (1) • That may not be working (1) • Requires maintenance/recalibration (1) • Requires power//will not work if there is a power outage (1) Pedestrian controlled lights e.g. • Traffic flow only interrupted when a pedestrian requests to cross (1) • Smoother traffic flow (1) • Reduces accidents (1) • Fake pushes still stop traffic/no guarantee that anyone actually wants to cross the road (1) Conclusion • Overall, conclusive/evaluative statement about control technologies (1)	8

Question	Answer	Marks
11	FOUR from: • Calibration improves the accuracy of sensors after they have become inaccurate (1) • Can correct the sensor so original reading is accurate (1) • (OR) can apply an offset to take account of an inaccurate reading (1) • Only one measurement point is needed (1) • Appropriate for sensors that have to measure a single type of value/variable (1) • Does not need to be calibrated against any other range of parameters (1) • Often needed for sensors being used at extreme values (1) • Reading is taken (from the sensor) and compared with a known value (1st) – in the range being measured (1) • Sensor reading is subtracted from the known value (1st) – which gives the offset (1) • The algorithm uses the offset by adding it to every reading in the parameter range being measured. (1)	4

Question	Answer	Marks
4	One mark per bullet point to a maximum of six marks • They can cause a loss of manual skills (1) • They can stop people from performing simple exercise (1) • They can lead to people becoming lazy (1) • They could become over reliant on microprocessor-controlled devices (1) • The system will not function if there is a problem with the Wi-Fi/internet (1) • The system will not function if there is a power cut (1) • They can be vulnerable to hackers (1) • All devices have to be on the same network (1) • Can increase electricity use (1) • Automatic running of routines may have undesired affect (1)	6

Question	Answer	Marks
7(a)	One mark per bullet point to a maximum of six marks. • A moisture/humidity sensor/tensiometer (1st) – is used to measure absolute and relative humidity (1) – measures the amount of moisture in the air (1) • A wind sensor/anemometer (1st) – is used to measure the direction of the wind (1) – used to measure strength of wind (1) • A raindrop/raingauge/rain sensor (1st) – is used to measure rainfall (1) • A light sensor/photodetector (1st) – is used to measure/sense light (1) • A pressure/barometric sensor (1st) – is used to measure pressure values and height changes. (1) • A temperature/NTC thermistor/RTD thermocouple/IC sensor (1st) – is used to measure temperature (1)	6
7(b)	Six marks from: • The data will be sent to the computer for processing (1) • Via Analogue Digital Convertor (ADC) (1) • ADC will convert the analogue data to digital (1st) - So, the computer can read it (1) • It can create graphs/charts (1st) – To show monthly trends (1) • It can do calculations (1st) – Such as averages/max min (1) • It can produce predictions (1st) – Based on historical of weather data (1) • It can produce reports of weather trends (1st) – Such as global warming (1) • It can produce weather warnings (1st) – From its collected data. (1)	6

Question	Answer	Marks
5	One mark per bullet point to a maximum of five marks. - Computers can collect readings (from sensors) at more frequent intervals (1) - Computers are capable of accessing more than one variable simultaneously (1) - Computers work any time, day or night//24/7 (1) - Computers accept readings at precise/regular intervals (1) - Computers can be programmed to automatically (NOT immediately) take action on readings (1) - Computers will always accept the readings/do not forget (1) - It is safer to have a computer take readings in extreme conditions (1) - No need to input the data manually//avoids data input/recording mistakes (1) - Improved processing speed (1) - Improved accuracy of processing (1)	5

Question	Answer	Marks
8	Max one from: - An actuator converts energy into action//mechanical motion (1) - Performs a movement//physical action (1) - An actuator reacts to a signal to make something move (1) Max one from: EXAMPLE - linear - rotary - soft - hydraulic - pneumatic - electric - thermal - magnetic - mechanic.	2

Question	Answer	Marks
9	One mark per bullet point to a maximum of five marks. • A vehicle passing over the loop causes a change in inductance (1) • The change in inductance is used to detect the vehicle (1) • The metal in the vehicle causes a change in the magnetic field (1) • The change in the magnetic field causes a current to flow (1) • Loop sends back data which is converted to digital (1) • Computer compares data to pre-set value (1) • If computer detects a change, (1st) – it sends signal to the actuator (1) • The actuator will raise/lower the barrier. (1)	5

Question	Answer	Marks
5(a)	Two from: • Sensor accuracy changes with time/use because of wear and tear from environmental circumstances/constant use (1) • Sensors might not be adapted to the required environment because the conditions can change (1) • Calibration maintains the accuracy of the sensors to ensure they produce reliable/accurate/correct readings (1) • New sensors need to be setup/calibrated to ensure they meet the appropriate requirements because they will be set to factory settings//have preset parameters (1) • Other components that are part of the system/impact on the reading may affect the reading so making it inaccurate (1)	2
5(b)	Five from: e.g. • Only one measurement point/reading with one-point calibration//with multipoint calibration at least three readings are taken (1) • One-point calibration is used with sensors that measure a single value// multipoint calibration is used with sensors that measure a range of values that have a non-linear relationship (1) • One-point calibration requires a simple calculation to take account of the error//multipoint calibration requires a complex calculation (1) • They both mitigate/reduce impact of structural errors in readings//inaccurate readings (1) • They both need to be done regularly (1) • They both compare values with known/correct/accurate measurements (1)	5

Question	Answer	Marks
5(a)(i)	ONE mark available: - The force applied is measured (1) - Uses the piezoelectric effect (1) - The force applied to its surface creates an electrical signal (1) - Electronic resistance changes when a weight is applied (1)	1
5(a)(ii)	ONE mark available: - Affected by BOTH temperature and moisture (1) - Gathers changes in electrical impedance/capacitance/temperature (1) - A metal oxide strip resistance changes with humidity (1) - Impedance changes with ions in salts (1)	1

Question	Answer	Marks
5(b)	EIGHT marks available: - The standard pressure (without a car) is measured/has a preset value (1) - Data is captured constantly (do not award sensors <u>send</u> data for this mark point) (1) - Converted to digital (1st) by an ADC (1) - The computer/processor compares the readings to the standard pressure reading//a preset value (1) - If there is a pressure increase (from the standard), a car is present (in the space) (1st) This is recorded/reflected in the database (1) - If there is no change from the standard pressure reading, the space is empty (1st) This is recorded/reflected in the database (1) - Number of free spaces is calculated (1) - The number of free spaces/vehicles present is displayed (DO NOT ACCEPT alternative outputs that are not to do with finding out how many free spaces are available) (1)	8
5(c)(i)	ONE mark available: An adjustment to a sensor (reading)//tuning a sensor to make it accurate (1)	1
5(c)(ii)	TWO marks available: - Sensors become inaccurate (1) - Calibration takes account of inaccuracy (1) - Increases the active/working life of the sensors (as long as they are sending a reading) (1) - So that readings are (always/remain) accurate (1)	2

Question	Answer	Marks	Guidance
8(a)	TWO marks available: • Temperature sensor/Thermostat (1) • Keypad (1) • Microphone (1) • Touch screen (1) • (Push) Switch (1) • Remote control (1)	2	Do Not Accept: • Smart Phone etc. • Button • Heat sensor
8(b)	TWO marks available: • The actuator converts a signal <u>from the microprocessor</u> into an action/to perform an action (1) • To turn the pump on/off (1)	2	

Question	Answer	Marks	Guidance																								
8(c)	<table><tr><td colspan="2">EIGHT marks available:</td></tr><tr><td>IF temperature >28</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)</td><td>1</td></tr><tr><td>(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u></td><td>1</td></tr><tr><td>Send signal to turn on indented</td><td>1</td></tr><tr><td>ENDIF correctly positioned within the algorithm</td><td>1</td></tr><tr><td>IF temperature <19 with matching ENDIF</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u></td><td>1</td></tr><tr><td>Any use of ENDIF when in capitals and not split</td><td>1</td></tr><tr><td>UNTIL SYSTEM SWITCHED OFF</td><td>1</td></tr><tr><td>UNTIL statement correctly positioned and indented</td><td>1</td></tr><tr><td>ALL IFs and ENDIFs correctly indented</td><td>1</td></tr></table>	EIGHT marks available:		IF temperature >28	1	THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1	(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1	Send signal to turn on indented	1	ENDIF correctly positioned within the algorithm	1	IF temperature <19 with matching ENDIF	1	THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1	Any use of ENDIF when in capitals and not split	1	UNTIL SYSTEM SWITCHED OFF	1	UNTIL statement correctly positioned and indented	1	ALL IFs and ENDIFs correctly indented	1	8	
EIGHT marks available:																											
IF temperature >28	1																										
THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1																										
(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1																										
Send signal to turn on indented	1																										
ENDIF correctly positioned within the algorithm	1																										
IF temperature <19 with matching ENDIF	1																										
THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1																										
Any use of ENDIF when in capitals and not split	1																										
UNTIL SYSTEM SWITCHED OFF	1																										
UNTIL statement correctly positioned and indented	1																										
ALL IFs and ENDIFs correctly indented	1																										

4 Algorithms and flowcharts

Questions

9626/12 Feb/March 2022 · Question 7

Mark scheme

- 7 The average (mean) of a set of numbers can be found by adding the numbers together and dividing the resulting total by how many numbers there are in the set.

Write an algorithm in pseudocode to enter and find the average of a set of numbers using a REPEAT...UNTIL loop. Your algorithm must work for different sets of numbers.

[8]

- 6** Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

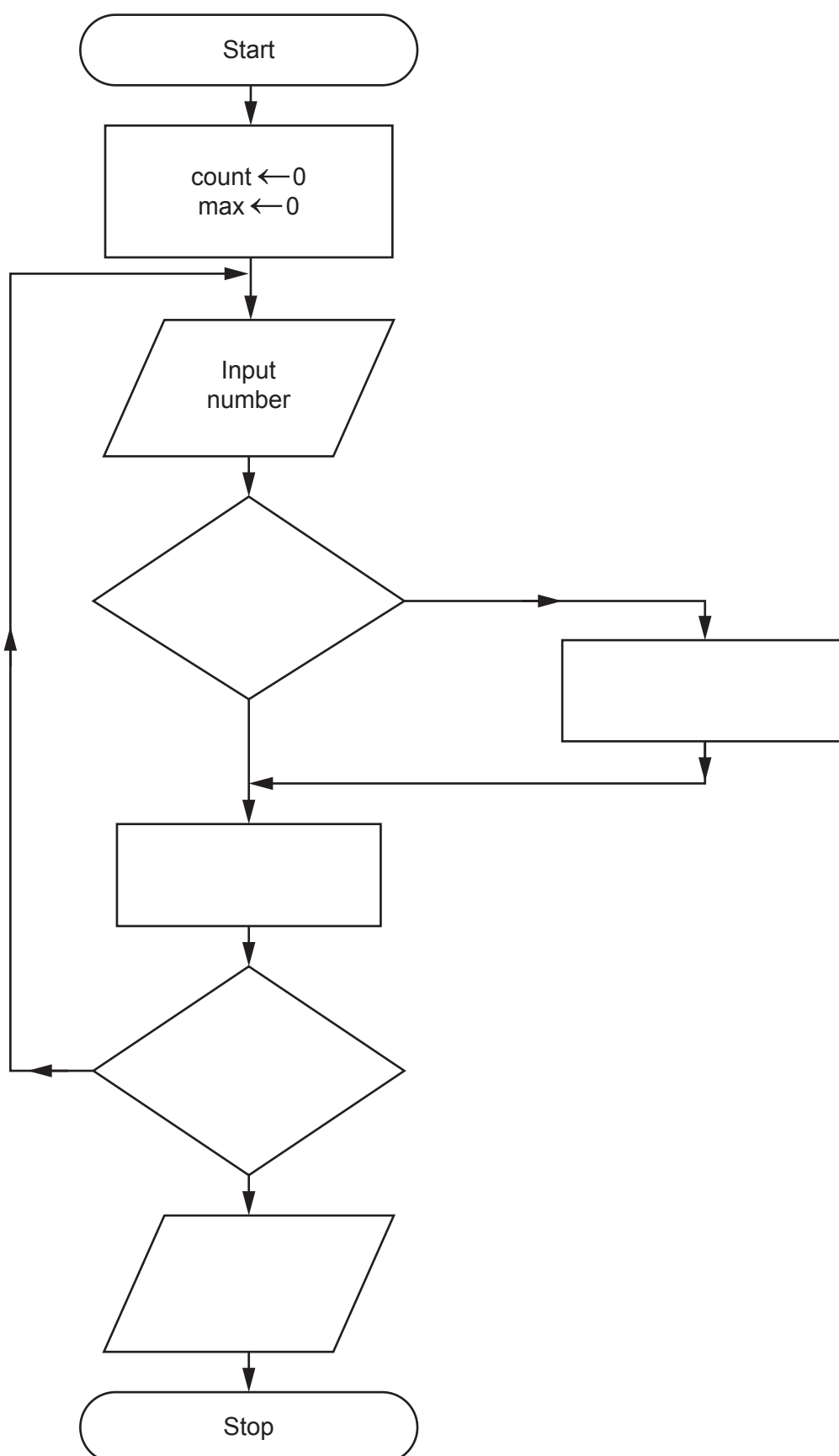
Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
  IF temperature < preset
    THEN
```

[6]

- 7 Josefine has started to draw a flowchart which inputs 10 numbers and outputs the largest value.

Complete the flowchart by filling in the empty boxes and by placing yes and no as appropriate in the diagram.



[7]

9 Many companies use computers to process their payroll.

A transaction file is produced once a week. It consists of the WorkerID and HoursWorked fields. The master file contains the WorkerID, RateOfPay and WagesSoFar fields as well as other details. The WagesSoFar field represents how much each worker has earned so far this year.

Each worker's wage is calculated by multiplying the HoursWorked by the RateOfPay. When the wage for that week is calculated, it is added on to the WagesSoFar field.

Every week the master file has to be updated to include the new wage earned so far this year.

(a) Complete the following pseudocode algorithm which shows the process of updating the master file. You can assume no other transactions are being carried out.

```

READ first record in transaction file
READ first record in old master file
WHILE not end of transaction file
    IF transaction file WorkerID = master file WorkerID
        THEN

```

[6]

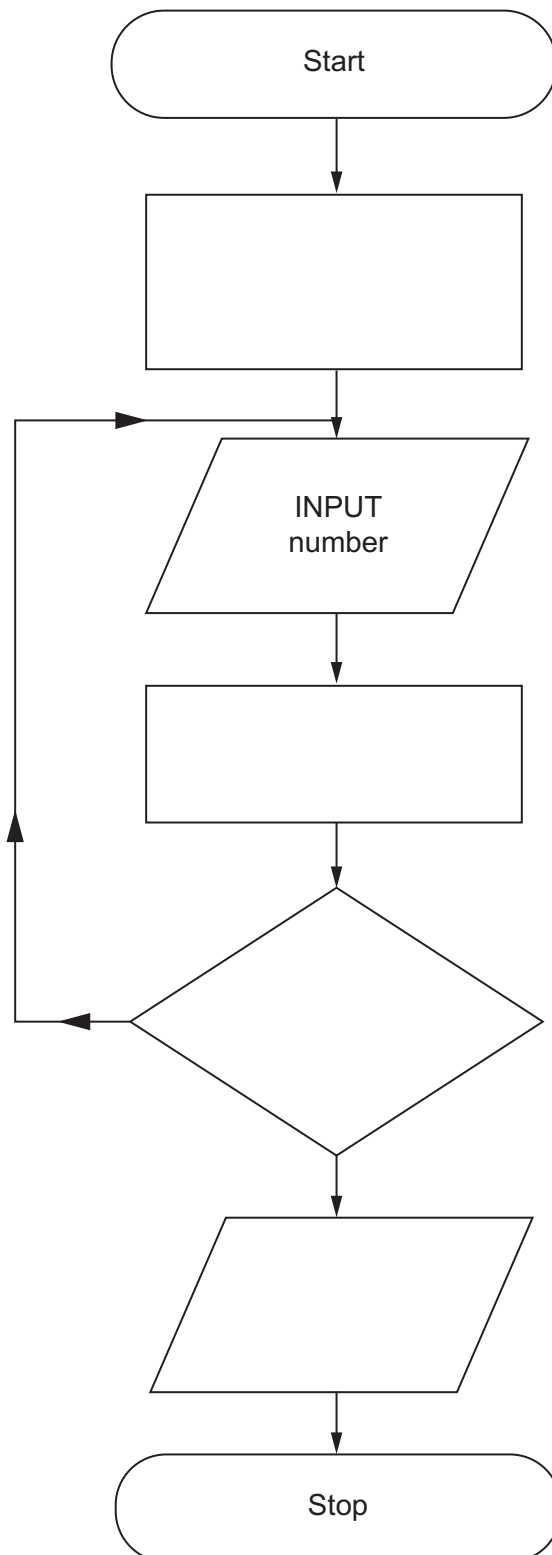
- 8 A microprocessor-controlled car park barrier is used to allow cars to enter a car park. When a car approaches the barrier, the barrier automatically rises to allow the car to enter. It does not lower until the car is safely clear of the barrier system. There is a light sensor and light beam source immediately before the barrier. No tickets are issued by the system. The system operates continuously while it is switched on.

Complete this pseudocode algorithm to describe the processing involved. Your pseudocode algorithm should include IF...THEN statements.

```
REPEAT
    INPUT reading from induction loop
```

..... [7]

- 11 Complete the flowchart below to add up 6 numbers.
Use the variable *total* to store the sum of the numbers and the variable *count* to control the number of times the loop is repeated.



[7]

- 9 Jose has a greenhouse to grow his plants. He lives in a country which has a warm climate so the greenhouse does not need a heater but has microprocessor controlled windows and a temperature sensor.

(a) Describe how the sensor represents the temperature so the computer can read the data.

.....
..... [1]

(b) Complete this pseudocode algorithm to show the processing which takes place to control the temperature of the greenhouse.

```
WHILE system switched on
  INPUT temperature
  IF temperature > preset
    THEN
      IF window closed
        THEN
```

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [8]

- (c)** Discuss the advantages and disadvantages, to Jose, of using microprocessor control in this scenario.

[8]

- 9 A programmer is writing an algorithm in pseudocode to represent a company's payroll system. The system calculates a worker's wages before tax by multiplying the number of hours worked by the rate of pay per hour. A procedure within the main algorithm representing this could be:

```
PROCEDURE BeforeTax(Hours, Rate)
    WagesBeforeTax ← Hours * Rate
ENDPROCEDURE
```

There are two stages involved in calculating the wages after tax:

- the amount of tax paid by the worker is calculated by multiplying the WagesBeforeTax by the rate of tax (35%);
- the amount of tax paid is then subtracted from the WagesBeforeTax.

- (a) Write a procedure for calculating the wages after tax, assuming the value of WagesBeforeTax is passed to it.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) The programmer's algorithm will use the `BeforeTax()` procedure and the procedure created in part (a) to calculate and output the wage after tax for each worker.

Complete the algorithm. The statements have been numbered to help you.

```
1  count ← 0
2  INPUT NumberOfWorkers
3  REPEAT
4  .....
5  .....
6  .....
7  .....
8  .....
9  UNTIL ..... [6]
```

- 8 A teacher wishes to produce a computer program to output the grades awarded for **all** of her students.

If a student scores:

- more than 60 marks, they are awarded a grade A
- 50–60 marks, they are awarded a grade B
- 40–49 marks, they are awarded a grade C
- below 40 marks, they are awarded a grade D.

The teacher has written the following algorithm before writing the program. Unfortunately, there are errors and some lines have been left out (omitted). She has, however, managed to include the correct number of ELSE statements.

```

1.      INPUT mark
2.      IF mark > 60
3.          THEN
4.              PRINT "A"
5.          ELSE
6.              IF mark < 50
7.                  THEN
8.                      PRINT B
9.                  ELSE
10.                     IF mark >= 40
11.                         THEN
12.                             PRINT "C"
13.                         ELSE
14.                             PRINT "D"
15.                     ENDIF
16.             ENDIF

```

Identify each error or omission and how these could be corrected. Line numbers have been included to help you.

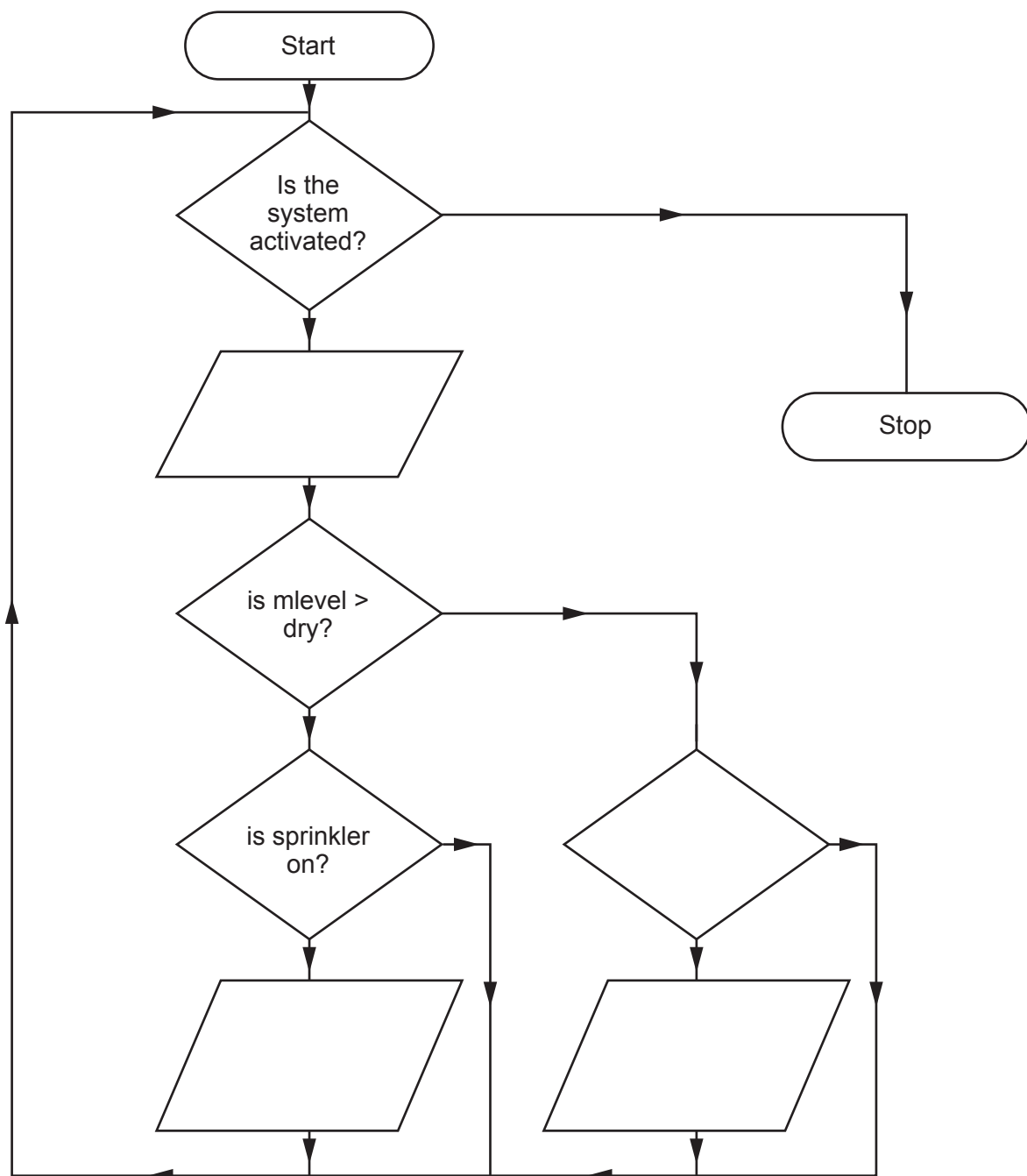
..... [6

- 4 Mark has a small glasshouse. He lives in a warm country where plants can die if they are left without water for too long. The glasshouse has a microprocessor controlled sprinkler system and a moisture sensor in the soil.

Complete this flowchart to show the processing required to control the level of moisture in the soil of the glasshouse. You may assume that a pre-set value, **dry**, has already been input so that the system can test if the amount of moisture in the soil has reached or fallen below a certain level.

The user activates the microprocessor system by pressing a switch.

- (a) Complete the flowchart which inputs the moisture reading in the glasshouse and switches the sprinkler on and off according to this value. It also checks if the sprinkler is already on or off before attempting to switch it on or off.



[6]

(b) Flowcharts often contain subroutines.

(i) Draw the symbol used to represent a subroutine in a flowchart.

[1]

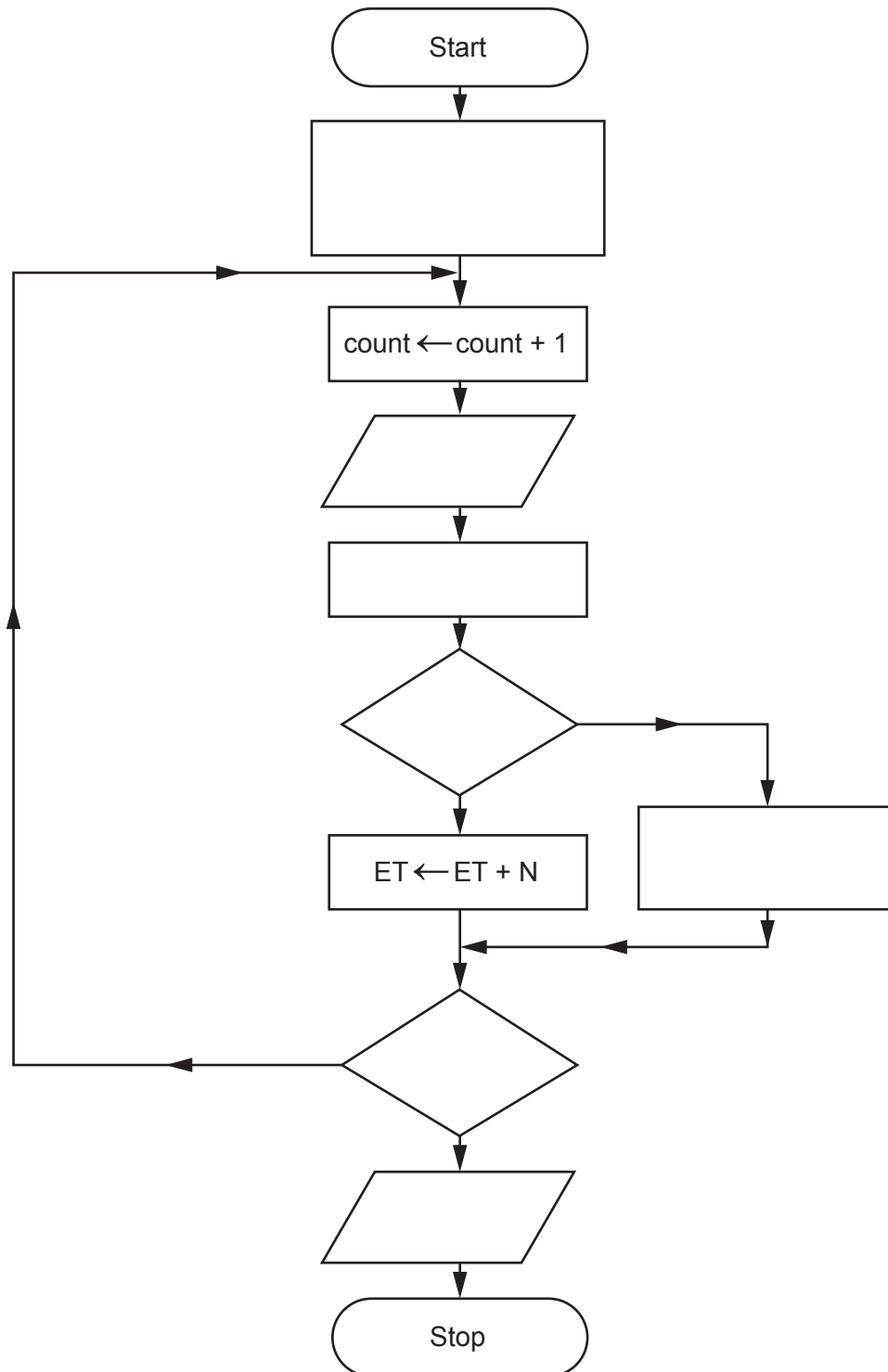
(ii) State what you would expect to see in the first box of a subroutine.

.....

..... [1]

8 Complete this flowchart which will allow the input of ten numbers.

- A number is even if the number divided by 2 is an integer.
- If a number (N) is odd, it will be added to the odd total (OT).
- If a number (N) is even, it will be added to the even total (ET).
- Both these totals will be output after the tenth number has been processed.



[8]

- 10 Gustav has started to write an algorithm using pseudocode. It will input 6 numbers, all less than 1000, and output the smallest value.

Complete the algorithm using pseudocode.

```
count ← 0
smallest ← 1000
WHILE count < 6
```

.....

.....

.....

.....

.....

.....

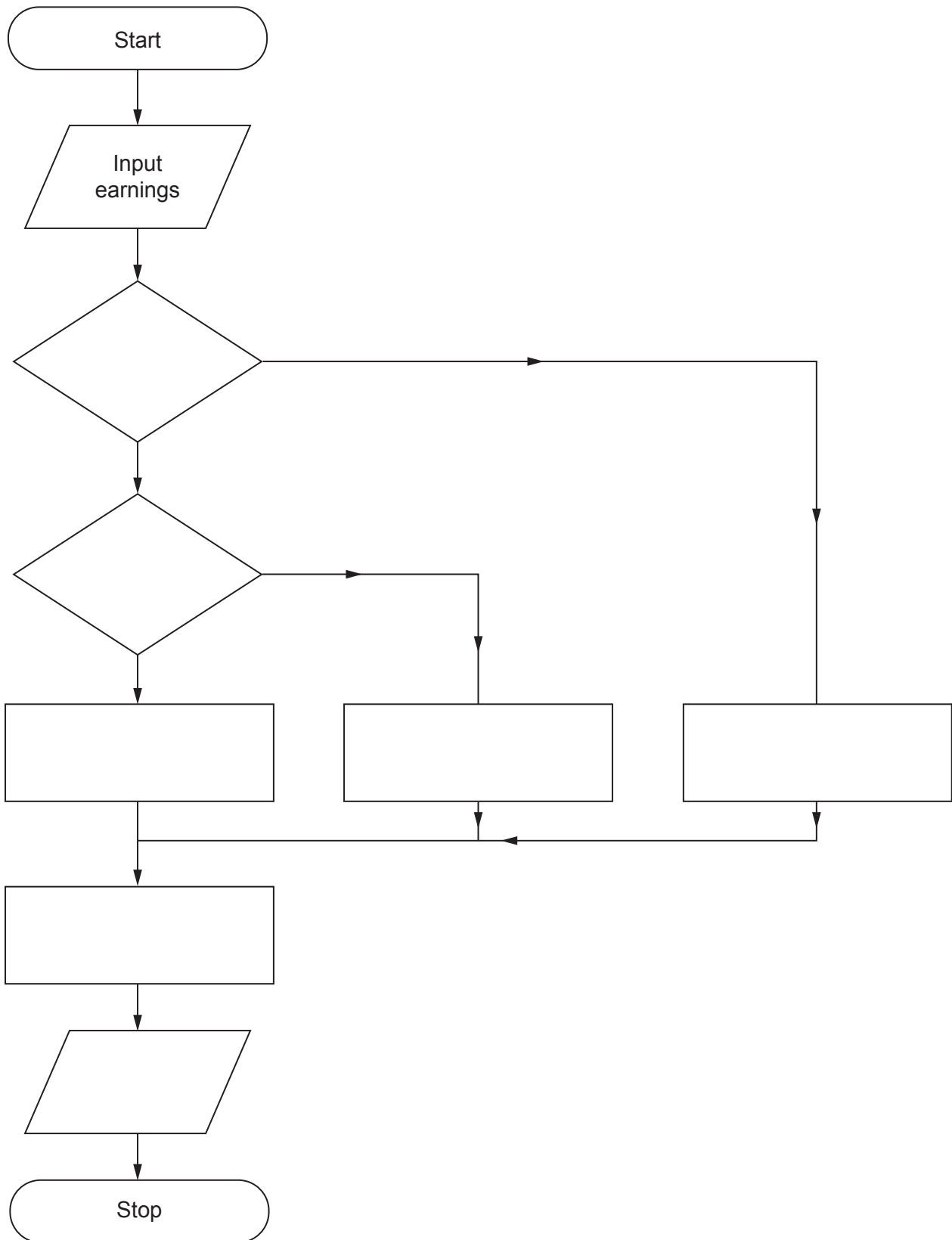
.....

..... [6]

- 11 In Ruritania these are the tax rules.

- People who earn up to \$10 000 pay no tax.
- People who earn between \$10 000 and \$40 000 pay 20% tax on their earnings **above** \$10 000.
- People who earn more than \$40 000:
 pay \$6 000 ($\$30\,000 \times 20\%$)
 and pay 40% tax on their earnings **above** \$40 000.

Complete this flowchart which shows how the net wage (the money earned after tax is paid) is calculated and is output for an individual in Ruritania.



[8]

7 A company employs workers as:

- labourers who are paid \$18 per hour
- office workers who are paid \$22 per hour
- managers who are paid \$30 per hour.

A computer programmer has written part of an algorithm in pseudocode.
The completed algorithm will calculate and output the wage for each worker.

The inputs will be:

- the number of workers
- the job code for each worker (L for labourer, O for office worker or M for manager)
- the number of hours each worker has worked.

If L, O or M is not entered for the job code then the message 'Invalid job code' will be output.
The algorithm will terminate when all the workers' details have been entered. It will use the CASE...ENDCASE construct to determine the rate of pay.

Complete the algorithm by filling in the missing lines.

.....
INPUT number

REPEAT

 INPUT jobcode, hours

 'L' : rate ← 18

 count ← count + 1

.....
[8]

- 9 Sukhjinder, a mathematics student, has defined a procedure using pseudocode. It calculates the area of a parallelogram by multiplying the base by the height. All parallelograms will have an area of less than 1000.

The procedure Sukhjinder has written is:

```
PROCEDURE parallelogram(base,height)
area ← base*height
ENDPROCEDURE
```

He has also begun writing an algorithm which calls the procedure and then passes back the area.

The main algorithm accepts inputs of the base and the height of 10 parallelograms, one at a time. It will output the areas of the largest and smallest parallelograms.

Complete the algorithm for Sukhjinder by filling in the missing lines.

```
count ← 0
smallest ← 1000
```

```
.....
WHILE count < 10
    INPUT base, height
```

```
.....
    IF area < smallest
        THEN
```

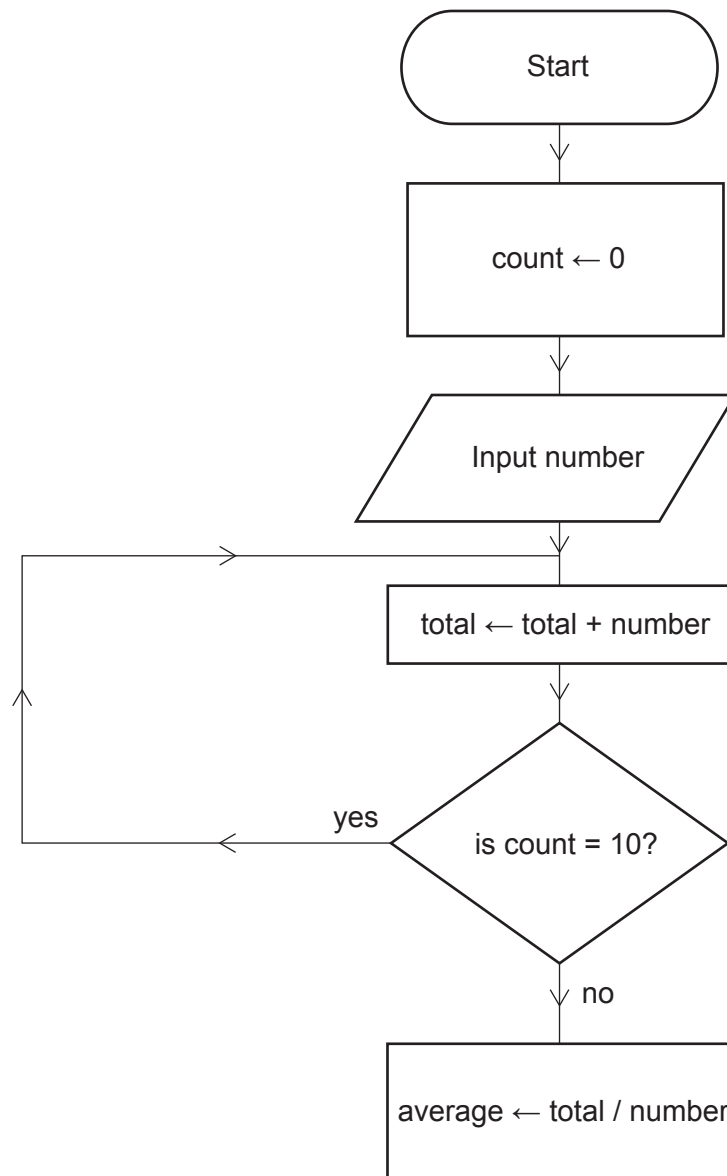
```
.....
        ENDIF
```

..... [8]

8 The average (mean) of a set of 10 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 10.

Juan has attempted to draw a flowchart of an algorithm which finds the average of 10 numbers. He has made some mistakes and omitted to include some boxes.



Suggest **six** improvements he should make to his flowchart.

- 1
- 2
- 3
- 4
- 5
- 6 [6]

- 9 A programmer is writing an algorithm in pseudocode to determine whether a number is a factor of another number.

Y is a factor of X if the result of X/Y is a whole number.

In pseudocode `INT (W)` returns the whole number part of W.

If W is the same as `INT (W)` that means W is a whole number.

Complete the algorithm, including the missing first line, in pseudocode. Each `THEN` or `ELSE` statement should be on a separate line to the action resulting from it. The algorithm uses:

- W to store the result of X divided by Y
- Z to store the value of `INT (W)`.

The algorithm must output the values of Y and X together with an appropriate message if Y is a factor of X or if Y is **not** a factor of X.

.....

`W ← X / Y`

`Z = INT (W)`

.....

.....

.....

.....

.....

..... [8]

- 8 Create a pseudocode algorithm to allow a number to be entered and to output a result when the number is:
- divided by 4 if it is less than 8
 - divided by 8 if it is between 8 and 12 inclusive
 - divided by 12 if it is any other number.

Write the pseudocode algorithm using CASE...ENDCASE.

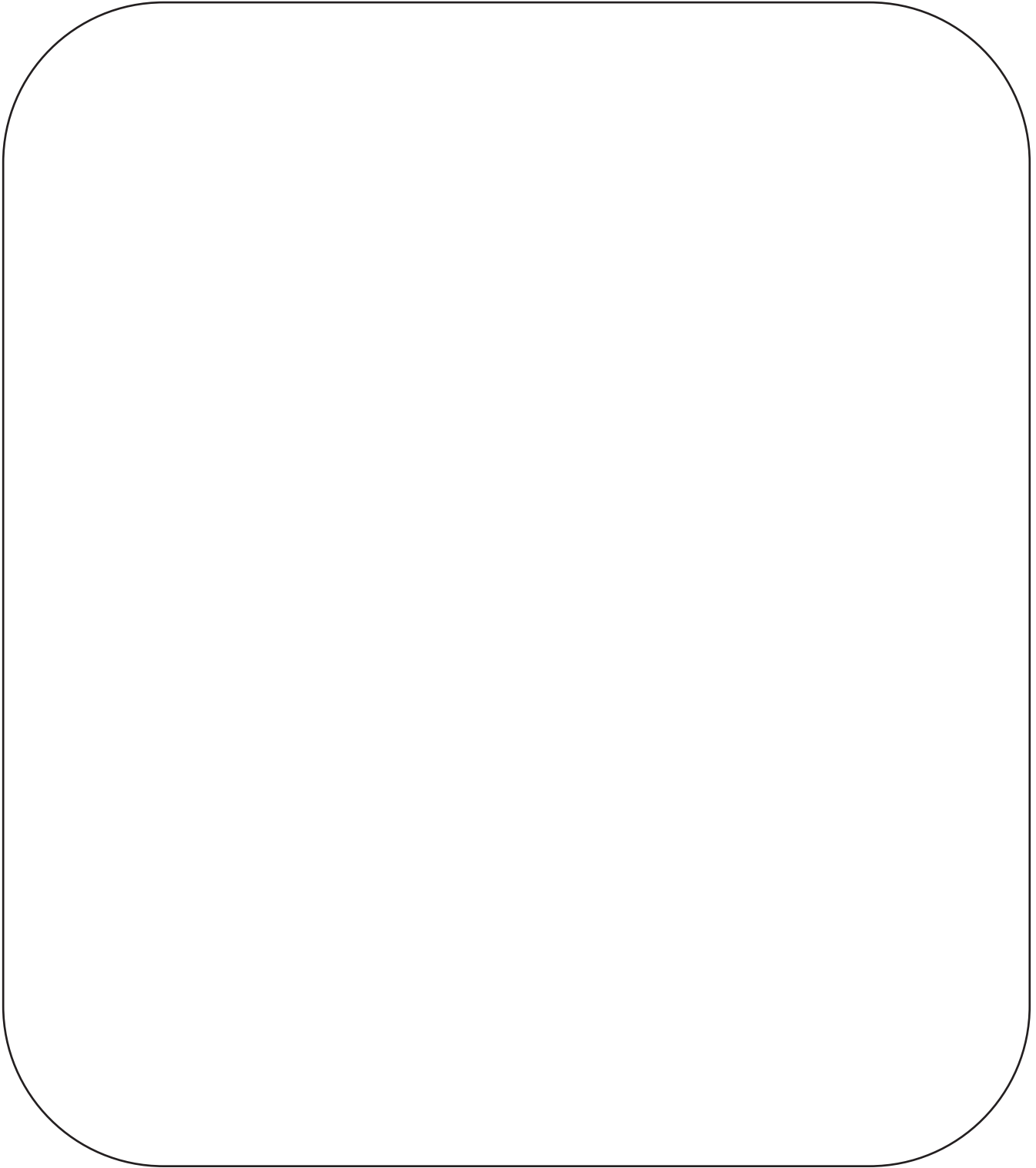
[6]

- 7 Draw a program flowchart, in the space provided, that will allow an examination mark to be entered and then output an appropriate message giving the correct grade, as shown in the table. All marks are integers.

Mark	Message
85 or more	Grade A
70 or more	Grade B
50 or more	Grade C
Any other mark	Grade U

4 Draw a program flowchart to represent this algorithm:

- Input an integer value.
- Repeat this until you have entered 20 values.
- At the end, output the largest value and the average of the values.



[7]

1 Different symbols are used in program flowcharts.

Draw, in the spaces provided, **four** different program flowchart symbols and for each one, state its purpose.

Symbol 1

Purpose
.....
.....

Symbol 2

Purpose
.....
.....

Symbol 3

Purpose
.....
.....

Purpose

.....

.....

[8]

12 Employees log in to their company's database with a User ID and password.

A login screen is displayed. A user enters their User ID.

If employees have forgotten their password, they click on a 'Forgot Password?' button.

Draw a program flowchart to represent the following process.

- If the 'Forgot Password?' button is clicked, a message box appears and asks them to enter their User ID.
- A check will be performed to see if the system recognises the User ID.
- If the User ID is **not** recognised, an error message is displayed and the user will be returned to the start of the process.
- If the User ID is recognised, a security question is asked.
- The user's response to the security question will be checked. If the user's response is correct, then an email containing the password reminder will be sent to the user's email address.
- If an incorrect response is given, an error message will be displayed and the user will be returned to the start of the process.

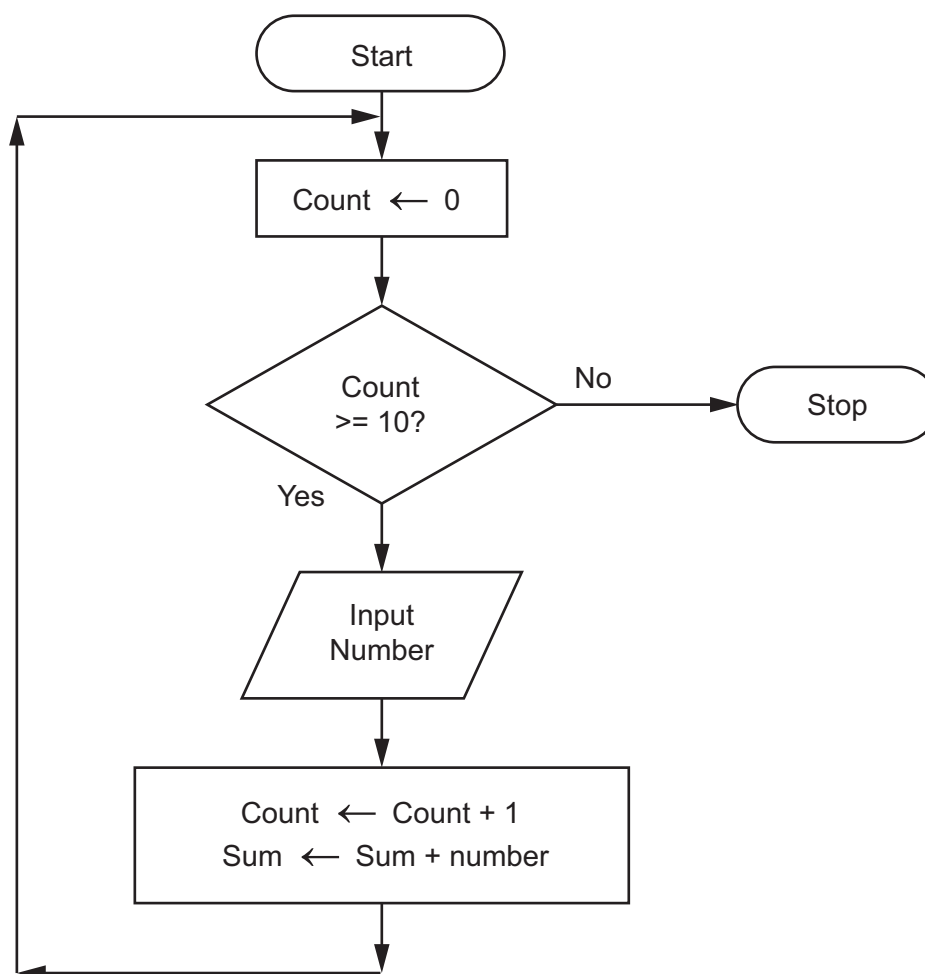
Draw your program flowchart on the opposite page.

[8]

Draw your program flowchart below.

- 9** The program flowchart produces the output of the sum of 10 numbers.

There are errors and omissions in the flowchart.



Explain **two** corrections that should be made to the flowchart.

[4]

7 A teacher is writing a program to process test marks for a class.

For marks above 70, the program will output “You have achieved a Distinction”.

For marks from 55 to 70, the program will output “You have achieved a Merit”.

For marks from 40 to 54, the program will output “You have achieved a Pass”.

For marks below 40, the program will output “You have not reached the Pass mark”.

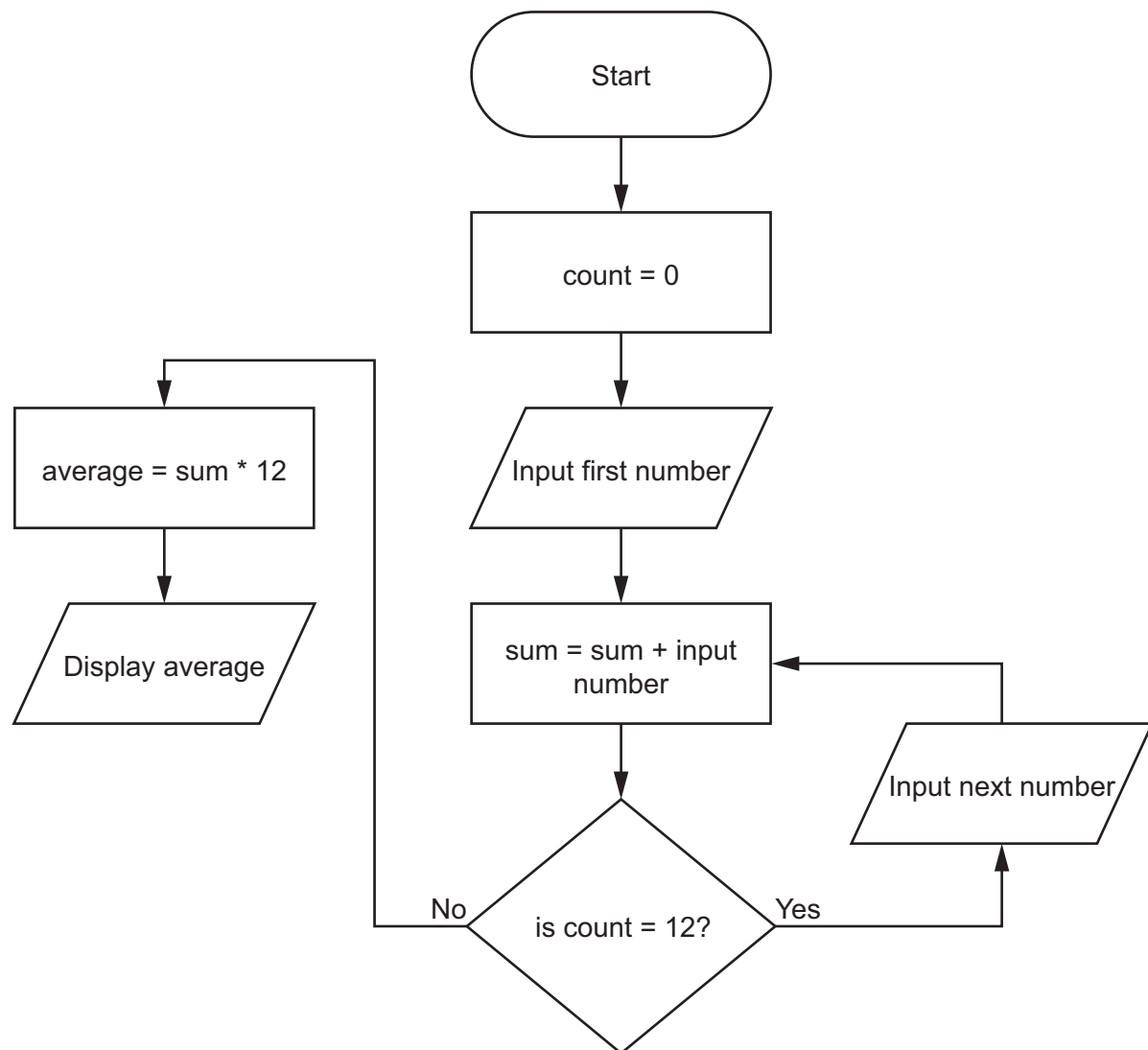
Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark.

[8]

13 The average (mean) of a set of 12 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 12.

A student has attempted to draw a flowchart of an algorithm which finds the average of 12 numbers. They have made some errors.



Redraw the flowchart, with the errors corrected.

[5]

- 5 Lucia has written an algorithm in pseudocode to determine whether a number (X) is a factor of another number (Y).

X is a factor of Y if the result of Y/X is a whole number.

- The algorithm uses a WHILE...ENDWHILE statement to check whether X is greater than zero and that X is less than Y.
 - If X is greater than zero and X is less than Y, the algorithm will output:
 - the values of X and Y
 - an appropriate message if X is a factor of Y
- OR
- an appropriate message if X is **not** a factor of Y.

In pseudocode `INT (W)` returns the whole number part of W.

```

INPUT X
INPUT Y
WHILE X=0 AND <=Y
W←Y/X
Z=INT(W)
IF W=Z
    THEN
        PRINT X, " is a factor of Y"
    ELSE
        PRINT X, " is not a factor of ",Y
ENDWHILE

```

Lucia has made **four** mistakes in her algorithm.

Identify **four** mistakes in the code. For each mistake, explain in detail what the code as written would achieve.

Stating that the code would not work is not sufficient to gain any marks.

Mistake 1

.....

.....

Mistake 2

.....

.....

Mistake 3

.....

.....

Mistake 4

.....

.....

[8]

7 A student has created an algorithm to calculate the average of a set of numbers.

- The user will be asked how many numbers they wish to enter
- The user will respond with a number
- The user will then enter that many numbers
- The algorithm will total the numbers entered and divide the total by how many numbers the user said they will enter.

The algorithm is shown. The line numbers have been included to help you.

```

10 total ← 10
20 PRINT "Input number of repeats"
30 INPUT n
40 count → 1
50 REPEAT
60     count ← + 1
70     PRINT "Input number ", count
80     INPUT number
90     total + number ← total
100 UNTIL count = number
110 average ← total / n
120 PRINT average

```

The student has made some mistakes in the algorithm.

(a) State **three** mistakes in this algorithm and suggest how each can be corrected.

mistake 1

.....

.....

.....

mistake 2

.....

.....

.....

mistake 3

.....

.....

.....

[6]

This algorithm uses a REPEAT...UNTIL loop. The student wishes to change the algorithm so that it uses a WHILE...ENDWHILE loop.

(b) Contrast a REPEAT...UNTIL loop with a WHILE...ENDWHILE loop.

.....

.....

.....

.....

.....

..... [2]

9626/11 May/June 2026 · Question 8

[Mark scheme](#)

- 8 A combined heating and air conditioning system has been installed in a house. The system measures the temperature in the house and reacts as programmed. The user can override the temperature settings of the system to suit their needs.

(a) Identify **two** input devices that will be used as part of this system.

input device 1

input device 2 [2]

(b) Explain, with the example of a pump, the purpose of an actuator in this system.

.....

.....

.....

.....

.....

..... [2]

The system has the following rules:

- If the temperature exceeds 28 degrees Celsius, a fan will turn on and the heating system will turn off.
 - If the temperature falls below 19 degrees Celsius, the fan will turn off and the heating will turn on.
- (c) Complete the pseudocode algorithm to show the processing that takes place in this combined system. The variable `temperature` represents the temperature of the house.

```
REPEAT  
    INPUT temperature
```

- 2 A partially completed algorithm is shown. Line numbers are given for reference. Some lines are incomplete.

The user will input five numbers, one number at a time.

After five numbers have been entered, the largest number and the smallest number will be displayed as part of a suitable message.

The algorithm uses a WHILE loop.

```

10  Count  ←    0
20  Number ←    0
30  Largest_Number ← 0
40  Smallest_Number ← .....
50  INPUT Number
60  Largest_Number ← Number
70  Smallest_Number ← Number
80  WHILE Count < .....
90      INPUT .....
100     IF Number ..... Largest_Number
110         .....
120             Largest_Number ← Number
130         .....
140             IF Number ..... Smallest_Number
150                 THEN Smallest_Number ← Number
160             ENDIF
170     .....
180     Count  ←    .....
190 ENDWHILE
200 PRINT "The largest number is "Largest_Number"
      and the smallest number is "Smallest_Number"."

```


- (a) Use the table to show how each line may be completed. Line 40 has been done for you as an example of how the question should be answered.

line	current code	completed code
40	Smallest_Number ←	Smallest_Number ← 0
80	WHILE Count <	
90	INPUT	
100	IF Number Largest_Number	
110		
130		
140	IF Number Smallest_Number	
170		
180	Count ←	

[8]

- (b) Explain why Largest_Number (line 60) and Smallest_Number (line 70) are both set to the same value collected in line 50.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

4 Algorithms and flowcharts

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Eight from: Input number of numbers Initialise count Initialise total Loop that works involving REPEAT... UNTIL (e.g. count = n) Input number within loop Increment count within loop Update total within loop Calculation of average output average Possible solution could be: <pre>INPUT n count ← 0 total ← 0 REPEAT INPUT number count ← count + 1 total ← total + number UNTIL count = n average ← total / n PRINT average</pre>	8

Question	Answer	Marks
6	Six from: <pre> REPEAT INPUT temperature IF temperature < preset THEN IF pump switched off THEN send signal to switch pump on ENDIF ELSE IF pump switched on THEN send signal to switch pump off ENDIF ENDIF UNTIL system switched off </pre> 1 mark for correct 1st IF statement 1 mark for correctly positioned first action 1 mark for correctly positioned 1st ENDIF 1 mark for correct 2nd IF statement 1 mark for correctly positioned second action 1 mark for correctly positioned 2nd ENDIF 1 mark for correctly positioned 3rd ENDIF 1 mark for correct UNTIL statement... ...1 for correct position of correct UNTIL statement	6

Question	Answer	Marks
7	<pre>graph TD; Start([Start]) --> Init[count ← 0
max ← 0]; Init --> Input[/Input number/]; Input --> Dec1{Is number
≤ max?}; Dec1 -- yes --> IncCount[count ← count + 1]; Dec1 -- no --> UpdateMax[max ← number]; UpdateMax --> Dec1; IncCount --> Dec2{Is count =
10?}; Dec2 -- yes --> PrintMax[/Print max/]; Dec2 -- no --> Input; PrintMax --> Stop([Stop]);</pre> Is number ≤ max in 1st decision box – 1 mark Yes/no in correct positions at 1st decision box – 1 mark max ← number in correct box – 1 mark increment count in correct box – 1 mark is count = 10? in 2nd decision box – 1 mark yes/no in correct positions at 2nd decision box – 1 mark Print max in correct box – 1 mark	7

Question	Answer	Marks
9(a)	<pre> READ first record in transaction file READ first record in old master file WHILE not end of transaction file IF transaction file WorkerID = master file WorkerID THEN WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) WRITE updated master file record to new master file READ next record in transaction file ENDIF READ next record in master file ENDWHILE </pre> Six from: Correct calculation – RateOfPay*HoursWorked – 1 mark Correct calculation - WagesSoFar ← WagesSoFar + (RateOfPay*HoursWorked) – 1 mark Correct update – 1 mark Read next transaction file record... – 1 mark ...immediately before ENDIF – 1 mark ENDIF– 1 mark Read next master file record... – 1 mark ... immediately after ENDIF – 1 mark ENDWHILE at end of algorithm – 1 mark	6

Question	Answer	Marks
9(b)	Eight from: <i>Advantages</i> Batch processing does not require constant use of the computer when it might be needed for other tasks... ...the payroll only needs to be run at the end of the week/month... ...whereas real time processing would require computer to be available all the time... ...batch processing/payroll can be run at convenient times for the company/when the computing resources are less busy/when there is little demand for computer resources/can be run overnight Real-time processing requires immediate processing of data which is unnecessary with payroll... ...as hours worked would only be known at the end of each day so no point in using real-time processing Real-time processing is not usually capable of processing large volumes of data such as running a payroll whereas batch processing is Batch processing systems only need lower specification computers... ...whereas real-time processing needs more complex computer systems Online processing requires more hardware making it more expensive than batch processing Batch processing only needs a few employees to enter the data... ...so total wage bill is less... ...whereas online processing systems can need employees to be constantly using the system In batch processing systems there are fewer/no transcription/update/data errors... ...than human operators would produce in an online system <i>Disadvantages</i> With batch processing there is a time delay between collecting the input data and getting an output/data is not processed until the time specified With batch processing the information is only up to date after the master file has been updated by the transaction file... ...whereas in real-time processing information is always up to date With batch processing systems errors are only revealed during processing/after batch process is complete which can be overnight and so cause delays/errors cannot be corrected until the batch process is complete... ...in online processing errors are revealed, and can be acted upon, immediately Users cannot terminate a process during execution/have to wait until execution is completed Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Question	Answer	Marks
8	REPEAT <i>INPUT reading from induction loop</i> IF reading = pre-set value THEN send signal to raise barrier REPEAT IF light sensor activated THEN send signal to lower barrier ENDIF UNTIL light sensor activated ENDIF UNTIL system switched off Seven from: Appropriate IF statement (comparison of reading) Send signal to raise barrier after first IF Appropriate other IF statement (checking if light sensor activated) Send signal to lower barrier after 2nd IF Appropriate position and matching 1st ENDIF Appropriate position and matching 2nd ENDIF Appropriate UNTIL system switched off statement Appropriate loop checking that light sensor has been activated	7

Question	Answer	Marks
11	<pre>graph TD; Start([Start]) --> Init[count ← 0
total ← 0]; Init --> Input[/INPUT
number/]; Input --> Loop[count ← count + 1
total ← total + number]; Loop --> Decision{is count = 6?}; Decision -- yes --> Print[/PRINT total/]; Print --> Stop([Stop]); Decision -- no --> Loop;</pre> The flowchart starts with an oval 'Start' terminal. It proceeds to a rectangular process box containing 'count ← 0' and 'total ← 0'. This is followed by a parallelogram input box labeled 'INPUT number'. Then, a rectangular process box contains 'count ← count + 1' and 'total ← total + number'. This leads to a diamond decision box asking 'is count = 6?'. A 'yes' path leads down to a parallelogram output box labeled 'PRINT total', which then leads to an oval 'Stop' terminal. A 'no' path leads left and then up, looping back to the entry point of the 'INPUT number' box. Initialise count – 1 mark Initialise total – 1 mark total ← total + number – 1 mark count ← count + 1 – 1 mark is count = 6 – 1 mark PRINT total – 1 mark yes and no in correct positions – 1 mark	7

Question	Answer	Marks
9(a)	One from: - Electrical resistance is changed/voltage is generated (depending on the temperature) (1) Electrical signals are generated (which are converted into values (to represent temperature)) (1).	1
9(b)	<pre> WHILE system switched on INPUT temperature IF temperature > preset THEN IF window closed THEN send signal to motor to open window ENDIF ELSE IF window open THEN send signal to motor to close window ENDIF ENDIF ENDIF ENDWHILE </pre> - send signal to open//open window (NOT 'output') (1) - send signal to open//open window as 1st statement (1) - ENDIF closes the IF statement (1) - ALL three ENDIFs in same vertical plane as relevant IFs (within reason) (1) - ELSE (1) - ELSE in same vertical plane as THEN (1) 'IF window open' (or equivalent) ... (1) ... IF window open condition closed with ENDIF (1) - Close window instruction (1) - Send signal to motor to close window appropriately positioned (1) - IF temperature > preset value condition closed with ENDIF (1) - Whole of IF Window open section indented correctly (allow follow through on close window instruction position) (1) - ENDWHILE as final statement (1)	8

Question	Answer	Marks
9(c)	Eight from: Advantages max 6 • Reduced chance of plants dying/wilting etc (1) • Jose could forget to (e.g.) open window//be so busy that he is unable to (e.g.) open window ... (1) ... computers can be permanently switched on and take action regularly/will not forget (1) • Jose can do other tasks (1) • Computers can take action immediately if the temperature gets too hot ... (1) ... there would be a delay before Jose can take action (1) • Readings taken by computers tend to be more accurate ... (1) ... and action taken is therefore more likely to be the correct one (1) (ACCEPT CONVERSE ARGUMENTS ABOUT JOSE'S INACCURACY) • If Jose was physically disabled, he may find it difficult to open window ... (1) ... whereas computer can do it automatically (1) Disadvantages max 6 • Computers can be expensive to buy ... (1) ... whereas it would not cost Jose any money to look after the plants himself (1) • Computer maintenance costs can be expensive ... (1) ... whereas with Jose that is not a problem (1) • It is quite difficult to program computers to take correct action ... (1) ... but Jose would know instinctively//will have expertise (1) • Sensors can deteriorate//become inaccurate ... (1) ... whereas Jose's actions will tend to be more consistent (1) ... and will need recalibration (1) • A computer-controlled system will not be able to function if there is a problem with the computer/there is a power cut/load shedding (1) Max. six marks if bullets/list of points Must have at least expansions or comparisons to achieve six marks in either section	8

Question	Answer	Marks
9(a)	<pre>PROCEDURE AfterTax(WagesBeforeTax) TaxPaid ← WagesBeforeTax * 35% WagesAfterTax ← WagesBeforeTax - TaxPaid ENDPROCEDURE</pre> • PROCEDURE and Appropriate name as first statement (ignore any gaps) (1) • <code>WagesBeforeTax</code> in brackets on first line (1) • Calculations and assignments within the procedure result in the correct numerical outcome for the question (Ignore errors with variable names, as long as it is clear which variable is being referred to) (1) • Sensible variable names with no gaps (<code>WagesBeforeTax</code> has been given, must be at least TWO extra variables) (1) • ENDPROCEDURE as last statement (1)	4

Question	Answer	Marks
9(b)	<pre>count ← 0 INPUT NumberOfWorkers REPEAT</pre> • <code>INPUT Hours, Rate</code> (1) (Candidate may do this in two steps) • <code>INPUT Hours, Rate</code> as first line/before BeforeTax is called (allow follow through) (1) • (CALL) BeforeTax(Hours, Rate) (1) • (CALL) AfterTax(WagesBeforeTax) (1) • <code>BeforeTax</code> before <code>WagesAfterTax</code> (1) • <code>PRINT/OUTPUT WagesAfterTax</code> (1) (Must be in correct place) • <code>count ← count + 1</code> (1) (can be anywhere) • UNTIL <code>count = NumberOfWorkers</code> (1)	6

Question	Answer	Marks
8	Six marks available: - Any indication that an ENDIF is missing/needs to be added (1) Line 6 should be <code>>=</code> (1) Line 8 B should be in quotes//it should be 'B' (1) - This only allows one student's marks to be entered (1) - There should be an INPUT of the number of students ... (1) ... as the 1st line/before INPUT mark (1) - Any indication that a COUNT is required ... (1) ... Count should be initialised at the second line/3rd line (1) - Any indication that iteration is required ... (1) ... A loop should be set up as the third statement/ after the 1st input and count has been initialised (1) - The count should be incremented within the loop (1) - The loop should be closed as the last statement (1) - Numbers between 50 and 60 would not have a grade//no B grades would be awarded (1) Where candidate has not identified the location of the error/improvement required they can include direct quotes from the code to make it clear to which section they are referring. Where candidate has done so, if their answer is the corrected version of the code, award the mark.	6

Question	Answer	Marks
4(a)	<pre> graph TD Start([Start]) --> D1{Is the system activated?} D1 -- no --> Stop([Stop]) D1 -- Yes --> I1[/Input mlevel/] I1 --> D2{is mlevel > dry?} D2 -- no --> D3{is sprinkler off? R} D2 -- yes --> D4{is sprinkler on?} D4 -- yes --> O1[/send signal to actuator to switch sprinkler off/] D4 -- no --> D3 O1 --> J1(()) D3 -- yes --> O2[/send signal to actuator to switch sprinkler on/] D3 -- no --> J1 O2 --> J1 J1 --> D1 </pre> SIX marks available: • Yes/No in correct position off first decision box (1) • Two other Yes/No used (off a decision) (1) – A further Yes/No used (off a decision) (1) • <u>Input MLevel</u> in correct place (1) • <u>Is sprinkler on/is sprinkler off</u> added in correct place (1) MLevel>dry NO branch: • Is Sprinkler OFF – YES – switch <u>Sprinkler ON</u> in correct place (1) OR • Is Sprinkler ON – NO – switch <u>Sprinkler ON</u> in correct place (1) MLevel >dry YES branch • Is Sprinkler ON -YES- switch <u>Sprinkler OFF</u> in correct place (1) OR • Is Sprinkler OFF – NO – switch <u>Sprinkler OFF</u> in correct place (1) For final four mark points, marks may only be awarded where the logic leading up to the decision matches that given here and is shown in full.	

Question	Answer	Marks			
4(b)(i)	<table border="1" style="width: 100%; height: 80px;"><tr><td style="width: 15%;"></td><td style="width: 70%;"></td><td style="width: 15%;"></td></tr></table>				1
4(b)(ii)	ONE from: - The name of the subroutine (1) - Start (1)	1			

Mark scheme · 9626/11 May/June 2023 · Question 8

[Back to question](#)

Question	Answer	Marks
8	<pre> graph TD Start([Start]) --> Init[OT ← 0 ET ← 0 count ← 0] Init --> LoopStart(()) LoopStart --> CountInc[count ← count + 1] CountInc --> InputN[/Input N/] InputN --> XCalc[X ← N / 2] XCalc --> IsInt{Is/If X = INT(X)?} IsInt -- No --> OTAdd[OT ← OT + N] IsInt -- Yes --> ETAdd[ET ← ET + N] OTAdd --> Count10{count = 10?} ETAdd --> Count10 Count10 -- No --> LoopStart Count10 -- Yes --> Print[/Print OT, ET/] Print --> Stop([Stop]) </pre>	8

Question	Answer	Marks
10	Six marks available <pre> count ← 0 smallest ← 1000 WHILE count < 6 INPUT number IF number < smallest THEN smallest ← number ENDIF count ← count + 1 ENDWHILE PRINT smallest </pre>	6

Question	Answer	Marks
11	<pre> graph TD Start([Start]) --> Input[/Input earnings/] Input --> D1{earnings >= 10 000} D1 -- no --> D2{earnings > 40 000} D1 -- yes --> D2 D2 -- no --> Tax0[tax ← 0] D2 -- yes --> Tax1[tax ← 6000 + ((earnings - 40 000) * 40%)] Tax0 --> Net[net ← earnings - tax] Tax1 --> Net Net --> Print[/Print net/] Print --> Stop([Stop]) </pre>	8

Question	Answer	Marks
7	Eight from: 1 mark for each correct statement <pre> count ← 0 INPUT number REPEAT INPUT jobcode, hours CASE (1 - present and location) OF jobcode (1 - allow on FT) 'L' : rate ← 18 'O' : rate ← 22 'M' : rate ← 30 OTHERWISE (1): PRINT/OUTPUT "Invalid job code" ENDCASE CASE and ENDCASE correctly indented wage ← hours * rate statement outside of CASE statement PRINT/OUTPUT wage (present) PRINT/OUTPUT wage (outside of CASE statement) count ← count + 1 UNTIL count = number UNTIL count = number as final statement </pre>	8

Question	Answer	Marks
9	Max Eight: <pre> count ← 0 smallest ← 1000 largest ← 0 WHILE count < 10 INPUT base, height CALL (must NOT have arrow, or colon) (1) parallelogram (base, height) (may award as FT if error in CALL or in wrong place) IF area < smallest THEN smallest ← area ELSE (appears here even if the THEN statement is not completed/is wrong) IF area > largest THEN (appears here (IF test may be wrong)) largest ← area ENDIF IF and ENDIF indented correctly ENDIF count ← count + 1 (if present) count ← count + 1 (if within while statement but not if within IF statement (unless in both) (allow FT for structure from previous mark point)) ENDWHILE PRINT/OUTPUT smallest (1) largest (Do not accept OUTPUT smallest and largest) ANY PRINT/OUTPUT statement as final statement </pre>	8

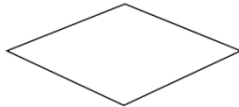
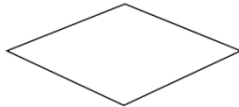
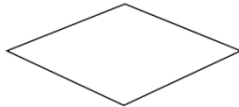
Question	Answer	Marks
8	Six from: - Immediately before/after/inside $\text{count} \leftarrow 0$ box insert $\text{total} \leftarrow 0$ (1) - Immediately before/after/inside $\text{total} \leftarrow \text{total} + \text{number}$ box insert $\text{count} \leftarrow \text{count} + 1$ (1) $\text{average} \leftarrow \text{total}/\text{number}$ should be changed to $\text{average} \leftarrow \text{total}/10$ (1) OR $\text{average} \leftarrow \text{total}/\text{count}$ (1) - yes/no should be reversed (1) OR $\text{count}=10$ changed to $\text{count}<10$ (1) - Loop line should return to flowchart above Input number box (where drawn on the flowchart, no need to delete the original line or add arrow heads) (1) - Print/output average should be included after $\text{average} \leftarrow \text{total}/10$ box (1) - Stop box/end should be included at end of flowchart (1) - Add instructions to input box (1)	6

Question	Answer	Marks
9	<pre> INPUT X, Y W←X/Y Z=INT(W) IF W=Z THEN PRINT/OUTPUT Y, " (e.g.) is a factor of ", X ELSE PRINT/OUTPUT Y, " (e.g.) is not a factor of ", X ENDIF </pre> Eight from: - INPUT X (1) Y (1) - Correct IF statement (1) - IF and ENDIF indented correctly (1) - Use of THEN (1) - Valid output statement (1) with both commas (1) with speech marks (1) - Use of ELSE (1) - THEN and ELSE indented correctly (1) - Valid output statement after ELSE (1) Commas between values and text in second statement text within speech marks (1) - Correct use of ENDIF (1) - No further code after ENDIF statement (1)	8

Question	Answer	Marks
8	Six from: e.g.: INPUT number CASE number >12 : X ← number / 12 >8 : X ← number / 8 OTHERWISE: X ← number / 4 ENDCASE OUTPUT X • INPUT number (name of variable is not important) (1) • INPUT number as first item (1) • CASE number (any variable, but must match MP1 above) in correct place (1) • Action if >12 correct plus calculated output (1) • Action >=8<=12 correct (1st) plus calculated output (MUST be >= or <= structure) (1) • Action <4 correct plus calculated output (1) • ENDCASE in correct position (1) • CASE and ENDCASE correct indentation (1) • ALL other correct aspects indented from CASE and ENDCASE (1)	6

Question	Answer	Marks
7	Eight from: Example of program flowchart <pre> graph TD Start([Start]) --> Input[/INPUT
Mark/] Input --> D1{Mark < 85?} D1 -- No --> P1[/PRINT
"Grade A"/] D1 -- Yes --> D2{Mark < 70?} D2 -- No --> P2[/PRINT
"Grade B"/] D2 -- Yes --> D3{Mark < 50?} D3 -- No --> P3[/PRINT
"Grade C"/] D3 -- Yes --> P4[/PRINT
"Grade U"/] P1 --> Stop([Stop]) P2 --> Stop P3 --> Stop P4 --> Stop </pre> 1 START and STOP terminators 1 mark 2 Input Mark 1 mark 3 A, B and C grade decisions logically correct (check order) 2 marks OR 4 Two of A,B or C decisions logically correct (check order) 1 mark 5 All decision box used have Yes and No 1 mark 5 A, B and C grades output from relevant logic 1 mark 6 U Grade output from correct logic AND correct yes/no decision 1 mark 7 All flow lines correctly located 1 mark 8 All lines have arrow heads with direction of flow 1 mark 9 All correct symbols/shapes 1 mark	8

Question	Answer	Marks
4	Example solution: <pre>graph TD Start([START]) --> Input[/INPUT Num/] Input --> SetMax[SET MAX to Num] SetMax --> SetTotal[Set Total to Num] SetTotal --> SetCount[Set Count to 1] SetCount --> Decision1{In Count = 20?} Decision1 -- YES --> SetAvg[Set Avg to Total/20] SetAvg --> OutputMax[/Output MAX, Avg/] OutputMax --> End([END]) Decision1 -- NO --> Input2[/INPUT Num/] Input2 --> SetTotal2[Set Total to Total + Num] SetTotal2 --> Decision2{Num > MAX?} Decision2 -- YES --> SetMax2[Set MAX to Num] SetMax2 --> SetCount2[Count = Count + 1] Decision2 -- NO --> SetCount2 SetCount2 --> Decision1</pre> SEVEN marks available: 1 Start and End (accept STOP) terminators (1) 2 TWO Correctly drawn decision boxes with YES/NO decisions (1) 3 All YES/NO decisions (used) go to correct outcomes (i.e. all logic is correct) (1) 4 Input box and output box are both parallelograms (1) 5 Process box(es) are rectangles (1) 6 Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1) 7 Input value from user (attempt at) (1) 8 Total set to first input value (1) 9 Inputs limited to 20 values (attempt at use of logic to find) (1) 10 Sum all values input (1) 11 Set new MAX when input value > MAX in loop 12 Increment COUNT (1) 13 Correctly calculate average (must include process/calculation) (1) 14 Output MAX and average (1) (This may happen anywhere, but if in loop, must be after any new number is added).	7

Question	Answer	Marks																
1	EIGHT from: 1 mark for each correct symbol (1st mark) (max 4) 1 mark for each correct matching purpose stated (max 4) Taken from syllabus: <table><tr><th><u>Purpose</u></th><th><u>Symbol</u></th></tr><tr><td>Input/output</td><td></td></tr><tr><td>Decision</td><td></td></tr><tr><td>Terminator (Start/Stop)</td><td></td></tr><tr><td>Process box</td><td></td></tr><tr><td>Subroutine</td><td></td></tr><tr><td>Connector</td><td></td></tr><tr><td>Flow line</td><td></td></tr></table>	<u>Purpose</u>	<u>Symbol</u>	Input/output		Decision		Terminator (Start/Stop)		Process box		Subroutine		Connector		Flow line		8
<u>Purpose</u>	<u>Symbol</u>																	
Input/output																		
Decision																		
Terminator (Start/Stop)																		
Process box																		
Subroutine																		
Connector																		
Flow line																		

Question	Answer	Marks
12	One mark per bullet point to a maximum of eight marks. - Start and stop correct shape (1) - Two relevant questions asked in diamond shape (1) - Third relevant question asked in diamond shape (1) - Any use of YES and NO leading from any relevant question (1) - Any example of Yes and No logic BOTH correct (logic must match the question being asked) (1) - User ID input message (1) - Password reminder input message (1) - User ID Error message (1) - Answer Error message (1) - Email/message sent (1) - Loop back to start (from any error) (1) - Inputs and outputs in parallelograms (may be 1) (1)	8

Question	Answer		Marks
9	One mark per bullet point to a maximum of four marks.		4
	Error (MAX 2)	Answer – MUST identify what is wrong with the code OR what the new code now does, and justify the change	
	Count is incorrect for the logic – this error identified (1st)	e.g. - Count > 10 means that the code ends on first run through AND - Change to Count < 10 OR - Change to Count = 9 (1) - Candidate may answer by arguing that Count should have been set to 1. This is acceptable, but this expansion may only be marked once.	
	SUM variable not initialised (1st)	- SUM ← 0 added to process box AND - SUM + calculation will not work, as no initial value for SUM (1)	
	No output (1st)	- The flowchart will not output a value for the sum of the numbers - AND - OUTPUT added for SUM along NO path (1)	
	Loop goes to wrong place (1st)	- Flowchart will run without end OR - Loop goes to wrong part of the flowchart AND - Loop changed to go after initialisation of COUNT/before decision box (1)	

Question	Answer	Marks
7	<i>Example solution:</i> <pre> Count ← 0 REPEAT INPUT mark IF mark > 70 THEN OUTPUT "You have achieved a Distinction" ENDIF IF mark >= 55 AND <= 70 THEN OUTPUT "You have achieved a Merit" ENDIF IF mark >=40 AND <= 54 THEN OUTPUT "You have achieved a Pass" ENDIF IF mark <40 THEN OUTPUT "You have not reached the Pass mark" ENDIF Count ← Count+ 1 UNTIL Count = 20 </pre> Eight marks from: • Initialise Count before Mark variable is input (1) • REPEAT in correct logical location (1) • Input Mark (1) • Correct Distinction grade criterion//correct fail criterion (1) • Correct Merit grade criterion//correct Pass grade criterion (1) • Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1) • Use of one IF...ENDIF (1) • All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)	8

Question	Answer	Marks
13	Max five marks from: • Add sum = 0 (sum initialisation) (1) • Add count = count +1 (iteration of count) (1) • Replace average calculation from * to / (1) • Reverse Yes/No correctly (1) • Add end shape (1) Corrected flowchart: <pre>graph TD; Start([Start]) --> Sum0[sum = 0]; Sum0 --> Count0[count = 0]; Count0 --> Input1[/Input first number/]; Input1 --> SumAdd[sum = sum + input number]; SumAdd --> CountInc[count = count + 1]; CountInc --> Decision{is count = 12?}; Decision -- Yes --> AvgCalc[average = sum / 12]; AvgCalc --> Display[/Display average/]; Display --> End([End]); Decision -- No --> Input2[/Input next number/]; Input2 --> SumAdd;</pre>	5

Question	Answer	Marks														
5	Eight available. Two per error and improvement. ONLY consider a maximum of four errors. <table><tr><td>Location (1st mark)</td><td>What would be achieved</td></tr><tr><td>WHILE X=0</td><td>WHILE loop would continue if X is equal to zero/would not continue if x is greater than 0</td></tr><tr><td>AND <=Y</td><td>Logic error as <Y is not being compared to anything</td></tr><tr><td><=Y</td><td>If X = Y, algorithm will continue.</td></tr><tr><td>PRINT X, "is a factor of Y"</td><td>Would not show value of Y / would show Y as the value</td></tr><tr><td>Lines of code following the WHILE statement are NOT indented</td><td>Code would run (regardless if conditions are met) ONCE/no loop</td></tr><tr><td>No ENDIF</td><td>The IF statement will only consider one line/will not compute as no close detected</td></tr></table>	Location (1st mark)	What would be achieved	WHILE X=0	WHILE loop would continue if X is equal to zero/would not continue if x is greater than 0	AND <=Y	Logic error as <Y is not being compared to anything	<=Y	If X = Y, algorithm will continue.	PRINT X, "is a factor of Y"	Would not show value of Y / would show Y as the value	Lines of code following the WHILE statement are NOT indented	Code would run (regardless if conditions are met) ONCE/no loop	No ENDIF	The IF statement will only consider one line/will not compute as no close detected	8
Location (1st mark)	What would be achieved															
WHILE X=0	WHILE loop would continue if X is equal to zero/would not continue if x is greater than 0															
AND <=Y	Logic error as <Y is not being compared to anything															
<=Y	If X = Y, algorithm will continue.															
PRINT X, "is a factor of Y"	Would not show value of Y / would show Y as the value															
Lines of code following the WHILE statement are NOT indented	Code would run (regardless if conditions are met) ONCE/no loop															
No ENDIF	The IF statement will only consider one line/will not compute as no close detected															

Question	Answer	Marks
7(a)	SIX marks available, TWO per identified and corrected error: <pre> 10 total ← 10 (1) (Incorrect initial value//value should be 0//value does not need to be set) (1) 20 PRINT "Input number of repeats" 30 INPUT n 40 count → 1 (1) (Count arrow pointing in wrong direction) (1) (Count value should be 0 (this mark may be claimed at line 100)) (1) 50 REPEAT 60 count ← + 1 (1) (Incorrect statement for increment) (1) 70 PRINT "Input number ",count 80 INPUT number 90 total + number ← total (1) (Incorrect order//setting wrong value to variable) (1) 100 UNTIL count = number (1) (Incorrect statement for comparison/should compare to n) (1) 110 average ← total / n 120 PRINT average </pre>	6

Question	Answer	Marks
7(b)	TWO marks available: • A WHILE...ENDWHILE loop will only be executed if the condition is true (1) BUT a REPEAT...UNTIL loop will happen even it is not true. (1) • A REPEAT...UNTIL loop is guaranteed to happen, (1) but a WHILE...ENDWHILE loop may not happen. (1) • For a WHILE...ENDWHILE loop, the condition is tested before the statement runs (1) for a REPEAT...UNTIL loop, the condition is tested after the statement runs. (1) • A WHILE...ENDWHILE loop is a pre-condition loop, (1) and a REPEAT...UNTIL loop is a post-condition loop. (1) • A REPEAT ...UNTIL loop ends when a condition is met (1) but a WHILE...ENDWHILE loop ends when a condition is not met. (1)	2

Question	Answer	Marks	Guidance
8(a)	TWO marks available: • Temperature sensor/Thermostat (1) • Keypad (1) • Microphone (1) • Touch screen (1) • (Push) Switch (1) • Remote control (1)	2	Do Not Accept: • Smart Phone etc. • Button • Heat sensor
8(b)	TWO marks available: • The actuator converts a signal <u>from the microprocessor</u> into an action/to perform an action (1) • To turn the pump on/off (1)	2	

Question	Answer	Marks	Guidance																						
8(c)	EIGHT marks available: <table><tr><td>IF temperature >28</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)</td><td>1</td></tr><tr><td>(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u></td><td>1</td></tr><tr><td>Send signal to turn on indented</td><td>1</td></tr><tr><td>ENDIF correctly positioned within the algorithm</td><td>1</td></tr><tr><td>IF temperature <19 with matching ENDIF</td><td>1</td></tr><tr><td>THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u></td><td>1</td></tr><tr><td>Any use of ENDIF when in capitals and not split</td><td>1</td></tr><tr><td>UNTIL SYSTEM SWITCHED OFF</td><td>1</td></tr><tr><td>UNTIL statement correctly positioned and indented</td><td>1</td></tr><tr><td>ALL IFs and ENDIFs correctly indented</td><td>1</td></tr></table>	IF temperature >28	1	THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1	(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1	Send signal to turn on indented	1	ENDIF correctly positioned within the algorithm	1	IF temperature <19 with matching ENDIF	1	THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1	Any use of ENDIF when in capitals and not split	1	UNTIL SYSTEM SWITCHED OFF	1	UNTIL statement correctly positioned and indented	1	ALL IFs and ENDIFs correctly indented	1	8	
IF temperature >28	1																								
THEN (on separate line from the IF) (award if EITHER THEN fits this mark point)	1																								
(Send signal to) turn on fan ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE >28 <u>ONLY</u>	1																								
Send signal to turn on indented	1																								
ENDIF correctly positioned within the algorithm	1																								
IF temperature <19 with matching ENDIF	1																								
THEN (on separate line from the IF) (Send signal to) turn heating on ONLY AWARD IF THIS IS AS A CONSEQUENCE OF TEMPERATURE <19 <u>ONLY</u>	1																								
Any use of ENDIF when in capitals and not split	1																								
UNTIL SYSTEM SWITCHED OFF	1																								
UNTIL statement correctly positioned and indented	1																								
ALL IFs and ENDIFs correctly indented	1																								

Question	Answer	Marks	Guidance																											
2(a)	EIGHT marks available: <table><tr><th>Line</th><th>Item to be added</th><th></th></tr><tr><td>80</td><td>4</td><td>1</td></tr><tr><td>90</td><td>Number (accept lower case)</td><td>1</td></tr><tr><td>100</td><td>></td><td>1</td></tr><tr><td>110</td><td>THEN (accept lower case)</td><td>1</td></tr><tr><td>130</td><td>ELSE (accept lower case). Do not award ELSEIF</td><td>1</td></tr><tr><td>140</td><td><</td><td>1</td></tr><tr><td>170</td><td>ENDIF (DO NOT accept lower case or if split)</td><td>1</td></tr><tr><td>180</td><td>Count + 1 (accept lower case)</td><td>1</td></tr></table>	Line	Item to be added		80	4	1	90	Number (accept lower case)	1	100	>	1	110	THEN (accept lower case)	1	130	ELSE (accept lower case). Do not award ELSEIF	1	140	<	1	170	ENDIF (DO NOT accept lower case or if split)	1	180	Count + 1 (accept lower case)	1	8	Each answer is CAO ONLY considers answers on the table
Line	Item to be added																													
80	4	1																												
90	Number (accept lower case)	1																												
100	>	1																												
110	THEN (accept lower case)	1																												
130	ELSE (accept lower case). Do not award ELSEIF	1																												
140	<	1																												
170	ENDIF (DO NOT accept lower case or if split)	1																												
180	Count + 1 (accept lower case)	1																												
2(b)	FOUR from: - It is the only number that has been entered at that stage (1st) there is no comparison to be made (1) (award once only) - It must be both largest and smallest (1st) there is no comparison to be made//any comparison would be comparing a number with itself (1) (award once only) - Ensures that the highest/lowest mark recorded is valid (1) - Overwriting both set values with a value inputted by the user (1) - One mark for a numerical example of either lower or higher number being ignored (1)	4																												

Questions

Mark scheme

- Evaluate the different methods which can be used to prevent phishing.

[8]

- 9 Hackers sometimes commit fraud after causing spyware to be installed on other users' computers.

(a) Describe what is meant by the term 'spyware'.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Explain how a hacker uses the information gained from spyware to commit fraud.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

8 Pharming is a method used by hackers to obtain information from bank customers.

(a) Explain what is meant by pharming.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe **three** methods a customer could use to minimise the risk of becoming a victim of pharming.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 3 Wei is trying to minimise the effects of malware on her computer system.

Describe the types of software Wei would need and how she could use them to help prevent malware affecting her computer system.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 2 Sandeep works for a company which uses software to help protect its computer systems against malware. The company is looking for additional ways to minimise the effects of malware. Sandeep and the other workers have already been advised on good practice when logging on and how to use emails safely.

Describe other methods the company can instruct workers to use so that malware can be prevented from entering its computer systems.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 10** Companies who store the personal data of individuals use a number of methods to prevent illegal access to this data.

Apart from preventing illegal access, suggest other methods companies can use to protect the confidentiality of this data.

[5]

3 Phishing, smishing and vishing are methods of fraudulently obtaining personal data.

(a) Describe **two** differences between phishing and vishing.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe how smishing is used to obtain personal data.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 7 (a) Explain how malware can be used to carry out industrial espionage.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) A rootkit is a type of malware.

Describe the features of a rootkit.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 9** Hackers often use malware to attack the computer networks of large organisations.

Analyse the consequences of this for organisations.

[8]

11 Many companies store the personal data of their workers electronically.

(a) Describe what is meant by the term ‘personal data’.

.....

.....

.....

.....

.....

..... [2]

(b) Describe the methods that could be used to attempt to prevent unauthorised access to this personal data.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 3** Most computer systems have a variety of utility software including anti-virus software.

Discuss the benefits and drawbacks of anti-virus software.

[6]

- 7 Phishing is a method used by hackers to obtain banking information from computer users.

Explain the process of phishing.

[5]

9 Describe what is meant by the malware terms.

(a) worm

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Trojan

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 9** Vishing is a method used by fraudsters to obtain banking information from bank customers by phoning them.

Evaluate the methods that bank customers could use to prevent becoming a victim of vishing.

[8]

4 Some individuals post personal data on social media or in emails.

Discuss why personal data needs to be kept confidential.

[8]

8 Spyware is a type of malware.

Describe the consequences to the individual of the use of spyware on their computer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 Explain how each of these methods keeps data safe from unauthorised users.

(a) encryption

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) a firewall on a network

.....

.....

.....

.....

.....

.....

.....

..... [3]

6 Describe these types of malware.

- (a) Malicious bots
-
-
-
-
-
-
-
-
- [4]

(b) Rootkit
.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

(c) Virus
.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

2 Unauthorised people can access personal data in different ways.

Describe what is meant by the terms:

(a) pharming

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) smishing

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 7 Explain why personal information should be kept confidential. Your answer must include examples of how personal information can be used if it is **not** kept confidential.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9 Describe the following types of malware:

(a) spyware

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) worm

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

3 Describe the following types of malware:

(a) adware

[6]

(b) ransomware.

[4]

3 Describe **four** ways a firewall increases security of personal data.

Way 1

.....

.....

.....

Way 2

.....

.....

.....

Way 3

.....

.....

.....

Way 4

.....

.....

.....

[4]

- 6 Computer systems can be affected by malware. A network manager has already installed malware prevention software.

Describe additional methods the network manager can use to help prevent malware from entering the computer systems.

[6]

- 4** Describe what is meant by the malware term *trojan*.

..... [3]

3 An unauthorised person can try to obtain login details to bank accounts.

(a) Describe how vishing can be used to obtain these login details.

.....

.....

.....

.....

.....

..... [2]

(b) Identify **two** ways of preventing login details from being obtained by vishing.

.....

.....

.....

.....

.....

..... [2]

6 Malware can be used to gather information to enable people to commit fraud.

(a) Identify **two** other uses for malware.

1

.....

2

.....

[2]

(b) Identify and explain **three** disadvantages of using anti-virus software to protect your computer from malware.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

1 Irene would like to post a photograph of her family on a social media website.

- (a) Describe **two** ways in which Irene can ensure that the personal data in the photograph can be protected after it has been posted online.

way 1

.....

.....

way 2

.....

.....

[2]

Irene has installed a firewall on her home computer network.

- (b) (i) Describe how a firewall works to prevent unauthorised access to a computer network.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii) Describe **one** drawback of using a firewall on a computer network.

.....

.....

..... [1]

7 A company has been hit by a ransomware attack.

(a) Describe the possible effects of this ransomware attack on the company.

[2]

(b) Evaluate the use of physical methods to prevent a second ransomware attack on the company.

[6]

- 2 Matthew has money in his bank account. Hackers are trying to gain access to his account and steal his money.

Identify **three** items of personal data that hackers could use to access this bank account.

1

2

3

[3]

5 eSecurity

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 8

[Back to question](#)

Question	Answer	Marks
8	Four matched pairs from examples such as: It is important to use anti-phishing software often found within web browsers or email software to prevent emails from even arriving... ...but not all web browsers provide this facility so it is very important to use one that does... ...but unless the software is updated at regular intervals it may not be able to prevent attacks Look for grammatical and/or spelling mistakes in emails as this will cause receivers to be reluctant to reply... ...however, it may be possible that users could see minor grammatical and/or spelling mistakes in genuine emails and users may delete them unnecessarily Users should never trust emails that come from people whose names they do not recognise so if any email looks suspicious it is important to delete it... ...but sometimes, emails may come from people whose names they do not recognise but they may still be genuine If an email starts 'Dear customer' rather than using the receiver's name, this can be viewed with suspicion and it is important to delete it... ...however, some companies use this method of communication for general emails so if the emails is not asking for personal information there may be no need to delete it. Emails requesting personal or financial information are likely to need deleting as it is important to know that personal and financial information should never be sent in an email... ... check with sender/company if email asks for personal information as it may be genuine If the email contains a message that the receiver is entitled to a large amount of money it is likely to be fake and deletion is probably needed... ...however, it is possible that a distant relative has left the user a lot of money so it is important to read these messages carefully before considering deletion Users need to treat links within emails with suspicion... ... but if a user decides never to click on links within emails this can lead to important emails not being acted upon Must be a proper evaluation to obtain full marks Max. four if points not matched Must have expansions to be a proper evaluation	8

Question	Answer	Marks
9(a)	Three from: Spyware is a type of malware that collects information about a computer user's activities without their knowledge... ...but a user might notice a reduction in processing power and bandwidth Data such as web browsing habits are collected Data such as user's personal data are collected... ...such as the user's credit card numbers/usernames and passwords A keylogger can be used to collect a record of the user's keystrokes Data is sent to the hacker It does not replicate like a virus or worm	3
9(b)	Four from: Hackers can sell the personal/financial information to other criminals Hackers can impersonate the user/commit identity theft Hackers can use the user's financial data to ask the bank for a new PIN Hackers can ask the bank for an extra bank card Hackers can buy goods through internet shopping using the credit card details they have obtained Hackers can withdraw large sums of money from the user's bank account	4

Question	Answer	Marks
8(a)	Four from: Hacker <u>creates</u> a fake website which looks like the actual bank's website/legitimate-looking website Fools the user into thinking they have gone to the correct website/have not been re-directed Hacker installs a piece of malicious software/code on customer's computer/ user downloads a piece of malicious software/code without realising it Software corrupts the Hosts file on the user's computer Software adds the URLs of banks together with IP address of the fake website Fraudster redirects genuine website's traffic to own website/ when user enters a bank's URL they are redirected to hacker's websiteuser is now sending personal details to fraudster's website unknowingly / website asks customer to enter banking information/hacker gains <u>personal</u> information from website /customer types in personal details on website Hacker uses the acquired information from the website to log on to the user's bank account/make purchases/commit identity theft/withdraw cash	4
8(b)	Three from: Use up to date anti-virus software to prevent the downloading of malicious software Use firewalls to prevent malicious software entering the computer Use trusted DNS servers / don't enter personal data into sites without https The latest browsers alert users when a fake website has been loaded/users need to use the latest <u>browser</u> software updates Use a trusted, legitimate Internet Service Provider Check the digital certificate to make sure the site is legitimate Be suspicious and careful not to use <u>websites</u> which contain poor grammar or spelling Avoid using public Wi-fi systems	3

Question	Answer	Marks
3	Four from: Use anti-virus software to scan for malicious content/virus as it enters the computer/detect any unusual behaviour in the computer/system... ...anti-virus can remove / quarantine viruses/alert/recommend the user to take action Keep anti-virus software up to date/running continuously/set up frequent scans Only use one anti-virus program as two different programs can fight against each other Use a firewall to filter incoming traffic... ... if it detects malicious software trying to enter the system it can prevent it from entering Use anti-spyware software if the anti-virus software does not include it Use a spam filter to minimise the effect of spam if the user's email software does not use one	4

Question	Answer	Marks
2	Four from: - Follow anti-malware policies. (1) - Report malicious websites to the e.g. network manager/compliance(1) - Maintain up to date knowledge of risks (from malware) (1) - Be suspicious about clicking on links (1) - Know that they must not connect removable media/memory sticks//personally owned device to their computer//the network (1) - Report strange/unexpected system behaviour to the e.g.network manager/compliance (1) - Only download software from a trusted source//do not download software onto the computer (1) - Disconnect computer from network when not making use of network facilities (1)	4

Question	Answer	Marks
10	Five from: - Follow data protection act rules (1) - Use only the minimum necessary confidential personal information (1) - Use anonymised information wherever possible (1) - Securely dispose of confidential personal information when it is no longer needed (1) - Only share/disclose/publish confidential personal information where it is lawful to do so (1) - Only allow authorised workers to view the data (1) - Encrypt the data (1) - Ensure workers have a duty of confidence/sign a confidentiality agreement (1) - Use aggregated data wherever possible (1) - Remove geotags from photos/images (1)	5

Question	Answer	Marks
3(a)	Two matched pairs from: - Phishing involves sending an email (to gather personal details) ... (1) ... whereas vishing involves a phone/voice call (to gather personal details) (1) (Candidate MUST have referred to personal data/details being gathered by either method within the body of the answer) - Phishing involves sending a website link/email address, asking the receiver to email back/click on link ... (1) ... whereas vishing involves getting the victim to give out their details verbally / leaves a number on their answerphone / tells them over the phone the bank's supposed phone number the victim can call to check the call is authentic (1) - With phishing the link takes the victim to a (fake) website ... (1) ... whereas with vishing the phone number causes the victim to be put through to the hacker's own phone/answerphone (1) - Vishing involves the customer having to respond immediately to the caller ... (1) ... whereas phishing allows the user more time to check whether the email is authentic (1) One mark may be awarded for unmatched statement Two marks may be awarded for one unmatched statement from each side	4
3(b)	Three from: - Fraudster sends a (cell phone) <u>text message</u> (or VERY close alternative) to persuade customer to divulge their personal information (accept any reference to personal information/data in the body of the answer) ... (1st) ... includes a link to fraudster's/fake/replicated website (1) ... Example of message that could be used (must be more than 'please send information') (1) ... the target's immediate attention/action is demanded (accept any reference to peril or scaring the target into immediate action) (1) - They include a telephone number in the message that connects to an automated voice response system (1)	3

Question	Answer	Marks
7(a)	Four marks available: • Malware can be downloaded without realising when an employee (e.g.) clicks on adware (1) • Spyware/Trojan is an example of malware that can be used ... (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Malware can gain access to computers/network/database (NOT – the business) to get files/secrets ... (1) ... such as (e.g.) intellectual property/research projects/future business plans (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Malware is used to exploit weaknesses in software to gain access ... (1) ... the data collected is sent back to the hacker (Allow once only) (1) • Types of spyware target specific computers for industrial espionage ... (1) ... to spy on a particular company (1)	4
7(b)	Four marks available: • (Allows a hacker to) install a set of tools (in a computer) (1) • Allows (continuous) remote access (to the computer) (1) • It allows the hacker/third party to gain administrator access to a victim's computer//gain (full) control of a computer (1) • It's buried deep within the operating system (1) • The user is unaware that their computer has been infected//secretly gains access (1) • Designed to stay hidden/to avoid detection by anti-malware software (1) • The attacker/a third party can access/steal the information/data/files on the computer (1) • It changes any anti-virus software so it thinks the rootkit is not inside the computer (1) • It can remove anti-virus software (1) • It can hide illegal files on a computer (1) • It can be difficult to get rid off (1)	4

Question	Answer	Marks
9	Eight marks available: e.g. • The procedures an organisation uses to secure its computer network and systems can be wrecked (1) • It can disturb their business operations ... (1) ... leading to (e.g.) loss of income (1) • Hackers/malware can corrupt/destroy/steal organisation's data (1) • Stolen/leaked data can provide a (competitive) advantage to others/competitors (1) • Hackers can obtain personal information/data (may exemplify) from organisations ... (1) ... leading to (e.g.) legal action/fine (1) • Stolen data can be used to blackmail organisations (1) • <u>Ransomware</u> can be used to hold organisations to ransom (1) • Infected computers have to be disconnected from the network ... (1) ... so other devices are not infected (1) ... fewer machines available (or related answers) (1) • Infected computers have to be treated with anti-virus software (1) • Costs to an organisation will increase ... (1) ... extra expertise may need to be brought in to clean computers (1) • Undermines confidence in the organisation ... (1) ... leading to (e.g.) loss of customers/business (1) • Impact on computer/network (slowing down etc) (1) • <u>Rootkit</u> allows the computer to be taken over and commit illegal acts (1) Max six marks if bullets/list of points Max six marks if no attempt at expansion	8

Question	Answer	Marks
11(a)	Two from: • Information that relates to an identified//identifiable living individual (1) • Data that can identify a particular person (1) • Any data that can be reconstituted so that it identifies an individual (1) • Two examples (1)	2

Question	Answer	Marks
11(b)	Four marks available: Physical protection – identify and describe the protection e.g. • Security Guard to prevent access to unauthorised people (1) • Swipe cards (1) Electronic protection – identify and describe the protection e.g. • Biometric data – unique features of an individual need to be presented to the system which a hacker would not have (1) • Passwords using passwords which are difficult for hacker to guess and so gain access (1) • Firewall/whitelisting – network has a list of allowed IP addresses/computers any computer/device not on the list will be denied access (1) • Access levels – different groups of users of networks have access to different data so even if hackers gain access to network wouldn't necessarily have access to personal data (1) • Software updates – remove vulnerabilities in the system so hackers cannot exploit any weaknesses (1) • Use a VPN – as its more secure that transferring data over a public network (1) Any form of e-safety advice that could be applied to a business context (one mark per method described) Provide training (1)	4

Question	Answer	Marks
3	SIX marks available: Benefits (MAX 4) • Protects from computer viruses/malicious code (accept either ‘finds’ or ‘stops new infections’ (1) • Prevents data/files (in files) becoming corrupted/deleted (1) • Can prevent passwords from being changed (1) • Can prevent user’s email account sending out spam (1) • Can prevent computer from crashing (1) • User can select how regularly the anti-virus scan is to take place (1) • User can select which files to scan (1) • Quarantines any infected files (removes BOD) (1) Drawbacks (MAX 4) • Anti-virus software can cost//has to be purchased (any awareness of cost or expense) (1) • Only purchase for one year (1) • Anti-virus needs to be kept up to date ...(1) – ...because there always new virri to be countered (1) • Anti-virus can only detect a virus if it is listed in the AV database (1) • Anti-virus background scans can slow a computer down//slow down download process (1) • Can give a false sense of security/does not give 100% coverage (1) • Can result in false positives (award as a concept, not necessarily these words) ...(1) – ...what appears to be suspicious activity turns out to be harmless (1) • Needs expertise to personalise (1) Accept examples such as – “Can take a long time to customise account/settings”. • Takes up space on the hard drive (1) • Can only delete the whole file that is infected (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
7	FIVE from: • Uses email to collect/gather/persuade to give <u>personal information/data</u> (award any concept of 3rd party getting access to personal data) (1) • Any statement that email or website look legitimate/real, but is/are in fact fake/from a hacker/criminal (1) • The email usually contains something that demands the target's (immediate) attention (1) • Any suitable example of a message/scenario demanding attention (1) – EITHER • The email may ask customer to <u>reply</u> (1) • Target asked to enter their information (1) – OR • The email <u>includes</u> a website URL/link (inviting the receiver to go to the site) (1) • <u>Clicking on/opening the link</u> takes them to a <u>fake/malicious</u> website (1) • Target asked to enter their information (1)	5

Question	Answer	Marks
9(a)	FOUR from: • A (standalone) malicious computer program/code (1) • A program that replicates itself in order to spread to other computers (1) • Does not need to attach itself to an existing program/file (1) • Worms consume bandwidth, slowing down the (traffic in a) network (1) • Worms consume processing power, therefore (e.g.) slowing down computer operation/system//stopping the computer from operating (1) • A worm exploits (security) holes in networks/computers (1) • They replicate themselves occupying (more) space (1) • Can modify/delete files//bring other (malicious) software//install backdoor (1)	4
9(b)	FOUR from: • A malicious computer program which is used to hack into a computer (1) • Misleads users of its true intent/disguises itself as authentic software (1) • Enables the person who created it to take control of the computer it has infected (1) • Trojans (generally) do not attempt to infect other files (1) • They do not attempt to self-replicate themselves (1) • Has to be/is activated by (e.g.) the target person (1) ONE mark for any further action a Trojan may perform (1) e.g. • Provide a backdoor onto your computer so a 3rd party can take control • Delete files from the hard disk • Copies data/files • Can corrupt data (damage = TV) • Can be used to download other malware	4

Question	Answer	Marks
9	MAX five from: • Call back/phone the number (1) • Do not call back on the same number (1) • Do not call back on the same phone (1) • Ask to speak to the SAME person as just spoke to you (1) • TWO FROM: – Never give out PIN – Never give out password – Never give out Login ID (1) • Never share security question AND answer (1) • Do not answer calls from unknown callers//hang up calls from unknown caller//set phone to only accept calls from numbers in contact list (1) • Use number checking//suspected spam checking software (1st) – to block suspected vishing numbers (1) • (Customers) block numbers suspected of being vishing (1) • Keep blocking software updated regularly (1) Any generic evaluative comment on methods as a whole (1) e.g. These could also block genuine numbers. MAX five from ANY evaluative comment as an extension to any point from above. e.g.to check that the call was a genuine call from the bank (1)so that vishing calls can't be placed/reduce the chance of you accidentally answering the call (1)so that list of numbers is up to date (1) MAX six marks if a list of bullet points	8

Question	Answer	Marks
4	Eight from: • (because) It is sensitive information about yourself (1) • an example of personal data that could be given (1) plus how this would be detrimental//how used to (e.g.) gain (your) trust (1) Fraudsters could use the data to: e.g. • commit identity theft (1) to (e.g.) buy products (may exemplify) (1) • get past bank security (1st) withdraw large sums of money from bank accounts (1) (for this second mark accept any situation where identity could be proven by details shared online – credit card/inheritance etc) • By sharing (e.g.) phone number may lead to phishing (1) Burglars could use the data to e.g. • target homes of people who announce they are going on away from their home (holiday etc) (1) Employers could e.g. • use comments about politics/religion/their current job//embarrassing photos to decide on future job opportunities (accept any example of an area of opinion, but must be at least one stated) (1) Insurers/employers could e.g. • make decisions based on evidence of risky/dangerous activities (may exemplify about decisions) (1) Max. six marks if bullet list of points	8

Question	Answer	Marks
8	Four from: - Spyware enables hackers to gain/steal personal information (1) - One mark for TWO items of information that could be gained, E.g. bank account number, date of birth, email address and password User ID Spyware can identify patterns of computer use (1) - Spyware can access computer camera (1) - The individual's computer system can slow down (1) - Personal data can be used to commit identity theft// for fraud (1) - Login details used to withdraw//transfer money from the bank account (1) - Login details to make unauthorised purchases (1) - Target computers for specific marketing purposes/ can send pop-up adverts (1) - Can use data to blackmail users (1)	4

Question	Answer	Marks
10(a)	Three from: - Data is scrambled (1st) - so it is not understandable (1) - of no use to a hacker/third party (1) Can only be unencrypted with a key//need the key to unencrypt//access the data (1) - Use of/access to encryption key/software known only to authorised user (1)	3
10(b)	Three from: - Acts as 'barrier' between internal and external networks/internet (1) - Reads (packet) sender's IP address/reads domain of sender (1st) - checks against database/list of IP addresses (1) - checks against database/list of network ports (1) - Permits access/passage of packet if not forbidden/if on allowed list/rules (1) - Drops packet if on forbidden/not on allowed list (1) - Can create logs for analysis (1) - Can create alerts. (1)	3

Question	Answer	Marks
6(a)	Four from: • (Self) replicates (1) • A malicious bot infects a host/series of computers (not 'a system') (1st) – takes control (1) – connect back to a central server//bot herder (1) • Automates tasks/provide services that would otherwise be conducted by a human (1) • Bots are used to gather/steal/copy/gather information (1) • Attackers can launch broad-based//flood-type attacks//brute force attacks (against their targets) (1) • They can launch DDoS attacks/spam attacks/relay spam/open back doors on the infected network (1)	4
6(b)	Four from: • Installs a set of tools (1st) – such as e.g. (one of) key stroke logger/web monitoring (1) • Software designed to provide continued privileged/administrator access//remote control (NOT remote <u>access</u>) to a computer (1st) – example of action that can be carried out by the administrator once they have access (e.g.) change system configurations/delete files/change passwords (1) MUST CLEARLY STATE THAT THIS IS ACHIEVED BY THE ADMINISTRATOR/hacker • A rootkit can access log files//spy on the computer owner's usage (for example, key stroke monitoring) (1) • Hides its presence//user is unaware of its presence//installed without computer user's knowledge (1) • Activated/loads before a computer's operating system has (completely) booted up (1) • Root refers to the admin account on Unix and Linux systems (1) • Kit refers to the software components that implement the attack (1)	4
6(c)	Four from: • Spreads from one computer to another (1st) – by e.g. the internet/(infected) external storage device (1) • They can replicate/duplicate (1) • They can delete/corrupt/change data/files/program (1) • They attach/insert themselves to/into another program/file (1) • They can often lie dormant until a particular situation arises (1st) – which causes the computer to execute the code (1) – This can be a specific date/time (1) • Viruses can cause the computer to crash//slow down processing speed/computer system (accept associated concepts) //change a user's homepage/change user's password preventing them from logging on. (1)	4

Question	Answer	Marks
2(a)	Four from: • Uses a fake website/illegitimate website/website pretends to be real (1) • Installs a piece of software/code on user's computer (1st) • unintentionally/unknowingly/without consent (1) • Software corrupts the host(s) file (1st) • by adding URLs and corresponding IP addresses (1) • User thinks they are on a genuine//non fraud website (1st) • customer types in the response to question (1).	4
2(b)	Four from: • Involves using text/SMS messages (1) • The message may invite the receiver to click a link (1st) • looks just like the/an actual company's website (1) • but is a fake website (1) • The message may invite the receiver to phone a telephone number (1st) • that connects to an automated voice response system (1) • The smishing message usually contains something that demands the target's immediate attention (award exemplification) (1) • User thinks they are on a genuine//non fraud website (1st) • customer types in the response to question (1). (accept phone call equivalent)	4

Question	Answer	Marks
7	FOUR marks available: • Definition of personal information. – Personal information is any item of information that can make the process of identifying you easier. (1) • General statement about access to personal information being used for an illicit purpose (e.g., "it can cause identity theft/theft from bank/blackmail") (1) • General statement about legal requirement for personal information to be kept confidential (1) ONE mark per identified negative impact from an identified breach of confidentiality e.g. • Identity theft because of leak of (e.g.) home address and name (1) • Unauthorised withdrawals/transfers after leak of (e.g.) bank account number and PIN (1) • Future/current job affected because shared (e.g.) political affiliation (1)	4

Question	Answer	Marks
9(a)	FOUR from: - Spyware is software that is installed/operates on a computer without the user's knowledge (1st) - often the first indication is a noticeable reduction in processor speeds. (1) - Records/tracks what a user does on their computer (1) - Can gather information like e.g. web browsing habits (1) - Can capture email messages//a user's personal data/information (can exemplify) (1) - Transmits this data to a third party over the internet (1)	4
9(b)	FOUR from: - A program (1) that replicates itself (1st) - until it fills all of the storage space (on a drive or network) (1) - It searches for more computers (1) - Exploits security holes in networks (1) - Worms can slow down a network (1st) - by consuming bandwidth (1) - Worms often originate from email attachments (1) - Worms then spread to a user's contacts via their email account/address book. (1)	4

Question	Answer	Marks
3(a)	One mark per bullet point to a maximum of six marks. ADWARE - To generate income/revenue for an identified 3rd party (the software developer/creator/user sending it) (1) <u>Generates/places</u> adverts (1st) - usually within the user's default browser. (1) - may be in the form of pop up or adverts on a website (1) - Installs itself/without user's knowledge (1st) - when installing a (different) program app (1) - Can create open doors for malicious programs (1) - Can track user's search/browsing history (1st) - to better target ads/create targeted marketing (1) - user's information can be sold to third parties. (1)	6

Question	Answer	Marks
3(b)	One mark per bullet point to a maximum of four marks. RANSOMWARE • Locks/encrypts/blocks access to files/devices/systems (1) so they cannot be used (1) • Works at root level to disable access to files/devices/systems (1st) – May also target backups. (1) • Hacker contacts the victim and demands a ransom (1) • Software usually can't be removed/access cannot be regained until the attacker receives a ransom payment. (award the concept of regaining access via payment here) (1)	4

Mark scheme · 9626/11 May/June 2025 · Question 3

[Back to question](#)

Question	Answer	Marks
3	One mark per bullet point to a maximum of four marks. • Blocks threats to data//potential attackers (1) • Only allows allowed IP addresses (1) • Checks allowed IP addresses in allowed/forbidden table/list (1) • Blocks/allows ports for data transmission (1) • Can be set to stealth mode (1st) – so that packets are not bounced back (1) • Blocks suspicious activity to/from the network (1) • Creates a log of issues for analysis (1) • It can manage IPsec clients (1) • Can act as authenticator for remote access (1).	4

Mark scheme · 9626/12 May/June 2025 · Question 6

[Back to question](#)

Question	Answer	Marks
6	One mark per bullet point to a maximum of six marks. • Publish anti-malware policies for staff (1) • Teach employees to report malicious websites//suspicious emails to e.g. the network manager/compliance (1) • Instruct employees to be suspicious about clicking on links (1) • Instruct employees that they must not connect removable media/memory sticks/personally owned device to their computer/the network (1) • Provide a dedicated stand-alone/sandbox computer to check for viruses (1) • Keep operating systems/browser software/anti-virus software/the software up to date (anti-malware sw is the focus of the question) (1) • Install/maintain a firewall (any awareness) (1) • Block employees from downloading software/files onto the computer (1)	6

Question	Answer	Marks
4	Three marks from: • A computer program/code/file/application (1) • It looks like a legitimate computer program/file (1st) – Intended to get victim/target to open them (1) • When program/sw is run, malware is activated (1) • It creates a back door to gain access to a computer (1) • Trojans do not attempt to infect other files/replicate themselves (1) • It can allow the third party to: (any 2 of the following) (1) – Download content onto the computer – Copy data from the computer – Gain control of the computer – Set up a DDOS attack.	3

Question	Answer	Marks
3(a)	Two from: • The unauthorised person phones the victim pretending it is a call from the bank (1st) – The unauthorised person claims that there is an issue/peril/fault with the victim's account (1) – The unauthorised person persuades the victim to give them their details (1) – The unauthorised person convinces/tricks the victim into believing they are talking to a real bank (candidate may exemplify) (1)	
3(b)	Two from: • Do not answer the phone to unknown numbers (1) • Do not give login/personal details (over the phone) (1) • Contact the bank with a different device (1) • Contact the bank on a known number (1) • Block the number (via phone settings/provider) (1) • Report the call (to phone provider/bank) (1) • Don't use any phone number provided by the caller (1)	2

Question	Answer	Marks
6(a)	Two from: e.g. • To commit theft (1) • To commit industrial espionage (1) • To commit sabotage (1) • Reasons of extortion/financial gain (1) • To track behaviour/actions/keystrokes/inputs (1)	2
6(b)	Six available for three matched pair answers. Max two for each matched pair: e.g. • System slow-down/sluggish/lack of response (1) anti-virus software uses up resources/processing power/memory/space (1) • Disrupts user experience/annoys user (1) with popups/system restarts. (1) • Limited to known viruses/database/software has to be updated regularly (1) new viruses are developed all the time (1) (only award if either of previous two marks given) • False sense of security (1st) – new viruses are developed all the time (1) – unless file is updated (1) • False positive/identification (1st) – unnecessarily quarantined/blocked files (1) Max two from: • Limited check/Only checks for viruses (1st) – it does not remove other type of malware (1) OR • It does not check all types of malware (1st) – such as (e.g.) rootkits (1)	6

Question	Answer	Marks
1(a)	TWO marks available: - Remove geotags//metadata (from the photograph) (1) - Restrict/flag photograph as not for public//change privacy settings (1) - Make access to the social media private (1) - Description of any edit to the photograph to keep personal data secure e.g. - ONE from - Obscure people's faces/features (1) - Obscure features which allow the location to be identified (1)	2
1(b)(i)	FOUR marks available: - Firewall only allows <u>authorised</u> traffic (DO NOT accept converse – mark as REP) (1) - TWO from: - Use of Packet filtering (1) - Use of Proxy service (1) - Use of Stateful inspection (1) - The firewall controls what can come in AND go out (1) - The firewall operates based on rules (1) - TWO from: IP addresses/IPs of known threats are blocked//only allows access to allowed IP addresses (allow converse) (1) - Only a restricted list of (communication) protocols are allowed (allow converse) (1) - Packages from certain (communication) ports are accepted/denied (allow converse) (1)	4
1(b)(ii)	ONE mark available: - Can block legitimate traffic (1) - Can slow down computer performance (accept any description) (1) - IP addresses can be masked/spoofed (1)	1

Question	Answer	Marks	Guidance
7(a)	TWO marks available: e.g. • Will lose access to (all) data/devices/files/system (1) • Financial cost (e.g.) of ransom//checking data when it comes back (1) • Data could be released/leaked/sold (any awareness of data being passed onto a 3rd party) (1) • Business disruption/reduced productivity (1) • Loss of revenue (1) • Liability/compliance costs/risk of prosecution (1) • Reduction in customer confidence (1)	2	

Question	Answer	Marks	Guidance
7(b)	SIX marks available. MAX FOUR if only first mark answers. - Definition of what is meant by physical method of prevention//identifies ONE methods of physical protection (1) Positives MAX FOUR e.g. - Cannot physically access the computers (1) - Physical defences can e.g. list malicious sites//give restrictions on how sites may be used (1st) Therefore, users know (which) sites to avoid (1) - Physical defences can state how external devices can be used (may exemplify) (1st) Therefore, users know if they can be used//how to make them safe (1) - A standalone dedicated computer can be set up for virus checking (1st) So, devices can be checked (1) Negatives MAX FOUR e.g. - Digital attacks can occur even if there is an (e.g.) guard (1) - Ransomware attacks are more likely to be software based//happen over a network//digital (1st) Therefore physical methods are ineffective (1) - Physical defences require the user to follow them//may be a change from working practice (1st) And so may be ignored (1) - New staff/temporary staff may not be aware of the restrictions (1) - Physical restrictions are not automatic//will require staff to make decisions/take actions (1) - Training may not take place at busy times (1st) Therefore, it becomes irrelevant (1) Conclusion (may only be awarded if positive(s) and/or negative(s) are included) - One mark for a value judgement/conclusion based on two arguments (1) E.g. it can increase consumer confidence	6	Do Not Award 'Cannot access' on its own

2	THREE from: • ANY item of information if clearly identified as a password /his password (1) • ANY item of information if clearly identified as the answer to a security question (1) • Name of the bank account owner (1) • Address of the bank account owner (1) • Date of birth of the bank account owner (1) • Personal Identification Number (PIN) (1) • Bank account number (1)	3	
---	---	---	--

6 The digital divide

Questions

9626/11 May/June 2022 · Question 9

Mark scheme

- 9** One reason for the digital divide is that people living in rural areas are less likely to have high speed broadband compared to people in cities.

Explain why this particular aspect of the digital divide exists.

..... [6

- 6 The digital divide affects many people. There can be a divide between older and younger people. Describe how this digital divide could be reduced.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

10 Describe, listing the groups affected, what is meant by the digital divide.

[6]

Please turn over for Question 11.

- 8 Describe how people in developing countries can be disadvantaged when trying to use online shopping.

[5]

4 The digital divide can exist between people in less industrially developed nations and those in more industrially developed nations.

(a) Explain why this digital divide exists.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe the effects of this form of digital divide.

.....

.....

.....

.....

.....

.....

.....

7 Explain why there can be a digital divide between different socioeconomic groups.

..... [5]

7 Explain the reasons for the digital divide between the old and the young.

[6]

- 6** There can be a digital divide between people who live in rural areas and people who live in urban areas. The divide can be caused by the lack of fast broadband in rural areas, resulting in limited access to the web.

(a) Explain the effects on people who live in rural areas of having limited access to the web.

[4]

(b) Describe ways to reduce this digital divide.

[3]

- 12** A digital divide exists between the more technologically aware and the less technologically aware nations.

Analyse this type of digital divide.

[6]

3 A government is concerned about the digital divide amongst its citizens.

Describe how the government could reduce the effects of the digital divide and improve access to its services.

[6]

- 5** Explain the reasons why there is a digital divide between people with high levels of education and people with lower levels of education.

[6]

8 Describe how access to information technology has created the digital divide.

[8]

6 Describe how a local government can help reduce the digital divide in rural areas.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

6 Describe, by referring to **two** affected groups, what is meant by the digital divide.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

14 Identify **two** groups of people who can be negatively affected by the digital divide. For each group, suggest a way the digital divide can be reduced.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

13 Describe the effects of the digital divide on health services.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

7 A young adult and his grandfather use modern technology in different ways.

Discuss possible reasons why the young adult may use modern technology to a greater extent than his grandfather.

[8]

6 The digital divide

Mark schemes

Mark scheme · 9626/11 May/June 2022 · Question 9

[Back to question](#)

Question	Answer	Marks
9	Six from: Infrastructure for broadband is less developed in rural areas than in cities... ...any existing infrastructure tends to use a lower level of technology making it difficult to install high speed broadband In rural areas people tend to be further away from a cabinet causing longer cables to be used leading to slower speeds Rural areas can have difficult terrain which make it difficult for companies to develop the infrastructure/lay fibre cables Rural parts tend to have a lower concentration of people/population so providers cannot recuperate costs quickly/less willing to invest in infrastructure More expensive for providers to offer services in rural areas and the extra costs are passed onto the customers Cities can have more providers competing to provide broadband than rural areas making it cheaper for the customer In rural areas people can use satellite broadband but it tends to be slower than cabled broadband connections Satellite broadband can be more expensive than cabled broadband connections Can use smartphones but signal quality depends on the number of towers and in rural areas there are fewer	6

Mark scheme · 9626/11 Oct/Nov 2022 · Question 6

[Back to question](#)

Question	Answer	Marks
6	Five from: e.g. • Younger people could train older people (1) • The government could subsidise internet access for older people (1) • Government could encourage libraries/community centres to offer free internet access to older people/those who do not have access (1) • Retirement villages could be built incorporating technologies (1) • Information providers should be required to produce content in suitable formats so that people with weaker eyesight can easily access the content (1) • Grants can be made to organisations to deliver digital literacy training programmes to older people (1) • Schools could allow older people access to technology facilities in the evenings//during school holidays (1) • Schools could provide teaching and training to older people to improve their skills (1)	5

Question	Answer	Marks
10	Six marks available: • The gap between people who have access to and the resources to use new information/communication technology and those who don't ... (1) ... This technology can include the telephone, television, personal computers and the Internet (1) • The quality of connection to the Internet may vary between groups/countries (1) • Differing levels of infrastructure between (e.g.) countries (1) • The price of connection to the Internet may vary between groups/countries//ability to afford access to the internet (1) • May refer to inequalities between individuals, households, businesses, or geographic areas (minimum of two mentioned) (1) • The divide between those who have the skills/knowledge/ability to use the technologies and those who do not (1) • The digital divide can exist between those living in rural areas and those living in cities (1) • The digital divide can exist between the educated and uneducated (1) • The digital divide can exist between socioeconomic groups/rich and poor (1) • The digital divide can exist between old and young people (1) • The divide between more and less industrially developed/technologically aware nations (1) • The divide between countries or regions of the world is referred to as the global digital divide (1)	6

Question	Answer	Marks
8	FIVE from: • Some people in developing countries have no actual postal address making the delivery of goods which are purchased online difficult (1) • Many people in developing countries have no access to broadband so speed of purchase is slower (1) • Many people in developing countries have no PCs/laptops/smartphones/tablet computers so need to borrow a PC/go to a library (1) • In developing countries power outages can mean that connection is lost during the process and so have to start again/lose the sale (1) • Many people in developing countries do not have bank accounts and credit cards so have to organise for someone else to buy the product on their behalf (1) • In many developing countries, goods bought using the internet are not covered by consumer-protection laws so people have no protection from unscrupulous sellers (1)	5

Question	Answer	Marks
4(a)	Three from: • The quality of connection to the Internet may vary (1) • Speed of access to the Internet may vary (1) • Some countries may not have a well-developed broadband (infrastructure)/rely on dialup connections (1) • Lack of incentive to invest in the country/nation (as a whole) (1) • In less developed countries they might have less infrastructure (may exemplify) (1) • Not all countries are able to keep up with the constant changes in technology (1)	3
4(b)	Three from: • The digital divide gets wider (1) Award ANY three implications of a digital divide: For example • higher/lower quality on education (1st) • higher/lower quality on health (1st) • higher/lower quality on access to resources (1st) • higher/lower quality on lifestyle (1st) • higher/lower quality on opportunities (1st) • higher/lower quality on work/employment opportunities (1st) • higher/lower quality on trade (1st) • reduced access to social media (1st) • higher/lower access to information (1st) A further mark can be awarded for an explanation of the impact (1) A final mark may be given for the alternative comment – such as MP3 in the example below. Note: can ONLY be given as a 3rd mark. E.g., reduced access to education (1) means lower skill level (1) whilst those in industrial societies have better access (1)	3

Question	Answer	Marks
7	Five from: • Different socioeconomic levels imply differing levels of (disposable) income (1) • Different incomes/socioeconomic groups impact on what people can buy/access/afford (1) such as 'things' ... (must be an extension linked to technology) (1) such as acquisition of skills/paying for training (1) • This impacts on what people can commit to renting (1) • New equipment is costly (1) • Old equipment is not as advanced (1) • People from lower socioeconomic groups tend to have lower digital skills (1) • No/reduced incentive for companies to develop cheaper systems ... (1) ... because the profits would be lower (1) • No/reduced incentive to provide infrastructure to poor areas (1) ... because reduced demand in the area (1)	5

Question	Answer	Marks
7	Six from: e.g. • Young people have grown up with the technology/have only ever known the use of smart phones/internet //its seen as part of their everyday life (1) • Older people may feel they are too old to learn about new technology (1) • Younger people are more likely to take risks/older people are more risk averse (1) therefore young buy new technology/old do not buy new technology (1) • Many older people already have established means of communication and so do not see the need to explore new technology (1) • IT businesses tend to aim new technologies at younger people ... (1) ... companies make few modifications to help older people use them more easily (1) • Technology seen as something for the young (1) • Older people can be awkward with their hands and fingers making the use of smartphones a problem (1) • Elderly people also find it difficult to read text on a screen or use a keyboard (due to visual impairments) (1)	6

Question	Answer	Marks
6(a)	Four from: A general statement of impact: • Inability to find information due to inability to access search engines (or other example of cause. Example must be more than “cannot access the web”) (1) • Tasks take longer to complete, which wastes time (1) Up to four marks for four effects and explanations. Areas include: Reduced health care, e.g. • Inability to get onto a waiting list as cannot access doctor’s website (1) • Not aware of imminent health risks as do not have access to a national news service (1) Reduced ability to work remotely, e.g. • Unable to have face-to-face communication over the web, and so cannot work remotely (1) Impact on education, e.g. • Cannot access online learning resources, and so quality of education is lower than could be (1) • Unable to do wider research and so will lack wider understanding of the topic (1) Impact on purchasing, e.g. • Inability to compare prices of products as cannot access search engines (1) • Inability to check on availability of products as can not access search engines (1) Other areas include, but are not limited to: • Impact of limited access to online facilities, banking • Impact of limited access to communication • Impact of limited access to mental health • Impact of limited access to online entertainment	4
6(b)	Three from: • Build/improve access to the internet/faster broadband (in rural areas) (1) • (Governments/industry can) provide/improve the needed/broadband infrastructure (for fast broadband in rural areas) (1) • Government provides incentives/subsidise to industry to build infrastructure (1) • Government investigates alternative means of distribution (1) • Create cyber/internet cafes (in rural areas) (1) • Deliver access using satellite technology (1)	3

12	Six available. One for a definition of the digital divide: - The digital divide is the gap between those who have access to//use technology and those who do not Max six from: - In this case, the gap is caused by differing levels of knowledge about what technology is available (1) AND differing levels of knowledge about what technology can achieve//benefits to be gained (1) - Not everyone is able to keep up with/keep aware of the constant changes in technology - A person who is unaware of what is available//what can be achieved will not use //not know how to use technology - A person who is aware of what is available//what can be achieved will use technology - Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1) (accept reverse argument for digital unawareness) This divide may be reduced by education/research/experience - This divide may exist within any socioeconomic bracket - This divide may exist within any age group.	6
----	--	---

Question	Answer	Marks
3	Six from, e.g.: One mark for a definition of the digital divide: e.g. - The digital divide is the gap between those who have access to//use modern/ICT technology and those who do not (1) Improve communications coverage/access to the internet/access to broadband (1) Improve infrastructure (1) such as (e.g.) invest in satellite technology/build more mobile phone masts/lay fibre cabling (1) - Provision of public access to internet facilities in, e.g. (ONE OF) government buildings/libraries/schools and internet kiosks/cafes (1) - Provide/subsidise laptops/PCs/technological equipment to, e.g. (ONE OF) the elderly/the poor/school children (must be a specific group likely to be negatively affected by digital divide) (1) - Provide/subsidise of mobile or cell phones to, e.g. (ONE OF) the elderly/the poor/school children must be a specific group likely to be negatively affected by digital divide) (1) - Provision of training services/sessions (1) - Subsidise access to a named service (1)	6

Question	Answer	Marks
5	SIX from: - Highly educated people are often employed in jobs that offer exposure to IT (1) - Highly educated people will often undertake IT training as part of their education//have IT skills as a result of their education (1) - Highly educated people are often aware of IT/importance of IT/benefits of IT/possibilities of IT (1) - Highly educated people have more access to the higher paid jobs (1st) therefore can afford/access the (e.g.) higher speed internet/most up to date IT equipment (1) (accept converse argument) - People with lower levels of education often work longer hours, therefore have less time to access IT (1) - Lower educated people are less like to invest in IT (1)	6

Question	Answer	Marks
8	EIGHT from, e.g.: Award ANY description of differing access to Information Technology leading to digital divide. ONE mark per description e.g. - Access to fast broadband can be better for those people living in urban areas than for those living in rural areas (1) - The quality of connection to the internet may be poorer in underdeveloped countries (1) - Buying televisions/personal computers/smartphones can be more expensive in rural areas (1) - Richer people are more/poorer people less able to afford the latest versions of televisions/personal computers/smartphones computers (1) - Lower paid workers may not be able to afford the hardware to be able to access the IT (1) - Uneducated people are often not aware of IT/importance of IT/benefits of IT/possibilities of IT (1) - Young people have grown up with the technology/older people may feel they are too old/unwilling to learn about new technology (1) - Disabilities of some users may inhibit the use of smartphones/computers. (1)	8

Question	Answer	Marks
6	One mark per valid description to a maximum of four marks. E.g.: • Expand broadband access/improve infrastructure. (1) • Establish/support community centres with computers/internet access. (1) • Offer workshops/courses to teach residents computer skills/ICT (1st) – such as – accept any suitable use of ICT (online or offline) (1) • Educate residents on//raise awareness of any identified aspect e.g. cybersecurity/online safety/technology. (1) • Establish equipment loan programs (1) • Equip schools with improved/better/latest technology (1) • Subsidise access to internet (1) • Subsidise purchase of computer equipment (1)	4

Question	Answer	Marks
6	Two marks from: • The gap between people who have access to//resources (computer hardware) to use new information//communication technology and those who don't (1) • The gap between people who have access to information and those who don't (1) Max 2 from: People with/from: • Varying ICT skills (1) • Living in rural areas and those living in cities (1) • Different education levels (1) • Different socioeconomic groups (1) • Different age groups (1) • Different industrially developed/technologically aware nations (1)	4

Question	Answer	Marks
14	Four marks for two matched group-remedy pairs: e.g. Groups (1st) • Old people (1) • People in rural areas (1) • Low/different levels of education (1) • Lower socioeconomic group/people who cannot afford to purchase (1) • People with learning difficulties (1) • People with physical impairments (1) • People with sensory impairments (1) • People in less industrially developed nations (1) Remedies (1) • Education/training (1) • Drop-in centres support (1) • One to one support (1) • Invest in infrastructure (1) • Provide specialist interfaces (1) • Provide (cheap) loans/subsidies/price ceiling/free (1) • Recycling/refurbish/repurpose equipment (1)	4

Question	Answer	Marks
13	Six from: e.g. • Can give differing levels of access to health services (1) • Increased risk of death/length of illness extended (1) • May not have online health services (1) – Reasonable implication (1) • Delayed access to results (1) • May need to attend doctor in person/travel to hospital (1) • Not everyone may be able to book online (1st) – need to visit/telephone (1) – which means time is wasted/used less efficiently (1) • Not everyone can order/reorder medicine/prescriptions online (1) • Doctors can work remotely from the patient (1st) – increased access to doctor/patient (1) – ambulances are less frequently used (1) • Not everyone can submit/receive test results online (1) • Not everyone can be monitored remotely (1)	6

Question	Answer	Marks	Guidance
7	EIGHT from: - Because there is a digital divide (between them) (1) Possible themes include: - Different locations (1) with different access (1) - Lower disposable income/affordability/socio-economic difference (1) affecting access (1) - Different levels of fear (of any type)/comfort (1) however, this fear will quickly disappear with use (1) if issue is not addressed, this problem will get worse (1) - Different day to day experience/training (1) - Different jobs make different use of IT (1) - Disability (1) - Lack of awareness of possibilities (1) - Any example of choice or lifestyle (absence of or different need lack of interest, lack of online friends (1) Answers may be awarded for: Identification of issue (1) Explanation/description of issue (NOT just another version of the same issue) (1)	8	

7 Expert systems

Questions

9626/12 Feb/March 2022 · Question 5

[Mark scheme](#)

- 5 Some expert systems use a mixture of forward chaining and backward chaining. Some use forward chaining only.

Describe the process of forward chaining.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 9 Insurance consultants have to make decisions about whether or not to accept a customer’s application for an insurance policy. They often use expert systems to help them assess the risk involved.
- (a) Such a system would have a knowledge base, which consists of a database of facts and a rules base.

Describe the other components of this expert system.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) Explain how this insurance expert system would be used by the consultant and how it would assess the risk.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

5 A delivery company uses an expert system to help produce route schedules for its vehicles.

(a) Without describing its components, explain how an expert system would produce these schedules.

[4]

(b) Explain what is meant by backward chaining.

..... [3]

- 9 A doctor uses an expert system to help the doctor diagnose the illnesses of her patients.

Explain how an expert system is used to help the doctor produce the diagnosis of an illness. You may assume only forward chaining is involved.

[8]

- 10** An expert system is being trialled for use as part of a study on animal identification. It has been set the following task.

The goal is to decide the colour of a bird named Hortense.
Two facts are provided:

- she sings
- she is 15 cm tall.

The rule base contains the following three rules:

- IF X is an ostrich – THEN X has a long neck
- IF X sings and X is 15 cm tall – THEN X is a canary
- IF X is a canary – THEN X is yellow

The inference engine has concluded that Hortense is yellow. It has taken the facts as data and used the rules base to decide:

- 1 Hortense is a canary
- 2 Hortense is yellow

- (a)** Identify which type of chaining the inference engine used.

.....
..... [1]

- (b)** Explain the reasoning the inference engine used.

.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [4]

8 Explain what is meant by the term ‘data driven’ within the context of expert systems.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

5 A bank uses an expert system to help customers with financial planning.

(a) Describe the use of **four** components of an expert system. Your answers **must** relate to financial planning.

1

.....

.....

2

.....

.....

3

.....

.....

4

.....

.....

[4]

..... [6]

(c) State **three** drawbacks, to the bank, of using an expert system.

1

2

3

463

- 3** In an expert system, the inference engine uses reasoning that consists of backward or forward chaining.

Explain the differences between backward and forward chaining.

..... [6]

- 10** A garage uses an expert system to diagnose faults in car engines.

Describe the components of this expert system.

[8]

10 Scientists are using an expert system to help them identify species of animals.

Describe **four** benefits to the scientists of using an expert system for this purpose.

1

.....

.....

2

.....

.....

3

.....

.....

4

.....

.....

[4]

7 Describe the term goal driven as used in expert systems.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

- 4 A delivery company uses an expert system to suggest a schedule of deliveries to be followed by its drivers.

(a) Describe the advantages to the company of using an expert system for this purpose.

[3]

(b) Describe how such an expert system would be created.

[6]

3 There is a fault with a car engine. An expert system is used to help diagnose the fault.

Describe how the following components of an expert system are used in the process of diagnosing the fault.

(a) User interface

.....

.....

.....

.....

.....

..... [2]

(b) Inference engine

.....

.....

.....

.....

.....

3 Describe the reasoning used by an inference engine in an expert system.

[6]

4 A knowledge base is one component of an expert system.

(a) Describe what is meant by the term knowledge base.

.....

.....

..... [1]

(b) Describe how a knowledge base is used in an expert system.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

3 An expert system contains an inference engine and a knowledge base.

(a) Describe how the inference engine performs its role.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe how a knowledge base is created.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

6 An expert system may be data driven.

Describe, with the use of an example, what is meant by the term data driven.

[5]

- 2 A company employs drivers to deliver parcels in a city. The company uses an expert system to plan the routes each driver will follow each day.

Describe how an expert system can be used to provide route scheduling in this scenario.

[8]

7 A mechanic uses an expert system.

This system has a user interface and an explanation system.

(a) Identify and describe **two** other components of an expert system.

component 1

.....

.....

.....

.....

component 2

.....

.....

.....

.....

[4]

(b) Evaluate the use of this system.

[8]

10 An expert system is used to help identify different types of flowers.

(a) Describe the role of an inference engine in this system.

.....

.....

.....

.....

.....

..... [2]

(b) Describe the role of a knowledge base in this system.

.....

.....

.....

.....

.....

..... [2]

7 Expert systems

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 5

[Back to question](#)

Question	Answer	Marks
5	Three from: An inference engine using forward chaining searches the inference rules... ...until it finds one where the IF statement is known to be true When such a rule is found, the inference engine uses the 'THEN' part to cause the addition of new information/facts The inference engine will repeat this process until a goal is reached The data entered determines which rules are selected and used and so this method is called data-driven	3

Mark scheme · 9626/13 May/June 2022 · Question 9

[Back to question](#)

Question	Answer	Marks
9(a)	Four from: The shell contains the explanation system, inference engine, user interface and knowledge base editor (must have 3) The knowledge base editor/knowledge base editing software enables the knowledge engineer uses to edit the rules and facts within the knowledge base Explanation system explains to a user the chain of reasoning used to arrive /why they arrived at a particular conclusion The user interface is how the computer interacts with the user/ allows the user to interact with the expert system The inference engine is the reasoning/decision-making part of the system/ provides the reasoning of an expert system	4

Question	Answer	Marks
9(b)	Four from: Details of the individual would be entered into the user interface... ...such as the applicant's age, gender, whether they smoke and general health The user interface will ask questions (of the consultant/user) The inference engine will compare the responses... ...with the facts in the knowledge base... ...using the IF...THEN rules The inference engine will produce a recommendation for how much the applicant will pay each month for their policy... ...or recommend whether the application is rejected May inform them of the details of the policy The recommendation is output through the user interface The consultant will use the expert system's recommendations to make a final decision	4

Question	Answer	Marks
5(a)	Four marks available: • An expert system would store the locations of each distribution point (1) • An expert system would store the type of the vehicle being used (1) • An expert system would store the working hours (1) • The scheduler would type in the destinations/quantity of parcels for the day ... (1) ... the expert system would match those against the available types of vehicle (1) ... match against locations (1) • The expert system would suggest how many drivers would be needed (1) • The expert system would suggest the kind of vehicle needed for each driver (1) • The expert system suggests the allocation of orders to each vehicle (1) • The expert system suggests a list of goods in reverse order (1) so that each vehicle had the first order loaded on to the vehicle last (1) • The expert system would take account of (e.g.) fuel and time//traffic and road conditions (1) • Outputs the most efficient route (1)	4
5(b)	Three marks available: • Backward chaining starts with a list of goals/hypothesis ... (1) ... and tries to find support for their hypothesis (1) • An inference engine would search the inference rules until it finds one which has a THEN part that matches the goal/hypothesis (1) • If the IF part of that rule is known to be true, then it is added to the list of goals (1) • This method is called goal-driven (1)	3

Question	Answer	Marks
9	Eight from: • Identification of three of: knowledge base, explanation system, knowledge base editor, inference engine, user interface, rules base (1) • Inference engine is able to find possible diagnoses by using a form of reasoning (1) • The inference engine uses the data or facts in the knowledge base to reason through the symptoms (1) • (User interface/expert) system asks questions/for symptoms about the illness (1) • Answers//symptoms/required data are typed in (1) • Inference engine compares symptoms to those in the knowledge base (1) • Any awareness of use of IF...THEN... rules/comparisons//rules base consists of IF...THEN rules... (If and THEN both used in the answer) (1) • Searches the inference rules to find one where the IF statement is known to be true (1) • If answer not found, further questions are asked (1) • When such a rule is found, the inference engine uses the 'THEN' part ... (1) ... to find the illness (1) • Possible/suggested/a range of diagnoses are output (to user interface) (1) • Explanation system produces reasons for suggestions/diagnoses (1) • Reasons for suggestions output using the user interface (1) • Doctor can choose to use or reject answer (1)	8

Question	Answer	Marks
10(a)	Forward chaining	1
10(b)	Four from: • An inference engine searched the rules until it found one where the IF statement was true (1) • An inference engine rejects any rule that is not TRUE (1) • The IF Hortense is an Ostrich is NOT TRUE (1) • The IF Hortense sings and Hortense is 15 cm is TRUE (1) • The 'THEN' part caused the addition of new information (1) • Inference engine looked for another matching IF... (1) • The IF X is a canary - THEN X is yellow is TRUE (1) • Inference engine stopped because the goal was reached (1)	4

Question	Answer	Marks
8	Four from: • One mark for identification of all of: Rule Base, Knowledge Base, Inference Engine (1) • Uses forward chaining to find a solution (1) • Makes strategic decisions based on data analysis and interpretation (1). • The data entered into the system determines which rules are selected and used (1) • The inference engine/process starts with the facts/data of a problem and progresses to a goal . • Any explicit awareness of IFTHEN being part of the process (1) • An inference engine searches the rule base (1st) • until it finds one where the IF statement is known to be true (1) • When a suitable rule is found, the inference engine uses the 'THEN' part to cause the addition of new information/facts (to the knowledge base) (1) • This process repeats until a goal is reached (1)	4

Question	Answer	Marks
5(a)	Four from: • A user interface so questions/queries about (e.g.) financial details appear (1) • A user interface so answers to questions can be input (1) • Knowledge base editor to enter/amend (e.g.) current financial facts and figures (1) • Database of facts/knowledge base to hold/store current financial details (1) • Rules base to hold/store financial rules, (1) • Inference engine to interrogate knowledge base//uses forward/backward chaining to come to a conclusion (1) • Explanation system to describe why/say why/justify how the conclusions were arrived at. (Do not award “explains”) (1)	4
5(b)	Six from: • Reduced staff costs (1) • Saves staff time in writing (e.g.) documentation (1) • Makes bank more competitive (1) • Expert systems give better advice than humans (1) • Increase customer satisfaction (1) • Makes the bank look more reliable (1) • Expert system, does not need e.g. breaks/holidays (1) • Cheaper to copy an expert system than train new staff (1) • Errors are reduced. (1)	6
5(c)	Three from: • It can lead to bad reviews/reputation (1) • It can lead to loss of customers (1) • Expert systems cannot apply common sense (1) • Expert systems cannot be creative in//cope with unusual situations (1) • Costs of creation//initial installation//initial purchase is high (1) • Knowledge base/rules require frequent updating//knowledge goes out of date (1) • Need to pay/employ a (e.g.) technician//pay maintenance costs for the expert system (1)	3

Question	Answer	Marks
3	Six from: Forward chaining MAX FIVE - Forward chaining starts with/uses the data (1) - Uses inference rules to extract more data. (1st) - until a goal is reached (1) - An inference engine searches until it finds an IF statement known to be true (1) - The inference engine uses the 'THEN' part of the IF THEN statement to cause the addition of new information (1) - Inference engines will iterate through this process (1st) - until a goal is reached (1) - This method is called data-driven (1) Backward chaining MAX FIVE - Backward chaining starts with/uses a list of goals/hypotheses (1) - Uses inference engine to search for data that fits a goal/hypothesis (1) - An inference engine searches until it finds a THEN part that matches a desired goal (1) - If the IF part of that rule is not known to be true (1st) - then it is added to the list of goals (1) - This method is called goal-driven (1)	6

Question	Answer	Marks
10	EIGHT from: Max five if only first points Knowledge base comprises a database of facts and a rules base (1st) - It is the information used to make the decision (1) - Database of facts gathered from experts (1) Rules base is a set of rules/statements (1st) - usually of the form IF . . . THEN//description of IF ...THEN in practice in context (outcomes based on what is seen/observed/measured) (1) The shell includes the user interface, explanation system, inference engine and knowledge base editor (THREE FROM) (1) Inference engine is the reasoning part of the system (1st) - That uses forward chaining or backward chaining (1) User interface is how the computer interacts with the user, (1st) - displays questions/information (1) - enables the user to type in answers to the questions (1) Knowledge base editing software/knowledge base editor enables the knowledge engineer to edit rules and facts (1) Explanation system explains to a user the chain of reasoning used to arrive at a particular conclusion. (1)	8

Question	Answer	Marks
10	FOUR from: • Contains the knowledge of many experts/zoologists/scientists (1) • Produce results quicker//reduces time taken to produce results (1) • No need to carry/have access to paper records/ information/ reference books/remember a great deal of information (1) • Outcome is consistent (1) • Fewer errors in identification of animal species (1) • Can enhance scientist's/zoologist's own knowledge (1) • Provide permanent record of identification of animal (1) • No need to check with other experts/zoologists/scientists if in doubt over findings. (1) • Expert systems provide justifications/explanations for decisions (1)	4

Question	Answer	Marks
7	FOUR from: • It is used in backward chaining (1) • It starts with a goal/set of goals (1st) – which establish the rules to be followed (1) – to identify initial conditions (1) • Inference engine uses the rules base (1) – to find one which has a THEN part that matches a required goal (1) (Award once only) • Inference engine explores the rules (1st) – until it finds one which has a THEN part that matches a required goal (1) (Award once only) • If the IF part of that rule is known to be true (1st) – it is added to the list of goals. (1)	4

Question	Answer	Marks
4(a)	One mark per bullet point to a maximum of three marks. • An expert system is able to consider various factors/inputs (1st) – such as distance/traffic/driver workload/customer preference (1) • It is a more efficient method of solving the task. (efficiency in production or of outcome – accept either, but , production of outcome OR quality of outcome must be stated) (1) • Fewer scheduling/human errors (1st) – such as forgetting about time windows/overlooking optimal routes/drive availability (1) • The company will be able to keep their scheduling system active 24/7 (1) • Reduced expenses/fuel usage/wear and tear (1). • Chooses the quickest/most fuel-efficient route (1) • The company will be able to scale their operations//An expert system will allow for improved business scalability/no need to employ new staff as business grows (1) • Able to produce a picking order (1) • Deliveries can be loaded in reverse order of delivery (award concept of) (1)	3

Question	Answer	Marks
4(b)	One mark per bullet point to a maximum of six marks. • <u>Gather information</u> about delivery processes. (1) • TWO marks for items of information that would be gathered (e.g.) – Distances (1) – Locations (1) – Speed limits (1) – Capacity of individual vehicles (1) – Road capacities (e.g. low bridge clearances) (1) • Consult with expert schedulers to understand decision-making processes/rules followed/schedule trade-offs. (1) • Decide how to encode/design/create//gather the acquired knowledge (1) turning it into a <u>knowledge base</u>. (1) • Define/create <u>rules base</u> (1) • Any example of a rule that may be in the rules base (1) • Design/create a <u>user interface</u> (1) that allows schedulers/users to interact with the system (1) • Create the inference engine (1) • Create the explanation system (1) • Test the <u>system</u> with archived data/real-world data/simulated scenarios (1) and improve as necessary (1)	6

Question	Answer	Marks
3(a)	One mark per bullet point to a maximum of two marks • Questions are displayed on the user interface (1) • The user types in the problems/answers the questions (1) • Outputs explanation of findings (1) • Outputs suggestions/probabilities of possible solutions. (1)	2
3(b)	One mark per bullet point to a maximum of two marks. • Acts as the reasoning system (1) • Compares the faults with those held in the knowledge base (1) • Uses/searches the rules base//set of rules (1) • Uses IF THEN (1) • Use backwards/forwards chaining (1) • Produces suggestions/solutions (1).	2

Question	Answer	Marks
3	One mark per bullet point to a maximum of six marks. • Inference engine compares inputs with rules in the knowledge base (1) • Uses IFTHEN reasoning (1) Forward chaining • The reasoning uses forward chaining (1) • Forward chaining starts with the available data (1st) – And uses inference rules to extract more data//ask more questions (1) – Until a goal is reached//Until it finds one where the IF statement is known to be true (1) – The THEN part adds new information (1) • Forward chaining can be called data-driven (1) Backward chaining • The reasoning uses backward chaining (1) • Backward chaining starts with a list of goals/hypotheses (1) • Backward chaining would search the inference rules (1st) – Until it finds one which has a THEN part that matches a desired goal (1) • If the IF part of that rule is not known to be true (1st) – The THEN part adds this to the list of goals (1) • Backward chaining can be called goal-driven (1)	6

Question	Answer	Marks
4(a)	One from: A database/store of facts and a rules base (1)	1
4(b)	Four from: - Knowledge base holds facts (1) gathered from an expert (1) - Knowledge base holds rules (1) that are used to interrogate the facts (1) - It is interrogated by the inference engine (1st) - Using relevant rules//rules that apply (1) - The rules base includes IF...THEN rules (1)	4

Question	Answer	Marks
3(a)	Four from: - Acts as a reasoning system/applies logic/makes decisions (1) - Searches the knowledge base/database (1) - Selects/applies/uses applicable/relevant rules (1) - Uses rules to interrogate the knowledge base/database (1) - Uses the rules to determine outcomes (1) - Uses IF THEN analysis (1) - Uses forward chaining/backward chaining to reason/arrive at solutions (1) Request clarification from the user if necessary/asks (further) questions from the user if necessary (1)	4
3(b)	Four from: - Data is collected from experts/relevant sources/data mining (any awareness of source) (1) Relevance of data is decided (1) - Data is entered (into knowledge base) (1) - Uses a knowledge base editor to edit/created by a knowledge base engineer (1) - Design/decide the rules base (1) Create the rules base (1) - Create the links between the rules and knowledge/data (1)	4

Question	Answer	Marks
6	FIVE marks available: • Use of an example (1) FOUR from: • It is part of/uses forward chaining (1) • A process of using/analysing data to find a solution (this mark may be awarded for a description of this process as part of the description of the context) (1) • Data entered into the system determines the rules to be used (1) • (The inference engine) starts with the data and progresses towards a goal (1) • (The inference engine) will try to match the (inference) rules//searches the rules (award any indication that the rules base is interrogated) (1st) Until one matches the IF statement (1) • The THEN part of the rule brings out new information/data/advice (may answer in context instead of saying information/data/advice) (1) • Process is repeated until a goal is reached//no further possible solutions can be found (1)	5

Question	Answer	Marks	Guidance
2	EIGHT marks available: • A user interface is used to input (1) • IF....THEN (any awareness of USE of) (1) • Any awareness of the use of rules (may exemplify) (1st) These rules are stored in the knowledge base (1) • An expert system takes prior learning/knowledge and applies it to a problem (1) • Locations/address (for delivery) are input (1) • Details of delivery are input//examples of details (weight/dimensions etc) (1) • The responses will be compared with the facts (1st) By the inference engine (1) In the knowledge base (1) • Suggested routes (1) and estimated times of delivery will be output (1) To the user interface//to a printed list (1)	8	Do Not Award • One word answers – must be the USE of. • ANY answers that are not in the scenario • GPS (references to real time processing, traffic delays etc may be taken as GPS) • Parcel collection The question states ‘in this scenario’ therefore, answers have to be in this scenario

Question	Answer	Marks	Guidance
7(a)	TWO from: - Knowledge base editor (1st) allows the engineer to update the knowledge base (1) allows the engineer to edit the rules base (1) - Knowledge base (1st) stores all the facts/information (1) stores the rules base (1) - Inference engine (1st) Applies logic (1) Uses IF THEN reasoning (1) (Uses reasoning to) identify possible solutions/conclusion (1) Uses forward chaining/backward chaining (1) - Rules base (1st) Contains the IFTHEN reasoning (that is relevant to the expert system) (1) - Contains the FORWARD/BACKWARD chaining to reach a solution/conclusion (1) Contains the rules that the expert system uses to draw conclusions (1)	4	Do Not Accept: - User interface systems - Explanation systems

Question	Answer	Marks	Guidance
7(b)	EIGHT marks available: The final decision to follow the advice given by the expert system rests with the mechanic (this may be presented as a positive or a negative aspect) (1) Benefits Max 6: e.g. - Provides a consistent analysis (1) - Knowledge is based on input from a range of experts (1st) therefore is based on people who have solved similar issues (1) ONE FROM as (1st) Uses a process of reasoning based on facts and rules Uses what-if analysis Uses IF THEN analysis so there is a structured process of reasoning (1) - Produces a range of possible solutions (awareness of more than one) (1st) based on answers to a set of questions (1) - Provides full coverage//covers all areas/issues (1) - Includes areas about which the mechanic does not know a great deal (1) - Provides an explanation for decisions (1st) therefore the mechanic can learn (1) - Available at all times (1st) therefore do not need to contact another mechanic at an inconvenient time (1) Drawbacks Max 6: e.g. - Can only deal with problems that are recorded in the expert system (1st) therefore anything outside of the scope/experience of the experts/knowledge base cannot be dealt with (1) - May need to create a new expert system//update current expert system (1) As new cars/engines come onto the market//new technology is used in cars (1)	8	

Question	Answer	Marks	Guidance
7(b)	- If <u>questions are badly structured</u>, will not have all the information needed to solve the problem (1st) therefore suggested solutions may be irrelevant/damage the car further (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) - One mark for a value judgement/conclusion after the discussion of at least two arguments (1)		

Mark scheme · 9626/13 May/June 2026 · Question 10

[Back to question](#)

Question	Answer	Marks	Guidance
10(a)	TWO from: - Uses IF THEN reasoning (1) - Uses forward/backward chaining (1) - Applies the rules to the data//interrogates the knowledge base (1) - To identify possible solutions/answers/results (1) - Application to identification of flowers (must be more than mentioning flowers and must be more than a repeat of question) (1)	2	
10(b)	TWO from: Holds all the facts (1) such as named item from identification of flowers (1) Holds all the rules (1) Holds all the heuristics (1)	2	

8 Spreadsheets

Questions

9626/13 May/June 2024 · Question 4

[Mark scheme](#)

4 Describe the terms absolute and relative cell replication, giving an example of each.

Absolute

.....

.....

.....

.....

.....

Relative

.....

.....

.....

.....

9626/13 Oct/Nov 2024 · Question 6

[Mark scheme](#)

6 In a spreadsheet, a user has entered the following formula into a cell:

=IF(NOT(OR(B6<5,B6>8)),"4th Quarter","")

Describe how the spreadsheet will calculate the result to be displayed.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 7 Test scores can be used to track a student’s progress over time. Describe **two** ways this can be done using a spreadsheet.

Way 1

.....

.....

.....

.....

.....

Way 2

.....

.....

.....

.....

.....

[4]

- Discuss the benefits and drawbacks of using a spreadsheet to undertake this task.

[6]

12 A spreadsheet is to be used to store students' examination results.

Each examination is marked out of 60, with the lowest mark being 0.

Only whole marks are allowed.

Each mark is validated using this validation rule ≥ 0 and ≤ 60

The developer is trying to test the validation rule.

Complete the table for the **three** types of test data. For each mark the developer enters, suggest how the validation rule deals with the mark.

Type of test data	Example of mark entered	How the validation rule deals with the mark entered
Normal		
Abnormal		
Extreme		

[6]

- 2** The spreadsheet shows the prices of various items in a clothes shop. The discounted price for a jumper is calculated by using the formula in cell F12.

=E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)								
	A	B	C	D	E	F	G	H
1								
2								
3				Discount code				
4								
5				A	25%			
6				B	20%			
7				C	10%			
8				D	5%			
9								
10	Item				Price	Discounted price		
11								
12	Jumper			B	\$36.00	\$28.80		
13	Skirt			A	\$25.00	\$18.75		
14	Blouse			D	\$30.00	\$28.50		
15	Trousers			A	\$40.00	\$30.00		
16	Jeans			C	\$33.00	\$29.70		
17	Dress			B	\$50.00	\$40.00		
18	Trainers			A	\$55.00	\$41.25		
19	Shoes			D	\$60.00	\$57.00		
20								

- (a)** The formula in cell F12 is =E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)

Describe how this formula works.

[5]

The formula in cell F12 is replicated to cells F13 to F19.

(b) In the formula, absolute cell referencing and relative cell referencing have **both** been used.

Explain why both types of cell referencing are needed.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 9 Eric has made a spreadsheet that he has shared with a number of friends. Some of his friends cannot open the spreadsheet.

Identify **three** possible reasons the spreadsheet will not open once received. For each reason, suggest a way Eric could have prevented the problem.

Reason 1
.....

Way 1
.....
.....

Reason 2
.....

Way 2
.....
.....

Reason 3
.....

Way 3
.....
.....

[6]

- 5 A student created a spreadsheet that requires users to enter numerical data. This data must be within a set range.

Identify and justify **three** types of test data that can be used to test this numerical data.

[6]

3 A teacher is creating a spreadsheet.

(a) Describe **two** ways in which the spreadsheet can validate data at the input stage.

.....

.....

.....

.....

.....

..... [2]

The spreadsheet will include both absolute cell referencing and relative cell referencing.

(b) Define the term absolute cell referencing.

.....

.....

..... [1]

(c) Identify and explain **one** benefit of using relative cell referencing when creating a spreadsheet.

.....

.....

.....

.....

.....

..... [2]

(d) The teacher has used the following formula in a cell:

```
=IF(C2="Dog",(IF(D2="Male","Block A","Block B")),(IF(D2="Male","Block C","Block D")))
```

Describe how the spreadsheet will produce the result to be displayed.

[6]

3 A student has created a spreadsheet to store data.

(a) Describe **two** features of spreadsheet software that can be used to make the process of entering data into the spreadsheet easier for the student.

feature 1

.....

feature 2

[2]

The student needs to analyse the information she has entered.

(b) Describe **four** features of spreadsheet software that can be used to analyse the information that has been entered into the spreadsheet.

[4]

8 A teacher has created a spreadsheet to store marks and grades achieved by students in an examination. Part of this spreadsheet is shown.

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
2	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

(a) Describe the formatting that has been applied to the contents of cell B2.

.....

.....

.....

.....

.....

..... [2]

The teacher needs to test the spreadsheet. When testing a spreadsheet, it is important to use appropriate test data.

(b) Describe **one** defining characteristic of each of:

- normal test data
- extreme test data
- abnormal test data.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Complete the table below to identify **three** items of test data that can be used to test the mark range for a grade A.

test type	test data
normal	
extreme	
abnormal	

[3]

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
2	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

The teacher will use the following formula in cell G3 to show the grade achieved by Kyle.

=IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6)))))))

- (d)** This formula has been tested and found not to work as intended.

Explain why this formula does not work as intended.

..... [5]

8 Spreadsheets

Mark schemes

Mark scheme · 9626/13 May/June 2024 · Question 4

[Back to question](#)

Question	Answer	Marks
4	Four from: Max two for each: <i>Absolute:</i> • Always copies the same cell reference without modification when replicated/a fixed reference to a cell (1) • Any example of an absolute value/formula/source cell not changing (1) <i>Relative:</i> • It copies the cell reference, but will increment the cell reference when replicated//cell reference changes as the formula location changes (1) • Any example of a relative value/formula/source cell changing (1)	4

Mark scheme · 9626/13 Oct/Nov 2024 · Question 6

[Back to question](#)

Question	Answer	Marks
6	THREE from: • It will check the contents of cell B6 (any awareness that outcome is based on contents of B6) (1) • If the contents are NOT less than 5 (1) OR NOT greater than 8 (1) (accept converse) – ONLY award following if both above stated (NB note effect if answer above is converse) – The cell displays 4th Quarter (1) – if not/otherwise the cell displays a null string. (1) (Answer is dependant) - OR • If condition not met, a blank cell is displayed (1)	3

Question	Answer	Marks
7	FOUR marks available. TWO per matched answer. - Identify a suitable feature of a spreadsheet (1) - State how this feature can be used for the purpose (1) E.g.: Award first answer in each section only, unless no answer written in Way 2. - Manually store/save data about a student's scores/progression (1) to create a record of the student's performance. (1) - Automation/Formulas could be implemented in a spreadsheet (1) to calculate/predict averages of a given/specific time frame. (1) - Conditional formatting could be used in a spreadsheet (1) to highlight grades that meet a set criteria/condition. (1) - A graph/chart could be created from data (1st) - To represent/visualise a student's progression overtime in any given time/test. (1) - A complex/advanced spreadsheet with formulas that function across sheets could be created (1st) - To compare results/scores/progression/trends across different students/groups/classes/tests. (1) - Sort scores into date order (1) so progress over time can be seen (1)	4

Question	Answer	Marks
2	Max 4 marks: MUST be a comparison <i>Benefits:</i> • Easier to create a storage systems//multiple tables/worksheets (1) • Easier to calculate (1) • Easier to create complex formulae//carry out complex calculations (e.g. use functions (if state functions, this is equivalent to complex)) (1) • Repeated data can be easier to enter (1) • Charts are easier to produce (1) • Includes conditional formatting to (e.g.) highlight data easier (1) Max 4 marks: <i>Drawbacks:</i> • Information may exist across many sheets (1) • Large spreadsheets are more difficult to use/read (1) • Spreadsheets are best suited to single focus (e.g. per year) (1) • Formulae may need to be modified as spreadsheet grows (1) • Harder to create complex queries (1) • Harder to create a report (1) • Validation is easier in a database (1) • Repeated data exists in spreadsheet//databases are normalised (1) • Referential integrity reduces (e.g.) redundant data (1)	6

Question	Answer	Marks												
12	One mark per bullet point to a maximum of six marks. <i>Normal</i> • Data is accepted by the system • Any integer between 0 and 60(inclusive) <i>Abnormal</i> • Data outside of the range which will be rejected/not accepted by the system • Any data that is not an integer between 0 and 60 <i>Extreme</i> • Data that is at the boundaries of the acceptable range • 0 or 60 E.g. <table> <tr> <th>Type of test data</th><th>Example of mark entered</th><th>How the validation rule deals with the mark entered</th></tr> <tr> <td>Normal</td><td>41</td><td>Data is within the range data and will be accepted by the system</td></tr> <tr> <td>Abnormal</td><td>68/-2/pink</td><td>Data outside of the range which will be rejected/not accepted by the system</td></tr> <tr> <td>Extreme</td><td>0/60</td><td>Data that is at the boundaries of the acceptable range and will be accepted by the system</td></tr> </table>	Type of test data	Example of mark entered	How the validation rule deals with the mark entered	Normal	41	Data is within the range data and will be accepted by the system	Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system	Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system	6
Type of test data	Example of mark entered	How the validation rule deals with the mark entered												
Normal	41	Data is within the range data and will be accepted by the system												
Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system												
Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system												

Question	Answer	Marks
2(a)	Max 4 from: VLOOKUP - It takes the B from cell D12 (1) - It looks for B in cells D5:E8 (1) - Returns/finds the corresponding value//20%//discount percentage (1st) - from column E/2nd column from the range (1) - It looks for an exact match//If it does not find an exact match it will return an error (1) Max 2 from: CALCULATION - It multiplies the percentage discount by the Price (1) - Subtracts discount from the original price. (1)	5
2(b)	Four marks from: Both are needed because some cell references need to remain the same (1) whilst others change (1). Absolute cell referencing always refers to the same cell therefore does not change (1st) - Valid example of a cell reference that must not change (1) Relative cell referencing changes depending on the position in the spreadsheet//may exemplify new positions (1st) - Valid example of a cell reference that must change. (1)	4

Question	Answer	Marks
9	Max 6 from three matched pairs - Different software on Eric's computer than his friends' computers (1st) - E.g. use generic file type (1) - Saved in a different/wrong (file) format (1st) - E.g. use generic file format (1) - Anti-Virus does not allow file to open/File has a virus (1st) - E.g. Eric needs to virus check the file before sending it (1) - File is password protected/encrypted and Eric didn't share the password/decryption key (1st) - E.g. Eric needs to share the password (1). - Wrong access rights/permissions assigned to the file (1st) - E.g. grant access rights/permissions (1)	6

Question	Answer	Marks
5	One mark per bullet point to a maximum of six marks. Valid/Normal//Data that is from within the acceptable range (1st) • To test that the output is as expected/valid (1) • To test that it is accepted (1) Invalid/Erroneous//Data//Abnormal that is outside of the acceptable range (1st) • Data is of the wrong type/invalid (1) • To test that it is rejected (1) Boundary/Extreme//Data that is at the boundary (of the acceptable range) (1st) • To test that the boundaries are set correctly (1) • To test that the outcome is as expected – accepted and correct output (1)	6

Question	Answer	Marks	Guidance
3(a)	TWO marks available: • Use a drop-down menu which only allows selection from the list (1) • Accept only whole number e.g. between two extremes/greater than X (1) • Accept only decimal e.g. between two extremes/greater than X (1) • Accept only date e.g. between two extremes/equal to a set date (1) • Accept only time e.g. between two extremes/before a set time (1) • Restrict text length e.g. between two extremes/greater than X (1) • Set allowable value to be equal to that in another cell (1) • Use a (run on change) macro to check the value (1)	2	Do Not Award • 'Use validation tools' • Length check on its own, but allow description of length check
3(b)	ONE mark available: • A cell reference that will not change when the formula is dragged down//copied to another cell/replicated (1)	1	MUST have both parts

Question	Answer	Marks	Guidance
3(c)	TWO marks available for one complete explanation Identify benefit (1st) state why it is a benefit/state the beneficial implication (1) • Identification of benefit (1st): e.g. Fewer errors Reduced time to create the spreadsheet/formula • Explanation (1) e.g. Because formula only has to be typed once Because can enter formula in one cell and can (e.g.) drag down Because cell references automatically change as copied/dragged/replicated	2	Do Not Award • Makes data more accurate, however, if this is further explained – such as fewer errors, you may award fewer errors • Enter numbers and drag down – has to be formulas
3(d)	SIX marks available: • Checks value from cell C2 (1) • If the contents are Dog (1) • The first IF statement is run (1) • It will check the contents of cell D2 (1) • If the contents are Male (1st) It will display “Block A” (1) If not, it will display “Block B” (1) • If C2 not Dog (1) • It will check contents of cell D2 AND if the contents are Male (1st) It will display Block C otherwise it will display Block D (1)	6	

Question	Answer	Marks	Guidance
3(a)	TWO marks available: • (Data) import from a different source/spreadsheet/database (1) • User forms which provide input boxes/radio buttons (1) • Macros to run input boxes (1) • Drop down lists from which can select an answer (1) • Input boxes linked to specific cells (1) • Copy and paste from another source/spreadsheet/database (1) • Autofill populate cells based on a set pattern/repeated data (1) • Fill series populate cells based on a set pattern (1) • Radio buttons to select options (1)	2	

Question	Answer	Marks	Guidance
3(b)	FOUR from: • Spreadsheet software allows what-if analysis to be carried out (award any mention of what if analysis) (1) • Can sort data into a rank order (concept of ordering) (1) • Can use macros which can create routines /subroutines (1) • Macros/subroutines can carry out fixed routines//e.g. goal seek (1) • Conditional formatting can show/highlight data ONE FROM that matches//is greater than//less than criteria (1) based on rules/conditions (may exemplify) (1) • Carries out calculations such as subtraction (accept any example of a calculation) (1) • Formulae can be created for calculations (1) • Functions can perform set tasks OR (one mark in total) Uses Functions such as OR (one mark in total) Identifies a function by name and states what it does (1) • Charts/graphs to present values//allow for comparison//visualise data (1) • Pivot Tables to analyse large quantities of data in a visual form (1) • Filtering based on set criteria (may exemplify e.g. for a certain class) (1)	4	Award first four attempts only Do Not Award: Features which are NOT directly to do with analysis

Question	Answer	Marks	Guidance
8(a)	TWO from: - In bold (1) Centrally aligned//centred horizontally/vertically (1)	2	Do Not Accept Border – not a content
8(b)	THREE marks available: Normal test data (max ONE) - Data that should produce expected results (1) - Data that is within acceptable boundaries//correct range (1) - Should be accepted by the cells (1) - Sensible data (1) Extreme test data (max ONE) - Data that is at the boundary (1) - At the limit of acceptability (1) Abnormal test data (max ONE) - Falls outside acceptable limit/range (1) - Data that is invalid (1)	3	
8(c)	THREE marks available, one per type of test: Normal - Any single number between 44 and 50 (inclusive) (1) Extreme 44//50 (ONE ONLY – NOT BOTH) (1) Abnormal - Any other single number beyond range 44 - 50 (1) - Any data other than numeric (1)	3	Accept real values between 44 and 50 Answer must be actual data (e.g. 50) and not description of data (e.g. 'a value between 44 and 50'). If descriptions given, do not accept this as non-numeric for abnormal test data

Question	Answer	Marks	Guidance
8(d)	FIVE from: =IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6))))))) - The formula will not produce the correct grade for all scores (1) at the (lower) boundary of each grade (1) it will produce the grade below (1) Wrong inequality symbol marks (route 1) - The formula only uses an inequality symbol (1) and should have used an inequality (accept equivalent statement, such as > has been used) AND an equals statement (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) Wrong number marks (route 2) - The wrong value has been used as the lower boundary (1) - The formula uses a greater than inequality (1) the value from the top of the lower boundary should be used (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) One mark may be awarded for: - One mark for showing how the formula may be improved:	5	Marks may be awarded from route 1 OR route 2 only. Where candidate mixes both explanations, mark to the candidate's best advantage.

Question	Answer	Marks	Guidance
8(d)	e.g. • IF(F3 > B3 – 1 • IF(F2 = B3 etc. IF(F2 > 43		

9 Modelling

Questions

2 Flight simulators are used to train pilots to fly modern jet aircraft.

Describe a typical flight simulator.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 7 Automatic re-calculation is one of the characteristics of spreadsheet models. This means it is not necessary to evaluate a formula every time you change it.

Describe other characteristics of spreadsheets that make them suitable for creating models.

[6]

- 7 A company has been asked by a government to construct a nuclear reactor. A computer model will be created before the reactor is built.

Explain why a computer model is needed in this scenario.

[5]

7 Spreadsheet models can be used to forecast the population growth of humans.

Analyse the effectiveness of using a spreadsheet model for this purpose.

[8]

- 4 Here is a spreadsheet created by the organisers of a concert. They have assumed that 400 people will attend the concert. They have also assumed that each member of the audience will purchase 2 snacks and 2 drinks during the concert. If these assumptions are correct, their profit will be \$710.

	A	B	C	D	
1		Estimated audience size		400	
2					
3		Price	Quantity per concert goer	Income	
4	Tickets	\$40	1	\$16,000	
5	Drinks	\$2	2	\$1,600	
6	Snacks	\$3	2	\$2,400	
7	Souvenirs	\$20	1	\$8,000	
8	Programmes	\$15	1	\$6,000	
9			Total income	\$34,000	
10					
11				Paid out	
12	Electricity			\$650	
13	Performers			\$22,000	
14	Drink supplies			\$860	
15	Snack supplies			\$880	
16	Venue hire			\$5,000	
17	Souvenir supplies			\$2,400	
18	Programme supplies			\$1,500	
19			Total outgoings	\$33,290	
20					
21			Profit/Loss	\$710	
22					
23					

A number of factors could affect the amount of profit.

- The audience size might be higher than the organisers estimated.
- Members of the audience might not buy 2 snacks and 2 drinks.
- The cost of souvenir supplies might change.

- (a)** Explain how the organisers could use what-if analysis to see the effects of these three factors on their profit. You should refer to specific cells from the spreadsheet in your answer.

..... [3]

(b) The organisers wish to achieve a larger profit.

By explaining what each step requires them to do, describe how the organisers would use goal seek to find the size of the audience needed to make a profit of \$5000. You should refer to specific cells from the spreadsheet in your answer.

Include in your answer a reason why the result might not be satisfactory.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

9 Computer models are often used to run simulations when designing a nuclear power plant.

(a) Describe the benefits of using simulations in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Describe the drawbacks of using simulations in this scenario.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

11 Weather forecasting makes use of computer models.

Analyse the effectiveness of computer models for producing weather forecasts.

[8]

- 1** Spreadsheet software can be used for computer modelling.

Describe the characteristics that make spreadsheet software suitable for computer modelling.

[6]

6 An airline company uses a flight simulator to train its pilots.

Discuss the advantages and disadvantages of using a flight simulator in this scenario.

[8]

6 Describe the reasons why computer models are used.

[6]

- 11** A driving school is using computer simulations to improve its lessons.

Discuss the benefits and drawbacks of using simulators for this purpose.

[8]

11 Evaluate the use of spreadsheets for financial modelling.

[8]

13 A car company uses simulations to test its cars.

Discuss the benefits and drawbacks of using computer simulations in this context.

[8]

4 Identify and explain **two** drawbacks of using a spreadsheet for computer modelling.

1

.....

.....

.....

.....

2

.....

.....

.....

.....

[4]

8 Identify and describe **three** characteristics of modelling software.

characteristic 1
.....
.....
.....
.....

characteristic 2
.....
.....
.....
.....

characteristic 3
.....
.....
.....
.....

[6]

4 Computer models make use of what-if analysis.

(a) Describe, with the use of relevant examples, how what-if analysis can be used as part of a model of population growth.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

6 (a) Justify the use of a computer simulation as a suitable method to train helicopter pilots.

[8]

(b) Identify and explain **one** drawback of using a simulation as the only method to train helicopter pilots.

[2]

9 Modelling

Mark schemes

Mark scheme · 9626/11 May/June 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Four from: Simulates difficult situations/take off scenarios/speed of the aircraft/turbulence/different/extreme weather conditions/can control weather conditions Include software and specialised equipment/hardware... ...that can cost hundreds of thousands of pounds... ...a cabin containing screens, flight controls, control wheel, throttle, joystick, instrumentation, pilot seat, sound system, computer hardware, altitude gauges, fuel gauges (must have two) Software includes the rules of the model that define how the aircraft reacts... ...includes a database containing information about each airport... ...allowing pilots to practice landing at unfamiliar airports... Control panel in a cockpit mimics/is identical to one in a real plane/equipment is designed to react in a similar way to real aircraft Pilot uses joysticks/levers/switches/buttons/sliders to control movement/speed Displays are generated by computers on the screens in the cockpit/screens display runway/sky... ...computer graphics produce display using... ...projectors/back-projection screen/panoramic spherical mirror	4

Mark scheme · 9626/11 May/June 2022 · Question 7

[Back to question](#)

Question	Answer	Marks
7	Six from: Replication/copying means you don't have to enter the same formula/data several times/the rules of a model can be changed easily using replication Absolute and/or relative cell referencing makes sure you only increment the parts of a formula you need to when replicating a formula Allows what-if questions to be asked Cell protection makes sure that the cells you want do not change by accident User interface forms makes it easier to input values into the model Conditional formatting allows you to highlight certain values that match specific criteria Graphs/charts to show trends/predictions The use of input forms makes it easy to enter data Macros make it easier to create more complex formulae or functions/used to record a sequence of actions Goalseek can be used to determine which variables need to be changed to achieve a target or goal Pre-written functions allow users to create complex formulae Values can be changed to ask whatif questions	6

Question	Answer	Marks
7	Five from: Testing nuclear reactor designs using computer models avoids safety problems... ...such as explosions/meltdowns <u>Testing nuclear reactor designs</u> using computer models cuts costs... ...as don't have to pay as much money for workers/materials to replace damaged reactors Can obtain results in a short period of time regarding reactions that take a long time in real life Re-designing computer models is cheaper than re-building a nuclear reactor Researchers will have the tools to simulate scenarios that are hard to observe in operating reactors It may be too dangerous to modify the actual nuclear reactor You can test predictions more easily/model can make predictions more accurately	5

Question	Answer	Marks
7	Eight from: • Spreadsheet models can do complex calculations (1) • Formulas are recalculated automatically ... (1) ... speeding up the use of the model (1) • Graphs that are produced to show population growth ... (1) ... will (automatically) change as new values are added/old values altered (1) • (Reliable because) provide consistent results from the same inputs //will always produce the same answer (1) • Spreadsheets can automatically download data to provide data in the fields (1) • Cannot account for every possible variable when forecasting population growth ... (1) ... therefore model can give misleading results (1) • To try to account for as many variables as possible requires complex formulae ... (1) ... which may need experts who are expensive to hire (1) • Producing an effective model may be time consuming (1) • The model can have formulae replicated (1) • Complex population predictions may need additional software/modules ... (1) ... which is expensive to buy (1) • Data can be entered accurately because of computer-based validation and verification (1) Max. six marks if bullets/list of points Max six if no expansions	8

Question	Answer	Marks
4(a)	Three from: • They could increase/change (the number of customers by increasing/changing) the value in <u>cell D1</u> ... (1) • They could increase/change (the number of snacks by increasing/changing) the value in <u>cell C6</u> ... (1) (accept decrease answer) ... • They could increase/change (the number of drinks by increasing) the value in <u>cell C5</u> ... (1) (accept decrease answer) ... • They could increase/change (the cost of souvenir supplies by increasing) the value in <u>cell D17</u> ... (1) (accept decrease answer) The value in D21/profit would change (showing the new profit/loss//profit would change) (1)	3
4(b)	Four marks available Max three from: • The organisers would select the cell containing the formula that returns the result they want/D21, and select goal seek (1) They input the value they want the formula to return/5000 (1) They enter the cell reference of the input value that the spreadsheet can change to reach the target/D1 (1) Max one from: • The number could be to several decimal places and needs to be a whole number of concert goers (1) • If other assumptions are incorrect (e.g. the number of snack/drinks bought per concert goer), the number produced by goal seek would be inaccurate (1) • Venue may not be able to hold that many people (not 'number too large' on its own) (1)	4

Question	Answer	Marks
9(a)	Four from: • Definition of a simulation e.g. Simulations are a digital representation of a situation as it would happen in real life (1) • Using computer models avoids safety problems (1st) • such as e.g. explosions/meltdowns (1) • Using computer models cuts costs (1st) • as don't have to pay as much money for workers//materials to replace damaged reactors (1) • Can obtain results in a short period of time regarding reactions that take a long time in real life (1) • Re-designing computer models is cheaper than re-building a nuclear power plant (1) • Can simulate scenarios that are hard to observe in operating power plants//in real life (1)	4
9(b)	Four from: • Simulation may not be 100% accurate (1) • Models can't always recreate exactly the real-world experience/may miss variables (1) • Very complex to develop a simulation (1) • Specialist knowledge is required to build model (1st) • which requires specialists which costs money (1) • Specialist knowledge is required to analyse results (1st) • which requires specialists which costs money (1) • There may be too many variables to include in the model (1st) • to exactly simulate the environment/design/working of the nuclear reactor (1) • leading to inaccurate results (1)	4

Question	Answer	Marks
11	Eight from: • Weather forecasting used to involve several weather forecasters and days of calculations (1st) • today one model can manage the task (1) • Computer models help forecasters to make more reliable predictions (1) • Models can be used to predict future patterns (of weather) (1) • More complex models create more accurate forecasts (1st) • which allow forecasts further into the future (1) • Human weather forecasters are still needed to evaluate the results (1) • Weather forecasting requires the use of a lot of data (1) • To be effective models must consider many variables (1st) • such as (TWO Variables identified): atmospheric pressure, humidity, rainfall, temperature, wind speed, wind direction (1) • as well as calculations such as (ONE of): vertical air motions, temperature changes, moisture processes (1) • A lot of computing power is needed to run a computer model so supercomputers have to be/are used (1) • Supercomputers are (very) expensive (so makes the process an expensive one) (1) • Models can be used to compare current conditions with previous similar conditions/shown trends over a period of time (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
1	Six available. - Spreadsheet software allows what-if analysis to be carried out (award any mention of what if analysis) One per instance of a suitable characteristic: e.g. - Can sort data into a rank order - Can protect cells from accidental change - User interface forms to allow data input Can use macros which can create routines /subroutines - Macros/subroutines enable automation//can be executed by a button - Conditional formatting can show/highlight data that matches//is greater than//less than criteria - Lookup tables to store values - Automatic recalculation changes output when new value inputted/data updated - Allows values to input (in bulk) from an external source - Carries out calculations such as subtraction (accept any example of a calculation) - Formulae can be created for calculations - Functions can perform set tasks in formulae - Goal seek finds a suitable starting input for a desired output - Charts to present values//allow for comparison - Charts automatically update	6

Question	Answer	Marks
6	Eight from: Advantages: MAX SIX • The running/maintenance costs of a flight simulator are less than an actual aircraft (1) • Avoids repair costs for real aircraft (1) • Different/various/a range of situations can be simulated in a flight simulator (1st) – planned at the start of the simulation (1) – during the simulation (1) • Events can be simulated without waiting for them to happen in real life (1st) – such as (suitable named example) (1) • Situations can be repeated multiple times until the pilot gets it right (1) • (Safer because) Situations can be practiced in a flight simulator without putting the humans/trainee/flight training instructor in danger (1) • Flight simulator does not need to be at an airfield//can be in a convenient location (1) • Can concentrate training on specific controls (1) • The environment benefits from flight training in a simulator (1st) – as there is no pollution created by a flight simulator (1) • The level of noise in a flight simulator is not as loud as in an aircraft (1st) – makes the communication between the instructor and trainee much easier (1) Disadvantages: MAX FIVE • Simulators are not as realistic/real as a real aircraft (1) • Pilots get overconfident and may compromise safety (1) • It is difficult to recreate boredom/fatigue of a real flight (1) • Available simulated scenarios dependant on software packages (1) • Pilot may “freeze” when faced with a real flight/actual passengers (1)	8

Question	Answer	Marks
6	SIX marks available e.g. • Computer is used to create the rules behind a simulation (1st) so that the simulation matches the real world (1) • Easier to try out a range of different scenarios//allows what-if analysis (1) • It is less costly than if a mistake is made in real life/it is costly to remedy mistakes (1) • More difficult to construct several designs in real life (1) • Can be used to check on/make predictions for complicated scenarios//scenarios with a lot of variables (1) • Can observe aspects that would be difficult to observe in real life (1) • Can save time (1st) more tests can be done in a shorter amount of time (1) a much greater number of tests can be performed (1) • Simulation modelling can be used to plan for infrequent/one off events (1) • Computer models are much safer than experimenting in real life (1) • Faster, can speed up variables that would impact on results e.g. climate change (1)	6

Question	Answer	Marks
11	EIGHT marks available. Benefits (max 7): e.g. • Can provide results that are generally difficult to measure (1st) – such as e.g. reaction times. (1) • Can simulate a range of scenarios (1) • Some scenarios do not happen frequently in real life (1) • Can simulate driving in different vehicles. (1) • Reduces anxiety level//improves confidence of (new) learners. (1) • Can be repeated any number of times. (1) • Can be customised/personalised to the learner's needs. (1) • Simulation can be recorded (1) • Can analyse mistakes (1) and suggest solutions (1) • Simulation can be frozen to allow for questions/discussion (1) Drawbacks (max 7) e.g. • May not have enough data/information to create some scenarios. (1) • May not accurately predict the occurrence/effects of human behaviour. (1) • Might provide misleading scenarios if information was incorrectly/erroneously inputted/added. (1) • Can require high-end hardware for optimal performance increasing price. (1) • May give drivers a false sense of security regarding their ability to drive. (1) • Not a real-world experience (1)	8

Question	Answer	Marks
11	Eight marks from: MAX four for 1st marks only <i>Evaluations: E.g.:</i> • Can interrogate/interact with databases to extract data (1st) – So data only has to be typed/entered in once (1) • Can only use the features of the spreadsheet (1st) – Therefore model may be limited (1) • Spreadsheet software may include unneeded features (1st) – Which could be a distraction//which need to be hidden//increase file size (1) • Spreadsheets include formulae//functions (1st) – So complicated calculations can be created more easily (1) – Reduced chance of error (1) • Calculations can be performed more quickly/more easily/recalculated automatically (1st) – so spreadsheet can be used by someone who does not spreadsheet editing skills (1) • What-if statements can be asked/edited (1st) – without rebuilding the model/spreadsheet from scratch each time (1) • Creates graphs (1st) – Which allows data to be more easily interpreted (1) • Graphs that are produced will automatically change (1st) – as new values are added/old values altered (1) • Models/spreadsheets provide objective results (1st) – not affected by user's bias (1) • The use of templates provides consistency over time (1st) – therefore do not need to relearn the model (1) • Valid conclusion drawing together/based on arguments presented in the answer (1)	8

Question	Answer	Marks
13	Max one for: - Definition of computer simulation (1) Max six from: <i>Benefits</i> - Identified cost saving (1) - Drivers are not put in physical danger/can be safer/less risk of being hurt/no risk to life (1) - Different/extreme conditions can be simulated so no need to wait for natural occurrence (1) - Conditions/scenarios can be repeated//each simulation can be reset to same starting point/values/circumstances (1) - Simulation can be recorded for later playback/analysis (1) - Feedback can be visual overlays of previous test (1) - Real time data/telemetry produced (1) - Computer can more accurately measure performance metrics (1) - Greater number of testers could be used (1) - Testing can occur at multiple points in the development process (1) - Can try different (possible) features of the car without having to actually build it (1) Max six from: <i>Drawbacks</i> - Can be difficult to simulate all conditions found on roads (1) - Can be expensive to create a realistic simulation (1) - Lack of a realistic experience of the car (1) - Drivers may be more reckless in simulations because of lack of real danger (1) - Could produce phantom/ghost/erroneous problems (1) - Significant analysis/expertise may be required to make sense of data (1) - Simulated scenarios may be inaccurate/wrong (1)	8

Question	Answer	Marks
4	Four available. Max two for each matched pair: e.g. • Impossible/difficult/improbable to include all the possible variables (1st) – too many/unknown/unexpected variables (variables in widest sense) (1) • Spreadsheet software is limited size/capacity/scalability (1st) – plus suitable explanation of an issue this causes (1) • May not have the required formulae/functions (NOT “options”) (1st) – plus suitable explanation of an issue this causes (1) • Need training to use the software (properly)/requires an expert/complicated for new users (1st) – plus suitable explanation of an issue this causes e.g. cost of expertise (1) – therefore takes time to create (do not award as extension for “requires an expert”) (1) • Spreadsheets can be edited by anyone who has the software (1st) – So different versions may develop (1) – So different outcomes arrived at (1)	4

Question	Answer	Marks
8	SIX marks available: ONE mark per identified characteristic, ONE mark for the description/purpose of that characteristic. • Can carry out WHAT-IF analysis (1) So that scenarios can be explored (may exemplify scenario being changed)(1) Allows for complexity (1) • Has inputs/outputs (1st) So that values can be set by the user//so that results can be seen by the user (1) • Can create rules (1st) These define how the inputted data/values are used//the calculations that take place (1) • Can edit rules (1st) To take account of changing parameters/conditions (1) • Cell protection (1st) Protects cells from being (accidentally) changed by the user (1) • User forms (1st) Allows the easy input of data (1) Allows data to be entered in an organised way (1) • Can create macros (1st) To automate processes (1) • Automatic recalculation (1st) Outputs automatically updated (1) • Can hold/handle large amounts of data (use of Multiple – award as BOD).(1st) To allow for different possible outcomes/scenarios (1) Allows for complexity (1) (may be repeated from MP 1) Note: – NOT DEPENDANT • Includes function/formulae (1) to perform calculations/carry out equations (1)	6

Question	Answer	Marks	Guidance
4(a)	FOUR marks available, two marks per matched and described example. • What–if analysis explores what will happen if something else occurs/didn't occur/a variable is changed (1) • Rate of birth can be changed (1) to see impact of increased babies (1) • Rate of mortality can be changed (1) to see impact of increased//decreased teenagers//young adults/adults on number of births (1) • Life expectancy can be changed (1) to see impact of people living longer//shorter on overall population. (1) • Change structural constructs (e.g. schools, hospitals, access to water) (1) to see impact on e.g. mortality rates (1)	4	

Question	Answer	Marks	Guidance
4(b)	EIGHT marks available: Simulations can model/represent what happens in the real world/a natural disaster (1) Positives (max 6) • Simulation can be used in advance of a disaster (1) • Do not have to wait for the disaster to happen//simulations can run now (1) • Simulations can show (possible) impacts/damage of the disaster (1) • Identifies aspects that can be planned e.g. evacuation routes (1) • (e.g.) buildings/infrastructure can be checked to check resilience (1) • Structural/environmental changes can be made (based on the results) (1) • The replacement cost of a e.g. collapsed bridge exceeds the cost of a simulation (1st) Therefore money is saved by using on a simulator (1) • Simulations can repeat scenarios//allow for trial and error//can test different plans out (1st) e.g. So the response team can explore different responses//check if their response works (1) Learn from their mistakes (1) • There is no risk to human life in a simulation (concept of reduced risk/not being in a risk situation) (1st) Therefore the simulation is safer (1) • Small changes can be made to the scenario//different scenarios can be checked/explored/used (1st) To research into marginal effects (concept of) (1) To allow the team to practice in changing conditions/explore the impact of small changes (1) • Simulations can be recorded (1st) for (e.g.) future analysis/training (1)	8	

Question	Answer	Marks	Guidance
4(b)	Negatives MAX 6 • Simulations may not include all possible events (1) Simulation is not realistic (1) • Simulations assume fixed responses/fixed rules (1st) People may respond differently in real life (1) • Initial cost of purchase of software may be expensive (1) • Interpretation of the outcomes requires experts//team members from different groups (1) • Interpretation of the outcomes requires time//has a cost implication (1) Conclusion (may only be awarded if positives(s) and/or negatives are included) • One mark for a value judgement/conclusion based on two arguments (1)		

Question	Answer	Marks	Guidance
6(a)	Eight from: • Means does not use a (real) helicopter (1st) avoids wear and tear//maintenance costs on the helicopter (1) • Simulations copy the real world (1st) • So, pilots can experience relatively rare/dangerous (1) A range of real-life situations (1) Which may not otherwise occur//may have to wait for (1) <u>For dangerous ONLY and if safety not already awarded</u> – “avoids the risk” (1) • Concept of injury being avoided (1st) Therefore, the simulation is safer (1) • The running cost of a simulation is less than the running cost of a helicopter (1st) And so, money is saved//is a cost-effective solution/is cheaper (1) • It replicates the experience of being in a real/different helicopter (1st) by having the same layout/controls (as a real helicopter) (1) • The replacement cost of a (crashed) helicopter exceeds the cost of a simulation (1st) Therefore, money is saved by training on a simulator (1) • Simulations can repeat scenarios (1st) So, the trainee pilot can master key skills//learn from their mistakes (1) • Small changes can be made to the scenario (1st) To allow the pilot to practice in changing conditions (1) • Different locations can be trained for (1st) Without the need to go there (1) • Pilots can train on their own in the simulator (1st) So, no need for a trainer to be present (1) More lessons can take place (at the same time) (1) • Lesson can be saved (1st) And content analysed afterwards (1) MAX six marks if the candidate only gives first mark answers.	8	

Question	Answer	Marks	Guidance
6(b)	TWO marks available for an explanation of an identified disadvantage: - Pilot may have a false sense of security (1st) And so take more risks in a real helicopter (1) - Simulator does not risk anyone's life, but a real flight does (1st) So pilot may freeze when faced with really being in charge (1) - Concept that may not be totally accurate or (fully) represent real life or realistic (1st) Therefore not a value practice (1) May not inspire//give confidence in the real world (1) Cause problems when they experience things that were not practiced (1)	2	ONLY mark first drawback given/attempted Do Not Award - Training is inefficient – TV - Isn't up in the air - The pilot could get scared on a real flight - Is a less pressured experience (TV) - A simulation is not training on real life (as in, it is a simulation, not the real thing). This is different to “a simulation may not be realistic”. - It doesn't feel like flying a helicopter

10 Database and file concepts

Questions

9626/12 Feb/March 2022 · Question 10

Mark scheme

10 (a) Define the following database terms.

(i) One-to-one relationship

.....
.....
..... [1]

(ii) One-to-many relationship

.....
.....
..... [1]

(iii) Compound key

.....
.....
..... [1]

(b) Explain how a many-to-many relationship can be created in a database.

.....
.....
.....
.....
.....
.....
..... [3]

10 Normalisation is used when designing relational databases.

Discuss the advantages and disadvantages of normalisation.

[8]

4 Describe what is included in a management information system (MIS).

[6]

- 10** Describe what is meant by the term open source file format. Include in your answer an explanation of why open source file formats are needed.

[5]

- 8** Peter wants to create a database containing the details of all the books he owns and the shops he bought them from. Mary suggests that he should create a relational database. Peter will need to use normalisation.

(a) Describe what is meant by normalisation.

[4]

(b) Describe the benefits of creating a relational database rather than a flat file database.

[6]

1 (a) Describe the features and the role of a primary key in a database.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(b) Describe the features and the role of a foreign key in a database.

.....

.....

.....

.....

.....

..... [2]

- 5** Compare static parameter queries with dynamic parameter queries. You must include similarities and differences.

..... [5]

- 10** A school stores all the records of its students in alphabetical order. Records could be found using sequential access.

(a) Explain what is meant by sequential access.

.....

.....

.....

..... [1]

The school decides to use the indexed sequential method to search for the records of individual students. An index table contains all the letters of the alphabet in alphabetical order.

(b) Describe how the details of a student called Johnsen would be found.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- 7 Louisa, a teacher with limited IT knowledge, has created a simple database of her students. The database contains the names of her students and the results of their latest test.

She prints out the names of the students who have passed the latest test by gaining a mark of 40 or more.

When she created the database, she considered creating a dynamic parameter query to produce her printout. However, she decided to create a static parameter query.

Explain why this was a good decision.

[5]

8 Students often transfer files between different word processing and database packages.

Justify the need for generic file formats for the transfer of files between different packages.

[8]

- 2** Irene has produced a document using a word processing package. She has saved it as a text file and also as a separate word processed file.

Give **four** advantages of saving the document as a text file compared with saving the document as a word processed file.

[4

- 6** When searching a database it is possible to use different types of query.

Describe the differences between simple and complex queries.

[4]

- 2 A data dictionary is required when creating a relational database.

Identify and describe **four** components that should be included in a data dictionary.

1

.....

.....

.....

2

.....

.....

.....

3

.....

.....

.....

4

.....

.....

.....

[8]

11 Explain what is meant by the term open-source file format.

.....

.....

.....

.....

.....

.....

.....

..... [3]

4 Describe the term indexed sequential access.

[8]

9 A company in Nairobi, Kenya, uses a database to store all its customer details.

Evaluate the use of static and dynamic parameters in a query to search the database for customers in Nairobi, Kenya.

[8]

1 A building company has a relational database. The database has two tables: CUSTOMERS and QUOTATIONS.

(a) The QUOTATIONS table has a foreign key.

Describe what is meant by the term foreign key.

.....

.....

.....

.....

.....

..... [2]

(b) (i) Define the term referential integrity.

.....

.....

.....

..... [1]

(ii) Explain why referential integrity is important.

.....

.....

.....

.....

.....

.....

.....

..... [3]

(c) Describe what is meant by the term relational database.

[5]

1 A relational database has been created for customer orders.

The entities and attributes in the relational database are:

CUSTOMER	ORDER	PRODUCT
Cust_ID	Order_code	Product_ID
First_name	Cust_ID	Description
Family_name	Date	Price
Address_1	Product_ID	
Town	Quantity	
Zip_code		

(a) Identify the primary key in the CUSTOMER entity.

..... [1]

(b) Identify the primary key in the ORDER entity.

..... [1]

(c) Identify **one** foreign key in the ORDER entity.

..... [1]

(d) Identify the relationship between the CUSTOMER and ORDER entities.

..... [1]

10 Describe the benefits and the drawbacks of using the csv file format to save data.

[8]

3 Describe how a company can use management information system (MIS) software to help run its business.

[6]

[6]

4 Relationships are used in relational databases.

(a) Describe the following database relationships:

(i) one-to-one

..... [1]

[1]

(ii) one-to-many

..... [1]

[1]

(b) Describe how a many-to-many relationship could be created in a database.

.....

.....

.....

.....

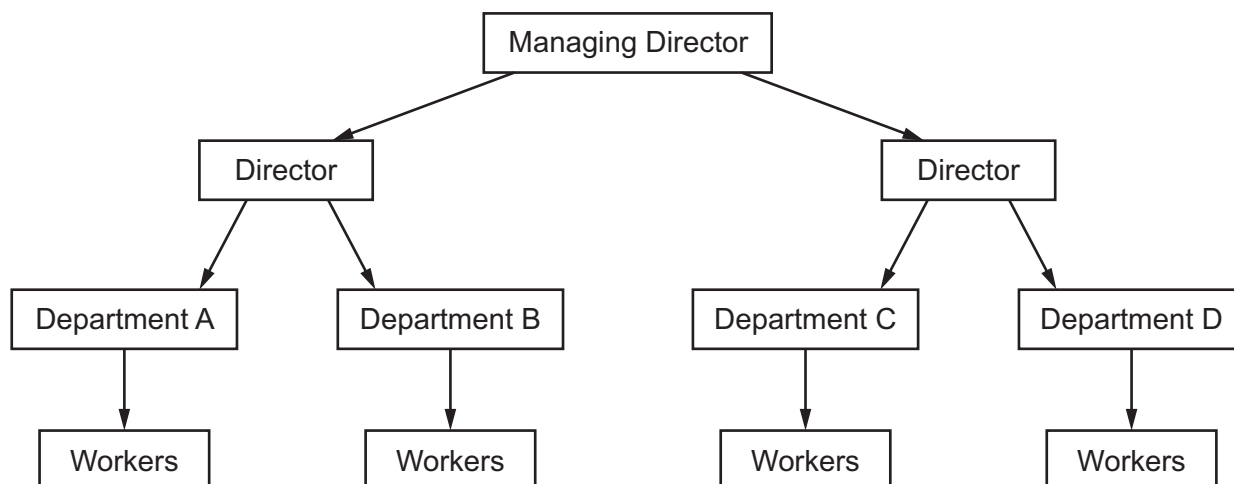
.....

.....

.....

..... [3]

- 1 A company uses the following hierarchical staffing structure.



The company stores all its staff records in a hierarchical database.

- (a)** Explain what is meant by a hierarchical database management system.

[3]

- (b)** Describe how the company's hierarchical database would be used to store all the staff records.

..... [3]

8 A furniture company uses databases in its warehouse to help with stock control.

(a) The company has a Product database which contains the fields shown in the table.

Field Name	Data Type	Example of Data
Number_in_Stock		199 (Number_in_Stock can vary between a minimum of 0 and a maximum of 250.)
Location		07.44.03
Reorder_Level		250
Price		\$203.50
Supplier_Telephone_No.		07700900819

Complete the Data Type column in the table by writing the most appropriate data type for each field. For any numeric fields, specify the numeric data type. [5]

2 Describe the features of a management information system (MIS).

[6]

3 Normalisation is used in a relational database.

(a) State what is meant by the term second normal form (2NF).

[2]

(b) Discuss the benefits and drawbacks of normalising data.

[illegible]

10 Describe the following types of database query:

(a) summary

.....

.....

.....

.....

.....

..... [2]

(b) complex

.....

.....

.....

.....

.....

..... [2]

(c) nested

.....

.....

.....

.....

.....

..... [2]

- 7 A school stores records of its students in alphabetical order of family name. It uses indexed sequential access to locate a student record.

Explain what is meant by indexed sequential access.

[5]

- 9 Eric has made a spreadsheet that he has shared with a number of friends. Some of his friends cannot open the spreadsheet.

Identify **three** possible reasons the spreadsheet will not open once received. For each reason, suggest a way Eric could have prevented the problem.

Reason 1
.....

Way 1
.....
.....

Reason 2
.....

Way 2
.....
.....

Reason 3
.....

Way 3
.....
.....

[6]

- 12 Define the following types of database key fields:

(a) Composite key
.....
..... [1]

(b) Compound key
.....
..... [1]

7 (a) Describe the term dynamic parameter query.

.....

.....

.....

.....

.....

..... [2]

(b) Describe the benefits of using dynamic parameter queries when using a database.

.....

.....

.....

.....

.....

.....

.....

..... [3]

15 Discuss the importance of referential integrity.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

7 A company wishes to gather some information from employees.

(a) Describe the process of organising face-to-face interviews.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Justify the use of a relational database to store the information gathered.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 A list of some field names from a database is shown. For each field name, identify the most appropriate data type.

(a)

Field name	Most appropriate data type
employee_name	
hourly_rate_of_pay	
was_employee_born_in_france?	

[3]

(b) A presence check has already been run on each field of the database.

For each of the following fields, identify and describe **one** other type of validation that can be used. You should only use each type of validation once.

(i) hourly_rate_of_pay
.....
.....
.....
..... [2]

(ii) date_of_birth
.....
.....
.....
..... [2]

(c) Justify why a field holding the age of each employee has been set as a calculated field.

.....
.....
.....
.....
.....
.....
..... [3]

5 A school has paper-based records of all students that have attended the school since 1993.

These records include:

- home addresses and contact details
- the grade each student achieved in each of their end of year exams
- the years when they joined and left the school.

The school is going to create and use a single database to store these records.

(a) Compare the use of flat file database structures and relational database structures to store these records.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

The database will include validation rules.

(b) Identify and describe **one** validation rule that could be applied to the database. State why it is important.

.....

.....

.....

.....

.....

.....

.....

..... [3]

9 A student is planning a relational database.

(a) Define the term relational database.

.....

.....

..... [1]

The relational database could include both primary keys and compound keys.

(b) (i) Describe the characteristics of a primary key in a relational database.

.....

.....

.....

.....

.....

..... [2]

(ii) Describe the role of a primary key in a relational database.

.....

.....

..... [1]

(c) Describe how a compound key differs from a primary key.

.....

.....

..... [1]

The database the student is planning needs to be normalised.

(d) Describe the conditions that must be met for the database to be in first normal form (1NF).

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(e) Describe **four** benefits of normalising the relational database.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 Database and file concepts

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 10

[Back to question](#)

Question	Answer	Marks
10(a)(i)	Both tables can have only one record on either side of the relationship	1
10(a)(ii)	A table contains individual records that each relate to many records in the related table	1
10(a)(iii)	A compound key is a primary key that combines more than one foreign key to make a unique value	1
10(b)	Three from: A many-to-many relationship has to be broken down into two/several one-to-many relationships... ...using a third table, a “join table”... ...each record in the “join table” would have the foreign key fields... ...linked to the primary keys of the two tables it is joining together	3

Question	Answer	Marks
10	Eight from: <i>Advantages:</i> Any change to one record which is needed can be instantly made to any related records The database does not have data duplicationmaking the file size smallerso less money needs to be spent on storage ...so there are fewer errors in the data... ...so there is less chance of storing incorrect copies of the data Modifying a table is easier as there is less data to modify <i>Disadvantages:</i> A larger numbers of tables requires more relationships to be designed... ...taking more time Making data atomic may not always be the best solution such as date of birth can be separated into day, month and year... ...but this may serve no purpose Data may be stored as codes rather than meaningful data... ...making it difficult for humans to interpret/level of detail can be lost With more tables, setting up queries can become more difficult... ...the more complex the database queries need to be so can take longer to run You can end up with more tables than an unnormalised database making it difficult to keep track of data May require greater expertise/knowledge and understanding... ...which may need to come from outside and be paid for Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions to be a proper discussion	8

Question	Answer	Marks
4	Six from: • Data resources/(a) database(s) (Accept any indication of the type of data being held e.g. Employee data/information) (1) • Hardware resources of a system (1) • People management and project management applications (1) • Decision support systems//computer program applications used by managers to gather information to support problem solving//decision making (may exemplify) (1) • Executive information systems/a reporting tool that provides quick access to summarised reports (coming from all departments) (1) • Marketing Information Systems/systems designed specifically for managing the marketing aspects of the business (1) • Accounting information systems for accounting functions (1) • Human resource management systems//systems used for analysing personnel aspects of an organisation (1)	6

Question	Answer	Marks
10	Max four from: • A format used for storing data, for anyone to use (1) • An open-source file format can be used by both proprietary and (free and) open-source software (1) • Open source formats are often called free file formats if they are not covered by any copyrights/patents (1) • Anyone may use them at no cost (1) for any desired purpose (1) • Code is open to inspection (1) • Code can be amended without breach of copyright (1) Max three from: • Needed because not everyone can afford proprietary software (1) • Needed so users can work on files in different locations without the need for the same software (1) • Provides a standard file type (1)	5

Question	Answer	Marks
8(a)	Four from: • One mark for identification of any three forms (1) • A method of organising data tables (1) • Attempts to avoid data loss (1) • Attempts to avoid data redundancy//data duplication (1) • Attempts to improve/improves data base efficiency (1) • Breaks data from non-atomic structure to atomic//description of atomisation (takes an example of non-atomic data and states how it becomes atomic) (1) • Breaks data into smaller/other units/tables of data (1) • The units/tables are related to each other (may exemplify) (1) • Only related data is stored in a table (1) • It is an iterative process// It is a multi-step/stage process (1) ... (where) each step/stage improves the efficiency of the database (may be given as a standalone mark) (1)	4
8(b)	Six from: • Searching a large database is quicker to locate records (1) • A relational database does not have duplicated/redundant data (1) • Uses reduced storage space (DNA 'memory') (1) • When data in one field of a table is modified the data in the corresponding field//related field is updated (1) • Integrity constraints ensure that table relationships are/remain valid (1) • It is easier to maintain security of data (1) • Use of related tables reduces need for (as much) data entry (1) • Reduced chance of data entry errors (1) • Easier for users to produce reports (1)	6

Question	Answer	Marks
1(a)	Three from: - Contains unique data/values (1) - Cannot contain blank data/be empty (1) - Helps to provide quicker searches (1) - It enables relationships/links to be formed between tables (1st) - if a relational database is to be created (1) - There can only be one primary key in a table (1) - Provides a unique identifier for each record (1)	3
1(b)	Two from: - A field that contains data found in a primary key field in/from another table (1) - Links to the primary key of another table (1) - Used to form a relationship between the two tables (1) - Referential integrity is usually implemented through the use of foreign keys (1) - There can be many foreign keys in a table (1)	2

Question	Answer	Marks
5	Five marks available: Similarities Max 4 - Both search for the data that the user/query requires//that matches the (search) criteria (1) - Both return records containing that data (1) - Both can be used with complex queries (1) Differences Max 4 - A dynamic (parameter) query can be used to search for different values/data each time it is run (1) - In a static query every time that the query is run it will search for the same data/values (1) - Dynamic query needs a value to be entered every time (1) - Static query has the criteria hard coded/does not need to be added each time (1) - Static parameter query would save the time of editing the query every time a different criteria/value was used//Dynamic parameter query would waste the time of editing the query every time a different criteria/value was used (1) - Dynamic parameter query requires more technical knowledge of the user//are more complicated to create (1)	5

Question	Answer	Marks
10(a)	One from: Records are accessed in the order they were entered (1) - Each record is read one by one until a match is found (1)	1
10(b)	Three from: - The index table would contain the corresponding part of the disk (1st) - where names beginning with each letter of the alphabet can be found (1) - For names beginning with J the part of the file containing all the As to I's can be ignored/the disk/records is/are accessed where the J's begin (1) - All the records beginning with J will be read one by one until Johnsen is found (1)	3

Question	Answer	Marks
7	Five from: - She doesn't have to spend time responding to the prompt/typing in the pass mark, (1st) - which Louisa would have to with a dynamic parameter query (1) - because the pass mark is hard coded (1) - The <u>pass</u> mark is always the same/fixed so there is no need for a dynamic parameter query//to type in the (same) number (1) - Louisa would need greater technical knowledge to set up a dynamic parameter query (rather than a static parameter query) (1) - It would have taken Louisa longer to set up a dynamic parameter query (1) - Louisa would have been more likely to have made mistakes setting up a dynamic parameter query (1) - There is an increased chance of entering the wrong number as the pass mark, as she must enter it every time (1)	5

Question	Answer	Marks
8	Eight from: - Files created in one software package are sometimes difficult to use/open in other software (1st) - particularly if the packages are incompatible (1) - Generic file formats are compatible with different software packages (1) - When transferring a file to a different computer/device it might not load properly (1st) - since generic file formats are needed to enable different operating systems to read the file/because the particular file format will not operate on that OS (1) - Two marks plus extensions for two examples of the use of generic file types (for a total of 4 marks). e.g. - Word processing files should be saved as a .txt file type (1st) - so that database software can understand it (1) - Database files should be saved/exported as a .csv/.txt/.rtf file type (1st) - so that word processing software can understand it (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
2	Four from: File size is smaller (1) - Takes up less storage (space) (1) - Text format doesn't need to buy a licence/software (1) - As the file size is smaller the processing time is shorter/less processing power is required (1) - Does not need to be opened by a WP package (1) - Can be opened by more applications (1) - Data can be exchanged between different computers/operating systems (1) - Word processing software has different versions and one version may not open documents produced by a different version (1)	4

Question	Answer	Marks
6	Four from: - Simple query is on one field (1) - Complex query is on more than one field (1) - Complex query can use Boolean operators (1) - One mark for TWO examples of Boolean operators: AND, OR, NOT (1) - Simple query use a single criterion only (1) - Complex query uses more than one criterion (1)	4

Question	Answer	Marks
2	Eight available. Max two for each matched pair: Table name/title (1st) ...a unique/meaningful name for each table (in the database)//to give context (1) Field name/title (1st) ...a unique/meaningful name to identify each field//to give context (1) (Candidate may describe a meaningful name by example – e.g. ‘full name to hold first and surname’) Data type (1st) ...Such as (e.g.) integer/alphanumeric Field length/size (1st) ...specify the number of characters (in each field) (1) Field default value (1st) ...the value automatically entered when a new record is created (1) - Description of each field (1st) ...explanation of the content (of each field) (1) - Required/null field (1st) ...whether a field can be left blank (1) - Validation (1st) ...test to be applied to data that is entered (1) ...to ensure data entered is sensible/reasonable (candidates may describe this instead of expressly stating it) (1) - Table security (1st) ...who has access to edit (accept examples, such as write or delete) the table - Primary Keys/Foreign keys (1st) ...to identify unique records (1) ...to link tables (1) - Relationships (1st) ...types of relationship between tables are stated (1)	8

Question	Answer	Marks
11	Three from: • It is a format for storing (digital) data • Defined by a publicly published specification • Specification maintained by a standards organisation • Can be used and implemented by anyone • An open-source file format can be used by both proprietary AND free/open-source software • Also called free file formats if they are not covered by any copyrights/patents • They can be used at no monetary cost (for any desired purpose). • Example of open-source format (CSV/TXT/JPG)	3

Question	Answer	Marks
4	Eight from: • Allows records to be accessed in the order they were entered (1) • Allows records to be accessed randomly (1) • Each record is given index (1) • A database may have several indexes (1st) – based on the information required (1) • Each record is given index (1) • It is a method of allowing fast retrieval (1) • Data is organised into records which consist of fixed length fields (1) • Uses a set of hash tables (1st) – contain "pointers" into the records (1) • Individual records can be retrieved without having to search the entire file (1) • Can be searched quickly (1) • Searches use an index (1st) – which will narrow down the records/data to be searched (1) – that section of the file is then searched sequentially to find the record required. (1)	8

Question	Answer	Marks
9	Max six if concentrates on static or dynamic only Positive Max 6 from: - Static parameter query is fixed/doesn't change/always uses the same criteria (1st) - Which is set when the query is (first) written (1) - saves time when running the query (1) - less likely to have an error because no user input required (1) - A dynamic parameter query can be used to search for different values (1st) - each time it is run (1) - which makes it more flexible (search for other cities) (1) - uses a dialogue box for the user to enter the data required (1) - For dynamic query saves time in designing the query as you don't need to create similar queries (1) - Static queries are more user friendly (1st) - Because it's attached to a button/can click on a button to make it run/don't have to enter search criteria (1) Negative Max 6 from: - In a static query every time that the query is run it will search for the same data (1st) - cannot be changed/search for another city (1) - can end up having a lot of different static queries (1) - making it difficult to find the query you want (1) - If different data is to be searched for the user would need to open up the query in design view (1st) - and change the data in the criteria to that required (1) - Dynamic parameter query requires less technical knowledge of the user. (1) - Static queries require increased storage space/file size (allow converse, but award once only) (1) Two from: - Statement of when static parameter queries are the better option (1) - Statement of when dynamic parameter queries are the better option (1)	8

Question	Answer	Marks
1(a)	Two from: - It is a primary key in/from another table (1st) - A field in CUSTOMERS table (1) - OR - a primary key that is referenced/used in another table (1st) - A field in CUSTOMERS table (1) - Is used to link/create relationships between tables (1) (may be awarded where candidate has already identified that different tables exist)	2
1(b)(i)	One mark available: - Referential integrity forces table relationships to be consistent (1) Enforces table relationships (1) - Ensures that both sides of the relationship exist (1) - Ensures that any foreign key references an existing primary key (1)	1
1(b)(ii)	Three from: - It avoids/eliminates redundant/unnecessary data (1) - It avoids orphan data (1) (Ensures that) every foreign key value has a matching value in the corresponding primary key// Referential integrity can be used to ensure foreign keys are valid (1) - Referential integrity prevents//warns of the deletion of related records (1) - Value/data within the related fields cannot be changed (1)	3
1(c)	Five from: - A relational database consists of more than one table (1st) - Each table holds data/information about one area/that is related (1) - Tables are linked to each other//relationships are established between tables...(1st) - using primary key (fields) (1) foreign key (fields) (1) compound key (fields) (1) - Using one to one//one to many relationships (1) - Can create queries using more than one table (1) - Can create reports using more than one table (1)	5

Question	Answer	Marks
1(a)	Cust_ID (1)	1
1(b)	Order_code (1)	1
1(c)	One from: - Cust_ID (1) - Product_ID (1)	1
1(d)	One to many//one CUSTOMER to many ORDER(s) (1)	1

Question	Answer	Marks
10	Eight from: - CSV means comma separated value (1) Benefits MAX SIX: e.g. - csv is human readable (1) - can be edited in a text editor (1) - csv is processed by multiple//almost all applications (1) - csv is small in file size//takes up little storage space (1) Drawbacks MAX SIX: e.g. - csv includes data//values only (1) - Does not include formatting (1) - There is no distinction between text and numeric values (1) - No standard way to represent binary data (1) - Problems with importing csv into SQL (no distinction between NULL and quotes) (1) - Poor support of special characters (1) - No standard way to represent control characters (1) - Lack of universal standard. (1)	8

Question	Answer	Marks
3	SIX from: • MIS is used in decision making (1) • MIS is used to coordinate/control data in a business. (1) • MIS software can provide past, present and predictive (TWO FROM) information (1) • MIS software can be used with project management (1) • MIS system may be based on a database (1st) • which can be queried (1) • Can be used to gather/analyse/assess information (1st) • e.g. personnel/sales/inventory/production/employee performance (1) • MIS software could be used to create reports. (1) • By comparing reports to previous reports, trends can be identified (1) • Can produce graphs/charts (1)	6

Question	Answer	Marks
4(a)(i)	ONE mark: • Each record in one table relates to only one record in another table (1).	1
4(a)(ii)	ONE from: • Each record in one table can relate to many records in another table (1)	1
4(b)	THREE from: • Need to create a link table (1) • Is created using two one-to-many relationships (1) • Each primary key must exist as a foreign key (1st) • in the link table (1) (Only award once) • The primary key(s) link(s) to the foreign key(s) (1st) • in the link table (1) (only award once) • Not possible to create a single many-to-many relationship (1) • Only a theoretical type of relationship//cannot exist (1).	3

Question	Answer	Marks
1(a)	THREE from: - Structure defines levels of access to records/data (1) - Uses <u>one-to-many</u> relationships (1) - Uses a structure (1) - Links are based on dependencies (1) - Links a number of records to one owner/parent (1) - The system is inflexible (1) - It is limited by using only one type of relationship (1st) so is confined to some very specific uses. (1)	3
1(b)	THREE from: - Records would be stored in a fixed/structured format (1st) that allows different levels of access (1) Up to TWO marks for examples (must be specific examples from the example) e.g. - The managing director can see/access all records/departments (1) - One director can ONLY see/access Department A and Department B (1) Up to THREE marks: - The managing director can only be a parent record (1) - A worker can be a child record (1) - A director/department can be both a parent and a child record (1)	3

Question	Answer	Marks																		
8(a)	<table> <tr> <th>Field Name</th><th>Data Type</th><th>Example of Data</th></tr> <tr> <td>Number_in_Stock</td><td>Integer (1)</td><td>199</td></tr> <tr> <td>Location</td><td>String/alphanumeric/text (1)</td><td>07.44.03</td></tr> <tr> <td>Reorder_Level</td><td>Integer (1)</td><td>250</td></tr> <tr> <td>Price</td><td>Currency (1)</td><td>\$203.50</td></tr> <tr> <td>Supplier_Telephone_No.</td><td>String/alphanumeric/text(1)</td><td>07700900819</td></tr> </table>	Field Name	Data Type	Example of Data	Number_in_Stock	Integer (1)	199	Location	String/alphanumeric/text (1)	07.44.03	Reorder_Level	Integer (1)	250	Price	Currency (1)	\$203.50	Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819	5
Field Name	Data Type	Example of Data																		
Number_in_Stock	Integer (1)	199																		
Location	String/alphanumeric/text (1)	07.44.03																		
Reorder_Level	Integer (1)	250																		
Price	Currency (1)	\$203.50																		
Supplier_Telephone_No.	String/alphanumeric/text(1)	07700900819																		

Question	Answer	Marks
8(b)	SIX from: Three matched pairs: - Number_in_Stock – Type check (1st) must be numeric (1) - Number_in_Stock – Range check (1st) between 0 and 250 (1) - Price – Limit check (1st) must be more than zero (1) - Price – Type check (1st) must be numeric/currency (1) - Location – Format check (1st) Must be three pairs of digits separated by a full stop/NN.NN.NN (1) - Location – Length check (1st) must be 8 characters (1) - Reorder_Level – Type check (1st) must be numeric (1) - Supplier_Telephone_No – Length check (1st) must be 11 characters/digits (1).	6
8(c)	FIVE from: - Any awareness of normalising data (1) - Import the (two) tables into the relational database structure (1st) as separate tables (1) - Create/use a linking table (1) - Create/use/it needs Primary Keys for both/all tables (1) - Create/use/it needs Foreign Keys. (1st) These are needed to act as the link field between the tables (only award if BOTH of the previous two answers has been given) (1) - In the linking table (1) Link the Primary Key(s) to the Foreign Key(s) (1) - Any awareness of one-to-many relationships (1)	5

Question	Answer	Marks
2	SIX from, e.g.: - MIS is a computer-based system that provides managers with the tools to organise/evaluate/efficiently manage departments within an organisation (1) - It provides past, present and prediction information (answer must include TWO from) (1) - It is software based (1) - It includes a database/databases (1) - It includes a number of decision support systems (1) - It can be used to produce reports (1) - Can produce graphs/charts (1)	6

Question	Answer	Marks
3(a)	TWO from: - A database is in 1NF (accept list of 1NF principles) (1) - All the non-key attributes depend entirely on the primary key/it has no partial dependencies. (1)	2
3(b)	EIGHT from, e.g.: Benefits MAX 6 - Atomised data is easier to search on specific criteria (1) - Any change to one record which is needed can instantly be made to any related records//data editing is straightforward (1) - The database does not have redundant data (1) - File size is small/less storage needed (1) - There is no data duplication (1st) - so there are fewer errors in the data (1) - so there is less chance of storing incorrect copies of the data (1) Drawbacks MAX 6 - It can take longer to set up the database (1) - It is a more complex database (1st) - due to the relationships between tables (1) - Making data atomic may not always be the best solution (1ST) - such as (E.G.) date of birth can be separated into day, month and year but this may serve no purpose - Has more/multiple tables (1) - therefore setting up queries/reports/input forms can become more difficult (1) - making it difficult to keep track of data (1) - May require greater expertise (which may need to be bought in) (1).	8

Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. • Are used to summarise the contents of a table//summarise field values (1) • Also called Group-By queries/aggregate queries (1) • use aggregate functions (1) • They are used to extract aggregate of data items for a group of records • Uses functions such as//calculates (TWO of) SUM, AV(ERA)G(E), MIN, MAX, COUNT	2
10(b)	One mark per bullet point to a maximum of two marks. • Searches using more than one parameter value (1) • Searches on more than one field (1) • It often uses Boolean operands (1) • It is made up of (TWO of) AND, OR or NOT operands or a mixture of these (1)	2
10(c)	One mark per bullet point to a maximum of two marks. • A nested query is a query within another query (1) • It can be often referred to as a subquery (1) • Use the result of one query (1st) – as an input parameter of another (1) • Innermost subquery is executed first, then next level, until the main query is reached. (1)	2

Question	Answer	Marks
7	One mark per bullet point to a maximum of five marks. • Indexed sequential access is a mixture of sequential access and direct/random access (1) • A set of hash tables known as indexes contain pointers into the records/index value is generated (1) • The index is a pointer to whereabouts on the disk the record is stored (1) • Individual records can be retrieved without having to search the entire file (1) • Data is organised into records which consist of fixed length fields (1) • Records can be found randomly using an index (1) • Indexes can be searched quickly, thereby allowing the database to access only the records it needs (1) • Searches use an index which will narrow down the records/data to be searched//focuses on a subset of records (1st) – Then that section of the file is searched sequentially to find the record required. (1)	5

Question	Answer	Marks
9	Max 6 from three matched pairs • Different software on Eric's computer than his friends' computers (1st) – E.g. use generic file type (1) • Saved in a different/wrong (file) format (1st) – E.g. use generic file format (1) • Anti-Virus does not allow file to open/File has a virus (1st) – E.g. Eric needs to virus check the file before sending it (1) • File is password protected/encrypted and Eric didn't share the password/decryption key (1st) – E.g. Eric needs to share the password (1). • Wrong access rights/permissions assigned to the file (1st) – E.g. grant access rights/permissions (1)	6

Question	Answer	Marks
12(a)	• A composite key is a primary key that uses the contents of two or more fields from a table to create a unique value. (1)	1
12(b)	• A compound key is a primary key that combines more than one foreign key to create a unique value. (1)	1

Question	Answer	Marks
7(a)	Two from: • A query that prompts the user for an input (1) • Allows the parameter/criteria to be changed each time (1) • Allows to search based on specific parameters/criteria the user needs at any given time (1)	2
7(b)	Three from: • Efficient way to search a database (1) • Reduces the number of queries saved//number of queries through which to search (1) • Reduced file size (1) • Requires little technical knowledge by end user (1) • Allows the end user flexibility/control (1) • Saving time in not creating a new query every time (1) • It provides a user-friendly way to interact with the database (1)	3

Question	Answer	Marks
15	Four from: • It is used in database design (1) • Describes the relationship between two tables (1) • It ensures that all data in a database remains consistent (1) • It ensures that it is all up to date (1) • It ensures that all foreign keys are dependent on the matching primary key (1) • Gives user a warning if they try to delete parent data (1) • Helps to reduce redundant data (1)	4

Question	Answer	Marks
7(a)	Four from: • Decide on what you want to learn/know (1) • Identify/prepare questions (1) • Decide on interviewer (1) • Identify population/decide which group of staff to interview (1) • Choose suitable date/time (1) • Choose suitable location (any indication of choice OR criteria) (1) • Select/invite attendees/employees (1) • Choose suitable timetable/length (1) • Prepare suitable recording methodology (1)	4
7(b)	Four from: Analysis • You can create complex reports (1st) – allows to combine data from multiple tables (1) • You can create complex queries (1st) – ... (1) • You can create dynamic queries (1st) – ... (1) Organisation • Quicker to search (1st) – Because data is held in tables (1) Consistency • Data is not repeated/reduced redundant data/only entered once (1st) – Quicker/easier to modify the data (1) – File size can be smaller/frees up (NOT “creates”) storage space (1) – Reduced chance for errors (1)	4

Question	Answer	Marks								
10(a)	THREE marks available: <table><tr><th>Data title</th><th>Most appropriate data type</th></tr><tr><td>employee_name</td><td>Text/alphanumeric (1)</td></tr><tr><td>hourly_rate_of_pay</td><td>Currency/numeric (1)</td></tr><tr><td>was_employee_born_in_france?</td><td>Boolean (1)</td></tr></table>	Data title	Most appropriate data type	employee_name	Text/alphanumeric (1)	hourly_rate_of_pay	Currency/numeric (1)	was_employee_born_in_france?	Boolean (1)	3
Data title	Most appropriate data type									
employee_name	Text/alphanumeric (1)									
hourly_rate_of_pay	Currency/numeric (1)									
was_employee_born_in_france?	Boolean (1)									
10(b)(i)	TWO marks available: First mark for identifying a suitable validation rule/test (1st) Second mark for a description of that test (1) • Range (1st) – to check value is BETWEEN two figures (1) • Type – to check data is in numeric/currency • Format check – to check entered as (e.g.) \$3.14//description of • Lookup check – to check against a list of values • Limit check – to check that is not below//above a certain figure • Character check – to check that is a decimal/real	2								

Question	Answer	Marks
10(b)(ii)	TWO marks available: First mark for identifying a suitable validation test (1st) Second mark for a description of that test (1) • Range – to check value is BETWEEN two figures • Type – to check data is in numeric/date • Format check – to check entered as (e.g.) dd/mm/yyyy • Limit check – to check that is not below/above a certain figure • Character check – to check that is an integer	2
10(c)	THREE marks available: • Age is based on the calculation between date of birth and today (1) • Age constantly increases with time//age changes every year (1) • A calculated field allows for a changing/dynamic figure (1) • The result shown//age is constantly accurate (1st) Reduced chance of human error resulting in the wrong number being stored (1)	3

Question	Answer	Marks	Guidance
5(a)	FOUR from: STRUCTURE - In a flat-file database, there will be one table// in a relational database there will be more than one table (1) ACCURACY (accept converse for each) - A relational database is less likely to have data entry errors (1) (For the same data,) a relational database will be smaller//save storage space (1) THE DATA - A relational database will have less duplicated data/redundant data (1) OR - a flat-file database can have the same data repeated (many times)//have lots of redundancy (1) - Each record in a relational database only has to be entered once//in a flat-file database, the same data must be entered every time it is needed (1) - Any mention of data integrity being applied to a relational database (1) TIME (accept converse for each) - Relational databases take less time to maintain/edit the data (1) - Relational database takes longer to edit the structure (1) - Querying a relational database is faster than querying a flat-file databases (1) COMPLEXITY (do not accept converse) - Relational database is more complex (1) CAN ALSO AWARD IF STATED SEPARATELY - Relational database can create complex queries (1) - Relational database can create complex reports (1)	4	Do Not accept - Less time/more time to plan/make - Easier to create/update - Any reference to files rather than tables

Question	Answer	Marks	Guidance
5(b)	Three marks available: Structure: • Identifies the validation type (1st) • Describes a rule (1) • States why it is important//why is it used//what error does it stop (1) e.g. • Presence check (1st) Each field has to contain an item of data//cannot be left empty (1) To check that data has been entered into the field//because the data is necessary (1)	3	

Question	Answer	Marks	Guidance
9(a)	ONE mark available: • A database that stores data in more than one linked table (1) • It has linked tables (1)	1	Do Not award • Entities • Multiple databases • Multiple locations • Answers such as ‘two tables with relationships’ Use of relationship here – mark as REP
9(b)(i)	TWO from: • Contains unique data (1) • Cannot contain a null value//must contain a value (1) • Can be based on one cell OR composite/based on two cells (1) (whole answer required) • Only one per table (1)	2	
9(b)(ii)	ONE from: • It enables links to be formed between tables (1) • It uniquely identifies a record (1) • It may be used in a different table as a foreign key (1) • It maintains accuracy/consistency of data (1) • It allows for quick access to individual records (1)	1	

Question	Answer	Marks	Guidance
9(c)	ONE mark available: • A compound key is two or more pieces of data//a primary key (for an individual record) is a single piece of data (1) • A compound key is based on more than one column//a primary key is based on one column (1) • A compound key gives uniqueness from two columns of data (or more)//a primary key gives uniqueness from one column of data (1)	1	
9(d)	FOUR from: • All data must be atomic (1) • There must be a primary key (1) • There must be no duplicated rows/columns (1) • There must be only one value for each row of each column (1)	4	
9(e)	FOUR from: • Any change to one record which is needed can instantly be made to any • related records/improved data consistency (1) • Provides referential integrity (1) • The database does not have redundant data (1) • Makes the file size small/less storage needed (1) • There is no/less data duplication/repeating data (1st) so there are fewer errors in the data (1) so there is less chance of storing incorrect copies of the data (1) • There is less data to modify (1) • Change (to data) can be made more quickly (1)	4	Do Not Award • No repeat fields

11 Video and audio editing

Questions

- 6** Back-up software is a type of utility software. A network manager wants to back up the data stored on a server.

(a) Describe, with reasons, the steps the manager takes when setting up this process.

[5]

- (b)** Some of the data to be backed up consist of audio files. The network manager will compress these before starting the back-up process. Different methods of compression can be used.

Describe these methods giving the effects on the audio files of using each method.

[4]

- 10** The quality of sound produced from the data stored on an optical disk depends on the sampling resolution.

Describe what is meant by sampling resolution.

[5]

10 Explain why the following techniques could be needed when making videos.

(a) trimming a video clip

.....

.....

.....

.....

.....

..... [2]

(b) adding subtitles to a video

.....

.....

.....

.....

.....

..... [2]

(c) adding sound to a video

.....

.....

.....

.....

.....

..... [2]

7 Sound editing software has a number of features.

Describe what is meant by:

(a) fade in and fade out

.....

.....

.....

.....

.....

..... [2]

(b) lossless compression

.....

.....

.....

.....

.....

..... [2]

(c) lossy compression.

.....

.....

.....

.....

.....

..... [2]

10 Sound editing software can be used to alter the sampling rate in audio files.

(a) Explain what is meant by the term sampling rate.

.....

.....

.....

.....

.....

..... [2]

(b) Explain how the sampling rate can affect the audio files.

.....

.....

.....

.....

.....

..... [2]

8 Sound editing software has a number of features that can be used to edit audio files.

(a) State **two** reasons why an audio clip may need to be trimmed.

Reason 1

.....

.....

Reason 2

.....

.....

[2]

(b) State **one** reason why separate audio clips may need to be spliced together.

.....

.....

.....

.....

.....

.....

[1]

(c) Describe the term equalisation when applied to audio clips.

.....

.....

.....

.....

.....

.....

[3]

8 Explain why the following techniques could be used when editing a video.

(a) Creating credits

.....

.....

.....

.....

.....

..... [2]

(b) Extracting a still image

.....

.....

.....

.....

.....

..... [2]

(c) Exporting a video clip in a different file format

.....

.....

.....

.....

.....

..... [2]

11 Describe the effects of the different methods of compression on audio quality.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 Audio editing software has many features. These can be used when editing an audio clip.

Describe the following terms when applied to audio clips:

(a) normalisation

.....

.....

.....

.....

.....

..... [2]

(b) noise reduction.

.....

.....

.....

.....

.....

..... [2]

6 (a) Give reasons why a person editing a video would use fading effects.

.....

.....

.....

.....

.....

..... [2]

(b) Give reasons why a person editing a video would use captions.

.....

.....

.....

.....

.....

..... [2]

8 Sound files can be edited.

Describe the following features. For each feature, give a reason why it is used.

(a) overdubbing

.....

.....

.....

.....

.....

..... [2]

(b) equalisation

.....

.....

.....

.....

.....

..... [2]

(c) reverberation

.....

.....

.....

.....

.....

..... [2]

8 There are two methods of file compression, one is Lossy the other is Lossless.

(a) Describe what is meant by Lossy compression.

..... [1]

(b) Describe what is meant by Lossless compression.

..... [1]

(c) Compare the impact on quality of video of using each of these methods of compression.

..... [4]

5 A video has been created with an aspect ratio of 16:9.

(a) Describe what is meant by an aspect ratio of 16:9.

.....

.....

..... [1]

(b) Explain **two** reasons why an aspect ratio of 16:9 rather than an aspect ratio of 4:3 has been chosen.

.....

.....

.....

.....

.....

..... [2]

9626/11 May/June 2026 · Question 6

6 A student has to record an audio clip of him delivering a speech and wants to save this file. He can use MP3 or MP4a file formats.

Compare the use of MP3 and MP4a file formats to export this audio clip.

.....

.....

.....

.....

.....

..... [2]

11 Video and audio editing

Mark schemes

Mark scheme · 9626/11 May/June 2023 · Question 6

[Back to question](#)

Question	Answer	Marks
6(a)	Five from: • Manager checks the space on the target drive/location (1st) – to make sure there is sufficient space for the data (that will be moved) (1) • Manager selects the time they want the back-up to take place (1st) – manager can select to carry out the back up immediately... (1) – BECAUSE the back-up takes place at a time convenient to the manager/when the system is quiet (may explain why timing is convenient) (1) • Manager selects type of back-up (1st) – backing up only the specified data/files (1) – BECAUSE may want to back up whole device OR new files (1) • Manager selects how regularly the back-up is to take place (1st) – BECAUSE to minimise data loss (1) • Manager will decide where back-up is to be stored (1st) – BECAUSE to make sure back-up will be secure (1) • Manager selects whether to verify the back-up or not (1st) – BECAUSE to make sure the backup has been completed correctly (1) • Manager chooses whether or not to encrypt the back-up (1st) – BECAUSE adds an extra level of security (1) Must have at least TWO reasons to obtain full marks	5

Question	Answer	Marks
6(b)	MAX three from: • Can be either lossy or lossless. (1) • Lossless compression reduces number of bits (1st) – ...by identifying repeated patterns of data. (1) • No information is lost in <u>lossless</u> compression. (1) • <u>Lossy compression</u> reduces number of bits (1st) – ...by identifying unnecessary information. (1) Effects max three: • Quality of audio is the same when using <u>lossless compression</u> (1) • Quality of audio is lower when using <u>lossy compression</u> (1) • File size is smaller for lossy than lossless (accept converse) (1)	4

Question	Answer	Marks
10	Five from: • Sampling resolution is the number of bits (1st) – per sound sample (1) • The higher the sampling resolution (1st) – the more accurately the (final) outcome (of the process) will represent the sound made (accept converse argument for whole answer) (1) • The higher the sampling resolution/the greater the resolution the greater the size of the file (accept converse) (1) • Modern day digital audio is normally found in 16 bit/24 bit resolutions • 8-bit resolution can take one of 256 different values (1st) – which is not generally considered enough resolution to accurately represent music audio (1) • 16-bit resolution can have one of 65 536 different values (1) • 24-bit stores 16 777 216 different values (1) • the difference between 24 and 32- bit cannot be discerned by human hearing (1) • These values are all per channel (1)	5

Question	Answer	Marks
10(a)	Two marks for an explained reason: e.g. • To remove unwanted/unnecessary parts at start/end (1) to make it less boring (1) • To shorten the length (1) because it is too long//to reduce file size (1) • To bring video and audio into synch (1) so that video looks more natural (1) • To make it more exciting (1) by removing boring introduction (1)	2

Question	Answer	Marks
10(b)	Two marks for an explained reason: e.g. • What has been said is shown on the screen//can be read (1) so that nothing is missed (1) • Viewers of a video may have hearing difficulties (1) so subtitles help them understand what is being said (1) • There may be a lot of background noise where the video is being watched (1) so that nothing is missed (1) • Video may be being viewed in a quiet area (1) where having the audio on would bother other people (1) • People watching the video might speak a different language//accent//be muttering (1) so subtitles in other languages can be provided (accept subtitles could be presented in the same language, so words can be read) (1)	2
10(c)	Two marks for an explained reason: e.g. • It can invoke an emotional response from the viewer (1) to increase engagement/enjoyment (1) • To grab audience attention (1) by making the video more interesting (1) • A narrator can provide further content (1) to drive home the emotion/mood//can clarify a video (1) • To add sound effect (1) can heighten the impact of (e.g.) explosions/speeding cars (1) to add authenticity/context/reality (1) • To add a clear soundtrack (1) without background noise (1)	2

Question	Answer	Marks
7(a)	Two from: - Fade in occurs at the beginning of a sound clip when the sound gradually gets louder (1) - Fade out occurs towards/at the end of a sound clip when the sound gradually gets softer (1) - Volume stops increasing when the required sound level is reached (1) - Sound disappears altogether (1)	2
7(b)	Two from: - Lossless compression retains the quality of the original/does not reduce the quality of the audio file (1) - Data is not lost (1) - When lossless compression software sees a repeated sequence of bits it replaces the repeated sequences with a character (1st) - which indicates what is being repeated and how many times (1)	2
7(c)	Two from: - Removes some of the audio file's data (1) - Reduces the quality of the audio file (1) - Removes sounds at very low and high frequencies (which are outside the range of human hearing) (1) - Creates smaller file sizes than lossless compression (1) - Once this data has been removed from the files it can never be replaced (1)	2

10(a)	Two available: - The number of samples (1st) - within a given period of time (accept ANY time unit) (1) - usually within 1 second (1).	2
10(b)	Max two for one complete explanation: - The sampling rate can affect the quality of audio files/how the audio sounds (1st) - because it captures sounds at discrete points (and not the whole sound) (1) - A higher sampling rate sounds better than a lower rate (1st) - because a higher sampling rate more accurately represents the original sound (1) - The lower the sample rate, the smaller the file size (1st) - because there is less data stored (1)	2

Question	Answer	Marks
8(a)	Two from: - To remove unwanted parts of the clip//there are unwanted clips//parts (accept example) (1) - To reduce the length/time of the track//clip is too long (1) - To reduce file size//file size is too big//storage space required (1) - To make same length as an accompanying video clip//other example of fitting to an event (1)	2
8(b)	One from: - To make two (or more) audio clips flow into each other (1) - To combine the best parts of one clip with the best parts of another clip/combine the best parts of several recordings made from the same concert/act (1) - To reorder audio clips (1) - To give continuity/continuation//no gaps between them//sound files (1) - To create a single track/clip (1) - To make the file longer//a suitable length (1)	1
8(c)	Three from: - To boost/adjust the frequencies (in a sound signal) (1st) - By raising (1) or lowering (1) - Can use filters (1)	3

Question	Answer	Marks
8(a)	Two marks available for a full explanation: To identify the people involved in the creation of the video (1) so they can get recognition//acknowledge their role in creating it (1)	2
8(b)	Two from: A frame from the video to be used as a photograph/thumbnaill (1) used for (e.g.) marketing/publicity//to check for continuity//to be used as an icon (1)	2
8(c)	Two from: - Not all devices/platforms are able to play videos in all formats/the original format (1) - which means some videos are incompatible with specific devices/platforms - so this makes the video able to play/be accessed/used on a wider range of devices/platforms (1) - Videos may need to be converted to reduce their file size (1st) - to allow them to be attached to an email (1) - reduce storage space (1) - upload to video streaming website/social media (1)	2

Question	Answer	Marks
11	FOUR from: - Identifies Lossless and identifies Lossy compression (1) Max 3 for Lossless: - Does not reduce the quality of the audio file (1) - Audio file can be recreated exactly as it was when first created (1) - Algorithms used to perform this compression mainly look for repeating patterns (1) Max 3 for Lossy: - Reduces quality of the audio file (1) - Creates smaller file sizes than lossless compression (1) - The data that is removed can never be replaced (1) - Some lossy compression software will remove frequencies that are outside the range of human hearing. (1)	4

Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. - Sets the peak amplitude/loudest part of a track to a specific level. (1) - Makes multiple tracks have the same gain/peak/relative volume. (1) - Helps increase the volume of an entire audio track proportionally. (1) - Helps achieve consistent volume levels across different audio files. (1) - Helps maximising overall volume levels without compromising sound quality (1)	2
10(b)	One mark per bullet point to a maximum of two marks. - Eliminates/removes unwanted sounds. (1) - Makes audio track/signal clearer. (1) - Plays a mirror image of the original audio track (1st) - to partially cancel noise out. (1)	2

Question	Answer	Marks
6(a)	One mark per bullet point to a maximum of two marks. • To indicate time has passed between two scenes in a video (1) • To ease the emotions of the audience after a troubling scene (1) • To create suspense after/before a pivotal/important moment (1) • So that the audience are aware of the end/beginning. (1) • To provide a smooth transition between scenes (1)	2
6(b)	One mark per bullet point to a maximum of two marks. • To provide context • To provide information//better understanding (1) • Identifies author (1) • Identifies sound track (1) • Captions can help viewers engage with your video (1) • Search engines use metadata like captions to understand a video's content (1) • To provide translations (1) • To allow watchers to read what is being said//reason why what is being said may be inaccessible (1) Note This may be combined with “better understanding” to achieve two marks.	2

Question	Answer	Marks
8(a)	Max one for: - It means adding tracks/audio clips (1) Max one mark for any valid reason: e.g. - Add complexity to the music/audio (1) - Adds voiceover (1) - Add extra instruments (1) - Add sound effects to match what is happening on the screen (1)	2
8(b)	Max one for: - It means boosting/reducing/changing the levels of different frequency bands (1) Max one for any valid reason: e.g. - To pick out speech from background sounds (1) - To reduce the hiss (1) - To reduce/increase identified/intrusive noise frequencies (1) - To emphasise/highlight a guitar solo in a music track (1)	2
8(c)	Max one for: - Is an effect that adds echo to a sound (may describe sound being bounced off (e.g.) walls) (1) Max one for any valid reason: e.g. - To simulate being in a different environment (e.g. cavern/concert hall) (1) - To give depth/richness to a sound (1) - To create a dramatic effect in a radio/audio play/podcast (1)	2

Question	Answer	Marks
8(a)	One from: - Compressed data cannot be reconstructed (1) - It permanently removes data (1)	1
8(b)	One from: - Compressed data can be reconstructed (1) - Does not permanently remove data (1)	1

Question	Answer	Marks
8(c)	Four available. One per instance of a valid comparison: e.g. • Lossless will not affect the quality of video/lossy will affect the quality (1) • Lossy reduces the colour range (1) • Lossy can alter the frame rate (make the video jumpy/stutter) (1) • Lossy can desync the audio with video (1) • Lossy can affect the colour settings of the video (1) • Lossy can affect resolution of video (1) • Lossy can affect the bitrate/make incompatible with devices (1)	4

Question	Answer	Marks	Guidance
5(a)	ONE mark available: - It has a width of 16 units and height of 9 units (1) width to height ratio is 16:9/1.78:1 (1)	1	Do Not Award Length not allowed. Unless it is length and height
5(b)	TWO marks available for two complete explanations: - Ensures high-definition quality therefore final product will be of a high standard/clear to see (1) - Offers more content to the viewer so a better-quality product (1) - Is the more modern format, so product looks up-to-date (1) - Is the most widely used format in computer monitors/Television so the video will fit properly (1) 16:9 is the international/industry standard therefore it will be more widely used/meets international broadcast standards (1)	2	

Question	Answer	Marks	Guidance
6	TWO marks available: - Both use compression (1) - MP4a has a higher sound quality (1) - MP3 file sizes are larger (1) - MP4a has a higher bit rate (1) - MP3 is lossy/MP4a is lossless (1) - MP3 files can be used on more platforms than MP4a (may exemplify a hardware type) OR MP3 Files are more widely used OR MP3 files is the standard format to use for sound files, MP4a is less widely used for sound files (1) Allow converse answers where applicable	2	